

CPA

INTERMEDIATE LEVEL

**FINANCIAL MANAGEMENT
REVISION KIT
TOPICALLY ARRANGED**

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**Updated With
December 2023
Past Papers with Answers**

PAPER NO.8 FINANCIAL MANAGEMENT

GENERAL OBJECTIVE

This paper is intended to equip the candidate with knowledge, skills and attitudes that will enable him/her to effectively manage, control and optimize on institutional finances.

8.0 LEARNING OUTCOMES

A candidate who passes this paper should be able to:

- Explain the nature and scope of financial management
- Describe the nature and functions of financial institutions and markets
- Analyse the sources of finance for an organisation and evaluate various financing options
- Evaluate various investment decision scenarios available to an organisation
- Evaluate the performance of a firm using relevant financial tools
- Make appropriate capital structure decisions for a firm, Value financial assets and firms
- Make appropriate liquidity and dividend decisions for a firm
- Evaluate current developments in the field of finance and their impact on financial decisions

CONTENT

8.1 Overview of financial management

- Theoretical framework of financial management - The role and responsibilities of a finance manager towards shareholders, employees, society, government and other stakeholders.
- Goals of a firm and corporate strategy; financial and non-financial objectives, overlaps and conflicts among the objectives
- Agency theory, stakeholder's theory and corporate governance

- Measuring managerial performance, compensation and incentives
- Ethical issues in financial management

8.2 The financing decision

- Nature and objectives of the financing decision
- Factors to consider when making financing decisions
- Sources of finances for enterprises; internally generated funds and the externally generated funds, long term sources (equity, debt, hybrids, lease finance, venture capital, business angel finance, private equity, asset securitisation and sale, Islamic finance and initial coin offerings), medium term sources such as medium term loans and hire purchase financing and short term sources of finance such as overdraft finance, trade credit, issue of commercial papers, accruals, deferred income; characteristics of each source of finance, pros and cons of the various sources of finance.
- Evaluation of financing options
- Methods of issuing ordinary shares - Public issue, private placement, bonus issue, employee stock option plans (ESOPS) and rights issues

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8.3 Financial institutions and markets

- Overview of a financial system - Financial markets, financial institutions and financial instruments
- Nature and role of financial markets - primary and secondary securities market, money and the capital markets, over-the counter and organised market, derivatives market, mortgage market, forex market
- Nairobi securities exchange (NSE, or equivalent entity in other jurisdictions) - The role and functions of the securities exchange, securities exchange terminologies, security exchange listing and cross border listing, share indices, timing of investment at the securities exchange, Central depository system and automated trading system, de-mutualisation
- Stock market indices
- Central depository system and automated trading system

- Timing of investment at the securities exchange - Dow theory and Hatch system of timing
- The financial institutions and intermediaries: commercial banks, savings and loans associations and co-operative societies, foreign exchange bureaus, unit trusts and mutual funds, insurance companies and pension firms, insurance agencies and brokerage firms, investment companies, investment banks and stock brokerage firms, micro-finance institutions and small and medium enterprises (SMEs)
- Regulation of financial markets; Central bank of Kenya (CBK, or equivalent entity in other jurisdictions) - The role of the Central bank and the Monetary policy of the central bank; Capital market authority and the Insurance regulatory authority
- Factors responsible for the rapid development of financial institutions and markets
- Risks facing financial institutions

8.4 Time-value of money

- Concept of time value of money
- Relevance of the concept of time value of money in financial management
- Time value of money versus time preference of money
- Time line
- Real versus nominal cash flows
- Compounding techniques - Compound interest, Future value (FV) of a single cash flow and series of cash flows; Compounding of annuity cash flows
- Discounting techniques - Present value (PV) of a future cash flow and series of future cash flows and discounting of annuity cash flows
- Loan amortisation and sinking funds

8.5 Business/Financial asset Valuation models

- Concept of value; book value, going concern value, substitution value, replacement value, conversion value, liquidation value, intrinsic value and market value
- Reasons for valuing financial assets/business
- Theories on valuation of financial assets; fundamental theory, technical theory,

random walk theory and the efficient market hypothesis

- Valuation of redeemable, irredeemable and convertible debentures/ corporate bonds
- Valuation of redeemable, irredeemable and convertible preference shares
- Valuation of ordinary shares; net asset basis, price-earnings ratio basis, capitalisation of earnings basis, Gordon's model, finite earnings growth model, Super-profit model, Walter's model, discounted free cash flow, residual income model, Use of relative measures such as Economic Value added (EVA) and Market Value Added (MVA)
- Valuation of rights issues
- Valuation of unit trusts and mutual funds
- Valuation of private companies: income and market-based approaches

8.6 Introduction to capital structure decisions

- The meaning of capital structure and the factors to be taken into account when making capital structure decision.
- Cost of capital; Meaning of cost of capital, practical applications of cost of capital, component costs of capital
- The firm's overall cost of capital; Weighted average cost of capital (WACC), Weighted marginal cost of capital (WMCC) and factors influencing the firm's Cost of capital.
- Leverage and Risk; Operating leverage and Operating risk, financial leverage and financial risk and total leverage and total risk.
- Analysis of Operating profit (EBIT), Earning per share (EPS) at the point of indifference in the firm's earnings; Determination of the range of firm's operating profit (EBIT) within which each financing option will be recommended

8.7 Introduction to capital budgeting decisions

- The nature and importance of capital investment decisions
- Capital budgeting process
- Capital investment's cash flows - initial cash outlay, terminal cash flows and

annual net operating cash flows, incremental approach to cash flow estimation

- Capital investment appraisal techniques
- Non-discounted cash flow methods - payback period and accounting rate of return
- Discounted cash flow methods - net-present value, internal rate of return, profitability index, discounted payback period and modified internal rate of return (MIRR)
- Strengths and weaknesses of the investment appraisal techniques
- Expected relations among an investment's NPV, company value and share price
- Capital investment options - timing option, strategic investment option, replacement option and abandonment option
- Problems/difficulties encountered when making capital investment decisions in reality

8.8 Financial statements analysis and forecasting

- Scope of financial statement analysis, major financial statements and other information sources, financial statement analysis frame work and users of financial statements and their information needs
- Financial Statement analysis verses Business analysis
- Techniques of financial Statement analysis; Cross-Sectional analysis, Time series analysis and a combination of both techniques.
- Types of financial statement analysis
- Ratio analysis; nature of financial ratios, classification and calculation of financial ratios and limitation of financial ratios
- Common size statements - Vertical and horizontal analysis
- Financial forecasting; cash budgeting and percentage of sales method of forecasting

8.9 Working capital management

- Introduction and concepts of working capital
- Working capital versus working capital management
- Factors influencing working capital requirements of a firm

- Types of working capital
- Importance and objectives of working capital management
- Working capital operating cycle; the importance and computation of the working capital operating cycle
- Working capital financing policies: Aggressive, conservative and matching financing policy
- Determining the finance mix: basic approaches for determining an appropriate Working Capital finance mix; Hedging or matching approach, conservative approach, aggressive approach.
- Management of inventory: kinds of inventory, objectives of Inventory Management, Techniques of Inventory Management; stock levels, Economic Order Quantity (EOQ) and Just-In-Time (JIT).
- Cash Management; Motives of holding cash, techniques of managing cash, cash management models-Baumol model, Miller-Orr model and Orgler's model
- Management of accounts receivable and accounts payable; assessing creditworthiness, collecting amounts owing, trade credit

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8.10 Dividend decision

- Meaning and significance of dividend policy
- Factors influencing dividend policy
- Forms of dividend: script dividend, cash dividend, stock dividend, bond dividend, property dividend
- Dividend theories; Bird in hand theory, Clientele effect theory, Information signalling theory, Walter's model, Gordon Model, Tax differential theory, Modigliani and Miller dividend irrelevance theory
- Types of dividend policy: regular dividend policy, stable dividend policy, irregular dividend policy, zero dividend policy.
- The impact of dividend payout on share price

8.11 Introduction to risk and return

- Introduction - Risk-return trade off/relationship, Distinction between risk free and

risky assets

- Expected return, Standard deviation of returns and the Relative risk of an individual asset
- Expected return of a 2 asset-portfolio
- The Covariance and correlation coefficient of returns of assets.
- The actual portfolio risk of a 2-asset portfolio using the analytical and mathematical model and its interpretation

8.12 Islamic finance

- Justification for Islamic Finance; history of Islamic finance; capitalism; halal; haram; riba; gharar; usury
- Benefits and deficiencies of Islamic Finance
- Principles underlying Islamic finance: principle of not paying or charging interest, principle of not investing in forbidden items; ethical investing; moral purchases
- The concept of interest (riba) and how returns are made by Islamic financial securities
- Sources of finance in Islamic financing
- Types of Islamic financial products:- sharia-compliant products: Islamic investment funds; takaful the Islamic version of insurance Islamic mortgage, murabahah, Leasing - ijara; safekeeping - Wadiah; sukuk - islamic bonds and securitisation; sovereign - sukuk; Islamic investment funds; Joint venture - Musharaka, Islamic banking, Islamic contracts, Islamic treasury products and hedging products, Islamic equity funds; Islamic derivatives
- 12.7 International standardisation/regulations of Islamic Finance: case for standardisation using religious and prudential guidance, National regulators, Islamic Financial Services Board

8.13 Personal financial management

- Financial Problems encountered in managing individual financial affairs
- Savings and investment considerations
- Personal risk management

- Cost of credit
- Financial Alterations - Retirements; Estate and Tax planning and family budgeting

8.14 Contemporary issues and emerging trends

- Globalisation and growth of derivative markets and securitisation
- Cryptocurrency
- Block chain technology
- Cloud funding
- Digitisation of financial transactions
- Behavioural finance
- Big data project finance

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TOPIC 1

OVERVIEW OF FINANCIAL MANAGEMENT

QUESTION 1

December 2023 Question One A

Explain **FOUR** corporate objectives that may conflict with the financial objective of a firm. (8 marks)

ANSWER

Corporate objectives that may conflict with financial objectives of a company

Several corporate objectives can potentially conflict with a company's financial objectives, creating tension between short-term gains and long-term sustainability. Here are some key examples:

Short-term vs. Long-term Focus:

- Profit maximization: Prioritizing immediate profits might involve cutting costs on quality, research, or employee training, jeopardizing long-term competitiveness and brand reputation.
- Stock price manipulation: Engaging in short-term strategies to boost stock prices, like share buybacks or aggressive marketing, might divert resources from long-term investments and sustainable growth.

Social and Environmental Responsibility:

- Environmental sustainability: Implementing eco-friendly practices or reducing emissions might increase costs in the short term, potentially impacting profits.
- Social responsibility: Investing in ethical sourcing, fair labor practices, or community development programs might raise expenses and reduce immediate financial returns.

Stakeholder Interests:

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- Employee satisfaction: Providing higher wages, benefits, or training can increase costs and reduce short-term profits.
- Customer satisfaction: Prioritizing customer satisfaction through better service, warranties, or refunds might impact profits in the short term.

Strategic Growth:

- Market share expansion: Aggressive marketing campaigns or acquisitions to gain market share can be expensive and might not yield immediate financial returns.
- Innovation and research: Investing in research and development for future products or technologies might not show immediate financial benefits but is crucial for long-term growth and competitiveness.

Ethical Considerations:

- Bribery and corruption: Engaging in unethical practices for short-term gains can damage reputation, lead to legal issues, and ultimately harm financial stability.
- Tax avoidance: Aggressive tax strategies might save money in the short term but could result in penalties, reputational damage, and long-term financial risks.

QUESTION 2

August 2023 Question One A

Outline **FOUR** limitations of profit maximisation as a financial goal of a firm. (4 marks)

ANSWER

Limitations of profit maximisation

The profit maximization goal is the idea that a business should make as much profit as possible. This is a common goal for businesses, but it has some limitations. Here are some of the limitations of profit maximization:

- **Definition of Profit** - The precise meaning of the profit maximization objective is unclear. The definition of the term profit is ambiguous. Does it mean short or long-term, before or after tax, total or net profit? It is not clear.
- **Time Value of Money** - The profit maximization objective does not distinguish between returns received at different times. It does not consider the time value of

money; it values benefits received today and benefits received after a period as the same.

- Profit maximization can also lead to the production of **too many of a particular product that is of no use**. This will create wastage destroying the raw materials and stuffing the landfills. Therefore, producing efficiently is more acceptable than maximizing the profit only.
- Profit maximization has also been termed to be **vague**, and it ignores the risks and returns in a positive manner. The idea of profit maximization is not self-sustaining in nature and it is not built to target a market. That is why it does not consider the risks and returns as commonly as it should.
- **Ignores the Risk** - A decision solely based on the profit maximization model would decide in favor of profits. In the pursuit of profits, the risk involved is ignored, which may prove unaffordable at times simply because higher risks directly question the survival of a business. Between projects A and B, project A may be more profitable; however, if it is substantially more riskier, project B may be preferable.
- **Long-Term Sustainable Goals** - Profit maximization might be one of the top goals of financial management but this type of practice doesn't imply that short-term profit increases will help produce long-term sustainable goals for the company. While profit maximization in financial management has the potential to bring in extra money in the short-term, long-term earning could be drastically diminished.
- **Product Quality**- Another limitation of profit maximization in financial management is the potential to decrease product quality. Earning higher profits might be one of the goals of financial management but cutting corners, using lower quality materials, and sacrificing company values to earn a higher profit will affect the reputation of the company and potentially lose customers.

QUESTION 3

April 2023 Question One A

Management of a limited liability company is appointed to promote and protect shareholders' interest in the performance of their functions. The aim is to maximise

shareholders' value. The management, however, could have interest that might be in conflict with shareholders' interest.

Required:

In reference to the above statement:

- (i) Identify this type of conflict in modern day financial management of a firm. (1 mark)
- (ii) Explain THREE factors that could contribute to the conflict identified in (a) (i) above. (3 marks)
- (iii) As a financial management professional, explain FOUR strategies that could be used to manage or mitigate this conflict to protect shareholders. (4 marks)

ANSWER

(i) Type of conflict in modern day financial management of a firm.

Agency problem

(ii) Factors that could contribute to the conflict identified in (a) (i) above.

- Managements may use corporate resources for personal use
- Managements may take holidays and spend huge sum of company money.
- Creative accounting - involves manipulation of finances
- Empire building – managers may organize for mergers beneficial to themselves and not shareholders.
- Failure to declare dividend for no good reason.

(iii) Strategies that could be used to manage or mitigate conflict to protect shareholders.

1. **Performance based remuneration:** This involves remunerating managements for actions they take that maximizes shareholders wealth. Managements could be given bonuses, commission for superior performance in certain periods.
2. **Incurring agency cost:** Agency costs are those incurred by shareholders in trying to cut management behaviour and action and therefore minimize agency conflicts. They include: Monitoring cost, bonding assurance, opportunity cost, restructuring cost
3. **Direct intervention by shareholders:** These may be done in the following ways;
 - Making recommendations to the management on how the firm should be run
 - Threat of firing
4. **Use of corporate governance principles which specify the manner in which organizations are acted and managed.** The duties and rights of all stakeholders are outlined.
5. **Threat of hostile takeover (sell the business):** This may be arranged by shareholders to lock out managements who aren't responsible

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QUESTION 4

December 2022 Question One A

Ethical responsibilities arise not as a result of legal requirements but as a result of moral imperative for companies to operate in an ethical and fair manner.

In light of the above statement, summarise SIX elements of business ethics in management of companies. (6 marks)

ANSWER

Elements of business ethics in management of companies

Business ethics are the principles and values that guide the behavior of individuals and organizations in the business world. There are several elements of business ethics that are important in the management of companies, including:

1. **Honesty:** This involves being truthful and transparent in all business dealings.

2. **Integrity:** This involves acting with integrity and upholding ethical standards, even when it may be difficult or inconvenient to do so.
3. **Professionalism:** This involves exhibiting a professional demeanor and acting in a way that is respectful and responsible.
4. **Responsibility:** This involves being accountable for one's actions and taking responsibility for their impact on others.
5. **Fairness:** This involves treating all stakeholders, including employees, customers, and suppliers, fairly and justly.
6. **Respect:** This involves treating others with dignity and respect, regardless of their position or status.
7. **Compliance:** This involves following laws, regulations, and other rules that govern business operations.
8. **Corporate social responsibility-** Companies should strive to be good corporate citizens and strive to create positive social impact through their actions.

QUESTION 5

August 2022 Question Two B (i)

Distinguish between the following terms as used in finance:

- (i) “Agency cost” and “agency problem”. (2 marks)

ANSWER

Distinction between “Agency cost” and “agency problem” as used in finance

In finance, the term "agency cost" refers to the cost that a firm incurs as a result of having an agent (e.g., a manager) make decisions on behalf of a principal (e.g., the shareholders). These costs can include the cost of monitoring the agent's behavior, the cost of aligning the interests of the agent with those of the principal, and the cost of compensating the agent for their services.

The "agency problem" refers to the conflict of interest that can arise between the agent and the principal as a result of the different incentives that each party has. For example, a

manager may have an incentive to make decisions that benefit themselves rather than the shareholders, leading to a misalignment of interests. This can result in the agent acting in a way that is not in the best interest of the principal, which can lead to the agency costs mentioned above.

QUESTION 6

August 2022 Question Three B

Discuss three causes of conflict between shareholders and managers in relation to agency theory. (6 marks)

ANSWER

Causes of conflict between shareholders and managers in relation to agency theory

- **Creative accounting** - involves manipulation of finances
- **Empire building** – managers may organize for mergers beneficial to themselves and not shareholders.
- **Information asymmetry**: Managers often have access to more information about the inner workings of the company than shareholders do, which can make it difficult for shareholders to hold managers accountable for their actions.
- **Short-term versus long-term focus**: Shareholders may be more focused on short-term profits, while managers may be more concerned with the long-term stability and growth of the company.
- **Diversification of shareholder interests**: If a company has a large and diverse group of shareholders, it can be difficult to reach consensus on important decisions.
- **Managerial entrenchment**: Some managers may be resistant to change and unwilling to consider the interests of shareholders.

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QUESTION 7

April 2022 Question One A

Describe four ways of encouraging managers to achieve stakeholders' objectives.

(4 marks)

ANSWER

Ways of encouraging managers to achieve stakeholders' objectives.

- **Performance based remuneration:** This involves remunerating managers for actions they take that maximize shareholders wealth. Management could be given bonuses, commission for superior performance in certain periods.
- **Threat of hostile takeover (sell the business):** This may be arranged by shareholders to lock out managements who aren't responsible.
- Use of corporate governance principles which specify the manner in which organizations are acted and managed. The duties and rights of all stakeholders are outlined
- Threat of firing
- Bonding insurance: Insurance taken for managers who engage in harmful practice
- Restructuring cost: Are costs incurred in changing an organization structure so as to prevent undesirable management activities.
- Monitoring costs: Arise as a result of mechanism put in place to ensure interest of shareholders are met. These include; cost of hiring auditors

QUESTION 8

December 2021 Question One A

Describe four possible sources of conflict of interest between shareholders and bondholders. (4 marks)

ANSWER

Possible sources of conflict of interest between shareholders and bondholders.

- Dividend payments to shareholders could be very high
- Default on interest payment.
- Shareholders could organize mergers which aren't beneficial to bondholders.
- Shareholders could acquire additional debt that increases the financial risk of the firm.
- Manipulation of the financial statements so as to mislead creditors.

- Shareholders could dispose assets
- Shareholders could dispose assets which are security for credit given.
- Shareholders could invest in very risky projects.

QUESTION 9

September 2021 Question One B

Highlight four advantages of the wealth maximisation objective of a firm. (4 marks)

ANSWER

Advantages of the wealth maximisation objective of a firm

- Considers risks and uncertainties associated with the project.
- Takes into account the time value of money
- Consistent with the going concern of the firm
- Can be useful in case of joint ventures and limited companies
- The various resources are put to economic and efficient use
- The employees share in the wealth gets increased

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QUESTION 10

November 2019 Question One A

In the context of agency theory:

- (i) Explain the "principal-agent" problem. (2 marks)
- (ii) Explore two ways of addressing the principal-agent problem. (4 marks)

ANSWER

a) i) Explanation of Principal-agent problem

This is one conflict in priorities between a person or group and the representative authorized to act on their behalf. An agent may behave in a way that prejudices principal interest.

ii) Ways of addressing the principal-agent problem

1. Lenders may refuse to lend to the firm or charge higher than normal market interest rates to compensate for increased risk.
2. They could have covenants related to decision making on financing e.g they could require a representative in annual general meetings or board meetings where credit decisions will made.
3. Creditors can protect themselves by adopting restrictive covenants in the debt contract to restrict asset disposals, debt financing and company's dividend and remuneration capacity.
4. Incur agency cost such as :
 - Monitoring expenses such as audit fees, compliance fees.
 - Reorganization costs to structure the organization so that the possibility of undesirable management behavior would be limited.
 - Opportunity cost associated with Loss of profitable opportunities resulting from structure not to permit manager to take action on a timely basis as would be the case if the managers were also owners.
5. Threat of hostile takeover: under performance will lead to a situation whereby shares become undervalued in the stock market. This highlights the risk of a firm being taken over with the result that firm's current management will be replaced. The only cure for this is good performance.
6. Labour market actions whereby only professional managers with a track load of good performance are hired.
7. Shareholders intervention: hereby the shareholders vote out incompetent directors at the annual general meeting.
8. Use of corporate governance principles which prescribe how firms should be managed and controlled.

9. Stock option schemes for managers could be introduced. This forces managers to undertake that enhance shareholders wealth.

QUESTION 11

November 2017 Question Four B

Highlight three agency costs that might arise in the principal-agent relationship between shareholders and managers. (3 marks)

ANSWER

Agency costs that might arise in the principal-agent relationship between shareholders and managers

1. Monitoring expense such as audit fees and compliance fees
2. Reorganization costs to structure the organization so that the possibility of undesirable management behavior could be limited (cost of internal controls)
3. Cost of delaying decision

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QUESTION 12

May 2017 Question Five C

The goal of profit maximisation is considered to be a short-term objective with long-term survival. The firm's growth cannot be achieved without continuous profitability.

Required:

In relation to the above statement, summarise four arguments-in favour of and four arguments against profit maximisation as a business goal. (8 marks)

ANSWER

A. Arguments in favour and against profit maximization as a business goal

Against

1. Ignores risk and uncertainty

2. Ignores other participants
3. Does not account for time value of money
4. Regards profit. Ideally attention should be on cash flows
5. May lead to exploitation of workers and consumers
6. Creates immoral practices such as corrupt practices, unfair trade practices etc
7. It is vague

Arguments for

1. Main aim for business is earning profit
2. Profit is the parameter of the business operation
3. Profit reduces risks of the business concern
4. Profit is the main source of finance
5. Profitability meets the social needs

QUESTION 13

May 2016 Question Five C

- (i) The agency problem could be resolved using goal congruence.

Explain the term "goal congruence". (2 marks)

- (ii) One of the ways creditors could protect themselves against the inherent risk that might arise from agency conflict is through adopting restrictive covenants.

With reference to the above statement, describe three restrictive covenants in a debt contract. (6 marks)

ANSWER

- (i) Explaining the term "goal congruence"**

Goal Congruence refers to the alignment of the principles and agent. Thus agent strives to achieve goal set by the principal

ii) They are also called Negative covenants. They include:

1. **Liability related restrictive covenants:** These covenants prohibit some activities e.g. Borrowing more loans; issue of unsecured loans or reduction in promoters shareholding
2. **Cashflow related restrictive covenants:** These stipulate how the company's cashflow can be used e.g. Top management salaries limit; dividend payment and capital expenditure on new projects, expansion, modernization etc.
3. **Asset related restrictive covenants:** These contracts sets limits on how a company can use existing assets e.g. sale of fixed assets and using existing fixed assets to secure debt

QUESTION 14

September 2015 Pilot Paper Question Two A

In relation to financing of firm's activities, distinguish between the term —capital structure" and financial structure". (3 marks)

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ANSWER

Capital structure and financial structure

Capital structure refers to the long-term sources of finance that have been used by company e.g. long-term debt and ordinary share capital. Financial structure refers to the mix of long-term funds and short-term funds that have been used by the company to finance its operations

TOPIC 2

THE FINANCING DECISION

QUESTION 1

December 2023 Question One Five B

Summarise FOUR distinct features of commercial paper as a short term source of finance. (4 marks)

ANSWER

Features of commercial paper as a short term source of finance

Commercial paper is an unsecured, short-term debt instrument issued by corporations with high credit ratings to raise funds for their short-term needs. Here are some key features of commercial paper as a short-term source of finance:

- **Short-term maturity:** Typically, maturities range from one day to 270 days, with most issues maturing within one to six months. This makes it suitable for financing temporary needs or bridging funding gaps.
- **Unsecured debt:** Unlike loans secured by assets, commercial paper doesn't require collateral. This can be faster and less complex to arrange, but it also increases the risk for investors as they rely solely on the issuer's creditworthiness.
- **High credit quality:** Only companies with strong financial performance and credit ratings can issue commercial paper. This ensures a lower risk for investors compared to other unsecured debt options.
- **Negotiable rates:** Interest rates on commercial paper are usually lower than bank loans but can vary depending on the issuer's creditworthiness, market conditions, and maturity.

- **High liquidity:** Widely traded in the money market, commercial paper offers high liquidity, making it easy to buy and sell before maturity. This provides companies with flexible access to funds and investors with an option for short-term investments with readily available cash.
- **Cost-effective:** Issuing commercial paper can be less expensive than securing bank loans due to lower fees and simpler documentation. This can be advantageous for companies seeking affordable short-term financing.
- **Wide range of issuers:** While primarily used by large corporations, commercial paper can also be issued by municipalities, government agencies, and even some financial institutions.
- **Flexibility in issuance:** Companies can issue commercial paper in various denominations and maturities to match their specific financing needs.

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QUESTION 2

August 2023 Question One B and C

(b) Evaluate **FOUR** features of preference shares as a source of capital to a firm.

(4 marks)

(c) Maca Ltd. is considering raising funds for an expansion programme through a rights issue. The expansion programme is expected to cost Sh.20 million. The company has currently issued 2,400,000 ordinary shares which are currently selling for Sh.30 each. The board of directors have proposed an offer price of Sh.25 per share.

The funds to be raised will be invested in a project which is expected to generate net operating cash flow of Sh.6,000,000 each year over the project's useful life of five years. The salvage value of the project after five years is estimated at Sh.5,000,000. The cost of capital for the firm is 12% per annum.

Required:

- (i) Cum-right market price per share (MPS). (2 marks)
- (ii) Number of rights required to buy one new ordinary share. (1 mark)
- (iii) Theoretical ex-right market price per share. (2 marks)
- (iv) Theoretical value of each right. (2 marks)
- (v) Evaluate the impact of the rights issue on the value of wealth of an existing shareholder who owns 1,200 ordinary shares in the company and Sh.15,000 in his savings account under the various options available to him. (5 marks)

ANSWER**(b) Evaluate FOUR features of preference shares as a source of capital to a firm.**

Preference shares are a type of hybrid security that combines the features of debt and equity. They are issued by companies to raise capital, and offer investors a fixed dividend payment, as well as priority over common shareholders in the event of a liquidation.

Here are some of the features of preference shares as a source of capital to the firm:

- **Fixed dividend payments:** Preference shareholders are guaranteed to receive a fixed dividend payment, regardless of the company's profits. This can be attractive to investors who are looking for a steady income stream.
- **Priority in liquidation:** In the event of a liquidation, preference shareholders have priority over common shareholders in terms of receiving their investment back. This means that preference shareholders are less likely to lose money than common shareholders.
- **No voting rights:** Preference shareholders typically do not have voting rights, unlike common shareholders. This means that they do not have a say in how the company is run.

- **Can be convertible:** Preference shares can be convertible into common shares, which gives investors the option to convert their investment into equity if they so choose.
- **Can be callable:** Preference shares can be callable, which means that the company can buy them back at a predetermined price. This can be a way for the company to reduce its debt burden.

(c) Maca Ltd.

(i) Cum-right market price per share (MPS).

MPS before right issue + NPV per share

$$NPV = PVCIF - PVIOF$$

$$PVCIF = 6000,000 \times PVIFA_{12}^5\%$$

$$6,000,000 \times 3.6048 = 21,628,800$$

Add: PV of terminal benefit

$$5,000,000 \times PVIF_{12}^5\% = 5,000,000 \times 0.5674 = 2,837,000$$

$$\text{Total PVCIF} = 21,628,800 + 2,837,000 = 24,465,800$$

$$NPV = 24,465,800 - 20,000,000 = 4,465,800$$

$$NPV \text{ per share} = \text{Total NPV} \div \text{Number of shares}$$

$$\frac{4,465,800}{2,400,000} = 1.86$$

$$\text{Cum right MPS} = 30 + 1.86$$

$$= \text{Sh. } 31.86$$

(ii) Number of rights required to buy one new ordinary share.

$$\frac{\text{Current number of shares}}{\text{New shares to be issued}} = \frac{2,400,000}{800,000} = 3 \text{ shares}$$

$$\text{New shares to be issued} = \frac{\text{Amount to raise}}{\text{offer price}} = \frac{20,000,000}{25} = 800,000 \text{ shares}$$

(iii) Theoretical ex-right market price per share.

$$\begin{aligned} & \frac{\text{Current equity value} + \text{new funds to be raised}}{\text{Current number of shares} + \text{new shares to be issued}} \\ &= \frac{(2,400,000 \times 3) + 20,000,000}{2,400,000 + 800,000} = \text{Sh. } 28.75 \end{aligned}$$

(iv) Theoretical value of each right.

$$\begin{aligned} V_r &= \text{MPS} - P_{\text{ex}} \\ &= 30 - 28.75 \\ &= 1.25 \end{aligned}$$

(v) Evaluate the impact of the rights issue on the value of wealth of an existing shareholder

$$\text{Current ownership percentage} = \frac{1200}{2,400,000} \times 100\% = 0.05\%$$

Wealth status

$$= 1200 \times 30 = \text{Sh. } 36,000$$

Options available

1. Exercise the right

For 3 share – 1 new share

$$1200 - x$$

$$X = \frac{1200 \times 1}{3} = 400 \text{ shares}$$

$$\% \text{ of ownership} = \frac{1200+400}{2400,000+800,000} \times 100\% = 0.05\%$$

Wealth status	1,600 share × Sh. 28.75 = 46,000
	Less: cost of exercise 400 × 25 (10,000)
	Current wealth Sh <u>36,000</u>

2. Sell the right

$$\% \text{ of ownership} = \frac{1200}{3,200,000} \times 100\% = 0.0375\%$$

Wealth status

Remaining share	1200 × 28.75	= 34,500
Value of selling	400 × 28.75	11,500
Less: cost of buying	400 × 25	(10,000)
		Sh. <u>36,000</u>

3. Do nothing / ignore

$$\% \text{ of ownership} = \frac{1200}{3,200,000} \times 100\% = 0.0375\%$$

Wealth status

$$\text{Remaining share } 1200 \times 28.75 = 34,500$$

QUESTION 3

April 2023 Question Two A

Summarise FIVE factors that firms should consider when making financing decisions.

(5 marks)

ANSWER

1. The amount required
2. Purpose for which the capital is required
3. The length of time for which the money is required
4. The size, status and ability of the business to borrow
5. The business's current status of gearing
6. The business's level of profits and reserves
7. The cost of the source of finance

QUESTION 4

April 2022 Question Four A

Explain four methods that a company could use to issue ordinary shares. (4 marks)

ANSWER

Methods that a company could use to issue ordinary shares

1. **Private placement:** Shares are offered to a small group of investors as opposed to being offered to the public as a whole subscription is usually through invitation. The investors are mainly individuals such as pension funds and insurance companies.
2. **Prospectus issue:** In this case, the shares are issued to the members of the public, the subscribers are the general public who become owners of the firm.

3. **Tender offer:** The firm fixes a certain price which is a minimum and invites persons to tender for the purchase of the shares at the minimum price or at a higher price than the minimum. They are sold to the persons with highest price tendered.
4. **Bonus issue:** In this case, the shareholders receive their dividends in form of shares, the dividends are said to be capitalized to increase capital.
5. **Right issue:** Existing shareholders are given right to buy shares at a price lower than existing market price per share.

QUESTION 5

December 2021 Question One C

Outline four characteristics of ordinary share capital as a source of business finance.

(4 marks)

ANSWER

Characteristics of ordinary share capital as a source of business finance.

- They are not a legal obligation for the company to pay.
- Are raised by limited companies only.
- Carries variable returns where dividends vary with profitability.
- Entitled to voting power, one share for one vote.
- It is the largest single source of finance to the company.
- Have a residual claim on the profits and assets of the company.
- They are a permanent source, only refundable during liquidation.

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QUESTION 6

December 2021 Question Two A

The value of Faraja Ltd.'s ordinary share using the Gordon model (6 marks)

ANSWER

Factors that are likely to affect the success of a rights issue.

- Condition of investment market
- Effect of right issue on earnings per share.
- Size of the right issue
- Pricing of the right issue

QUESTION 7

May 2021 Question One D

Propose four factors to consider when choosing between long term loan capital and ordinary share capital as a source of finance. (4 marks)

ANSWER

Factors to consider when choosing between long term loan capital and ordinary share capital as a source of finance.

- Available interest rate
- Long term goals
- Borrowing requirements
- Current business structure
- Future repayment terms
- Access to equity markets

- The company's gearing level- highly geared companies will desist from raising further debts

QUESTION 8

May 2021 Question Two A

The role of a finance manager in a modern organisation is pervasive in all the activities of any business firm.

In light of the above statement, highlight four roles of a finance manager in an organisation. (4 marks)

ANSWER

The roles of a finance manager in an organisation

- Prepare financial statements, business activity reports and forecasts
- Investments (spending money) investing the firm's funds in projects and securities that provide high returns in relation to their risks.
- Financing(raising money),obtaining funding for the firm's operations and investment and seeking the best balance between debts (borrowed funds) and equity (funds raised through the sale of ownership in the business)
- Working capital management, working capital refers to that part of the firm's capital which is required for financing short term or current assets such as cash, receivables and inventories. It is essential to maintain proper levels of these assets financial manager is required to determine the quantum of such assets.
- Dividend policy decisions – this are decision that determine the portion of profit to be issued to shareholders and amount to be set aside as retained earnings
- Financial manager determines the optimal dividend rate.

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QUESTION 9

May 2021 Question Four A

Outline four factors that could hinder the success of a rights issue. (4 marks)

ANSWER

Factors that could hinder the success of a rights issue.

- Value of the right-when the subscription price is high and the value of the right is expected to be low this may be unattractive to shareholders
- Performance of the company- where a company does not have a stable liquidity position the shareholders may not exercise the right issue.
- The market price of the shares between the date right issue is declared and the date shareholders are expected to exercise their rights. If share price decreases the right issue could fail
- Returns from other investments - the shareholders will consider the opportunities foregone by investing in other investments and not in right issue.

QUESTION 10

November 2020 Question Two A

Jaribu Ltd. has been operating in the country for many years. The directors of the company wish to raise additional capital through a rights issue in order to explore opportunities in the region. The directors have decided to make a one - for-five rights issue at a discount rate of 30% on the current market value. The company's most recent financial statements are presented below:

Income statement for the year ended 31 March 2020

	Sh."million"
Sales	<u>1 400</u>
Net profit before interest and taxation	52
Interest payable	<u>24</u>
Net profit before taxation	28
Corporation taxation	<u>7</u>
Net profit after taxation	21
Ordinary dividends payable	<u>14</u>
Retained profit for the year	<u>7</u>

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Capital and reserves as at 31 March 2020

	Sh."million"
Sh.0.25 ordinary shares	60
Revaluation reserves	140
Accumulated profits	<u>320</u>
	<u>520</u>

Additional information:

The shares of the company are currently traded at the local Securities Exchange at a price to earnings (P/E) ratio of 16.

An investor holding 10,000 ordinary shares in the company has received the information on the forthcoming rights issue but cannot decide whether to take up the rights issue, sell the rights or allow the rights to lapse.

Required:

- (i) The theoretical ex-rights price of an ordinary share. (3 marks)
- (ii) The price at which the rights are likely to be traded. (1 mark)
- (iii) Evaluate each of the three options available to the investor with 10,000 ordinary shares. (6 marks)
- (iv) Comment on the wealth of the investor based on each of the options evaluated in (a) (iii) above. (2 marks)

ANSWER

- (i) The theoretical ex-rights price of an ordinary share.**

$$\text{Number of shares} = \frac{\text{Ordinary share capital}}{\text{Par value per share}} = \frac{60}{0.25} = 240$$

$$\text{Earnings per share (EPS)} = \frac{\text{Profit after tax and preference dividends}}{\text{No of shares}}$$

$$\frac{21}{240} = 0.0875$$

Market price per share

$$\text{EPS} \times \text{P/E}$$

$$0.0875 \times 16$$

Sh 1.4

Right issue price

$$= \frac{70}{100} \times \text{Sh } 1.4$$

$$= \text{Sh. } 0.98$$

Theoretical ex-right price

$$= \frac{(5 \times 1.4) + (1 \times 0.98)}{1 + 5}$$

$$= \frac{7 + 0.98}{6}$$

$$= \text{Sh. } 1.33$$

(ii) The price at which the rights are likely to be traded.

Value of right = cum right price – theoretical ex right price

$$1.40 - 1.33$$

$$= \text{Ksh } 0.07$$

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(iii) Evaluation of the three options available to the investor with 10,000 ordinary shares.

a. Impact of exercising the right issue

$$\text{New right issue shares} = \frac{1}{5} \times 10,000$$

$$= 2,000 \text{ shares}$$

$$\text{New number of shares} = 10,000 + 2,000$$

$$= 12,000$$

Value of wealth before right issue

$$\begin{aligned} &= (12,000 \times 1.33) - (2,000 \times 0.98) \\ &= 15,960 - 1,960 \\ &= \text{Ksh } 14,000 \end{aligned}$$

QUESTION 11**November 2020 Question Four B**

Upendo Ltd. has issued 5,000,000, Sh.20 par value ordinary shares which are presently trading at Sh.25 per share at the Securities Exchange. Upendo Ltd. has plans to issue rights to purchase one new ordinary share at a price of Sh.20 per share for every four shares held.

Required:

- (i) The theoretical ex-right price of Upendo Ltd.'s share. (4 marks)
- (ii) The theoretical value of a right of Upendo Ltd. before the shares sell ex-right. (2 marks)

ANSWER

Upendo Ltd.

- (i) The theoretical ex-right price of Upendo Ltd.'s share.**

$$\begin{aligned} &= \frac{(4 \times 25) + (1 \times 20)}{4+1} \\ &= \frac{100 + 20}{5} \\ &= \frac{120}{5} \\ &= \text{Sh. } 24 \end{aligned}$$

- (ii) The theoretical value of a right of Upendo Ltd. before the shares sell ex-right.**

$$\begin{aligned} \text{Value of right} &= \text{cum right price} - \text{theoretical ex-right price} \\ &= 25 - 24 = \text{Sh. } 1 \end{aligned}$$

QUESTION 12

November 2020 Question Five A

Donnat Ltd. has a capital structure that consists of Sh.150 million, 15% debentures and Sh.450 million in ordinary shares of Sh.20 par value.

The company adopts a 100% payout ratio as its dividend policy.

The finance manager of Donnat Ltd. intends to raise an additional Sh.20 million to finance an expansion programme and is considering two alternative financing options:

Option 1: Issue a 12% debenture stock.

Option 2: Issue additional ordinary shares of Sh.20 par value.

The corporation tax rate is 30%.

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Required:

Calculate the earnings before interest and tax (EBIT) and earnings per share (EPS) at the point of indifference in firm's earnings under financing option (1) and (2) above.

(8 marks)

ANSWER

a) Donnat Ltd.

Earnings before interest and tax (EBIT) and earnings per share (EPS) at the point of indifference in firm's earnings under financing option (1) and (2) above.

EPS at indifference point interest is 15% debentures

$$= \frac{15}{100} \times 150 = \text{Sh. 22.5million}$$

Number of ordinary shares

$$= 450 \div 20 = 22.5 \text{ million}$$

Financing option 1

Interest on 12% debenture

$$= \frac{12}{100} \times 20 = \text{Sh } 2.4 \text{ million}$$

$$\text{Total interest} = 2.4 + 22.5$$

$$= \text{Sh } 24.9 \text{ million}$$

EPS option 1

$$\text{EPS} = \frac{(\text{EBIT} - I)(1 - T)}{\text{Number of shares}}$$

$$= \frac{(\text{EBIT} - 24.9)(1 - 0.3)}{22.5}$$

$$\frac{(\text{EBIT} - 22.9)0.7}{22.5}$$

$$= \frac{0.7\text{EBIT} - 17.43}{22.5}$$

Option 2

New ordinary shares issued

$$\frac{20 \text{ million}}{20} = 1 \text{ million}$$

$$\text{New shares} = 22.5 + 1 = 23.5 \text{ million shares}$$

EPS option 2

$$\frac{(\text{EBIT} - I)(1 - T)}{\text{Number of shares}}$$

$$\frac{(\text{EBIT} - 22.5)0.7}{23.5}$$

$$\frac{0.7EBIT - 15.75}{23.5}$$

At point of indifference EPS of option 1 and 2 are similar

$$\frac{0.7EBIT - 17.43}{22.5} = \frac{0.7EBIT - 15.75}{23.5}$$

$$23.5(0.7EBIT - 17.43) = 22.5 (0.7EBIT - 15.75)$$

$$16.45EBIT - 409.605 = 15.75EBIT - 354.375$$

$$16.45EBIT - 15.75EBIT = - 354.375 + 409.605$$

$$\frac{0.7EBIT}{0.7} = \frac{55.23}{0.7}$$

$$EBIT = \text{Sh } 78.9\text{million}$$

Operating profit at indifference point is sh 78.9million.

Now, earnings per share at indifference point is:-

$$= \frac{(0.7 \times 78.9) - 15.75}{23.5}$$

$$= \frac{55.23 - 15.75}{23.5}$$

$$= \text{Sh } 1.68$$

QUESTION 13

May 2019 Question Two A

Propose four reasons that might make a firm use reserves to finance its operations.

(4 marks)

ANSWER

Reasons that make a firm use reserves to finance its operations

1. It's cheaper because company doesn't incur floatation costs
2. Readily available. A quick resolution can make it available for use to finance operations
3. Doesn't increase company's gearing ratio

QUESTION 14**May 2019 Question Five B**

The capital structure of Kara Kara Limited which is considered optimal is given as follows:

	Sh.“000”
Ordinary share capital (Sh. 10 par value)	90,000
Retained earnings	75,000
15% preference share capital (Sh. 100 par value)	45,000
16% debenture capital	90,000
	300,000

The company has total assets amounting to Sh. 360 million but it is expected the assets will rise to Sh.500 million by the end of the current financial year.

Additional information:

1. New equity shares sold will net 90% after floatation costs.
2. The current market price per share (MPS) of the ordinary shares is Sh.25.
3. New ordinary shares will be issued at the current market price subject to a floatation cost of 10% of the issue price.
4. New 16% debentures can be issued at par through the securities exchange.
5. The past and expected earnings growth rate is 10%. Dividend growth rate is expected to be matched with the earnings growth rate.
6. The current earnings yield is 24%.
7. The company adopts a constant dividend payout ratio of 50%.
8. New 15% preference shares can be issued at the current selling price of Sh. 120 each.
9. The retained earnings available for investment purposes is Sh.29,700.000.
10. The corporate tax rate is 30%.

Required:

The number of ordinary shares that must be sold in order to raise the required equity capital.

(6 marks)

ANSWER**Karakana Ltd**

Proportion of each source in current capital structure

Source	Amount(–000)	Weight
Ordinary share capital	90,000	30%
Retained earnings	75,000	25%
15% preference shares	45,000	15%
16% debenture capital	90,000	30%
	300,000	100%

Amount that needs to be raised

$$(500,000,000 - 360,000,000) = \text{Sh.}140,000,000$$

Amount to be raised from equity capital

$$140,000,000 \times (25\% + 30\%) = 77,000,000$$

Amount to be raised from share capital

$$(77,000,000 - 29,700,000) = 47,300,000$$

New ordinary shares issued

$$47,300,000 \div \left\{ \frac{90}{100} \times 25 \right\} = 2,102,222.22 \text{ shares}$$

QUESTION 15**November 2018 Question One B (ii)**

Differentiate between the following terms as applied in finance:

- (ii) —Cumright‘ and "ex-right” market price per share. (2 marks)

ANSWER

ii) **Cumright market price** allows a shareholder of record to take part in a right issue declared by a Company. Owners of shares that have cum rights are able to buy shares at a price lower than current market price per share. **Ex-right** is the price of share after the right issue.

QUESTION 16

November 2017 Question Two B and C

b) Maji Mazuri Ltd. an all equity financed company has an issued share capital of Sh. 10 million ordinary shares of Sh.10 par value. The company paid a dividend of Sh.0.4 per share last period and the market price per share is Sh.20 ex-dividend.

The company is contemplating raising additional funds through a rights issue. The management has proposed a 1 for 4 rights issue at an issue price of Sh. 15 per share. The funds raised are intended to be used to finance a major new project which is expected to increase the company's annual after tax cash flows by Sh.950,000 in perpetuity.

Required:

- (i) The cum-right market price per share (MPS) after the announcement of the rights issue. (4 marks)
- (ii) The theoretical ex-right market price per share. (2 marks)
- (iii) The theoretical value of each right. (2 marks)

(c) Evaluate the impact of the rights issue in (b) above on the value of wealth of an existing shareholder who holds 1,600 ordinary shares in Maji Mazuri Ltd. and Sh. 10,000 in his savings account assuming that this shareholder decides to:

- (i) Exercise all his rights. (3 marks)
- (ii) Sell all his rights. (3 marks)
- (iii) Ignore the rights issue. (2 marks)

ANSWER**b) Maji Mazuri Ltd**

- i. Cum right price per share (MPS) after the announcement of rights issue**

$$\text{Number of existing ordinary shares} = \frac{\text{Number of existing ordinary shares}}{\text{par value}}$$

$$= \frac{10,000,000}{10} = 1,000,000$$

$$\text{Cost of equity} = \frac{\text{dividend per share}}{\text{current market price per share}} \times 100\%$$

$$= \frac{0.4}{20} \times 100\% = 2\%$$

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Number of right issue shares

$$= \frac{1}{4} \times 1,000,000 = 250,000$$

Total amount raised from right issue

$$= 250,000 \times 15 = \text{Sh. } 3,750,000$$

NPV of new project

$$= \frac{\text{project after tax cashflows}}{\text{cost of equity}} - \text{initial outlay}$$

$$= \frac{950,000}{0.02} - 3,750,000$$

$$= 47,500,000 - 3,750,000$$

$$= \text{Sh. } 43,750,000$$

Cum right market price per share (P_o)

$$= \text{current MPS} + \text{NPV per share}$$

NPV per share

$$= \frac{\text{NPV of project}}{\text{Existing number of shares}}$$

$$= \frac{43,750,000}{1,000,000}$$

$$= \text{Sh. } 43.75$$

$$\text{Cum right market price per share} = 20 + 43.75 = \text{Sh. } 63.75$$

ii. The theoretical ex-right market price per share (P_x)

$$P_x = \frac{(P_o \times S_o) + (P_s \times S)}{S_o + S}$$

Where;

$$P_o = \text{cum right MPS} = 63.75$$

$$S_o = \text{existing shares} = 1,000,000$$

$$P_s = \text{right issue shares} = 250,000$$

$$S = \text{subscription price} = 15$$

$$P_x = \frac{(63.75 \times 1,000,000) + (15 \times 250,000)}{1,000,000 + 250,000}$$

$$= \text{Sh. } 54$$

iii. The theoretical value of each right

$$R = P_o - P_x$$

$$63.75 - 54 = \text{Sh. } 9.75$$

c) Impact of rights issue

i. Exercise all his rights

Amount owned before right issue

Value of shares + amount in savings account

$$(1600 \times 20) + 10,000 = 42,000$$

New right issue shares

$$\frac{1600}{4} \times 1 = 400 \text{ at } P_s \text{ of } 15$$

New number of shares

$$(400 + 1600) = 2,000$$

Value of wealth

$$(2,000 \times 54) + 10,000 - (400 \times 15) = \text{Sh. } 112,000$$

Prior to right issue value of wealth was Sh. 42,000 which has increased to Sh. 112,000 after right issue

ii. Selling all his rights

Value of wealth

$$= (1,600 \times 54) + 10,000 + (1600 \times 9.75)$$

$$\begin{aligned} &= 86,400 + 10,000 + 15,600 \\ &= \text{Sh. } 112,000 \end{aligned}$$

iii. Ignore the rights issue

Value of health

$$(1600 \times 54) + 10,000 = \text{Sh. } 96,400$$

QUESTION 17

November 2017 Question Five A

Describe three factors that have limited the growth of venture capital investment in most developing countries. (6 marks)

ANSWER

Limitations to the growth of venture capital investment in developing countries

Venture capital is a form of finance used to fund start-ups. The following are the limitations to its growth

1. We have a limited number of venture capitalists
2. Ignorance of venture capital existence
3. Inadequate financial systems to support venture capitalists in developing nations like Kenya
4. The high returns demanded by venture capitalists makes small business persons reluctant to explore use of venture capital
5. Startup businesses are risky ventures thus venture capitalists may be unwilling to support them
6. Inadequate government support
7. No supporting legal framework to support venture capitalists operations

QUESTION 18

May 2017 Question One A

Highlight four limitations of long-term debt finance to an organisation. (4 marks)

ANSWER

Limitations of long-term debt finance to an organisation

1. Issuing debt capital increases the company's hearing
2. Assets must be offered as security to get debt capital financing
3. Payment of interest is mandatory. Any failure to do so may lead to company being liquidated
4. Companies must abide by many conditions to keep enjoying the debt capital
5. Redemption may lead to liquidity problems due to significant cash out flows involved

QUESTION 19

May 2017 Question Two B

Nyadzua Limited is making a 1 for 4 rights issue costing Sh.6.40. The company has 4 million shares in issue with a price of Sh 10.00 per share. The new shares are expected to yield 5% earnings and price to earnings (P/E) ratio of 10.

Required:

- (i) The theoretical ex-right price. (4 marks)
- (ii) The value per share after the rights issue. (4 marks)

ANSWER

Nyadzua Ltd

- i. Theoretical ex-rights price (P_x)

$$\frac{P_m \times S_0 + P_s \times S_1}{S_0 + S_1}$$

Where; P_m is current market price per share = Sh. 10.8

S_o is existing number of shares = 4

P_s is right issue shares price = 6.40

S_1 is right issue shares = 1

$$P_x = \frac{4 \times 10.8 + 1 \times 6.40}{4 + 1} = \text{Sh. 9.92}$$

ii. The value per share after the rights issue

Value of right = cum right price – theoretical ex-right price

$$10.80 - 9.92 = \text{Sh. 0.88}$$

QUESTION 20

May 2016 Question Four A and B

- (a) Highlight four shortcomings of financial deepening. (4 marks)
- (b) (i) Define the term "franchising". (2 marks)
- (ii) Suggest four reasons why franchising could be considered as an alternative source of finance to a company. (4 marks)

ANSWER

a) Shortcomings of financial deepening

Financial deepening refers to the increasing provision of financial services. It can refer to both a wider choice of services and better access for different social economic groups.

Shortcomings or Limitations of financial deepening includes

- i. It can lead to informal intermediation
- ii. It will lead to domination by few large financial institutions

- iii. The coming up with different financial instruments to encourage financial deepening may heighten default risk.
- iv. A lot of documentation is required to facilitate financial transactions.
- v. Leads to asymmetrical information.

4 b) i) Meaning of the term "franchising"

A franchise is an authority that is given by an organization to someone, allowing them to sell its goods or services or to take part in an activity which the organization controls e.g. Almasi Beverages is a franchise of Coca-Cola

ii) Reasons why franchising could be considered as an alternative source of finance to a company

1. Products and services will have an existing market share. This means the company will save money that could have needed to be spent on market testing
2. You use a recognized brand name. This saves you cash that you could have spent on advertising or promotion to create product awareness.
3. You get support from franchisor like training. This cuts down on the amount you could have had to spend on the business. Such money can be utilized to expand the business.
4. Under the franchising agreement, the duty of doing research and product development is work of franchisor. Thus, saves business the heavy expenditure it could have had to make on research efforts.

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QUESTION 21

November 2015 Question Two A

Summarise four advantages of debentures over preference shares. (4 marks)

ANSWER**Advantage of debenture over preference shares**

1. Interest payments on debentures are tax allowable deductions whereas preference dividends are not allowable tax deductions
2. Debentures rank first to preference shares in case of liquidation.
3. Debentures have no voting rights
4. You can raise significant sum of cash using debentures as compared to preference shares
5. No formalities need to be complied with when raising money using debentures
6. No floatation costs for debenture unlike for preference shares
7. Debentures are secured source of financing while preference shares are unsecured financing source

QUESTION 22**September 2015 Pilot paper Question One A, B and C**

- (a) —Provision for depreciation is an internally generated source of finance to a company'-.
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Explain the basis upon which provision for depreciation is a source of finance to an organisation. (4 marks)

- (b) MM Company Ltd. is contemplating raising additional finance for an expansion programme. The company is considering Sh.50 million for this expansion programme. The company's existing capital structure is given below:

	Sh. "000"
Ordinary share capital (Sh.20 par)	60,000
10% debenture capital	25,000
12% preference share capital	15,000
Reserves	50,000
	150,000

Two alternative financing options available to the company are given as follows:

Option 1

Issue new ordinary shares at par to raise all the desired funds.

Option II

Issue new ordinary shares at par to raise Sh.30 million and the balance will be raised through the issue of 15% debentures.

The management are optimistic that this investment will enable the company to generate annual operating profit (EBIT) whose forecasted values in different states of nature and their probability of occurrence are given as follows:

State of Nature	Probability	Operating profit (EBIT)
Good	0.4	20,000
Moderate	0.25	15,000
Poor	0.35	10,000

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The firm pays corporation tax at the rate of 30%.

Required:

(i) Determine the level of expected operating profit (EBIT) and expected earnings per share at the point of indifference between the firm's earning under financing options I and II. (6 marks)

(ii) Determine the range of expected operating profit within which each financing option will be recommended

(Hint: a graph may be used to answer this question). (6 marks)

(c) With reference to (b) above, indicate the financing option you would recommend assuming that the company's expected operating profits are:

(i) As forecasted by the organisation. (1 mark)

- (ii) Sh.6,000,000 per annum. (1 mark)
- (iii) Sh.15,000,000 per annum. (2 marks)

ANSWER**A. Basis upon which provision for depreciation is a source of finance to an organization**

Depreciation is the systematic allocation of the cost of a business asset to expense over the useful life of the asset. The accounting for depreciation is a debit to Depreciation Expense and a credit to Accumulated Depreciation. As you can see, the entry does not involve the account Cash. Hence, depreciation expense is referred to as a noncash expense.

1. Depreciation is an expense as expired cost which is recovered in cash as the same is included with the cost of production. For this, it is quite logical that if sale of fixed asset is treated as a source of fund, why not depreciation.
2. When we ascertain 'Trading Profit (Adjusted)' or Funds from operation, the amount of depreciation is added back to net profit. The amount of depreciation should not be added to 'funds from operation' if the same is not considered as source.
3. Depreciation is an expense as expired cost which is recovered in cash as the same is included with the cost of production. For this, it is quite logical that if sale of fixed asset is treated as a source of fund, why not depreciation.
4. When we ascertain 'Trading Profit (Adjusted)' or Funds from operation, the amount of depreciation is added back to net profit. The amount of depreciation should not be added to 'funds from operation' if the same is not considered as source.
5. Funds normally mean (net) Working Capital. As such, amount received by way of profit and depreciation increase the amount of Working Capital although the respective figure cannot be ascertained. Thus, amount earned from net profit may be treated as a source so also the amount of depreciation.
6. There is no doubt about it that depreciation constitutes a major source of fund or inflow of fund for which depreciation plays a very significant role for financing the funds of a firm.

7. For ascertaining net profit, the amount of depreciation is deducted and consequently the resultant amount reduces. Now if that amount of net profit is distributed among the shareholders by way of dividend, the real figure of dividend is reduced by the amount of depreciation which remains in the firm and represents as a source of fund.
8. Depreciation is charged against revenue the object of which is to create a fund. It is not like other expenses (say Salary, Wages etc.) which are paid in cash i.e. for the amount of depreciation capital fund is not reduced. So, it (depreciation) can be treated as a source.

B. MM Company Ltd.

i. Level of expected operating profit and expected earnings per share

$$\text{Number of ordinary shares} = \frac{\text{Ordinary share capital}}{\text{par value}} = \frac{60,000}{20} = 3,000$$

$$10\% \text{ debenture interest} = \frac{10}{100} \times 25,000 = 2,500$$

$$\text{Preference dividends} = \frac{12}{100} \times 15,000,000 = 1,800,000$$

$$15\% \text{ debenture interest} = \frac{15}{100} \times 20,000 = 3,000,000$$

$$\text{Expected operating profit} = (0.4 \times 20) + (0.25 \times 15) + (0.35 \times 10) = 15.25$$

$$15.25 \times 1,000 = 15,250$$

ii. Range of expected operating profit

EPS option 1

$$\text{EPS} = \frac{[(\text{EBIT} - 2500)0.7 - 1800]}{3,000 + 2500} = \frac{0.7\text{EBIT} - 1750 - 1800}{5,500} = \frac{0.7\text{EBIT} - 3550}{5500}$$

EPS option 2

$$EPS = \frac{0.7(EBIT-5500)-1800}{3000+2500} = \frac{0.7EBIT-3850-1800}{4500} = \frac{0.7EBIT-5650}{4500}$$

At indifference point

$$\frac{0.7EBIT-3550}{5500} = \frac{0.7EBIT-5650}{4500}$$

$$4500(0.7EBIT - 3550) = 5500(0.7EBIT - 5650)$$

$$3150EBIT - 15,975,000 = -31,075,000 + 15,975,000$$

$$\frac{-700EBIT}{-700} = \frac{-15,100,000}{-700} \quad EBIT = 21,571.429$$

$$EBIT = 21,571.429 \times 1,000 = \text{Sh. } 21,571,429$$

Earnings per share at indifference point

$$\frac{(0.7 \times 21,571.429) - 3550}{5500} = 2.10$$

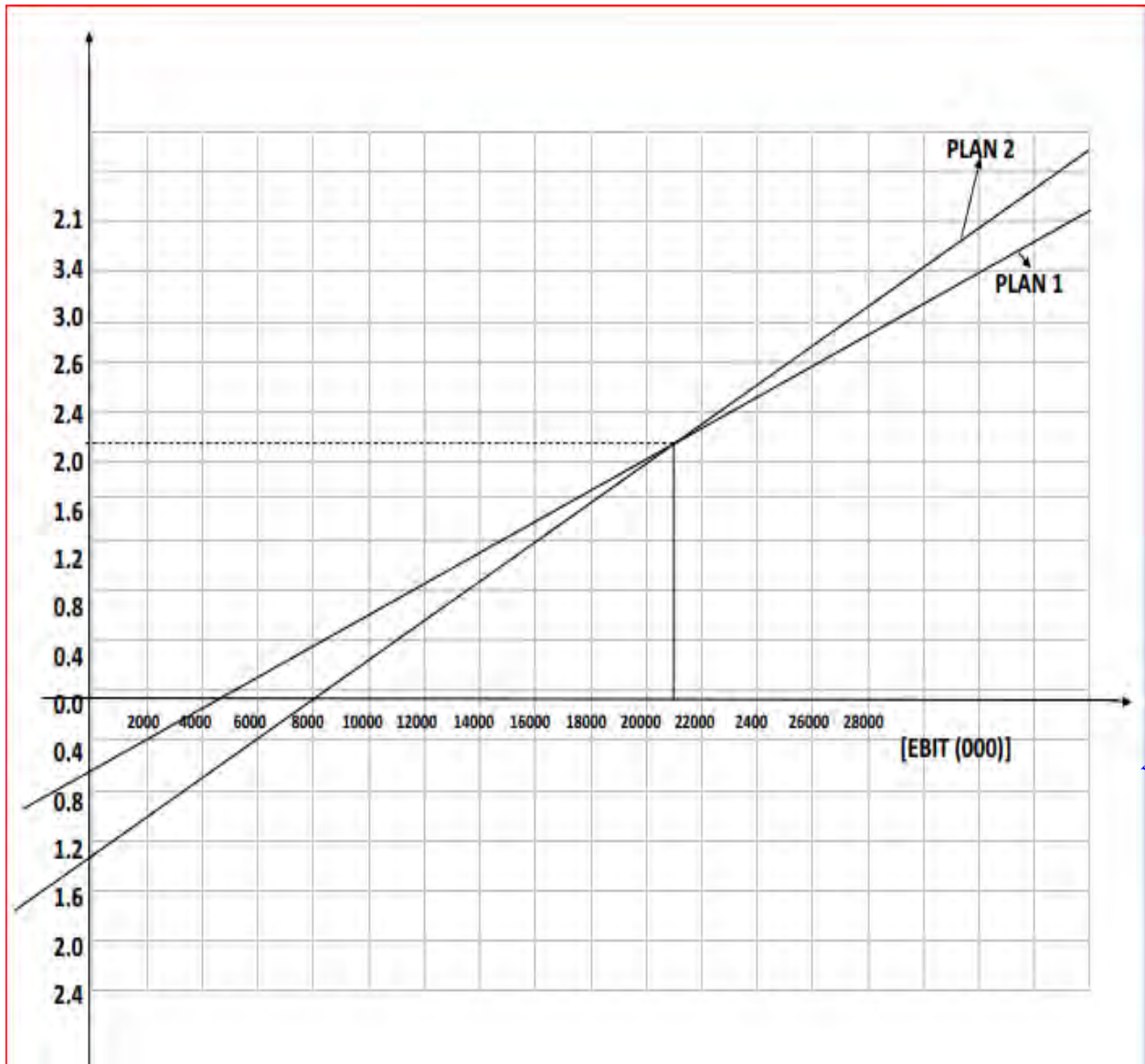
$$EPS_1 = \frac{0.7EBIT-3550}{5500}$$

Using above equation assume

EBIT	0	1000	21571.429
EPS	-0.65	-0.52	2.10

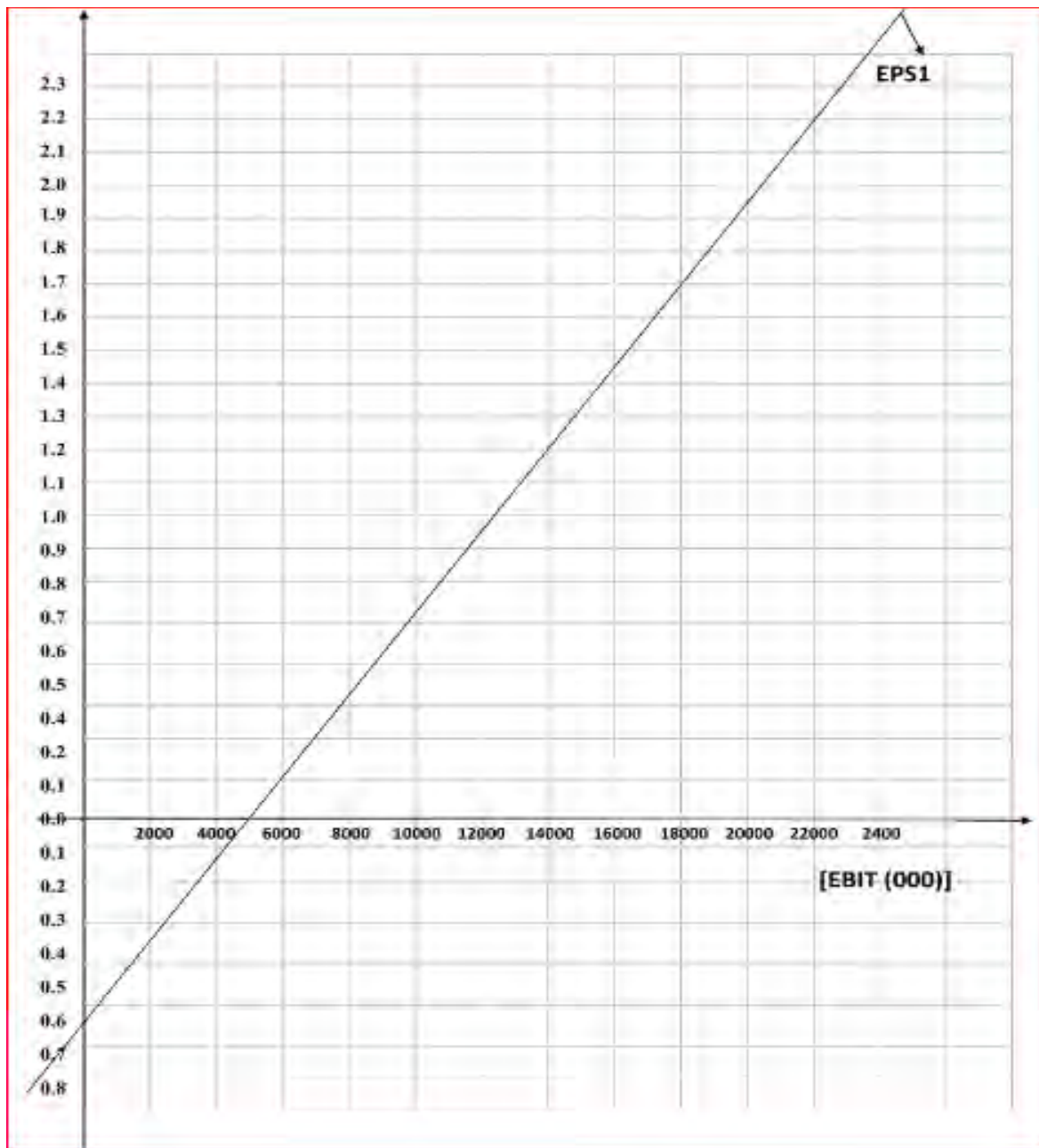
$$EPS_2 = \frac{0.7EBIT-5650}{4500}$$

EBIT	0	1000	21,571.429
EPS	-1.26	-1.10	2.10



Between an operating profit of 0 to 21,571,429 the best to adopt is plan 1

Above an operating profit of Sh. 21,571,429 plan 2 should be used.



C. With reference to (B)

i. As forecasted by the organization

$$\begin{aligned}
 &= (0.4 \times 20,000) + (0.25 \times 15,000) + (0.35 \times 10,000) \\
 &= 8,000 + 3,750 + 3,500 \\
 &= 15,250
 \end{aligned}$$

Earnings per share 1

$$= \frac{0.7EBIT-5650}{5500} = \frac{(0.7 \times 15250 - 5650)}{4500} = \text{Sh. } 1.12$$

If operating profit is as predicted best plan to adopt is plan 1

ii. Sh. 6,000,000 per annum

Earnings per share option 1

$$\frac{(0.7 \times 6000) - 5650}{4500} = \text{Sh. } -0.32$$

Plan 1 is the best plan of the two plans

iii. Sh. 15,000,000 per annum

$$\text{EPS plan 1} = \frac{(0.7 \times 15,000) - 3,550}{5,500} = \text{Sh. } 1.26$$

$$\text{EPS plan 2} = \frac{(0.7 \times 15,000) - 5,650}{4,500} = 1.07$$

Plan 1 is the best plan of the two plans

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QUESTION 23

September 2015 Pilot paper Question Five B

- (i) Define the term financial innovation (1 Mark)
- (ii) Highlight any three factors responsible for financial innovation (3 Marks)

ANSWER

Financial innovation

i. Definition of financial innovation

The improvement in the financial instrument as a result of unexpected changes; these improvements can be in terms of creating new securities or marketing the old ones

ii. Factors responsible for financial innovation

1. Technology advances – this has made possible to offer products globally where there is adversity of needs
2. Regulatory, legislative and tax changes in an attempt to overcome limitations imposed by various regulations and taxes, firms have had to innovate to water down the impact of such regulations
3. Academic work on market efficiency – research conducted has proved that financial innovation innovations make the market more efficient
4. Competition among market participants – this has prompted firms to innovate to set them apart from other firms
5. The level of economic activity – growth in economic activity has spurred the need for financial innovation all in an attempt to satisfy growing market and economy
6. Cross border listing in the different stock exchange
7. The need to reduce the agency costs
8. The high transaction costs have led to financial innovation to reduce them
9. Opportunities to increase the liquidity of assets for example the factory of debtors

TOPIC 3

FINANCIAL INSTITUTIONS AND MARKETS

QUESTION 1

December 2023 Question Three A

Outline **FIVE** eligibility requirements for public offering of shares and listing at the securities exchange in your country. (5 marks)

ANSWER

Eligibility requirements for listing shares in the Nairobi Securities Exchange

The Nairobi Securities Exchange (NSE) has specific requirements that companies must meet before listing their shares on the exchange. These requirements are divided into two main categories:

1. Issuer requirements:

- **Company type:** Must be a public company limited by shares and registered under the Companies Act (Cap 486) or any other corporate body.
- **Minimum authorized, issued, and fully paid-up capital:** Kshs. 50 Million.
- **Net assets:** Not less than Kshs. 100 Million immediately before the public offer or listing of shares.
- **Track record:** At least three years of audited financial statements.
- **Board of directors:** At least five directors, with at least a third being non-executive directors.
- **Management team:** Experienced and qualified management team with relevant industry experience.
- **Corporate governance:** Strong corporate governance practices in place.
- **Dispute resolution:** Clear mechanisms for resolving shareholder disputes.

2. Shares to be listed:

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- **Freely transferable:** The shares to be listed must be freely transferable and not subject to any restrictions on marketability or preemptive rights.
- **Minimum public holding:** At least 25% of the issued shares must be held by the public upon listing.
- **Minimum shareholding:** No single shareholder should hold more than 75% of the issued shares.
- **Dividend policy:** Clear dividend policy in place.

Additional requirements:

- The company must submit a comprehensive information memorandum or prospectus approved by the Capital Markets Authority (CMA).
- The company must pay listing fees and meet other ongoing listing requirements.
- The company must comply with all applicable laws and regulations.

QUESTION 2

April 2023 Question Four A and B

- (a) Distinguish between the following terms as used in financial institutions and markets:
- (i) —Disintermediation” and —intermediation”. (2 marks)
 - (ii) —Cdmarkets” and —contiuous markets”. (2 marks)
- (b) Outline FOUR roles of mutual funds as investment avenues. (4 marks)

ANSWER

(a) Terms used in financial institutions and markets:

(i) “Disintermediation” and “intermediation”.

Financial intermediation refers to the practice of linking an investor and borrower. Acting as a third party, an intermediary aims to meet the financial needs of both parties to mutual satisfaction.

In financial terms, disintermediation involves the removal of banks, brokers or other third parties, allowing individuals to transact or invest directly.

(ii) –**Call markets” and “continuous markets”.**

A call market matches and executes the grouped trade orders at specified times during an entire trading day. But, a continuous market matches and completes the grouped trade order throughout a trading day.

(b) **Roles of mutual funds as investment avenues.**

- Investment management
- Risk management
- Convenience
- Transparency
- Professional expertise

QUESTION 3

December 2022 Question Four B

Outline FIVE factors that influence the price of a listed company’s share. (5 marks)

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ANSWER

Factors that influence the price of a listed company’s share

- **Earnings:** If a company reports strong earnings, it is likely that its share price will increase. Conversely, if a company reports weak earnings, its share price may decline.
- **Dividends:** If a company announces an increase in its dividend payments, its share price may rise.
- **Market conditions:** External economic factors such as interest rates, inflation, and the overall performance of the stock market can all impact a company's share price.
- **Industry trends:** If the industry in which a company operates is performing well, it is likely that its share price will benefit.
- **Competition:** If a company faces increased competition, it may struggle to maintain its market share, which could lead to a decline in its share price.

- **Changes in management:** If there are changes in a company's management team, it could affect investor confidence and impact the share price.
- **Changes in regulation:** If there are changes in the regulatory environment in which a company operates, it could affect its business and impact its share price.
- **News:** Positive or negative news about a company can impact its share price.

QUESTION 4

August 2022 Question One A

Highlight four costs of issuing shares in the securities exchange in your country.

(4 marks)

ANSWER

August 2022 Question One A

Costs of issuing shares in the securities exchange in Kenya.

1. **Application fee:** This is a fee charged by the stock exchange to cover the costs of processing your application to list your shares.
2. **Listing fee:** This is a fee charged by the stock exchange to cover the costs of listing your shares on the exchange.
3. **Underwriting fee:** If you use an underwriter to help you issue your shares, you will need to pay them a fee for their services.
4. **Legal and accounting fees:** You may need to pay legal and accounting fees to help you prepare the necessary documents and ensure compliance with relevant laws and regulations.
5. **Printing and distribution costs:** You will need to pay for the printing and distribution of prospectuses and other materials related to the issue of your shares.
6. **Marketing and promotion costs:** You may need to pay for marketing and promotion to raise awareness of your shares and attract potential investors.

QUESTION 5

April 2022 Question Three A

- (i) Explain the term "cross border listing". (2 marks)

- (ii) Explain four benefits that would accrue to a firm that undertakes cross border listing of its shares. (4 marks)

ANSWER

(i) Term "cross border listing".

It occurs when securities of a domestic stock re listed in a foreign stock exchange or securities of a foreign stock exchange are listed in a domestic stock exchange e.g Umeme Ltd, Bank of Kigali, NMG and KCB

(ii) Benefits that would accrue to a firm that undertakes cross border listing of its shares.

- Listing in different stock exchanges leads to risk diversification. This enhances stability of the share price.
- It widens the company's reach hence increasing its market share.
- A company's goodwill is enhanced, since it attracts both foreign and local interests.
- Cross border listing increases the borrowing capacity and the access to funds is made easier
- It enhances mobilization of funds across regions.
- It increases the trading volume of the securities since they are traded in more than one stock exchange.

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QUESTION 6

April 2022 Question Four B

Highlight five functions of the Central Depository System (CDS). (5 marks)

ANSWER

Functions of the Central Depository System (CDS)

- Provision of book entry account i.e electronic exchange of ownership of securities and payment of cash.
- Effective distribution of dividends, interests, rights issue and bonus issues.

- Provision of detailed listings of investors according to the type of securities they hold e.g ordinary shares, preference shares.
- Dematerialization
- Immobilization of securities
- Simultaneous delivery and payment between the two parties exchanging or transferring securities

QUESTION 7

December 2021 Question One B

Securities exchanges across the world play a critical role in the economic development of countries.

In light of the above statement, summarise six roles of the Securities Exchange in the economic development of your country. (6 marks)

ANSWER

Roles of the Securities Exchange in the economic development in Kenya

1. Raising capital for business: It provides companies with a platform to raise capital through sale of shares to the interested public.
2. The fluctuation in prices facilitates measurement of the economic activities and performance of various firms.
3. It can help the government raise funds for development since the government sells its bond to the public through the stock exchange.
4. Improving corporate governance: The stock markets supervise the companies will not trade their shares efficiently. This makes managers improve their performance in order to attract customers for their shares.
5. Redistribution of wealth: By providing a wide spectrum for people to trade shares, any person who is able can purchase shares of a given company therefore owning it.
6. Mobilizing savings to be invested in productive enterprise as an alternative to making bank deposits

QUESTION 8

September 2021 Question Three A

An efficient and sound financial system of a country plays an important role in the economic development of that country.

In relation to the above statement, explain five functions of a financial system. (5 marks)

ANSWER

Functions of a financial system

- **Payment function:** The financial system offers a convenient mode of payment for goods and services.
- **Risk function:** The financial markets provide protection against health, life and income risks. These are accomplished through the sale of life, health, and property insurance policies. Overall, they provide immense opportunities for the investor to hedge himself / herself against or reduce the possible risk involved in various instruments
- **Policy function:** Most governments intervene in the financial system to influence macroeconomic variables like interest rates or inflation
- **Savings function:** Public savings find their way into the hands of those in production through the financial system. Financial claims are issued in the money

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QUESTION 9

September 2021 Question Four A

Summarise five features of the efficient markets hypothesis. (5 marks)

ANSWER

Features of the efficient markets hypothesis

- Security prices should be randomly determined
- Transaction costs should be minimal, so that they don't discourage the sale or purchase of securities

- The market should have free flow of information such that knowledgeable investors don't take advantage of others
- An efficient market allows investors an opportunity to outperform
- With disclosure of new information, the efficiency of market increases, diminishing opportunities for excess returns and arbitrage

QUESTION 10

May 2021 Question Two C

Examine five roles of the Capital Markets Authority (CMA) or a similar institution in your country. (5 marks)

ANSWER

Roles of the Capital Markets Authority (CMA)

- The CMA act as a government advisor through the ministry of finance regarding policies affecting the capital markets.
- The CMA has the responsibility of licensing and regulating stock brokers, investment advisor's security dealers and the authority depositories.
- The CMA is involved in the process of listing of new companies any company intending to be quoted in the NSE must apply through CMA
- CMA is involved in making of policies that would enhance the development of the capital market e.g.: policy regarding the selling and buying of securities, policy on admission of individual and institution to the capital market and generally policies on the introduction of securities and their regulation.
- The CMA act as a watchdog for the shareholders of listed companies'. This is through regulating the operations of the listed companies so as to protect investors against penalty, insider trading or suspensions.
- The authority assists in the development of new securities in the market. This is through research and evaluation of various recommendations of stakeholders in the NSE. It is the responsibility of the CMA to evaluate whether there is need of new securities and develop an appropriate policy.

QUESTION 11

November 2020 Question One B

In the context of financial markets:

- (i) Distinguish between "commodities markets" and "derivatives markets". (4 marks)
- (ii) Summarise four functions of financial markets. (4 marks)

ANSWER

a) In the context of financial markets:

(i) Distinction between "commodities markets" and "derivatives markets".

The commodities market is where traders and investors buy and sell natural resources or commodities such as corn, oil, meat and gold. A specific market is created for such resources because their price is unpredictable. There is a commodities futures market wherein the price of items that are to be delivered at a given future time is already identified and sealed today.

Derivative market involves derivatives or contracts whose value is based on the market value of the asset being traded

(ii) Functions of financial markets

- **Lowers the cost of transactions:** In financial markets, various types of information regarding securities can be acquired without need to spend.
- **Determines the price of securities:** Investors aim to make profits from their securities. However, unlike goods and services whose price is determined by the law of supply and demand, prices of securities are determined by financial markets

- **Makes financial assets liquid:** Buyers and sellers can decide to trade their securities anytime. They can use financial markets to sell their securities or make investments as they desire.
- **Immobilization of savings:** For an economy to be successful, it is crucial that the money does not sit idle. Thus, a financial market buyers and sellers can easily find each other without making too much efforts or wasting time. Also, since these markets handle so many transactions, it helps them to achieve economies of scale. This results in lower transaction cost and fees for the investors
- **They reduce risk** by having information publicly available to investors and traders.
- These markets **calm the economy** by instilling confidence in investors. Investor confidence stabilizes the economy
- helps in connecting those with money with those who require money
- **Saves time and money:** The financial market provides a platform where both the buyers and sellers can find each other easily without spending too much time, money or effort.

QUESTION 12

November 2019 Question Two A

- (i) Explain the term "cross-border" listing. (2 marks)
- (ii) Discuss two benefits of cross-border listing to a quoted company. (4 marks)

ANSWER

a) i) Explanation of the term "cross-border" listing

Cross borders meaning explanation. Cross bordering listing involves companies that trade on the stock exchange of their home country and also on the stock exchange of another country.

ii) Benefits of cross-border listing to a quoted company

- **Liquidity:** liquid stocks have lower expected returns and lower transaction costs. This increases total trade volume and decreases bid-ask spreads in home market.
- **Wider shareholder base:** Wider shareholder base are less risky, lower expected returns and higher prices.
- **Capital needs and growth opportunities:** Companies in emerging markets need to use cross-listing to raise capital to continue to grow beyond their home market.
- **Market integration:** securities of similar risk have same expected returns when trading in two markets. Global market integration is affected by direct and indirect barriers
- **Corporate governance signal:** Indirect barriers can be reduced through corporate governance. Entering markets with better investment protection reduces cost and bonding.

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QUESTION 13

November 2019 Question Five A

The concept of interest rate capping has generated considerable interest from players in both public and private sectors, particularly in developing economies.

Required:

- (i) Describe the link between interest rates and availability of credit to small and medium size enterprises (SMEs). (2 marks)

- (ii) Citing three reasons, summarise the case for and against interest rate capping in an economy. (6 marks)

ANSWER

i) The link between interest rates and availability of credit to small and medium size enterprises (SMEs)

There is clearly a relationship between the two. When the interest rate is set too low, commercial banks cannot be able to price in risk, this forces them to deny SMEs loans to finance their operations. When rates are left to supply and demand forces rather than capping them, commercial banks can be able to effectively price in risk which encourages lending more to SMES.

ii) Case for and against interest rate capping in an economy.

- Interest rate caps can be used to support a specific sector of the economy where market failure exists or where there is need for more financial institutions to differentiate between risky and safe clients
- Prices charged for access to credit can be erratic and anti-competitive and therefore be higher than the true cost of lending, setting a lower cap on interest rates will provide an excellent environment for lenders to operate.
- They can be used to protect consumers from exploitation by guaranteeing access to credit at reasonable rates.
- Interest caps protect the public interest by ensuring a fair and reasonable interest rate on loans.
- Protect borrowers from predatory lending and excessive interest rates
- Form of subsidy to specific groups.
- Decrease the risk – taking behavior of credit providers
- Help in managing increased inflation in the economy

- Instills discipline in the banking industry at it avails in formation to the public on the expected repayment value of credit.

Reasons against interest capping

- Interest rate capping undermines the independence of the central bank which is entrenched in the constitution of Kenya.
- Reduced transparency: an increase in the total cost of loans through additional fees and commissions – witnessed in America, Nicaragua, South Africa and Zambia.
- Decreased diversity of products for low income households – witnessed in France and Germany.
- Reduced banking competition witnessed initially.
- An increase in illegal lending – Evident in Japan and the United states
- Shifting to expensive loans:
- The riskier borrowers rationed out resort to borrowing from alternatives sources not regulated by the caps, which charge much higher rates.
- Impairment of monetary policy transmission
- Negative impact on credit growth
- Elevated risks to financial stability
- Reduced access to financial services
- It leads to unemployment e.g layoffs witnessed in the banking sector

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QUESTION 14

November 2018 Question Three B

Explain four roles that are played by insurance companies in the financial market of your country. (4 marks)

ANSWER

Roles that are played by insurance companies in the financial market of your country

1. Risk management
2. Asset management services .e.g. Britam and CIC money market fund
3. Financial intermediaries
4. Pooling of resources which they invest in shares of various companies
5. Financial advisory services

QUESTION 15

May 2018 Question Four B

Distinguish between the terms —“capital market” and —“money market”. (2 marks)

ANSWER

Distinguishing between the terms “capital market” and “money market”

- i. In the money market the goodwill of the borrower qualifies him for the finance while in the capital market the goodwill is not as important rather the ability of the borrower to raise securities is more important
- ii. In the money market finance is obtained in cash while in finance in the capital market its obtained in kind for example by use of fixed assets
- iii. Finance in the money market is more expensive than finance in the capital market which is secured against fixed assets.
- iv. Finance in the money market is usually unsecured as opposed to finance in the capital market which is secured against fixed assets
- v. In the money market finance takes a short time to be raised as on the other hand finance in the capital market takes a longer time to be raised due to the formalities involved

QUESTION 16

May 2018 Question Five B

Explain four benefits that might accrue from demutualisation of securities exchange of your country. (4 marks)

ANSWER

Benefits that might accrue from demutualisation of securities exchange of your country

1. It results in more flexible governance structure fostering decisive action in response to changes in the business environment
2. Demutualization leads to greater investor participation in the governance of the exchange
3. Demutualization yields an improved platform in response to potential competitors in the form of alternative trading system
4. Demutualization allows greater flexibility and access to global markets
5. Demutualization facilitates faster and more complete consideration of stock exchanges to enhance available synergies
6. It ensures increased access to resources for capital investment raised by way of equity issues

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QUESTION 17

November 2017 Question Four A

Describe four types of money market instruments. (4 marks)

ANSWER

Types of Money market instruments

1. Commercial paper – this refers to a written promise to pay some debts owed by a large credit worthy corporation like Safaricom
2. Treasury bills – short term securities issued by government to enable it finance budget deficits
3. Certificates of deposits also called fixed deposits - refers to written promises by a bank to pay a depositor

4. Bills of exchange – a written order to a person requiring them to make a specified payment to the signatory or to a named payee.
5. Repurchase agreement (REPO)

QUESTION 18

November 2016 Question Two A

Discuss three possible solutions to adverse selection. (6 marks)

ANSWER

Possible solutions to adverse selection

1. Producers of commodities should provide warranties, guarantees and refunds. This is prevalent in electronic business
2. Consumers and competitors should act as monitors for each other. Consumer reports, underwriters' laboratory, notaries public and online review services such as Yelp help bridge gaps in information.
3. For banks they should set higher interest rates to cover for adverse selection.

NB: Adverse selection refers to a scenario where buyer has more information than seller and vice versa.

QUESTION 19

November 2015 Question One A

Highlight three financial instruments that are traded in money markets. (3 marks)

ANSWER

Financial instruments that are traded in money markets

1. Commercial papers
2. Repurchase agreements
3. Treasury bills
4. Bills of exchange
5. Promissory notes

QUESTION 20

November 2015 Question Five C

Summarise six benefits of the integrated financial management information system (IFMIS). (6 marks)

ANSWER

A. Benefits of the integrated financial management information systems (IFMIS)

IFMIS is an automated system that is used for public financial management

Benefits of IFMIS implementation are as detailed below:

1. Improve budgets, planning and decision making
2. Reduce costs
3. Increase government revenue
4. Improve confidence through transparency
5. Improve efficiency and controls
6. Enable government reform
7. Enhances development partners' confidence
8. Accelerate the pace/scope of economic growth
9. Improves accounting, recording and reporting through timely, accurate financial data provision
10. Improves financial controls by availing reliable and timely financial information
11. Establishes effective links between key players in accounting financial management
12. Improves management information for decision making
13. Enables efficient resource allocation mechanisms

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TOPIC 4

TIME-VALUE OF MONEY

QUESTION 1

December 2023 Question Three C and D

(c) An investor plans to borrow a loan of Sh.3 million to purchase a piece of land for his family. The interest rate agreed upon is discounted at a rate of 10% per annum. The loan is repayable in 4 years of equal instalments.

Required:

- (i) Amount payable per instalment. (1 mark)
- (ii) A loan amortisation schedule. (4 marks)

(d) Jane Kiyo has the following investment options to choose from:

- Post Office MIS Scheme, monthly interest 8% per annum.
- ICD Bank Deposit, quarterly interest 8.25% per annum.
- HDFC Ltd., half-yearly interest 8.50% per annum.

Required:

Using effective interest rate (EIR) method, advise Jane Kiyo on the preferred investment option. (4 marks)

ANSWER

(c)

- (i) Amount payable per instalment.

$$P = \frac{r(PV)}{1 - (1+r)^{-n}}$$

P = payment

PV = Present value

r = Rate of period

n = number of periods

$$\frac{0.10 (300,000)}{1 - (1 + 0.1)^{-4}}$$

$$\frac{30,000}{0.316987}$$

$$= 94,641.24$$

(ii) A loan amortisation schedule.

Year	Balance B/D	Loan Installment	Interest@10%	Principal Repaid	Balance C/D
1	300,000.00	94,641.24	30,000.00	64,641.24	235,358.76
2	235,358.76	94,641.24	23,535.88	71,105.36	164,253.40
3	164,253.40	94,641.24	16,425.34	78,215.9	86,037.50
4	86,037.50	94,641.24	8,603.75	86,037.50	0

(d) Jane Kiyo

$$EAR = \left\{ \left(1 + \frac{i}{n} \right)^n - 1 \right\} \times 100$$

$$EAR \text{ Post Office MIS Scheme } \left\{ \left(1 + \frac{0.08}{12} \right)^{12} - 1 \right\} \times 100 = 8.30\%$$

$$EAR \text{ ICD Bank Deposit } = \left\{ \left(1 + \frac{0.0825}{4} \right)^4 - 1 \right\} \times 100 = 8.51\%$$

$$HDFC LTD = \left\{ \left(1 + \frac{0.085}{2} \right)^2 - 1 \right\} \times 100 = 8.68\%$$

The preferred investment option is the investment in HDFC Ltd because it has the greatest equivalent annual interest.

QUESTION 2**August 2023 Question Five B**

Joel Ouma needs Sh.9,000,000 in five years' time to purchase a house. He is determined to deposit a given amount of money each quarter in a sinking fund that earns interest at the rate of 12% compounded quarterly for five years.

Required:

Compute the amount of money that Joel Ouma should deposit in each quarter in a sinking fund so as to enable him realise Sh.9,000,000 in five years time. (5 marks)

ANSWER

Amount of money that Joel Ouma should deposit in each quarter

$$FV = A \left(\frac{(1+r)^n - 1}{r} \right)$$

$$9,000,000 = A \left(\frac{(1.03)^{20} - 1}{0.03} \right)$$

$$9,000,000 = A \left(\frac{1.03^{20} - 1}{0.03} \right)$$

$$\frac{9,000,000}{26,8704} = \frac{26.8704A}{26.8704}$$

$$A = \text{Sh. } 334,941.05$$

QUESTION 3**April 2023 Question Five D**

John Osoro has deposited Sh.700,000 into a savings account at an annual interest rate of 5% compounded monthly with additional deposits of Sh.10,000 per month (made at the end of each month).

Required:

Determine the value of the investment after 10 years. (4 marks)

ANSWER

Monthly interest rate = $5\% \div 12$

$$= 0.42\%$$

$$\begin{aligned}
 &700,000 (1.0042)^{120} + 10,000 \left(\frac{1.0042^{120} - 1}{0.0042} \right) \\
 &= 1,157,508.22 + 1\,556\,150.42 \\
 &= \text{Sh. } 2,713,658.64
 \end{aligned}$$

QUESTION 4

August 2022 Question Four C

Digital Ltd. has some computer industrial plant which it intends to replace four years from today. The company's director estimates that the cost of the plant to the company at that time will be Sh.30 million. To finance the operations, the Finance Director has decided to set up a fund with Golden Bank Ltd. Golden Bank Ltd. has assured the Finance Director that if he opts for this option, the rate of interest will be fixed at 8% per annum. The Finance Director intends to set aside a constant amount from his annual budgets to finance the plant. The rate of interest will be compounded semi-annually.

Required:

The amount that the Finance Director should deposit with Golden Bank Ltd. every year to achieve his objective. (2 marks)

ANSWER

The amount that the Finance Director should deposit with Golden Bank Ltd. every year to achieve his objective

$$V = \text{Annuity} \left[\frac{(1+r)^n - 1}{r} \right]$$

$$\text{Annuity} = FV \div \left[\frac{(1+r)^n - 1}{r} \right]$$

$$= 30,000,000 \div \left(\frac{1.04^8 - 1}{0.04} \right)$$

$$= 30,000,000 \div 9.2142$$

$$= \text{Sh. } 3,255,844.24$$

QUESTION 5**April 2022 Question Two A**

James Chege intends to accumulate Sh.10,000,000 in his bank account 15 years from now.

Required:

The amount he should deposit now assuming that the bank is offering a 5% interest rate per annum compounded semi-annually. (4 marks)

ANSWER

James Chege

The amount he should deposit

$$r = \text{Interest rate} = \frac{5\%}{2} = 2.5\%$$

$$n = \text{Period} = 15 \times 2 = 30$$

$$Fv = \text{Future value} = \text{Sh. } 10,000,000$$

$$\begin{aligned} PV &= FV(1 + r)^{-n} \\ &= 10,000,000 (1 + 0.025)^{-30} \\ &= 10,000,000 (1.025)^{-30} \\ &= 10,000,000 \times 0.4767 \\ &= \text{Sh. } 4,767,000 \end{aligned}$$

QUESTION 6**April 2022 Question Three B**

Outline four reasons behind investors' preference for current income as opposed to future income. (4 marks)

ANSWER

Reasons behind investors' preference for current income

- **Uncertainty:** Current income is preferred at present due to uncertainty of its availability in future.

- **Availability of investment opportunities:** Money investment at present can be invested to earn extra returns in future.
- **Inflation:** Volume of money will decline due to inflation and other macro-economic factors e.g interest rates
- **Transaction motive:** Money is required at present to meet immediate needs of the firms and individuals e.g money to buy food, raw materials for production.
- **Speculative motive:** Individuals / firms may require money at present to take advantage of cash discounts which may be available at present and not future.

QUESTION 7

December 2021 Question Three B

Siphony Ltd. borrowed Sh.15,000,000 from Baraka Bank at an annual interest rate of 16% repayable in four equal annual instalments at the end of each year.

Required:

Prepare a loan amortisation schedule for the company.

(6 marks)

ANSWER

Loan amortisation schedule for the company

Loan statement

$$\begin{aligned}
 &= 15,000,000 \div \frac{1 - 1.16^{-4}}{0.16} \\
 &= 15,000,000 \div 2.7982 \\
 &= \text{Sh. } 5,360,588.95
 \end{aligned}$$

Year	Bal. b/f	Installment	Interest 16%	Load repaid	Bal. c/d
1	15,000,000	5,360,588.95	2,400,000	2,960,588.95	12,039,411.05
2	12,039,411.05	5,360,588.95	1,926,305.77	3,434,283.18	8,605,127.87
3	8,605,127.87	5,360,588.95	1,376,820.46	3,983,768.49	4,621,359.38
4	4,621,359.38	5,360,588.95	739,417.50	4,621,171.45	187.94

QUESTION 8

September 2021 Question One A

The concept of time value of money lays a solid foundation upon which other finance concepts are developed.

In light of the above statement, explain three applications of the concept of time value of money. (6 marks)

ANSWER

Applications of the concept of time value of money.

- **Capital recovery problems:** A financial manager may also be interested to know the amount of equal instalments to be paid every year to discharge a specified amount of loan raised from some financial institutions at a given rate of interest and in a given period.
- **Compound growth problems:** Sometimes a finance manager may have to calculate the compound rate of growth over a period of time e.g for sales or profits. He can easily calculate such compound rate of growth by using compound factor tables.
- **Interest rate problems:** The time value technique of compounding and present value can also be applied to calculate the implicit rate of interest in certain situations
- **Valuation problems:** the techniques of time value of money are also applied in dealing with the valuation problems of bonds and debentures
- **Sinking fund problems:** A financial manager may have to determine the amount of annual payments so as to accumulate a specified sum of money on a future date to redeem an existing liability or provide funds for replacement of an existing asset.

QUESTION 9

November 2018 Question One B (iii)

Differentiate between the following terms as applied in finance:

(iii) —Time value of money” and —time preference for money”. (2 marks)

ANSWER

Time value of money refers to idea that a shilling is worth more than a shilling ten years from now. **Time preference of money** refers to the desire for present consumption as compared to future compensation. Therefore, to induce people to save, high interest has to be charged for money not consumed today.

QUESTION 10

May 2018 Question Four D

Maua Ltd. is in the process of completing construction of a green house.

The finance manager has estimated that the project's useful life is 15 years and shall generate the following cash flows:

Years	Cash flows (sh“000”)
1 - 5	5,000
6 - 10	9,000
11-15	<u>4,000</u>
	<u>18,000</u>

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The required rate of return for the company is 10%.

Required:

The total present value of the project.

(5 marks)

ANSWER

The total present value of the project

Year	Cashflows (sh.000)	Factor 10%	P.V (sh.000)
1-5	5,000	3.7908	18,954
6-10	9,000	$6.1445 - 3.7908 = 2.3538$	21,184.2
11-15	4,000	$7.6061 - 6.1446 = 1.4615$	5,846
		Sh.	45,984.2

TOPIC 5

BUSINESS / FINANCIAL ASSET VALUATION MODELS

QUESTION 1

December 2023 Question Three B

EFG Ltd. issued 300,000, 15% preference shares of Sh.100 each, redeemable at 10% premium after 20 years. Floatation costs amounted to Sh.3,000,000.

Required:

Determine the cost of preference share capital where the shares are issued at:

- (i) Par. (2 marks)
- (ii) A premium of 10%. (2 marks)
- (iii) A discount of 10%. (2 marks)

ANSWER

(i) Par.

$$K_p = \frac{D_p + \frac{(R_v - N_p)n}{R_v + N_p}}{\frac{2}{2}}$$

Where:

K_p = Cost of Preference Share

D_p = Dividend on preference share

N_p = Net proceeds from issue of preference share (Issue price – Flotation cost)

R_v = Redemption Value

N = Period of preference share

$$K_p = \frac{15 + \frac{(110 - 90)20}{110 + 90}}{\frac{2}{2}} \times 100\%$$

$$K_p = \left[\frac{15+1}{100} \right] \times 100\% \\ = 16.00\%$$

(ii) A premium of 10%.

$$K_p = \frac{D_p + \frac{(R_p - N_p)n}{2}}{\frac{R_p + N_p}{2}}$$

$$\frac{15 + \frac{(110 - 100)20}{2}}{\frac{110 + 100}{2}} \times 100\% \\ = \frac{15.5}{105} \\ = 14.76\%$$

(iii) A discount of 10%.

$$\frac{15 + \frac{(110 - 80)20}{2}}{\frac{110 + 80}{2}} \times 100\% \\ = \frac{15 + 1.5}{95} \times 100\% \\ = 17.37\%$$

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QUESTION 2

December 2023 Question Five D

The following financial information relates to Steps Ltd. for the year ended 30 September 2023:

1. Adjusted net operating profit after tax (NOPAT) is Sh.180 million.
2. Total capital is Sh.900 million (no debt).
3. Closing market share price is Sh.29.
4. Total number of shares outstanding is 86 million.
5. The cost of equity is 13%.

Required:

Compute the following relative performance measures for Steps Ltd.:

- (i) Economic value added (EVA). (2 marks)
- (ii) Market value added (MVA). (2 marks)

ANSWER

- (i) **Economic value added (EVA).**

$$\begin{aligned}
 \text{EVA} &= \text{NOPAT} - (\text{WACC} \times \text{TC}) \\
 &= 180 - (13\% \times 900) \\
 &= 180 - 117 \\
 &= \text{Sh.63 million}
 \end{aligned}$$

- (ii) **Market value added (MVA).**

$$\begin{aligned}
 \text{MVA} &= \text{Market Value of Shares} - \text{Book Value of Shareholders' Equity} \\
 &= [86 \times 29] - 900 \\
 &= 2494 - 900 \\
 &= \text{Sh.1,594 Million}
 \end{aligned}$$

QUESTION 3

August 2023 Question Four C

Zora Ltd., a securities market listed company has the following recent financial information:

	Sh.“000”
Profit after tax (earnings)	99,900
Dividends	60,000

Statement of financial position information:

	Sh.“000”	Sh.“000”
Non-current assets		892,500

Current assets	<u>187,500</u>
Total assets	<u>1,080,000</u>
Current liabilities	105,000

Equity:

Ordinary shares (Sh.1 par value)	120,000	
Reserves	<u>615,000</u>	735,000

Non-current liabilities:

6% bank loan	60,000	
8% bonds (Sh.100 par value)	<u>180,000</u>	<u>240,000</u>
Total equity and liabilities		<u>1,080,000</u>

Additional information:

- Financial analysts have forecast that the dividend of the company will grow in the future at a rate of 4% per year. This is slightly less than the forecast growth rate of the profit after tax (earnings) of the company, which is 5% per year.
- Considering the risk associated with expected earnings growth, earnings yield of 11% per annum can be used for valuation purposes.
- Zora Ltd. has a cost of equity of 10% per annum and a before tax cost of debt of 7% per annum.
- The 8% bonds will be redeemed at par value in six years' time and the company pays tax at an annual rate of 30% per annum.
- The ex-dividend market share price of Zora Ltd. is Sh.8.50 per share.

Required:

Calculate the value of Zora Ltd. using the following valuation methods:

- | | |
|------------------------------|-----------|
| (i) Net asset value method. | (2 marks) |
| (ii) Dividend growth model. | (2 marks) |
| (iii) Earnings yield method. | (2 marks) |

ANSWER

- (i) Net asset value method.**

$$\begin{aligned}\text{Value per share} &= \frac{\text{Total equity}}{\text{Shares outstanding}} \\ &= \frac{735,000}{120,000 \div 1} \\ &= \text{Sh. 6.125}\end{aligned}$$

(ii) Dividend growth model.

$$\begin{aligned}\text{Dividend per share} &= \frac{60000}{120000} = \text{Sh. 0.50} \\ V &= \frac{D_0(1+g)}{r-g} \\ &= \frac{0.50(1+0.04)}{0.10-0.04} \\ &= \frac{0.52}{0.06} \\ &= \text{Sh. 8.67}\end{aligned}$$

(iii) Earnings yield method.

$$\begin{aligned}\text{Earnings per share (EPS)} &= 99,900,000 \div 120,000,000 \\ &= 0.8325 \\ V &= \frac{0.8325(1+0.05)}{0.11-0.05} \\ &= 0.874125 \div 0.06 \\ &= \text{Sh. 14.57}\end{aligned}$$

QUESTION 4

April 2023 Question Two C

Umi Limited is contemplating to issue 8% bonds redeemable at Sh.100 par value in three years time. Alternatively, each bond may be converted on that date into 30 ordinary shares of the company. The current market price per share is Sh.3.30 and this is expected to grow at a rate of 5% per annum. The company's cost of debt is 6% per annum.

Required:

Compute the following:

- (i) Market value of the bond assuming conversion occur after 3 years. (2 marks)
- (ii) The floor value of the bond assuming redemptions occur at par. (2 marks)

(iii) The conversion premium per share.

(2 marks)

ANSWER

(i) Market value of the bond

$$\text{Annual interest} = \text{Sh. } 100 \times \frac{8}{100} = \text{Sh. } 8.00$$

$$\text{Redemption value} = \text{Sh. } 100$$

$$\begin{aligned} \text{Conversion value} &= \text{Sh. } 3.30 (1.05)^3 \times 30 \\ &= \text{Sh. } 114.60 \end{aligned}$$

Market bond value

$$\begin{aligned} &8 \times 2.6730 + 114.60 \times 0.8396 \\ &= \text{Sh. } 21.38 + 96.22 \\ &= \text{Sh. } 117.30 \end{aligned}$$

(ii) The floor value of the bond

This is price of a straight bond with same coupon rate of interest

$$\begin{aligned} \text{Floor values} &= \text{Sh. } 8 \times \text{PVIFA}, 3\text{years } 6\% \times 100 \times \text{PVIF } 3 \text{ years, } 6\% \\ &= \text{Sh. } 8 \times 2.6730 + \text{Sh. } 100 \times 0.8396 \\ &= 21.38 + 83.96 \\ &= \text{Sh. } 105.34 \end{aligned}$$

(iii) The conversion premium

Current market value – current conversion value

$$\text{Current market value} = 117.30$$

$$\begin{aligned} \text{Current conversion value} &= \text{Sh. } 3.30 \times 30 \\ &= \text{Sh. } 99 \end{aligned}$$

$$\text{Conversion premium} = \text{Sh. } 117.60 - 99$$

$$= \text{Sh. } 18.60 \text{ per share}$$

Or

$$\frac{18.60}{99} \times 100 = 18.79$$

QUESTION 5

December 2022 Question Two A

Dopco Ltd. has provided the following summary of statement of financial position for the year ended 30 September 2022:

	Sh. “000”	Sh. “000”	Sh. “000”
Non-Current Assets:			
Freehold land and buildings			320,000
Plant and machinery			160,000
Equipment			<u>40,000</u>
			520,000
Goodwill			40,000
			<u>560,000</u>
Current Assets:			
Inventory		160,000	
Trade receivables		120,000	
Short term investments		30,000	
Cash		<u>10,000</u>	
		320,000	
Current Liabilities:			
Trade payables	120,000		
Taxation	40,000		
Proposed ordinary shares dividends	<u>40,000</u>	<u>(200,000)</u>	<u>120,000</u>
			680,000
Equities and liabilities			
12.6% long-term bonds			(120,000)
Deferred Taxation			<u>(20,000)</u>
			<u>540,000</u>
Ordinary shares of Sh.10 each			160,000
Reserves			<u>280,000</u>
			440,000

6% preference shares	<u>100,000</u>
	<u>540,000</u>

Required:

- (i) Determine the value of each ordinary share using the Net Asset value basis of valuation. (5 marks)
- (ii) Outline THREE reasons for valuing securities of a company. (3 marks)

ANSWER

Dopco Ltd.

- (i) **Value of each ordinary share using the Net Asset value basis of valuation.**

Number of ordinary shares

Ordinary share capital ÷ number of shares

$$160,000,000 \div 10 = 16,000,000$$

$$\text{NAV} = \frac{\text{Total assets} - \text{Total liabilities}}{\text{Number of shares}}$$

$$= \frac{880,000,000 - (200,000,000 + 140,000,000)}{16,000,000}$$

$$= 540,000,000 \div 16,000,000$$

$$= \text{Sh. } 33.75$$

- (ii) **Reasons for valuing securities of a company.**

- **Investment:** One of the main reasons for valuing securities is to determine their potential as an investment. This includes assessing the risk and return potential of the securities.
- **Mergers and acquisitions:** Securities are often valued in the context of mergers and acquisitions to determine the fair value of the securities being exchanged.
- **Financial reporting:** Securities are valued for financial reporting purposes, such as in the preparation of balance sheets and income statements.

- **Taxation:** Securities may be valued for tax purposes, such as determining the cost basis for capital gains tax calculations.
- **Litigation:** Securities may be valued in the context of litigation, such as in the case of a dispute over the value of a company or its assets.
- **Risk Management:** Securities are valued to manage the risk of the portfolio, and help to determine the level of diversification needed for the portfolio.
- **Portfolio Management:** Securities are valued to help in the management of the portfolio by determining the weight of each security in the portfolio.
- **Executive Compensation:** Securities may be valued for the purpose of determining the value of stock options and other forms of equity-based compensation for company executives.

QUESTION 6

August 2022 Question Two B (ii)

Distinguish between the following terms as used in finance:

- (ii) —“Intrinsic value” and —“market value”. (2 marks)

ANSWER

Distinction between “Intrinsic value” and “market value” as used in finance:

Intrinsic value refers to the present value of expected cash flows over the expected life of a security while market value refers to the price of a security as determined by forces of demand and supply.

QUESTION 7

August 2022 Question Two C

Horizon Construction Limited wishes to increase the number of its branches in the country.

The board of directors of the company has decided to finance the expansion programme by raising funds from the existing shareholders through a one for four rights issue.

The published income statement of the company for the year ended 31 December 2021 had the following information:

	Sh.“000”
Turnover	<u>250,000</u>
Profit before interest and tax	9,000
Interest	<u>(500)</u>
Profit before tax	8,500
Corporation tax	<u>(2,550)</u>
Profit after tax	5,950
Ordinary dividend	<u>(2,950)</u>
Retained profit for the year	<u>3,000</u>

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The share capital of the company comprises of 10 million ordinary shares which have a par-value of Sh.10 per share.

The shares of the company are currently being traded on the securities exchange with a price-earnings (P/E) ratio of 20 times.

The firm's board of directors have decided to issue the new shares at a 25% discount on the current market price.

Required:

- (i) The theoretical ex-right price of an ordinary share of the company. (2 marks)
- (ii) The theoretical value of each right. (1 mark)

Assuming an investor held 5,000 ordinary shares of the company before the rights issue announcement and Sh.15,000 in his savings account. Calculate the following options and identify the best option to the investor if he:

- (iii) Exercises all his rights. (3 marks)
- (iv) Sells all his rights. (2 marks)
- (v) Ignores the rights issue. (2 marks)

ANSWER

Horizon Construction Limited

- (i) The theoretical ex-right price of an ordinary share of the company.

$$\text{Number of shares} = 10,000,000 \div 10$$

$$= 1,000,000 \text{ shares}$$

$$\text{Earnings per share} = \text{profit after tax} \div \text{Number of shares}$$

$$= 5,950,000 \div 1,000,000$$

$$= \text{Sh } 5.95$$

$$\text{Market price} = \text{Earnings per share} \times \text{P/E Ratio}$$

$$= 5.95 \times 20 = \text{Sh } 119$$

$$\text{Right issue price} = \frac{75}{100} \times 119 = \text{Sh } 89.25$$

Theoretical ex right price

$$= \frac{(4 \times 119) + (1 \times 89.25)}{4+1}$$

$$= \text{Sh } 113.05$$

(ii) The theoretical value of each right.

$$\text{Value of R} = 119 - 113.05$$

$$= 5.95$$

(iii) Exercises all his rights.

$$\text{New shares} = \frac{5000}{4} \times 1$$

$$= 1,250 \text{ shares}$$

$$\text{Total number of shares} = 5000 + 1250 = 6,250 \text{ Shares}$$

Value of wealth

$$= (6,250 \times 113.05) + 15,000 - (1,250 \times 89.25)$$

$$= 610,000$$

Value of wealth before right issue

$$= (5,000 \times 119) + 1500 = 610,000/=$$

Therefore, exercising right does not change value of investor's wealth.

(iv) Sells all his rights.

$$= (5000 \times 113.05) + 15,000 + (5000 \times 5.95)$$

$$= 565,250 + 15,000 + 29,750 = \text{Sh } 610,000$$

Value of wealth remains unchanged

(v) Ignores the rights issue.

$$= (5,000 \times 113.05) + 15,000 = 580,250$$

Decline in wealth:

$$610,000 - 580,250 = 29,750$$

The decline in wealth is equivalent to the value of right issue

Advice

The investor should exercise his right or sell all his rights as failure to act will see the value of his wealth by reducing by Sh 29,750.

QUESTION 8

August 2022 Question Five C

Koki Ltd. recently paid a dividend of Sh.2.50 per share. The dividend is expected to grow at a rate of 15% per annum for the first three years, then at a rate of 10% per annum for the next 2 years after which the dividend will grow at a rate of 5% per annum to perpetuity. The required rate of return for the ordinary share is 12%.

Required:

The intrinsic value of the ordinary share of Koki Ltd. (5 marks)

ANSWER

The intrinsic value of the ordinary share of Koki Ltd.

Year	DPS	D.F 12%	PV
1	$2.5 \times 1.15 = 2.875$	0.8929	2.567
2	$2.5 \times 1.15^2 = 3.306$	0.7972	2.636
3	$2.5 \times 1.15^3 = 3.802$	0.7118	2.706
4	$3.802 \times 1.1 = 4.182$	0.6355	2.658
5	$3.802 \times 1.1^2 = 4.600$	0.5674	2.610
6 to ∞	$\frac{4.600(1+0.05)}{0.12-0.05} = 69$	0.5674	<u>39.151</u>
Intrinsic value			<u>52.328</u>

QUESTION 9**April 2022 Question One D**

You are on attachment in an investment firm and the following information regarding investments has been provided to you by a company whose earnings per share (EPS) in the current financial year was Sh.8. The company adopts a 60% payout ratio as its dividend policy.

The company is considering two investment options as follows:

Option 1:

The firm has an investment opportunity which, if undertaken, the growth rate in dividends will be 10% per annum for the next 3 years. The growth rate will fall to 8% per annum for the next 2 years after the first 3 years and then stabilise at 6% per annum thereafter.

Option2:

The firm can continue with the current investments and the growth in dividends will continue at the rate of 10% per annum in perpetuity.

Required:

Assuming the required rate of return is 18%, advise the company on the option to take.

(5 marks)

ANSWER**Option to take**

$$\text{Current DPS} = \frac{60}{100} \times 8 = \text{Sh. } 4.8$$

Option 1

Year	DPS	PVIF@18%	PV
1	$4.8 \times 1.1 = 5.28$	0.8475	4.47
2	$4.8 \times 1.1^2 = 5.81$	0.7182	4.17

3	$4.8 \times 1.1^3 = 6.39$	0.6086	3.89
4	$6.39 \times 1.08 = 6.90$	0.5158	3.56
5	$6.39 \times 1.08^2 = 7.45$	0.4371	3.26
6 to ∞	$\frac{7.45 \times 1.06}{0.18 - 0.06} = 65.81$	0.4371	<u>28.77</u>
			<u>48.12</u>

Option 2

$$P = \frac{D_0 (1+g)}{r-g}$$

Where:

D_0 = Dividend per share = 4.8

g = perpetual dividend growth rate 10% or 0.1

r = Required rate of return = 18%

$$p = \frac{4.8 (1.1)}{0.18 - 0.10}$$

$$= \frac{5.28}{0.08}$$

$$= \text{Sh. } 66$$

Advice:

Go for option 2 as it could result to the highest intrinsic value for the company.

QUESTION 10

December 2021 Question One D

The following data was extracted from the financial records of Faraja Ltd:

Earnings before interest and tax (EBIT)	Sh.400,000,000
Dividend payout ratio	40%
Number of outstanding ordinary shares	100,000,000
Equity capitalisation rate	15%
Return on investment (ROI)	12%
Corporate tax rate	30%

Required:

The value of Faraja Ltd.'s ordinary share using the Gordon model. (6 marks)

ANSWER

The value of Faraja Ltd.'s ordinary share using the Gordon model.

$$P_0 = \frac{D_0 (1+g)}{K_e - g}$$

$$DPS = \frac{\text{Total dividends}}{\text{No. of ordinary shares}}$$

Total ordinary dividends = DPR × PAT

$$\frac{\text{Profit after tax}}{EBIT}$$

<i>Profit after tax</i>	400,000,000
<i>EBIT</i>	
Less: Tax 30%	(120,000,000)
EAIT	<u>280,000,000</u>

$$\begin{aligned} \text{Total dividends} &= 40\% \times 280,000,000 \\ &= 112,000,000 \end{aligned}$$

$$DPS = \frac{112,000,000}{100,000,000}$$

$$1.12$$

Growth rate

$$\begin{aligned} g &= R_e \times \text{return on investment} \\ &= 60\% \times 12 = 7.2 \end{aligned}$$

$$P_0 = \frac{1.12 (1+0.072)}{0.15-0.072}$$

$$= 15.39$$

QUESTION 11

December 2021 Question Two C

The following data shows the returns of two stocks, A and B and their respective probabilities of occurrence:

Probability	Returns of stock A (%)	Returns of stock B (%)
0.30	10	8
0.30	7	12
0.40	13	7

Required:

- (i) The expected returns of stock A and stock B. (2 marks)
- (ii) The standard deviation of the returns of stock A and stock B. (3 marks)
- (iii) Suppose that a portfolio is created composed of 70% of stock A and 30% of stock B. Determine this portfolio's risk in terms of portfolio standard deviation. (5 marks)

ANSWER

(i) The expected returns of stock A and stock B.

$$\begin{aligned}\bar{A} &= (0.3 \times 10) + (0.3 \times 7) + (0.4 \times 13) \\ &= 3 + 2.1 + 5.2 \\ &= 10.3\%\end{aligned}$$

$$\begin{aligned}\bar{B} &= (0.3 \times 8) + (0.3 \times 12) + (0.4 \times 7) \\ &= 2.4 + 3.6 + 2.8 \\ &= 8.8 \%\end{aligned}$$

(ii) The standard deviation of the returns of stock A and stock B.

$$\begin{aligned}\delta_A &= \sqrt{0.3(10 - 10.3)^2 + 0.3(7 - 10.3)^2 + 0.4(13 - 10.3)^2} \\ &= \sqrt{0.027 + 3.267 + 2.916} \\ &= \sqrt{6.21} \\ &= 2.49 \%\end{aligned}$$

$$\delta_B = \sqrt{0.3(8 - 8.8)^2 + 0.3(12 - 8.8)^2 + 0.4(7 - 8.8)^2}$$

$$\begin{aligned}
 &= \sqrt{0.192 + 3.072 + 1.296} \\
 &= \sqrt{4.56} \\
 &= 2.14 \%
 \end{aligned}$$

(iii) Portfolio's risk in terms of portfolio standard deviation.

P	A	A – $\bar{A}$	B	B – $\bar{B}$	P(A – $\bar{A}$)(B – $\bar{B}$)
0.3	10	- 0.3	8	- 0.8	0.072
0.3	7	- 3.3	12	3.2	- 3.168
0.4	13	2.7	7	- 1.8	- 1.944
					- 5.04

$$\bar{A} = 10.3 \% \text{ while } \bar{B} = 8.8 \%$$

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Portfolio Risk

$$\begin{aligned}
 \delta_A &= \sqrt{W_A^2 \delta_A^2 + W_B^2 \delta_B^2 + 2 W_A W_B \text{Cov}_{AB}} \\
 &= \sqrt{0.7^2 \times 2.49^2 + 0.3^2 \times 2.14^2 + 2 \times 0.7 \times 0.3 \times -5.4} \\
 &= \sqrt{3.04 + 0.41 - 2.12} \\
 &= 1.15 \%
 \end{aligned}$$

QUESTION 12

September 2021 Question One C

Belta Limited issued a 16% corporate bond with a par value of Sh.1,000. The bond will either be redeemed at 20% premium after 5 years or convertible into equity at a conversion rate of Sh.100 per 2 ordinary shares. Each ordinary share is currently trading at the Securities Exchange for Sh.50. It is expected that the share price will increase at a constant rate of 5% each year.

The minimum required rate of return by investors is 12%.

Required:

- i. The current value of the redeemable bond. (3 marks)
- ii. The current value of the convertible bond. (3 marks)
- iii. Propose four advantages that would accrue to Belta Limited by using commercial papers as a source of finance. (4 marks)

ANSWER

Belta Limited

- i. **The current value of the redeemable bond.**

$$\text{Annual bond} = \frac{16}{100} \times 1,000 = 160$$

Year	Cashflows	DF 12%	PV
1 – 5	160	3.6048	576.768
5	1200	0.5874	<u>704.88</u>
			<u>1281.648</u>

- ii. **The current value of the convertible bond.**

Share price in five years' time

$$= 50 \times 1.05^5$$

$$= 50 \times 1.2763 = \text{sh } 63.81$$

Number of shares in exchange

$$\text{Sh } 100 = 2 \text{ shares}$$

$$\text{Sh } 1,000 = ?$$

$$= 20 \text{ shares}$$

Value of shares

$$= 20 \times 63.81$$

$$= \text{Sh } 1,276.20$$

Year	Cashflows	D.F 12%	PV
1 - 5	160	3.6048	576.768
5	1,276.20	0.5874	749.640
			1,326.408

iii. Advantages of using commercial papers as a source of finance.

- They are cheaper than bank loans
- It provides the exit option to investors to quit the investment
- It is quick and cost effective way of raising working capital
- Best way to the company to take advantage of short term interest fluctuations in the market
- It has a wide range of maturity
- It is unsecured and thus does not create any liens on assets of the company
- As commercial papers are required to be rated, good rating reduces the cost of capital for the company
- Alternative source of raising short term credit

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QUESTION 13

September 2021 Question Three B

Hakika Ltd. is a newly listed company in the local Securities Exchange. The company has 1 million ordinary shares trading at Sh.49.50 per share. To finance a restructuring exercise, the company requires Sh.7,855,000. To raise the amount, the company intends to issue a one for five rights issue at a subscription price of Sh.39 per share. The finance manager has projected that upon restructuring, the company's annual cash inflows would increase by Sh.965,000.

In the previous financial year, the company paid a dividend of Sh.5 per share. The dividend and the company's earnings are expected to grow by 5% annually upon restructuring.

Required:

- The price of the shares after the rights issue but before they start selling ex-rights.

(4 marks)

ii. The theoretical ex-rights price of the shares. (3 marks)

iii. The theoretical value of the rights when the shares are selling cum-rights.

(2 marks)

ANSWER

- i. **The price of the shares after the rights issue but before they start selling ex-rights.**

Cum right market price per share = NPV per share + market price per share before right issue

$$MPS = 49.5$$

$$NPV = PVCIF - PVCOF$$

$$PVCIF = \frac{\text{Annuity (A)}}{\text{Cost of equity (Ke)}}$$

Cost of equity

$$Ke = \left(\frac{Do(1+g)}{Po-f} + g \right) \times 100$$

$$Do = 5$$

$$g = 5\%$$

$$Po = 49.5$$

$$= \left(\frac{5(1+0.05)}{49.5-0} + 0.05 \right) \times 100$$

$$= 15.6\%$$

$$NPV = PVCIF - PVCOF$$

$$PVCIF = \frac{A}{Ke} = \frac{965,000}{0.156} = 6,185,897.44$$

$$\text{Less PVCOF} \quad (7,855,000)$$

$$NPV \quad (1,669,102.56)$$

$$NPV \text{ per share} = \frac{\text{Total NPV}}{\text{No. of ordinary share}} = \frac{(1,669,102.57)}{1,000,000} = -1.669$$

$$\begin{aligned}\text{CUM right MPS per share} &= 49.5 - 1.669 \\ &= 47.831\end{aligned}$$

ii. The theoretical ex-rights price of the shares.

$$P_x = \frac{(P_o \times S_o) + (P_s \times S)}{S_o + S}$$

Where:

$$P_o = \text{cum right MPS} = 47.83$$

$$S_o = \text{Existing number of shares} = 1,000$$

$$P_s = \text{subscription price} = 39$$

$$B = \frac{1}{5} \times 1,000,000 = 200,000$$

$$P_x = \frac{(47.83 \times 1,000,000) + (39 \times 200,000)}{1,000,000 + 200,000}$$

$$\begin{aligned}&= (47,830,000 + 7,800,000) \div 1,200,000 \\ &= \text{Sh } 46.36\end{aligned}$$

iii. The theoretical value of the rights when the shares are selling cum-rights.

$$\begin{aligned}R &= P_o - P_x \\ &= 47.83 - 46.36 \\ &= \text{Sh } 1.47\end{aligned}$$

QUESTION 14

September 2021 Question Four D

The following financial data relates to Penguin Limited for the year ended 31 December 2020:

Statement of financial position**As at 31 December 2020**

Assets:	Sh. "million"
Non-current assets	250
Current assets	<u>150</u>
	<u>400</u>
Financed by:	
Debt	140
Equity	<u>260</u>
	<u>400</u>

Statement of profit or loss**For the year ended 31 December 2020**

	Sh. "million"
Net sales	650
Cost of sales	<u>(150)</u>
Gross profit	500
Operating expenses	<u>(200)</u>
Operating profit	300
Interest expense	<u>(30)</u>
Profit before tax	270
Corporation tax	<u>(81)</u>
Profit after tax	<u>189</u>

Additional information:

1. The firm's cost of equity is 12%.
2. Corporation tax rate is 30%.
3. Interest rate on debt is 10%.

Required:

Compute the Economic Value Added (EVA) of the company. (5 marks)

ANSWER**Penguin Limited****Economic Value Added (EVA) of the company.**

Weighted average cost of capital

$$\begin{aligned} &= \frac{140}{400} \times (10\% \times 0.7) + \left(\frac{260}{400} \times 12\% \right) \\ &\quad \left(\frac{140}{400} \times 7\% \right) + \left(\frac{260}{400} \times 12\% \right) \\ &= 2.45\% + 7.80\% \\ &= 10.25\% \end{aligned}$$

EVA = NOPAT – (capital invested × WACC)

$$\begin{aligned} &= (300 (1 - 0.3)) - (400 \times 10.25\%) \\ &= 210 - 41 \\ &= \text{Sh } 169 \text{ million} \end{aligned}$$

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QUESTION 15**May 2021 Question One C**

- (i) Describe four reasons for valuing financial assets. (4 marks)
- (ii) Hazi Limited expects to pay a dividend of Sh.5.40 per share in one year's time.

Additional information:

1. The company's dividend payout ratio is 60%.
2. The shareholders' required rate of return on equity is 15%.
3. Dividends have been growing at a constant rate in perpetuity.
4. The company's shares are currently trading at Sh.64.50 per share at the Securities Exchange.

Required:

Advise an investor who holds shares in Hazi Limited whether to buy more shares or to sell the shares. (4 marks)

ANSWER**(i) Reasons for valuing financial assets.**

- When a company wishes to sell part of a business holding in a subsidiary valuation is necessary to determine the price to sell the shareholding in subsidiary.
- Where the company wishes to go public and intend to issue shares or debentures through the stock market.
- In determining the purchase consideration in times of business combination e.g. mergers and take-over's
- Determine the price at which securities can be sold or acquired
- To facilitate such securities being used as collateral in securing a loan e.g. shares

(ii) Hazi Limited

Advice on whether to buy more shares or to sell the shares.

$$g = \text{ROE} \times \text{Retention ratio}$$

$$= 15 \% \times (1-0.6)$$

$$= 15 \times 0.4$$

$$= 6\%$$

$$v = \frac{D_0 (1 + g)}{r - g}$$

Where g = growth rate = 6%

$$r = 15 \%$$

$$\frac{5.4}{0.15 - 0.06} = 5.4 \div 0.09 = sh. 60.00$$

Advice:

An investor should not buy shares of Hazi limited since they are selling above their share value of sh. 60.

QUESTION 16

May 2021 Question Five D

Ork Limited has an outstanding Sh.2 million face value bond with a 14% coupon rate and 3 years remaining until maturity.

Interest payments are made semi-annually.

Required:

The value of this bond assuming the nominal annual required rate of return is 12%.

(2 marks)

ANSWER

Ork Limited

The value of the bond

$$\begin{aligned} \text{Coupon semiannually} &= \frac{14\% \div 2}{100\%} \times 200,000 \\ &= \text{sh. } 140,000 \end{aligned}$$

Period	Cash flows	D.F 6%	PV
1-6	140,000	4.9173	688,422
6	2,000,000	0.7050	1,410,000
	Bond value		2,098,422

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QUESTION 17

November 2019 Question Three C

Fairland Industries Ltd. has recently been listed on the securities exchange.

The company has a policy of paying out a gradually increasing dividend per share over the past five years, as indicated below:

Year	Earnings per share (EPS)	Dividend per share (DPS)
	Sh.	Sh.
2014	118	5.0
2015	125	5.5

2016	146	6.0
2017	135	6.5
2018	160	7.3

Additional information:

1. The company has recently paid the dividends for the year ended 31 December 2018. The shares are therefore quoted ex-dividend.
2. The management is considering a change in the financing policy whereby greater financing will be provided from internally generated funds. This is expected to reduce the dividend per share to Sh.5 in the year ending 31 December 2019.
3. The growth rate in earnings per share (EPS) and dividend per share (DPS) is expected to increase to 14% per annum from the year ending 31 December 2019.
4. The company's shareholders require a minimum return on investment of 16%.

Required:

- (i) Using the dividend growth model, determine the market price per share (MPS) as at 31 December 2018 prior to the change in the financing policy. (3 marks)
- (ii) The market price per share (MPS) as at 31 December 2018 under the new financing policy. (2 marks)
- (iii) The break-even growth rate in dividend per share (DPS) using the market price calculated in (c) (i) above. (2 marks)

ANSWER

- i) Market price per share (MPS) using the dividend growth model as at 31 December 2018 prior to the change in the financing policy**

Intrinsic value before change in financing policy

$$g = \left[\left(\frac{7.3}{5.0} \right)^{\frac{1}{4}} - 1 \right] \times 100\%$$
$$= (1.0992 - 1) \times 100\% = 9.22\%$$

$$\text{Value of share} = \frac{D_0(1 + g)}{r - g}$$
$$= \frac{7.3(1.0992)}{0.16 - 0.0992}$$
$$= \text{Sh. } 131.98$$

C) ii) The market price per share (MPS) as at 31 December 2018 under the new financing policy.

$$\text{Value of share} = \frac{5}{0.16 - 0.14}$$
$$= \text{Shs. } 2.50$$

iii) Break-even growth rate in DPS

$$P_0 = \frac{133}{0.16 - g} = 5$$
$$133.83(0.16 - g) = 5$$
$$\text{Thus, } g = 0.1226$$
$$= 12\%$$

QUESTION 18

November 2019 Question Five B and C

(b) With reference to valuation of securities, explain the distinctive features of the following valuations:

(i) Conversion value.

(2 marks)

(ii) Liquidation value. (2 marks)

(iii) Market value. (2 marks)

(c) Virgin Industries had issued 72 million ordinary shares as at 31 March 2019. The company had maintained an annual dividend payment of Sh.180 million including for the year ended 31 March 2019.

On 3 April 2019, the management of the company was awarded a four year tender that would cost Sh.720 million. The directors decided to finance the tender by issuance of ordinary shares at par.

The return on investment (ROI) was expected to be 25% per annum on the cost over the next four years ending 31 March 2023.

All earnings would continue to be paid out as dividends to the shareholders

The cost of capital is 20%

Required

(i) The value of an ordinary share as at 31 March 2019

(ii) The value of the company as at 3 April 2019 assuming that the management made a decision to undertake investment

ANSWER

a) i) Conversion value

- The term conversion value refers to the financial worth of securities obtained by exchanging a convertible security for its underlying assets.
- Convertibles are a category of financial instruments such as convertible bonds and preferred shares, which can be exchanged for an underlying asset, such as common stock

b) ii) Liquidation value

- Is the amount of money that could be realized if an asset or group of assets are sold separately from the operating organization
- Liquidation value is usually lower than book value but greater than salvage value.
- Liquidation value can also mean the total worth of a company's physical assets if it were to go out of business and the assets are sold.
- Intangible assets are exceeded when calculating liquidation value.

b) iii) market value

The value of security prevailing in the market as determined by forces of demand and supply

c) The value of an ordinary share as at 31 March 2019.

$$\text{Value of share} = \frac{\text{Dividend per share}}{\text{cost of capital}}$$

$$\text{Dividend per share} = \text{Dividend paid} \div \text{Number of shares}$$

$$= 180 \text{ million} \div 72 \text{ million}$$

$$= \text{Sh } 2.5$$

Therefore;

$$\text{Value of share} = 2.5 \div 20\%$$

$$= \text{Sh. } 12.50$$

c) ii) The value of the company as at 3 April 2019 assuming that the management made a decision to undertake the investment

$$\text{Profit after tax} = \frac{25}{100} \times 720\,000\,000$$

$$= \text{Sh. } 180,000,000$$

Present value of dividends payable in four years

$$= 180,000,000 \times \frac{1 - 1.2^{-4}}{0.2}$$

$$= 180,000,000 \times 2.5887$$

$$= \text{Sh } 465,966,000$$

Current value of company as at 3 April 2019

$$= (72,000,000 \times 12.5) + 465,966,000$$

$$= \text{Sh } 1,365,966,000$$

QUESTION 19

May 2019 Question One B and C

(b) Clare Mwatata is planning to invest in a long-term project. An investment banker has provided her with the following two investment options:

Option 1: To invest in a corporate bond selling at Sh.875. The bond would be redeemed after 5 years at Sh. 1,600.

Option 2: To invest in a 16% debenture with a face value of Sh. 100 quoted at Sh.95.

The debenture would be redeemed after 5 years.

The minimum required rate of return is 18%.

Required:

Using the yield to maturity (YTM) valuation model, advise Clare Mwatata in making the investment decision.

(6 marks)

- (c) An investor holds 1,000 ordinary shares in Upendo Ltd., a company quoted at the securities exchange. The company has been paying average dividends of Sh. 1.50 per share annually in recent years.

The firm's dividends are expected to grow at a rate of 10% per annum in the coming three years, then grow at a rate of 8% per annum over the next two years and thereafter grow at a constant rate of 5% per annum into perpetuity.

The minimum required rate of return is 12%.

Required:

Using the discounted cash flow valuation method, determine the current value of the 1,000 ordinary shares of Upendo Ltd. . (6 marks)

ANSWER

B) Advising Clare Mwatata in making the investment decision using the yield to maturity (YTM) valuation model

Option 1

$$YTM = \left[\left(\frac{1,600}{875} \right)^{1/5} - 1 \right]$$

$$= (1.1283 - 1) \times 100\% \quad \quad \quad \mathbf{12.83\%}$$

Option 2

$$YTM = \left[\frac{1 + \frac{MV - MPS}{n}}{\frac{MV + MPS}{2}} \right] (1 - t) \times 100\%$$

Where;

$$I = \frac{16}{100} \times 100 = 16\%$$

$$n = 5 \text{ years}$$

$$MV = 100$$

$$MPS = 95$$

$$T = 0.3 = 30\%$$

$$YTM = \left[\frac{16 + \frac{100-95}{5}}{\frac{100+95}{2}} \right] \times (1 - 0.3) \times 100\%$$

$$= \frac{17}{97.5} \times 0.7 \times 100\%$$

$$= 12.21\%$$

He should not invest in all the options

C.

Year	Dividends	Discount factor 12%	Present value
1	$1.5 \times 1.1^1 = 1.65$	0.8929	1.4733
2	$1.5 \times 1.1^2 = 1.815$	0.7972	1.4469
3	$1.5 \times 1.1^3 = 1.9965$	0.7118	1.4211
4	$1.9965 \times 1.08^1 = 2.1562$	0.6355	1.3703
5	$1.9965 \times 1.08^2 = 2.3287$	0.5674	1.3213
6 to ∞	$\frac{2.3287(1.05)}{0.12-0.05} \times 34.9305$	0.5674	<u>19.8196</u>
			<u>26.8525</u>

Current value of 1,000 Upendo shares

$$= 1,000 \times 26,8525$$

$$= \text{Ksh. } 26,852.50$$

QUESTION 20

May 2019 Question Two B

- (i) The management of Amani Limited is considering listing at the securities exchange and intends to undertake valuation of its shares.

The following information is provided:

1. The current earnings per share (EPS) of the firm is Sh.4.
2. The firm has in issue 10 million ordinary shares with a 40% dividend payout ratio.
3. The firm has an equity capital of Sh.200 million with a minimum required rate of return of 18%.

Required:

The current theoretical value of the firm's ordinary shares using dividend growth model.

(6 marks)

- (ii) XYZ Ltd. Has a net tangible assets worth sh.48 million and a return on assets (ROA) of 12%.

The company expects to receive a profit of sh. 10 million annually for the next 5 years.

The company has 2 million outstanding ordinary shares.

Required:

The theoretical value per share using super-profit model.

(4 marks)

ANSWER

Amani limited

- i. Book value per share = equity capital ÷ number of shares
- $$= 200,000,000 \div 10,000,000$$
- $$= \text{Sh. } 20$$

Return on equity

$$\frac{\text{Earnings per share}}{\text{Book value per share}} \times 100$$

$$= \frac{4}{20} \times 100 = 20\%$$

$$\begin{aligned} \text{Sustainable growth rate (g)} &= \text{Return on equity (1- payout ratio)} \\ &= 20 (1-0.4) \\ &= 12\% \end{aligned}$$

$$\text{Value of share} = \frac{D_0 (1+g)}{r+g} \quad \text{Where, } D_0 = \frac{40}{100} \times 4 = 1.6$$

$$\text{Value of share} = \frac{1.6(1.12)}{0.18-0.12} = \text{Shs. } 29.87$$

$$\text{ii. Theoretical value of share (P}_0\text{)} = \frac{\text{Net tangible assets+goodwill}}{\text{Number of shares}}$$

Net tangible assets = 48 million

Ordinary shares = 2 million

Where Goodwill = super profits $\times$ number of years of purchase

$$\begin{aligned} \text{Super profits} &= 10,000,000 - \left(\frac{12}{100} \times 48,000,000 \right) \\ &= 4,240,000 \end{aligned}$$

$$\text{Goodwill} = 4,240,000 \times 5 = 21,200,000$$

$$P_0 = \frac{21,200,000 + 48,000,000}{2,000,000} = \text{Shs. } 34.60$$

Note: Super profits = expected profits – expected return on assets

QUESTION 21**November 2018 Question Two C**

The earnings per share (EPS) and dividend per share (DPS) data for Maraba Ltd. over the last five years are provided below:

Year	Dividend per share (DPS) (Sh.)	Earnings per share (EPS) (Sh.)
2013	1.00	2.50
2014	1.10	2.70
2015	1.20	3.00
2016	1.50	3.20
2017	1.80	3.50

Additional information:

1. A prospective investor is considering buying the shares of this company which are currently selling at Sh.55 each at the securities exchange.
2. The investor's required rate of return is 20%.

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Required:

Advise the investor on whether to buy the shares of Maraba Ltd. using Gordon's model.

(8 marks)

ANSWER

$$g = \left[\left(\frac{D_n}{D_0} \right)^{\frac{1}{n}} - 1 \right] \times 100$$

Where;

$$D_n = 1.8$$

$$D_0 = 1$$

$$n = 4 \text{ years}$$

$$= \left[\left(\frac{1.8}{1} \right)^{\frac{1}{4}} - 1 \right] \times 100$$

$$= (1.1583 - 1) \times 100$$

$$= 15.83\%$$

Theoretical value of shares

$$V = \frac{D_0(1 + g)}{r - g}$$

Where;

$$D_0 = 1.8$$

$$r = 20\%$$

$$g = 15.83\%$$

$$V = \frac{1.8(1 + 0.1583)}{0.2 - 0.1583}$$
$$= 50$$

Advice:

Investor should not buy shares of Maraba limited since they are over-valued. Current market price of 55 is more than the intrinsic value of 50 shillings

QUESTION 22

November 2018 Question Three C

Bemunyonge Ltd. has just paid a dividend of Sh.4 per share. The company expects that the dividend will grow at the rate of 20% per annum for the first two years, then grow at the rate of 10% per annum for the next 2 years and thereafter grow at the rate of 6% per annum into perpetuity. The required rate of return for the company is 16%.

Required:

The theoretical value of the company's share.

(4 marks)

ANSWER

The theoretical value of the company's share

Year	Dividend per share	Discount factor 16%	PV
1	$4 \times 1.2^1 = 4.8$	0.8621	4.14
2	$4 \times 1.2^2 = 5.76$	0.7432	4.28
3	$5.76 \times 1.1^1 = 6.336$	0.6407	4.06
4	$5.76 \times 1.2^2 = 6.970$	0.5523	3.85
> 4	$\frac{6.970(1.06)}{0.16 - 0.06} = 73.88$	0.5523	40.80
			Sh. 57.13

Theoretical value of company's shares = Sh. 57.13

QUESTION 23

November 2018 Question Five B

William Mgunya intends to invest Sh.200,000 in a redeemable 12%, Sh.100 debentures for 3 years. The current market value of the debentures is Sh.80 per debenture.

The required rate of return on the debentures is 10% per annum.

Required;

Advise William Mgunya on whether to invest in the debentures.

(4 marks)

ANSWER

$$\text{Annual interest paid} = \frac{12}{100} \times 100 = \text{Sh. } 12$$

Year	Cashflow	D.F 10%	P.V
1-3	12	2.4869	29.8428
3	100	0.7513	75.1300
			104.9728

Since current market price of sh.80 is less than intrinsic value of debenture of shs.104.9728, then William Mgunya should invest in the debentures.

QUESTION 24

May 2018 Question One B

Chauringo Limited wishes to expand its business. The company is considering to issue Sh.50 million worth of redeemable bonds denominated in Sh. 1,000. The bond's rate of interest is 10% and will mature on 30 June 2028.

The bonds will be issued on 1 July 2018.

The cost of capital is 18% per annum for the whole period.

Required:

- (i) The current value of the bond. (3 marks)
- (ii) The par value of the bond. (2 marks)

ANSWER

$$\text{Interest annually} = \frac{10}{100} \times 1000 = \text{Sh. } 100$$

Year	Cash flows	D.F 18%	PV
1-10	100	4.4941	449.41
10	1000	0.1911	191.10
			640.51

Current Bond value

$$\frac{50,000,000}{1,000} \times 640.51$$

$$= \text{Sh. } 32,025,500$$

- ii. Par value of bond Sh. 1000

QUESTION 25

May 2018 Question Two A

Explain three assumptions of the Gordon's dividend model.

(6 marks)

ANSWER

Assumptions of the Gordon's dividend model

- i. The company has stable financial leverage
- ii. The company's free cashflow is paid as dividends
- iii. The company grows at a constant, unchanging rate
- iv. The company's business model is stable i.e there are no significant changes in its operations
- v. The rate of return (r) and cost of capital is constant (k)
- vi. The life of a firm is indefinite
- vii. Cost of capital (k) is greater than rate of return r
- viii. Retention ratio once decided remains constant
- ix. The firm is an all equity financed company.

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QUESTION 26

May 2018 Question Three C

The profit after tax of Muhendato Ltd. as at 30 April 2017 was Sh.6,500,000. The company is quoted at the securities exchange and its shares are currently selling for Sh.40 each.

The company's dividend policy is to pay out 40% of its earnings for the year as dividends on its 1,000,000 issued and fully paid up shares.

The company's profit after tax is expected to increase by 15% per year for three years and 8% per year thereafter. Dividends will grow at the same rate as profits of the company.

The shares of the company are expected to sell at Sh.64 per share at the end of five years from now.

The cost of capital for the company is 12% per annum.

Required:

Determine whether the shares of Muhendato Ltd. are currently fairly valued, undervalued or overvalued for an investor expecting to sell them after 5 years. (6 marks)

ANSWER

Determining whether the shares of Muhendato Ltd. are currently fairly valued, undervalued or overvalued for an investor expecting to sell them after 5 years

$$\text{Dividends paid} = \frac{40}{100} \times 6,500,000$$

$$= \text{shs. } 2,600,000$$

$$\text{Dividends per share} = \frac{2,600,000}{1,000,000} = \text{Sh } 2.6$$

Year	Dividend per share	D.F 12%	PV
1	$2.6 \times 1.15^1 = 2.99$	0.8929	2.67
2	$2.6 \times 1.15^1 = 2.99$	0.7972	2.74
3	$2.6 \times 1.15^1 = 2.99$	0.7118	2.81
4 to ∞	$\frac{3.95(1.08)}{0.12 - 0.08} = 106.65$	0.7118	<u>75.91</u>

Sh. 84.13

Since the price of sh.65 five years from now is less than the intrinsic value of sh.84.13 then muhendato shares are undervalued

QUESTION 27

November 2017 Question Two A

Explain four limitations of dividend growth model.

(4 marks)

ANSWER

Limitations of dividend growth model

1. Only applies for dividend paying firms. Useless for non-dividend paying firms
2. Assumes that only a firm is financed by equity whereas a firm can finance its operations by debt too
3. Can't work if cost of capital r is less than growth rate (g) in dividends
4. Does not consider non-dividend factors such as brand loyalty, customer retention and the ownership of intangible assets all of which increase the value of the company

QUESTION 28

November 2017 Question Three A

The ordinary shares of Kwekwe Ltd. are currently selling at Sh.60 each at the securities exchange. The company's price-earnings ratio is 6 times.

Kwekwe Ltd. adopts a 40% pay-out ratio as its dividend policy.

It is predicted that the company's earnings and dividends will grow at an annual rate of 10% for the first three years, 5% for the next two years and 4% thereafter in perpetuity.

The investors' minimum required rate of return is 12%.

Required:

- (i) The current intrinsic value of the shares. (6 marks)
- (ii) Advise the investors based on the result obtained in (a) (i) above. (2 marks)

ANSWER

The current intrinsic value of the shares.

i. **Intrinsic value of the shares.**

$$P/E \text{ ratio} = \frac{MPS}{EPS}$$

Where;

EPS = earnings per share

$$MPS = P/E \text{ ratio} \times EPS$$

$$EPS = \frac{MPS}{PE \text{ ratio}}$$

Where:

MPS is market price per share = 60

$$\frac{60}{6} = \text{Sh. } 10$$

Dividend per share = dividend payout ratio × earnings per share

$$\frac{40}{100} \times 10 = \text{Sh. } 4$$

Year	Dividend per share	D.F 12%	P.V
1	$4 \times 1.1^1 = 4.4$	0.8929	3.93
2	$4 \times 1.1^2 = 4.84$	0.7972	3.86
3	$4 \times 1.1^3 = 5.32$	0.7118	3.79
4	$5.32 \times 1.1^1 = 5.59$	0.6355	3.55
5	$5.32 \times 1.1^2 = 5.87$	0.5674	3.33
6 to ∞	76.31	0.5674	43.30
Dividend to perpetuity			61.76

$$6 \text{ to } \infty = \frac{D(1 + g)}{r - g}$$

$$= \frac{5.87(1 + 0.04)}{0.12 - 0.04}$$

$$= \text{Sh. } 76.31$$

Current intrinsic value Sh.61.76

ii. Advice to the investors based on result above

Shares of Kweke Ltd. are undervalued since current market price per share of Sh.60 is less than the intrinsic value of Sh.61.76 so investors should buy shares of Kweke Ltd

QUESTION 29

November 2017 Question Five B

Deye Ltd has provided the following financial results:

Year	Profit after tax (Sh. "million")
2014	6.0
2015	6.2
2016	6.3
2017	6.3

The firm's earnings yield is 12%.

Required:

The value of the firm based on the present value of the expected earnings approach.

(4 marks)

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ANSWER

The value of the firm based on the present value of the expected earnings approach

$$g = \left[\left(\frac{\text{profit after tax latest}}{\text{profit after tax earliest}} \right)^{1/n} - 1 \right] \times 100\% = \left[\left(\frac{6.3}{6.0} \right)^{1/3} - 1 \right] \times 100\%$$

$$= (0.0164 - 1) \times 100\% = 1.64\%$$

Earnings approach

$$\text{Value of firm} = \frac{\text{Profit after tax} (1 - g)}{\text{Earning yield} - g}$$

$$= \frac{6,300,000 (1.0164)}{0.12 - 0.0164}$$

$$= \text{Sh. } 61,808,108.11$$

QUESTION 30**May 2017 Question Two A**

The following information was extracted from the financial statements of Mwaka Limited:

Earnings per share (EPS)	Sh. 15
Capitalisation rate	12%
Retention ratio	40%
Internal rate of return	16%

Required:

The price per share under:

- (i) Gordon's growth model. (4 marks)
- (ii) Walter's model. (4 marks)

ANSWER

The price per share under

i. Gordon's growth model

Growth rate = internal rate of return $\times$ retention ratio

$$16\% \times 0.4 = 6.4\%$$

$$V = \frac{D_0(1 + g)}{r - g}$$

Where, D_0 = dividend per share $(0.6 \times 15) = 9$

$g = 6.4\%$ is growth rate

r = capitalization rate = 12%

$$\frac{9(1.064)}{0.12 - 0.064} = \frac{9.576}{0.056} = \text{Sh. } 171$$

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ii. Walter's model

$$\text{Price per share} = \frac{D + \frac{r}{k}(E - D)}{k}$$

Where: P = market price per share

D = dividend per share = $(100\% - 40\%) \times 15 = \text{Sh. } 9$

E = earnings per share = Sh. 15

K = capitalization ratio = 12%

r = internal rate of return = 16%

$$P = \frac{9 + \frac{0.16}{0.12}(15 - 9)}{0.12}$$

$$= \frac{9 + 8}{0.12}$$

$$= \text{Sh. } 141.67$$

QUESTION 31**May 2017 Question Four B**

Luri Limited has a bond that has 3 years to maturity. The bond's par value is Sh.1,000. Coupon payment for the bond is made annually. The current market value of the bond is 120% of par with a coupon of 12%.

Required:

The yield to maturity (YTM).

(4 marks)

ANSWER**A. Luri limited yield to maturity (YTM).**

$$\text{YTM} = \left[\frac{I + \frac{MV - MPS}{n}}{\frac{MV + MPS}{2}} \right] (1 - T) \times 100\%$$

Where:

$$I = \text{interest rate paid} = \frac{12}{100} \times 1,000 = 120$$

$$T = \text{tax rate} = 30\%$$

$$MV = \text{maturity value} = 1,000$$

$$MPs = \text{market price per bond} = 1,200$$

$$n = \text{number of years to maturity} = 3$$

$$\text{YTM} = \left[\frac{120 + \frac{1,000 - 1,200}{3}}{\frac{1,000 + 1,200}{2}} \right] (1 - 0.3) \times 100\%$$

$$= 3.39\%$$

QUESTION 32

May 2017 Question Five B

Summarise four assumptions of the efficient market hypothesis (EMH). (4 marks)

ANSWER

B. Assumptions of EMH

1. There are no speculations in the market
2. The stock exchange treats all people same and the knowledgeable investors do not use their knowledge for malicious gains
3. There is no insider trading and all investors receive information at the same time
4. There is continuous trading of the securities
5. Information once released in the market is incorporated in determining the security prices
6. If transaction costs such as the commission paid to the brokers are minimized so as not to increase the prices of securities and discourage the investors from purchasing

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QUESTION 33

May 2017 Question Five D

Downtop Ltd. has achieved earnings of Sh.6 million this year and the company intends to pursue a policy of financing all its investment projects from retained earnings. There are a number of investment opportunities available for Downtop Ltd., although if it does not undertake any of the projects, its annual retained earnings are expected to remain at Sh.6 million in perpetuity.

The following information is available for Downtop Ltd.:

Proportion of retained earnings (%)	Growth rate in earnings (%)	Required return on all investments by shareholders (%)
0	0	16
30	6	17
45	9	19

Required:

Using dividend growth model, determine the optimum retention policy for Downtop Ltd.

(6 marks)

ANSWER

C. Optimum retention policy for Downtop Ltd

$$\text{Value of firm} = \frac{\text{dividends}(1 + g)}{r - g}$$

1. Proportion of retained earnings 0%

$$V = \frac{6,000,000}{0.16} = \text{Sh. } 37,500,000$$

2. Proportion of retained earnings 30%

$$\text{So dividends paid} = \frac{70}{100} \times 6,000,000 = \text{Sh. } 4,200,000$$

$$V = \frac{4,200(1.06)}{0.17 - 0.06} = \text{Sh. } 40,472,727.27$$

3. Proportion of retained earnings 34%

$$\text{Dividends paid} = \frac{55}{100} \times 6,000,000 = \text{Sh. } 3,300,000$$

$$V = \frac{3,300,000(1.09)}{0.17 - 0.06} = \text{Sh. } 31,800,000$$

The optimum retention policy is 30%

QUESTION 34

November 2016 Question Four A and B

(a) Bundacho Ltd generated Sh.50 million profits after-tax in the previous financial year.

The firm adopts 40% payout ratio as its dividend policy. The total numbers of issued ordinary shares are 10,000,000.

The company has a potential investment opportunity. If undertaken, dividends are expected to grow at the rate of 10% each year for the first 3 years and then stabilize at the rate of 5% each year thereafter in perpetuity.

The investor's minimum required rate is 18%

Required;-

The current intrinsic value of the share. (6 marks)

(b) A firm issued 10% preference shares to raise funds. The shares have a par value of Sh. 100 each and are currently selling at Sh. 110 each.

The minimum rate of return by the investor is 8%.

Required;-

Explain whether the share is overvalued or undervalued by the market. (4 marks)

ANSWER

a) Current intrinsic value of a share:

$$\begin{aligned} \text{Dividends} &= \text{Dividend payout ratio} \times \text{profit after tax} \\ &= \frac{40}{100} \times 50,000,000 = \text{sh. } 20,000,000 \end{aligned}$$

$$\begin{aligned} \text{Dividend per share (DPS)} &= \frac{\text{Dividends paid}}{\text{Issued ordinary shares}} \\ &= \frac{20,000,000}{10,000,000} = \text{sh. } 2 \text{ per share} \end{aligned}$$

Year	Future cash flows	PVFn,18%	Perfect value
1	$2(1.1) = 2.2$	0.8475	1.86
2	$2(1.1)^2 = 2.42$	0.7182	1.74
3	$2(1.1)^3 = 2.662$	0.6086	1.62
4(perpetuity (W1))	21.5008	0.6086	13.08
Intrinsic value			18.30

$$\text{Dividends to perpetuity} = \frac{D_0(1 + g)}{r - g}$$

$$= \frac{2.662(1.05)}{0.18 - 0.05} = 21.5008$$

(b) The current intrinsic value of the preference share:

We simply capitalize the preference dividends

$$K_p = \frac{\text{Preference dividend}}{\text{MPS of preference}}$$

$$\text{MPS of preference} = \frac{\text{Preference dividend}}{K_p}$$

Preference dividend = 10% x 100 = 10

Therefore MPS = $10 / 0.08 = 125$

The share is currently undervalued by the market since it is selling at a price of ksh. 110 which is lower than its current intrinsic value/theoretical/realistic value of ksh.125.

Prospective investors are advised to buy the shares of the company since they are underpriced by the market forces.

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QUESTION 35

November 2016 Question Five A and B

(a) Explain the following terms as used in the bond market:

(i) Yield-to-maturity (YTM). (2 marks)

(ii) Yield-to-call (YTC). (2 marks)

(b) Kaoyeni Limited has issued a Sh. 10,000 par value 10-year bond with a coupon rate of 12% per annum. The bond is currently trading at Sh.8,830 and is callable at Sh. 10,500 after 5 years.

The company pays interest on its bonds semi-annually.

Required:

- (i) Yield-to-maturity of the bond. (3 marks)
- (ii) Yield-to-call of the bond. (3 marks)

ANSWER**a) Explaining the following terms as used in the bond market****(i) Yield to maturity (YTM)**

- It's the total return expected to be received on a bond if it is held until maturity.
- It considers both the interest income and capital losses or gains on bond.

(ii) Yield to call (YTC)

- Total return anticipated on a bond if the bond is called before maturity takes into account total interest received plus capital gains or losses.

(b) Yield-to-maturity of the bond.**(i) Calculation for YTM**

To compute YTM we will compute it using trial and error method.

Par value =sh.10,000

$$\text{Coupon rate} = \frac{1290}{2} = 6\%$$

$$YTM = \frac{\text{interest} + \frac{1}{n}(PV - MV)}{\frac{1}{2}(PV - MV)}$$

Note: tax rate is not given

$$YTM = \frac{6\% \times 10,000 + \frac{1}{10}(10,000 - 8,830)}{\frac{1}{2}(10,000 - 8,830)} \times 100 = 0.0699 = 6.99\% \text{ semi annual}$$

$$= 13.98\% \text{ per annum}$$

(ii) Calculation of yield to call (YTC)

$$YTC = \frac{\text{interest} + \frac{1}{n}(PV - CV)}{\frac{1}{2}(PV - CV)}$$

$$YTC = \frac{6\% \times 10,000 + \frac{1}{10}(10,000 - 10,500)}{\frac{1}{2}(10,000 - 10,500)} \times 100 = 0.054 = 5.4\%$$

$$= 10.8\% \text{ per annum}$$

QUESTION 36

May 2016 Question One C

Chigiri Ltd. is a private company which intends to be listed in the securities exchange. The company has recently made a dividend issue of Sh.3.20 per share. This dividend is expected to grow at the rate of 15% per annum for 2 years and then drop to 12% per annum for the next 3 years. Thereafter, the dividend will grow at 6% per annum indefinitely. The required rate of return is 11%.

Required:

The intrinsic value of the share.

(6 marks)

ANSWER

Calculating the intrinsic value of the share

The intrinsic value of the share

Year	Dividends	PVIF,12%	PV
1	$3.20 \times 1.15^1 = 3.68$	0.9009	3.32
2	$3.20 \times 1.15^2 = 4.23$	0.8116	3.43

3	$4.23 \times 1.12^1 = 4.74$	0.7312	3.47
4	$4.23 \times 1.12^2 = 5.31$	0.6587	3.5
5	$4.23 \times 1.12^3 = 5.94$	0.5935	3.53
6-∞	125.93		<u>74.74</u>
	Intrinsic value		<u>91.99</u>

Dividends to perpetuity

$$6 - \infty = \frac{D_5(1 + g)}{r - g}$$

Where,

$$D_5 = \text{Dividend last year} = \text{sh. } 5.94$$

$$g = \text{Growth rate} = 6\%$$

$$r = \text{required rate of return} = 11\%$$

$$= \frac{5.94(1 + 0.06)}{0.11 - 0.06}$$

$$= \text{sh. } 125.93$$

QUESTION 37

May 2016 Question Two C

James Chiwende is considering the purchase of a 4-year Sh.1.200.000 par value bond.

The bond has a coupon interest rate of 10% per annum.

The investor's required rate of return is 8%.

Required:

The current value of the bond.

(3 marks)

ANSWER

Value of bond

$$\text{Annual coupon payment} = \frac{10}{100} \times 1,200,000$$

$$= \text{sh. } 120,000$$

Year	Cashflows	D.F 8%	PV
1-4	120,000	3.3121	397,452.00
4	1,200,000	0.7350	882,000.00
			1,279,452

Current value of the Bond= sh. 1,279,452

QUESTION 38

May 2016 Question Four C

Ngoba Ltd. has just paid an annual dividend of Sh.38 per share. The management of the company has a target to increase the market share value to Sh.800 per share by considering appropriate investment policies.

Shareholders expect a return on investment of 12%.

Required:

The annual expected growth rate.

(5 marks)

ANSWER

The annual expected growth rate

Use dividend valuation model

$$V_0 = \frac{D_0(1 + g)}{r - f}$$

Where

V_0 = Current value of the share

g = growth rate

r = expected rate of return (cost of equity) = 12%

D_0 = current dividend per share = sh.38

$$800 = \frac{38(1 + g)}{0.12 - g} \Rightarrow 800(0.12 - g) = 38(1 + g)$$

$$96 - 800g = 38 + 38g$$

$$838g = 58$$

$$g = \frac{58}{838} = 0.069 \approx 6.92\%$$

QUESTION 39**November 2015 Question One B, C and D**

(b) Explain the following theories in relation to valuation of financial assets:

(i) Fundamental theory. (3 marks)

(ii) Random walk theory. (3 marks)

(c) Ngatata Limited has issued a 20-year bond with a nominal value of Sh. 1,000 and a coupon annual rate of 9%. Coupon payments are made semi-annually in arrears. The yield to maturity of the bond is 12% per annum.

Required:

(i) The value of the bond. (3 marks)

(ii) The new value of the bond, if yield to maturity goes down to 8% per annum. (2 marks)

(d) Rematex Limited's earnings have been growing at the rate of 18% per annum. This growth is expected to continue for 4 years, after which the growth rate will fall to 12% per annum for another 4 years.

Thereafter, the growth rate is expected to be 6% in perpetuity. The company's last dividend paid was Sh.2. The investors' required rate of return on the company's equity is 15%.

Required:

The intrinsic value of the share.

(6 marks)

ANSWER

A. Valuation of financial assets theories

I. Fundamental theory

As per theory, the intrinsic value of a security is the total present value of all expected cash flows from security in form of dividends and interests.

$$\text{Value of security} = \frac{\text{Annuity}}{\text{required rate of return}}$$

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II. Random walk theory

Asserts that share prices are unpredictable. Value of shares will keep fluctuating in line with change in information as pertains that security

B. Ngatana Limited

I. Valuation of the bond

Coupon payments semi annual

$$\frac{4.5}{100} \times 1,000 = \text{Sh.4.5}$$

Year	Cash flows	D.F 6%	PV
1 – 40	45	15.0463	677.0835

40	1,000	0.0972	97.2000
Sh. 774.2835			

II. New value of the bond, if yield to maturity goes down to 8% per annul

Year	Cash flows	D.F 6%	PV
1 - 40	45	19.7928	890.676
40	1,000	0.2083	208.300
Sh. 1098.976			

C. Ramatex limited

Year	Dividend per share	D.F 15%	PV
1	$2 \times 1.18^1 = 2.36$	0.8696	2.05
2	$2 \times 1.18^2 = 2.78$	0.7561	2.11
3	$2 \times 1.18^3 = 3.29$	0.6575	2.16
4	$2 \times 1.18^4 = 3.88$	0.5718	2.22
5	$3.88 \times 1.12^1 = 4.35$	0.4972	2.16
6	$3.88 \times 1.12^2 = 4.87$	0.4323	2.11
7	$3.88 \times 1.12^3 = 5.45$	0.3759	2.05
8	$3.88 \times 1.12^4 = 6.11$	0.3269	2.00
9 to ∞	71.96	0.3269	23.52
Intrinsic value Sh. 40.38			

$$\text{Dividends to infinity} = \frac{D(1+g)}{r-g} = \frac{6.11(1.06)}{0.15-0.06} = 71.96$$

QUESTION 40

September 2015 Pilot Paper Question Four C

The shares of Bidii Ltd. are currently selling at Sh.60 each at the securities exchange. Bidii Ltd.'s price earnings ratio is 6 times. The company adopts a constant 40% payout ratio as its dividend policy. It is predicted that the company's dividends will grow at an

annual rate of 20% for the first three years, 15% for the next 2 years and thereafter at a constant rate of 10% per annum in perpetuity. The investor's minimum required rate of return is 12%.

Required:

- (i) Current intrinsic value of the shares of Bidii Ltd. (6 marks)
- (ii) Advise a prospective investor whether or not to buy shares of Bidii Ltd. (2 marks)

ANSWER

A. Bidii Ltd

i. Intrinsic value

$$\text{Earnings per share} = \frac{\text{Market price per share}}{\text{P/E ratio}} = 60 \div 6 = 10$$

$$\begin{aligned} \text{Then dividend per share} &= \text{dividend payout ratio} \times \text{earnings per share} \\ &= 10 \times 0.4 = \text{Sh. } 4 \end{aligned}$$

ii. Advice to prospective investor

Year	Dividend per share	D.F 12%	P.V
1	$4 \times 1.2^1 = 4.8$	0.8929	4.29
2	$4 \times 1.2^2 = 5.76$	0.7972	4.59
3	$4 \times 1.2^3 = 6.91$	0.7118	4.92
4	$6.91 \times 1.15^1 = 7.95$	0.6355	5.05
5	$6.91 \times 1.15^2 = 9.14$	0.5674	5.19
6 to ∞	502.7	0.5674	285.23
Intrinsic value			309.27
Dividends to perpetuity			

$$\frac{D_0(1+g)}{r-g} = \frac{9.14 (1.1)}{0.12-0.1} = \text{Sh. } 502.7$$

An investor should buy shares of Bidii Ltd since current market price per share is less than the intrinsic value of Sh. 309.27

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TOPIC 6

INTRODUCTION TO CAPITAL STRUCTURE DECISIONS

QUESTION 1

December 2023 Question Four B

The following is an extract of statement of financial position for Oak Timber Ltd. for the year ended 30 September 2023:

Oak Timber Ltd.

Extract of statement of financial position for the year ended 30 September 2023

Equity and liabilities: Equity:	Sh.“million”	Sh.“million”
Share capital	17	
Retained earnings	<u>15</u>	<u>32</u>
Total equity		32
Non-current liabilities:		
Long term borrowings	13	
Current liabilities	<u>21</u>	<u>34</u>
Total liabilities		<u>34</u>
Total equity and liabilities		<u>66</u>

Additional information:

- The share capital of Oak Timber Ltd. consists of Sh.12 million of ordinary shares and Sh.5 million, 5% irredeemable preference shares.
- The ordinary shares of Oak Timber Ltd. have a nominal value of Sh.0.50 per share, an ex-dividend market price of Sh.7.07 per share and a cum dividend market price of Sh.7.52 per share. The dividend for the year 2023 will be paid in the near future.

Dividends paid in recent years have been as follows:

Year	2022	2021	2020	2019
Dividend per share (Sh.)	0.43	0.41	0.39	0.37

3. The 5% irredeemable preference shares of Oak Timber Ltd. have a nominal value of Sh.0.50 per share and an ex-dividend market price of Sh.0.31 per share.
4. The long-term borrowings of the firm consist of Sh.10 million of 7% loan notes and Sh.3 million bank loan. The bank loan has a variable interest rate.
5. The 7% loan notes have a nominal value of Sh.100 per loan note and a market price of Sh.102.34 per loan note. Annual interest has just been paid and the loan notes are redeemable in four years' time at a rate of 5% premium to nominal value.
6. The corporation tax rate is 30%.

Required:

Compute the following:

- (i) Cost of equity. (3 marks)
- (ii) Cost of preference shares. (2 marks)
- (iii) The cost of 7% loan notes. (3 marks)
- (iv) After tax weighted average cost of capital (WACC) for the firm using market value. (4 marks)

ANSWER

- (i) **Cost of equity.**

$$R_e = \frac{D_1}{P_0} + g$$

Where:

R_e = Cost of Equity

D_1 = Dividends/share next year

P_0 = Current share price → Sh.7.07

g = Dividend growth rate → 5.02%

$$g = \left[\left(\frac{0.45}{0.37} \right)^{\frac{1}{4}} - 1 \right] \times 100\%$$

$$= 5.02\%$$

$$R_e = \left[\frac{0.45 \times 1.0502}{7.07} + 0.0502 \right] \times 100\%$$

$$= 11.70\%$$

(ii) **Cost of preference shares.**

$$R_p = \frac{D \text{ (dividend)}}{P_o \text{ (Price)}} \times 100\%$$

$$D \text{ (dividend)} / P_0 \text{ (price)} \times 100\%$$

$$D = \frac{5}{100} \times 0.50 = 0.025$$

$$= \frac{0.025}{0.31} \times 100\%$$

$$= 8.06\%$$

(iii) **The cost of 7% loan notes.**

$$K_d = \frac{I + \frac{M - P_b}{N}}{\frac{M + P_b}{2}} \times (1 - T) \times 100\%$$

I is the annual interest

M is the Redemption value of the debenture

P_b is the current market value of the debenture n is the period to maturity

$$\frac{7 + \frac{105 - 102.34}{4}}{\frac{105 + 102.34}{2}} \times (1 - 0.30) \times 100\%$$

$$\frac{7.665}{103.67} \times 0.7 \times 100\%$$

$$= 5.18\%$$

(iv) **After tax weighted average cost of capital (WACC)**

Source	Amount	Weight	Cost of Capital%	Weighted cost %
Ordinary	$12 / 0.5 \times 7.07$	$169.6868 / 186.0208$	11.70%	10.65%

shares	= 169.6868	= 0.91		
5% Preference shares	$5/0.50 \times 0.31$ = 3.10	$3.10/186.0208$ = 0.02	8.06%	0.16%
7% Loan note	$10/100 \times 102.34$ = 10.234	$10.234/186.0208$ = 0.06	5.18%	0.31%
Bank loan	3	$3/186.0208$ = 0.02	0.00%	0.00%
	186.0208		WACC	11.12%

QUESTION 2

August 2023 Question Four D

Using the information in (c) above, calculate the weighted average after tax cost of capital (WACC) of Zora Ltd. using market values. (6 marks)

Refer to August 2023 Question Four C

ANSWER

After tax cost of 8% bond

$$= 8 (1 - 0.3) = 5.59\%$$

Market value of ordinary shares

$$(120,000 \div 1) \times 8.50 = \text{Sh. } 1,020,000$$

Market value of bonds = Sh. 180,000

Market value of bank loan = Sh. 60,000

Total value of capital

$$= 1,020,000 + 180,000 + 60,000$$

$$= \text{Sh. } 1,260,000$$

Cost of debt

$$K_d = r(1 - T) = 7 (1 - 0.3) = 4.9\%$$

WACC

$$\begin{aligned} & \left(10 \times \frac{1020000}{1260000}\right) + \left(3.6 \times \frac{180000}{1260000}\right) + \left(4.9 + \frac{60000}{1260000}\right) \\ & = 8.10\% + 0.80\% + 0.23\% \\ & = 9.13\% \end{aligned}$$

QUESTION 3

August 2023 Question Five C

Penda Ltd. deals with laboratory accessories. A dropper sells for Sh.500 per piece and has a variable cost equivalent to 50% of the selling price per piece of the dropper. The firm has a fixed operating cost of Sh.500,000 and fixed financing cost of Sh.750,000.

Further analysis of the firm reveals that if the firm sales increase by 10%, the firm's earnings before interest and taxes (EBIT) increase by 15% and if the firm's EBIT increase by 10%, the firm's earnings per share (EPS) increases by 12%.

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Required:

Calculate the following measures of leverage for the firm:

- (i) Break-even quantity of sales in units. (2 marks)
- (ii) Operating break-even quantity of sales in units. (2 marks)
- (iii) Degree of operating leverage (DOL). (2 marks)
- (iv) Degree of financial leverage (DFL). (2 marks)
- (v) Degree of total leverage (DTL). (2 marks)

ANSWER

- (i) Break-even quantity of sales in units.

$$\text{Variable cost per unit} = \frac{50}{100} \times 500 = \text{Sh. } 250$$

$$\begin{aligned} \text{Contribution per unit} &= \text{selling price per unit} - \text{variable cost per unit} \\ &= 500 - 250 \end{aligned}$$

Sh. 250 per unit

$$\begin{aligned}\text{Break-even point in units} &= \frac{\text{Fixed cost} + \text{Fixed financing costs}}{\text{Contribution per unit}} \\ &= \frac{500000 + 750000}{250} \\ &= 5000 \text{ units}\end{aligned}$$

(ii) Operating break-even quantity of sales in units.

$$\begin{aligned}&\frac{\text{Fixed operating cost}}{\text{Contribution per unit}} \\ &= \frac{500000}{250} \\ &= 2000 \text{ units}\end{aligned}$$

(iii) Degree of operating leverage (DOL).

$$\begin{aligned}\text{DOL} &= \frac{\% \Delta \text{ in EBIT}}{\% \Delta \text{ in sales}} \\ &= \frac{0.15}{0.10} \\ &= 1.50\end{aligned}$$

This implies that if revenue was to increase by 10%, operating profit will increase by 15%

(iv) Degree of financial leverage (DFL).

$$\begin{aligned}\text{DFL} &= \frac{\% \text{ change in earnings per share}}{\% \text{ change in EBIT}} \\ &= \frac{12\%}{10\%} \\ &= 1.20\end{aligned}$$

This means if operating profit was to increase by 10%, EPS will increase by 12%

(v) Degree of total leverage (DTL).

$$\begin{aligned}\text{DTL} &= \text{DFL} \times \text{DOL} \\ &= 1.50 \times 1.20 \\ &= 1.80\end{aligned}$$

QUESTION 4

April 2023 Question Two B

The following information was extracted from the financial statements of Rembo Ltd. for the year ended 31 December 2022:

Capital structure	Amount Sh.“000”
Ordinary shares: 750,000 shares of Sh.20 each	15,000
Retained earnings	48,900
8% preference shares (Sh.100 par value)	5,000
10% debentures	7,500
	76,400

Additional information:

1. The preference shares were originally sold in 2014 at Sh.104 per share. The current price is Sh.94 although a similar issue can be made at Sh.89 net.
2. Ordinary shares are currently selling at Sh.38.40 on the securities exchange.
3. The debentures were sold in 2015 and realised Sh.96 per unit. The current price is Sh.80 and it is anticipated that a similar issue would also sell at Sh.80 per unit.
Corporate tax rate is 30%.
4. Last year's dividend amounted to Sh.3,000,000 which was 70% of the net earnings. The company expects these dividends to grow at a rate of 6% per annum and the dividend payout ratio to remain the same.
5. New ordinary shares can be sold at Sh.40 but in order to guarantee success, they would have to be sold at Sh.35 per share.

Required:

- (i) Cost of preference shares. (2 marks)
- (ii) Cost of ordinary shares. (2 marks)
- (iii) Cost of debentures. (2 marks)
- (iv) Market weighted average cost of capital (WACC) of the firm. (3 marks)

ANSWER**(i) Cost of preference shares.**

$$K_p = \frac{D_p}{P_o} \times 100\%$$

K_p is dividend per preference share

$$= \frac{8}{100} \times 100 = \text{Ksh } 8$$

P_o = Current price = Sh. 94

$$\begin{aligned} K_p &= \frac{8}{94} \times 100\% \\ &= 8.51\% \end{aligned}$$

(ii) Cost of ordinary shares.

$$K_s = \left(\frac{D_0 (1+g)}{P_o} \right) + g \times 100\%$$

D_0 = Dividend per ordinary share = Sh 3

g = Annual growth dividends = 6% or 0.06

P_o = Market price per share = Sh. 38.40

$$\begin{aligned} K_s &= \frac{3(1+0.06)}{38.40} + 0.06 \times 100\% \\ &= 14.28\% \end{aligned}$$

(iii) Cost of debentures.

$$K_d = \frac{I(1-T)}{P_o} \times 100\%$$

$$I = \frac{10}{100} \times 96 = 9.60$$

$$= \frac{9.6(1-0.3)}{80} \times 100\%$$

$$= \frac{6.72}{80} \times 100\%$$

$$= 8.40\%$$

(iv) **Market weighted average cost of capital (WACC) of the firm.**

Source	Amount “000”	Weight	Cost of capital %	Weights
Ordinary shares	$750 \times 38.4 = 28,800$	$\frac{28,800}{39,750} = 0.72$	14.28%	10.28
Preference shares	$\frac{5000}{100} \times 94 = 4700$	$\frac{4700}{39750} = 0.12$	8.51%	1.02%
Debentures	$\frac{7500}{96} \times 80 = 6250$	$\frac{6250}{39750} = 0.16$	8.40%	1.34%
	39,750		WACC	12.64%

QUESTION 5**December 2022 Question One B**

Highlight FOUR factors that influence the choice of debt finance by a company.

(4 marks)

ANSWER

Factors that influence the choice of debt finance by a company.

- **Lower cost** - Debt finance is often a lower cost form of financing than other forms of finance such as venture capital or angel investor financing. This can help a company to save money and remain competitive.
- **Leverage** - Debt finance is often used to leverage a company's existing funds, allowing it to borrow additional funds in order to expand operations or purchase new assets.
- **Tax efficiency** - Debt finance is often more tax efficient than equity finance, as interest payments are often tax deductible. This can help a company to reduce its overall tax burden.
- **Risk tolerance** - Companies may opt for debt finance if they are willing to take on more risk than with other forms of finance. Debt finance often carries higher interest rates and more stringent repayment terms than other forms of finance.

- **Availability of funds** - A company will opt for debt finance when it has limited funds available. This could be due to the company not having enough equity to finance its operations, or not having access to other forms of finance such as venture capital or angel investor financing.

QUESTION 6

December 2022 Question Five C

Kanga Ltd. expects earnings before interest and tax (EBIT) of Sh.7,500,000 in the current financial year. The company pays interest of 8% per annum on a long term loan of Sh.25,000,000.

The company has 1,200,000 ordinary shares and the corporate tax rate is 30%. The finance manager is currently examining two options:

- Option I:** A case where earnings before interest and tax (EBIT) is 20% more than expected.
- Option II:** A case where earnings before interest and tax (EBIT) is 20% less than expected.

Required:

- (i) Determine the earnings per share (EPS) under option I and option II and where there is no change in the expected earnings before interest and tax (EBIT).

(6 marks)

- (ii) Degree of financial gearing for option I and option II.

(4 marks)

ANSWER

Kanga Ltd.

- (i) **EPS under option I and option II and where there is no change in the expected earnings before interest and tax (EBIT).**

Details	Option I	Option II	No change
EBIT	9,000,000	6,000,000	7,500,000
Less: interest	<u>2,000,000</u>	<u>2,000,000</u>	<u>2,000,000</u>
Profit before tax	7,000,000	4,000,000	5,500,000
Less tax 30%	2,100,000	1,200,000	1,650,000
Net income	4,900,000	2,800,000	3,850,000
Number of shares	1,200,000	1,200,000	1,200,000
EPS	4.083	2.333	3.208

(ii) Degree of financial gearing for option I and option II.

$$= \text{EBIT} \div \text{EBT}$$

$$\text{OPTION I} = 9,000,000 \div 7,000,000 = 1.29$$

$$\text{OPTION II} = 6,000,000 \div 4,000,000 = 1.50$$

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QUESTION 7

August 2022 question One C

The following information relates to the capital structure of Tamu Caterers Limited for the year ended 31 December 2021:

Capital source	Current market value
	Sh.“000”
Corporate bond	11,927
Ordinary shares	26,170
Preference shares	7,203

Additional information:

1. The corporate bond has a Sh.1,000 face value, pays interest at a rate of 11% annually and will mature in 10 years' time. The bond is currently trading at Sh.1,125 at the securities market and its yield-to-maturity is 9.05%.
2. The ordinary shares paid a dividend of Sh.1.80 last year and each share is selling at Sh.27.50 at the securities market. The firm's dividend on ordinary shares is expected to grow at a rate of 7% per annum to perpetuity.
3. The firm's preference shares pays a 9% dividend on a Sh.100 par value.
4. The corporation tax rate is 30%.

Required:

- (i) The weighted average cost of capital (WACC) for the firm. (6 marks)
- (ii) Explain two factors that will determine the cost of capital for Tamu Caterers Limited. (2 marks)

ANSWER

Tamu Caterers Limited

(i) The WACC for the firm.

Source	Market value(000)	Weight
Corporate bond	11,927	0.26
Ordinary shares	26,170	0.58
Preference shares	7,203	0.16
	45,300	1.00

Cost of ordinary shares (KS)

$$K_S = \frac{D_0(1+g)}{P_0} + g \times 100\%$$

$$= \frac{1.8(1.07)}{27.5} + 0.07 \times 100\%$$

$$=14\%$$

Cost of cooperative bond (Kd)

$$K_d = \frac{Int + (P_v - P_o) \div n}{(P_v + P_o) \div 2} \times (I - T) \times 100\%$$

$$\frac{110 + (1000 - 1125) \div 10}{(1000 + 1125) \div 2} \times 0.7 \times 100\%$$

$$= \frac{97.5}{1,062.5} \times 0.7 \times 100\%$$

$$= 6.42\%$$

WACC

$$= (0.2 \times 6.42) + (0.58 \times 14) + (0.16 \times 9)$$

$$= 1.6692 + 8.12 + 1.44$$

$$= 11.23\%$$

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(ii) Factors that determine the cost of capital.

- **Risk:** Higher risk projects typically require a higher return to compensate investors for the potential loss.
- **Inflation:** Inflationary expectations can increase the cost of capital as investors demand a higher return to compensate for the loss of purchasing power.
- **Competition for Capital:** Companies that are in high demand for investment capital may have to offer higher returns to attract investors.
- **Availability of Capital:** The cost of capital may be affected by the availability of capital in the market, with a higher cost of capital during periods of tight credit.
- **Taxation:** The tax treatment of different forms of capital, such as debt and equity, can affect the cost of capital.

- **Company's Financial Performance:** A company's financial performance, including its profitability, growth potential, and stability, can affect the cost of capital.
- **Market conditions:** Market conditions, such as interest rates and stock market performance, can impact the cost of capital.

QUESTION 8

April 2022 Question Three C

The following financial information relates to Panda Ltd.:

Statement of financial position extract as at 31 December 2021

	Sh."000"	Sh."000"
Equity:		
Ordinary shares (each Sh.5 par value)	16,000	
Reserves	<u>72,000</u>	88,000
Long term liabilities:		
4% preference shares (each Sh.10 par value)	12,000	
7%, 6 year redeemable bonds	12,000	
Long term bank loan	<u>4,000</u>	<u>8,000</u>
		<u>116,000</u>

Additional information:

1. The ordinary shares of Panda Ltd. have an ex-dividend market value of Sh.47 per share and an ordinary dividend of Sh.3.63 per share has just been paid.

Historic dividend payments have been as follows:

Year	2018	2019	2020	2021
Dividend per share (Sh.)	3.09	3.22	3.36	3.50

2. The preference shares of Panda Ltd. are not redeemable and have an ex-dividend market value of Sh.4 per share.
3. The 7% bonds are redeemable at a 5% premium to their nominal value of Sh.100 per bond and have an ex-interest market value of Sh.104.50 per bond.
4. The bank loan has a variable interest rate that has averaged 4% per year in recent years.
5. The corporate tax rate applicable to Panda Ltd. is 30% per year.

Required:

The market value weighted average cost of capital (WACC) of Panda Ltd. (10 marks)

ANSWER

The market value weighted average cost of capital (WACC) of Panda Ltd.

$$\begin{aligned}
 g &= \left[\left(\frac{D_n}{D_0} \right)^{\frac{1}{n}} - 1 \right] \times 100 \\
 &= \left[\left(\frac{3.63}{3.09} \right)^{\frac{1}{4}} - 1 \right] \times 100 \\
 &= (1.0411 - 1) \times 100\% \\
 &= 4.11\%
 \end{aligned}$$

Cost of ordinary shares (K_o)

$$K_o = \frac{D_1}{P_0} + g \times 100\%$$

Where: D_1 = Dividend paid = Sh. 3.63

g = growth rate dividends = 4.11%

P_0 = market price per ordinary share = Sh. 47

$$\begin{aligned}
 &= \left[\frac{3.63}{47} + 0.0411 \right] \times 100\% \\
 &= (0.0772 + 0.0411) \times 100\% \\
 &= 11.83\%
 \end{aligned}$$

Cost of 4% preference shares

$$K_p = \frac{D_p}{P_o} \times 100\%$$

Where: D_p = Dividend per preference share = $\frac{4}{100} \times 10 = \text{Sh. } 0.4$

P_o = Market price per share = Sh. 4

$$K_p = \frac{0.4}{4} \times 100\% = 10\%$$

Cost of 7%, 6 year redeemable bond

$$K_{rd} = \frac{\text{Int} + \frac{P_v - M_v}{n}}{\frac{P_v + M_v}{2}} \times (1 - T) \times 100$$

Where: Int: = Interest = $\frac{7}{100} \times 100 = \text{Sh. } 7$

P_v = par value = Sh. 100

M_v = market price per bond = 104.50

n = years to maturity = 6

$$\begin{aligned} &= \frac{7 + \frac{100 - 104.50}{6}}{\frac{100 + 104.50}{2}} \times (1 - 0.3) \times 100\% \\ &= \frac{6.25}{102.25} \times 0.7 \times 100\% \\ &= 4.28\% \end{aligned}$$

Cost of long-term bank loan

$$\begin{aligned} K_l &= r (1 - T) \\ &= 4 (1 - 0.3) = 2.80\% \end{aligned}$$

Market value ordinary shares

$$\frac{16,000}{5} \times 47 = \text{Sh. } 150,400$$

Market value 4% preference shares

$$= \frac{12,000}{10} \times 4 = \text{Sh. } 4,800$$

Market value 7% bond

$$= \frac{12,000}{100} \times 104.50 = \text{Sh. } 12,450$$

Total value of financing sources

$$\begin{aligned} &= 150,400 + 4,800 + 12,450 + 4,000 \\ &= \text{Sh. } 171,650 \end{aligned}$$

WACC

$$\begin{aligned} &= \left(11.83 \times \frac{150,400}{171,650} \right) + \left(\frac{4,800}{171,650} \times 10 \right) + \left(4.28 \times \frac{12,450}{171,650} \right) + \left(2.80 \times \frac{4,000}{171,650} \right) \\ &= 10.37 + 0.28 + 0.31 + 0.07 \\ &= 11.03\% \end{aligned}$$

QUESTION 9

December 2021 Question Three A

Mavazi Ltd. is a firm that operates in the textile industry. Over the last 5 years, the firm has experienced stiff competition that has significantly reduced its turnover. In order to remain profitable, the firm's management is considering diversifying its operations. This activity will require additional financing of Sh.50 million.

The firm's existing capital structure is given as follows:

	Sh. "000"
Ordinary share capital (Sh.20 par value)	60,000
Reserves	10,000
11% Debentures (Sh.100 par value)	20,000
13% Preference share capital (Sh.15 par value)	<u>10,000</u>
	<u>100,000</u>

Additional information:

1. The firm's existing capital structure is considered to be optimal.
2. The firm expects to raise Sh.5 million from internal sources in order to finance this diversification activity.

However, new ordinary shares will be issued at Sh.32 per share and incur a floatation cost of Sh.2 per share.

3. The most recent dividend paid by the company was Sh.2.5 per share which is expected to grow at a constant rate of 5% per annum in perpetuity.
4. New 12% redeemable debentures will be issued at Sh.110 each. A floatation cost of Sh.15 per unit will be incurred. The par value of each unit is Sh.100 and the debentures will have a maturity period of 10 years.
5. New 14% irredeemable preference shares will be issued at Sh.90 per share. The par value of each irredeemable preference share is Sh.100.
6. The corporation tax rate applicable is 30%.

Required:

- (i) The cost of ordinary share capital. (2 marks)
- (ii) The cost of retained profit. (2 marks)
- (iii) After tax cost of new 12% debenture capital. (4 marks)
- (iv) Cost of new 14% preference share capital. (2 marks)
- (v) Weighted marginal cost of capital (WMCC). (4 marks)

ANSWER

Mavazi Ltd

- (i) **The cost of ordinary share capital.**

$$K_d = \frac{D_0 (1+g)}{P_0 - F} + g \times 100\%$$

$$= \frac{2.5 (1+0.05)}{32-2} + 0.05 \times 100\%$$

$$= (0.0875 + 0.05) \times 100\%$$

$$= 13.75\%$$

(ii) The cost of retained profit.

$$K_r = \frac{D_o (1+g)}{P_o} + g \times 100\%$$

$$= \frac{2.5 (1+ 0.05)}{32} + 0.05 \times 100\%$$

$$= (0.082 + 0.05) \times 100\%$$

$$= 13.20\%$$

(iii) After tax cost of new 12% debenture capital.

$$K_d = \frac{Int + (P_v - P_o) \div n}{(P_v + P_o) \div 2} \times (1 - T) \times 100\%$$

Where: Int = Interest = $12/100 \times 100 = 12$

P_v = Par Value = Sh.100

P_o = Market price = Sh. 110 – 15

= Sh. 95

N = Maturity period = 10years

$$K_d = \frac{12 + (100 - 95) \div 10}{(100 + 95) \div 2} \times (1 - 0.3) \times 100\%$$

$$\frac{12 + 0.5}{97.5} \times 0.7 \times 100\%$$

$$= 8.97\%$$

(iv) Cost of new 14% preference share capital.

$$K_p = \frac{D_p}{P_o} \times 100\%$$

$$= \frac{0.14}{90} \times 100\%$$

$$= 15.56\%$$

(v) Weighted marginal cost of capital (WMCC).

Source	Proportion capital structure
Equity	$\frac{60,000}{100,000} \times 100\% = 60\%$
11% debentures	$\frac{20,000}{100,000} \times 100\% = 20\%$
13% per shares	$\frac{10,000}{100,000} \times 100\% = 10\%$
Reserves	$\frac{10,000}{100,000} \times 100\% = 10\%$

Amount to be raised from financing sources

1. 12 % debentures = $0.2 \times 50M = 10M$
2. 14% preference shares = $0.1 \times 50M = 5M$

- 3. Retained earnings = 5M
- 4. Ordinary share capital $0.6 \times 50M = 30M$

Total = 50M

Weighted marginal cost of capital

$$\begin{aligned} &= (30/50 \times 13.75) + (5/50 \times 13.2) + (10/50 \times 8.97) + (5/50 \times 15.56) \\ &= 8.25 + 1.32 + 1.794 + 1.556 \\ &= 12.92\% \end{aligned}$$

QUESTION 10

September 2021 Question Five B

The management of Biashara Ltd. are in the process of determining the optimal capital budget of the company for the year ending 31 December 2021.

The following information is available:

1. The profit after tax for the year ending 31 December 2021 is estimated to be Sh.22,500,000.
2. The retention ratio is 60%.
3. The ordinary shares of the company are currently trading on the securities exchange at Sh.80 per share.
4. Ordinary shareholders expect a dividend of Sh.6 per share for the year ending 31 December 2021.
5. The annual growth rate in dividend is 6% per annum.
6. Floatation costs amounts to Sh.8 per share issued.
7. The company could issue an unlimited number of 11% preference shares at Sh.96 per share. The par value is Sh.100.

8. The company could obtain a bank loan of upto Sh.24,000,000 at a pre-tax interest rate of 10% per annum.

Thereafter, an unlimited amount of bonds could be issued under the following terms:

- Coupon interest rate of 12% per annum.
 - Par value at Sh.1,000 per bond.
 - Discount of Sh.30 per bond.
 - Floatation cost of Sh.20 per bond.
 - Maturity period of ten years.
9. The optimal capital structure of the company comprises 15% debt, 40% preference shares capital and 45% equity.
10. Corporate tax rate is 30%.

Required:

- (i) The cost of capital for each source of finance available to Biashara Ltd. (6 marks)
- (ii) The break-point(s) in the marginal cost of capital (MCC) schedule with respect to retained earnings and debt. (4 marks)
- (iii) The marginal cost of capital (MCC) at each break-point identified in (b) (ii) above. (4 marks)

ANSWER

The management of Biashara Ltd.

- (i) The cost of capital for each source of finance available to Biashara Ltd.**

Cost of retained earnings (Kr)

$$K_r = \frac{D_1}{P_o} \times g \times 100\%$$

$$D_1 = \text{Expected dividend} = \text{Sh.6}$$

$$g = \text{growth rate dividends} = 6\% \text{ or } 0.06$$

$$P_o = \text{market price per share} = \text{Sh. 80}$$

$$\begin{aligned} &= \frac{6}{80} + 0.06 \times 100\% \\ &= (0.075 + 0.06) 100\% \\ &= 13.5\% \end{aligned}$$

Cost of ordinary shares (Ko)

$$K_o = \frac{D_1}{P-F} + g \times 100\%$$

Where:

D1 = Dividend per share = Sh. 6

P = Market price per share = Sh. 80

F = Floatation cost per share = Sh 8

g = Growth rate = 6% or 0.06

$$= \left(\frac{6}{80-8} + 0.06 \right) \times 100\% \\ = 14.33\%$$

Cost of unlimited 11% preference shares

$$K_p = \frac{D_p}{P_o} \times 100\%$$

Where:

Dp = Dividend per share = $\frac{11}{100} \times 100 = \text{Sh. } 11$

Po = Market price per share = Sh. 96

$$K_p = \frac{11}{96} \times 100\% = 11.46\%$$

Cost of bank loan (Kl)

$$K_l = r(1-T) \\ = 10(1-0.3) \\ = 7\%$$

Cost of unlimited bond

$$K_d = \frac{Int + (PV-MV) \div n}{(PV+MV) \div 2} \times (1-T) \times 100$$

Where:

Int = interest = $\frac{12}{100} \times 1000 = 120$

Pv = Par value = Sh 1,000

Mv = market value less discount, less floatation cost $1,000 - 30 - 20 = 950$

n = maturity period = 10 years

T = tax rate = 30%

$$\begin{aligned}
 K_d &= \frac{120 + (1000 - 950) \div 10}{(1000 + 950) \div 2} \times 0.7 \times 100 \\
 &= \frac{125}{975} \times 0.7 \times 100\% \\
 &= 8.97\%
 \end{aligned}$$

(ii) The break-point(s) in the marginal cost of capital (MCC) schedule with respect to retained earnings and debt.

$$\begin{aligned}
 \text{Breakpoint} &= \frac{0.4 \times 22,500,000}{0.45} \\
 &= \text{Sh. } 20,000,000
 \end{aligned}$$

Breakpoint debt 0 – 24,000,000

$$\text{Breakpoint} = \frac{24,000,000}{0.15} = \text{Sh. } 160,000,000$$

Breakpoint debt > 24,000,000

$$= \frac{\infty}{0.15} = \infty$$

(iii) The marginal cost of capital (MCC) at each break-point identified in (b) (ii) above.

Range	Debt 15%	Preference shares	Equity
0 – 20 million	7%	11.46%	13.5%
20 – 160million	8.97%	11.46%	14.33%
>160million	8.87%	11.46%	14.33%

$$\begin{aligned}
 \text{MCC} &= (0.15 \times 7) + (0.4 \times 11.46) + (0.45 \times 13.5) \\
 &= 1.05 + 4.584 + 6.075 \\
 &= 11.709\%
 \end{aligned}$$

$$\begin{aligned}
 \text{MCC2} &= (0.15 \times 8.97) + (0.4 \times 11.46) + (0.45 \times 14.33) \\
 &= 1.346 + 4.584 + 6.449
 \end{aligned}$$

$$= 12.379\%$$

QUESTION 11

May 2021 Question Three B

The following are extracts from Riziki Ltd.'s statement of financial position as at 30 March 2021:

Book values	Sh. "Million"
Ordinary shares (Sh.50 par value)	9,600
6% preference shares (Sh.100 par value)	7,900
4.8% debenture (Sh.100 par value)	6,400
	23,900

Additional information:

1. The ordinary shares of Riziki Ltd. are currently quoted at Sh.72 per share (cum dividend).
2. The most recently declared dividend was Sh.2 per share and will be paid in a years' time. The dividend growth rate is 5%.
3. The dividend will continue to grow at the rate of 5% into the foreseeable future.
4. The preference shares currently trade at Sh. 80 per share. There is no preference dividend owing at this point in time.
5. The debentures are irredeemable and currently trade at 120% of their nominal value.
6. The corporation tax rate is 30%.

Required:

- i. The cost of capital for each source of finance for Riziki Ltd. (4 marks)
- ii. The weighted average cost of capital (WACC) for Riziki Ltd. (4 marks)

ANSWER**Riziki Ltd****i. The cost of capital for each source of finance for Riziki Ltd.**

Cost of ordinary shares

$$K_e = \frac{D_0 (1 + g)}{p_0} + g \times 100\%$$

Where: D_0 = dividend per share = 2

g = growth rate dividends = 5 %

p_0 = market price per share ex dividend = 72 - 2 = 70

$$\begin{aligned} K_e &= \frac{2(1.05)}{70} + 0.05 \times 100\% \\ &= \frac{2.1}{70} + 0.05 \times 100\% \\ &= 8\% \end{aligned}$$

Cost of 6% preference shares

$$k_p = \frac{D_p}{P_0} \times 100\%$$

Where: D_p = Dividend per preference shares

$$\frac{6}{100} \times 100 = sh. 6$$

P_0 = market price per share = sh. 80

$$= \frac{6}{80} \times 100\% = 7.50\%$$

Cost of 4.8% debentures

$$K_d = \frac{Int(1 - T)}{P_0} \times 100\%$$

Where: Int - interest = $\frac{4.8}{100} \times 100 = 4.8$

T – Corporate tax rate = 30% or 0.30

P_0 = market price per debenture

$$\begin{aligned} \frac{120}{100} \times 100 &= sh. 120 \\ k_d &= \frac{4.8(1 - 0.3)}{120} \times 100\% \end{aligned}$$

$$= \frac{3.36}{120} \times 100\% = 2.80\%$$

Market value of ordinary shares

$$= \frac{9600}{50} \times 70 = 13,440 \text{ million}$$

Market value of 6% preference shares

$$= \frac{7900}{100} \times 80 = 6,320 \text{ million}$$

Market value of 4.8% debentures

$$\frac{6,400}{100} \times 120 = 7,680 \text{ million}$$

ii. The weighted average cost of capital (WACC) for Riziki Ltd.

Source	Amount	Weight	Cost of capital %	Weighted cost %
Ordinary shares	13,440	$\frac{13,440}{27,440}$ = 0.49	8.0%	3.920%
6% preference shares	6,320	0.23	7.5%	1.725%
4.8% debentures	7,680	0.28	2.8%	<u>0.784%</u>
	27,440		WACC	<u>6.429%</u>

QUESTION 12

November 2020 Question Three B

The following information was extracted from the books of Domingo General Merchants Ltd.:

Statement of financial position as at 31 December 2019:

	Sh."000"	Sh."000"
Non-current assets		10,115
Investments		821
Current assets	3,658	
Less: Current liabilities	<u>(1,735)</u>	<u>1,923</u>
Total assets		<u>12,859</u>
Financed by:		
Ordinary share capital: 3,000,000 shares each Sh.1		3,000
General reserves		<u>7,125</u>
Shareholders' funds		10,125
7% Bonds		1,300
Corporation taxation		<u>1,434</u>
Total equity and liabilities		<u>12,859</u>

Summary of profits and dividends:

Year ended 31 December	2015	2016	2017	2018	2019
	Sh."000"	Sh."000"	Sh."000"	Sh."000"	Sh."000"
Profit after interest and before tax	<u>1,737</u>	<u>2,090</u>	<u>1,940</u>	<u>1,866</u>	<u>2,179</u>
Less: Tax	(573)	(690)	(640)	(616)	(719)
Profit after interest and tax	1,164	1,400	1,300	1,250	1,460
Less: Dividends	<u>620</u>	<u>680</u>	<u>740</u>	<u>740</u>	<u>810</u>
Retained earnings	<u>544</u>	<u>720</u>	<u>560</u>	<u>510</u>	<u>650</u>

Additional information:

1. The current (1 January 2020) market value of ordinary shares is Sh.3 per share ex div.
2. The bonds are redeemable at par in ten years' time.
3. The current market value of the bonds is Sh.77.10 per Sh.100 of nominal value and the annual interest has just been paid on the bonds.
4. There have been no issues or redemptions of ordinary shares or bonds during the past five years.
5. The corporate tax rate is 30%. Assume that there have been no changes in corporate tax rate for the past five years.

Required:

The weighted average cost of capital (WACC) that the company should use as a discount rate when appraising new investment opportunities. (12 marks)

ANSWER

- a) The weighted average cost of capital (WACC) that the company should use as a discount rate when appraising new investment opportunities.**

$$\begin{aligned}
 \text{Growth rate of dividends} &= \left[\left(\frac{D_n}{D_o} \right)^{1/n} - 1 \right] \times 100\% \\
 g &= \left[\left(\frac{810,000}{620,000} \right)^{1/4} - 1 \right] \times 100 \\
 &= \left(1.3065^{1/4} - 1 \right) \times 100\% \\
 &= (1.0691 - 1) \times 100\% \\
 g &= 6.91\%
 \end{aligned}$$

$$\text{Dividend per share 2019} = \frac{\text{Dividends paid}}{\text{Number of shares}} = \frac{810,000}{3,000,000} = \text{Sh. } 0.27$$

Cost of ordinary shares (k_e)

$$K_e = \frac{D_o(1+g)}{P_o} + g \times 100\%$$

Where:

$$D_o = \text{Dividend paid} = \text{Sh. } 0.27$$

$$g = \text{growth rate dividends} = 6.91\%$$

$$P_o = \text{Market price per share} = \text{Sh. } 3$$

$$\begin{aligned} K_e &= \frac{0.27 (1.0691)}{3} + 0.0691 \times 100\% \\ &= (0.096219 + 0.0691) \times 100\% \\ &= 16.53\% \end{aligned}$$

Cost of 7% bond redeemable

$$K_{rd} = \left(\frac{I + \frac{P_v - P_o}{n}}{\frac{P_v + P_o}{2}} \right) (1 - T) \times 100\%$$

Where:

$$I = \text{interest paid} = \frac{7}{100} \times 100 = \text{Sh. } 7$$

$$P_v = \text{maturity value current} = \text{Sh. } 100$$

$$P_o = \text{market value current} = \text{Sh. } 770$$

$$n = \text{period to maturity} = 10 \text{ years}$$

$$T = \text{corporate tax rate} = 30\%$$

$$K_{rd} = \frac{7 + \frac{100 - 77.10}{10}}{(100 + 77.10) \div 2} \times (1 - 0.3) \times 100\%$$

$$= \frac{7+2.29}{88.55} \times 0.7 \times 100\%$$

$$= 7.34\%$$

Market value of ordinary shares = 3,000,000 × 3 = Sh 9,000,000

Market value of bond

$$= \frac{1,300,000}{100} \times 77.10$$

$$= \text{Sh } 1,002,300$$

Weighted average cost of capital

Source	Amount	Weight	Cost of capital	Weighted cost
Ordinary shares	9,000,000	1-0.1 = 0.9	16.53%	14.877
7% bond	1,002,300	$\frac{1,002,300}{10,002,300} = 0.1$	7.34%	0.734%
	10,002,300		WACC	15.611%

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QUESTION 13

November 2019 Question One C

The current capital structure of Ahadi Ltd. is given as follows:

	Sh."000"
Ordinary shares (Sh.10 each)	30,000
10% debentures	15,000
12% preference shares (Sh.20 each)	<u>5,000</u>
	<u>50,000</u>

Additional information:

1. The current market value of ordinary shares and preference shares is Sh.50 and Sh.30 respectively.

2. The debentures are irredeemable and have a market value of Sh.120 per Sh.100 nominal value.
3. The most recent earnings per share (EPS) of the company is Sh.6.
4. The company currently adopts a 60% dividend payout ratio as its dividend policy. However, the firm's future dividends are expected to grow at a rate of 7% each year Aar the-foreseeable future.
5. Corporate tax rate is 30%.

Required:

The company's weighted average cost of capital (WACC) using market value weights.

(7 marks)

ANSWER

The company's weighted average cost of capital (WACC) using market value weights

Cost of ordinary shares (K_e)

$$K_e = \frac{D_0(1 + g)}{P_0} + g \times 100\%$$

$$\text{Where } D_0 \rightarrow \text{Dividend per share} = \frac{60}{100} \times 6 = 3.6$$

$$g \rightarrow \text{growth rate dividends} = 7\%$$

$$P_0 \rightarrow \text{market price per share} = \text{Sh. } 50$$

Therefore;

$$\begin{aligned} K_e &= \frac{3.6(1.07)}{50} + 0.07 \times 100\% \\ &= (0.07704 + 0.07) \times 100\% \end{aligned}$$

$$= 14.7\%$$

Cost of preference shares (K_p)

$$K_p = \frac{D_p}{P_0} \times 100\%$$

Where; $D_p \rightarrow$ Dividend per preference share

$$= \frac{12}{100} \times 20 = 2.4$$

$P_0 \rightarrow$ market price per share = Sh. 30

Therefore,

$$K_p = \frac{2.4}{30} \times 100\% = 8\%$$

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Cost of 10% debentures (K_d)

$$K_d = \frac{I(1 - T)}{P_0} \times 100\%$$

Where : $I \rightarrow$ interest = $\frac{10}{100} \times 100 = \text{Sh. } 10$

$T \rightarrow$ Tax rate = 30% or 0.3

$P_0 \rightarrow$ market price per debenture = 120

Therefore;

$$\begin{aligned} K_d &= \frac{10(1 - 0.3)}{120} \times 100\% \\ &= 5.83\% \end{aligned}$$

$$\begin{aligned}\text{Number of ordinary shares} &= \text{Ordinary share capital} \div \text{Par value} \\ &= 30,000,000 \div 10 = 3,000,000\end{aligned}$$

$$\begin{aligned}\text{Market value ordinary shares} &= 3,000,000 \times 50 \\ &= \text{Sh. } 150,000,000\end{aligned}$$

$$\text{Number of debentures} = 15,000,000 \div 100 = 150,000$$

$$\text{Market value 10\% debentures } 150,000 \times 120 = 18,000,000$$

$$\text{Number of 12\% preference shares} = 5,000,000 \div 20 = 250,000$$

$$\text{Market value} = 250,000 \times 30 = 7,500,000$$

Source	Amount (000)	Weight	Cost of capital %	Weighted cost %
Ordinary shares	150,000	0.85	14.7	12.50%
10% debentures	18,000	0.10	5.83	0.58%
12% preference shares	<u>7,500</u>	0.05	8.00	0.40%
	<u>175,500</u>		WACC	13.48%

QUESTION 14

November 2019 Question Five B

Riverside Ltd. requires Sh.4,500,000 to finance an upcoming project. The firm's existing share capital constitutes 120,000 ordinary shares whose current market price per share is Sh.100. The management of the company has proposed to raise funds through a rights offering at a discount rate of 25% on current share price.

Required:

- The number of ordinary shares to be issued to raise the required capital. (2 marks)
- The number of rights required to subscribe for one new ordinary share. (2 marks)

(iii) The theoretical ex-right market price per ordinary share.

(2 marks)

ANSWER

i) The number of ordinary shares to be issued to raise the required capital

Subscription price right issue price

$$= \frac{75}{100} \times 100 = \text{Sh. } 75$$

Number of shares issued

$$= \text{initial outlay} \div \text{subscription price}$$

$$= 4,500,000 \div 75$$

$$60,000 \text{ ordinary shares.}$$

b) ii) The number of rights required to subscribe for one new ordinary share

$$120,000 \text{ ordinary shares} = 60,000 \text{ rights issue shares}$$

$$\text{Ordinary share} = ?$$

$$\text{Therefore; } \frac{2}{120000} \times 60000 = 0.5$$

Therefore 1 ordinary share is equivalent to 0.5 right issue shares

b) iii) The theoretical ex-right market price per ordinary share

$$= \frac{(P_0 + S_0) + (P_s \times S)}{S_0 + S}$$

$P_0 \rightarrow$ Cum right market price per share = Sh 100

$S_0 \rightarrow$ Existing number of shares

$P_s \rightarrow$ Offer price

$S \rightarrow$ New issued shares

Therefore;

$$P_x = \frac{1 \times 100 + 0.5 \times 75}{1 + 0.5}$$

$$= \frac{137.5}{1.5}$$

$$= \text{Sh. } 91.67$$

QUESTION 15

May 2019 Question Three A

The capital structure of Maweni Limited is given as follows:

	Sh. "000"
Ordinary share capital (Sh.20 par value)	50,000
Retained earnings	30,000
12% irredeemable debenture capital (Sh.20 par value)	25,000
14% preference share capital (Sh.25 par value)	<u>15,000</u>
	120,000

Additional information:

1. The current market price of the firm's ordinary shares is quoted at Sh.45 cum-dividend.
2. The firm paid a dividend of Sh.5 per share in the just ended year.
3. The firm adopts a 60% dividend payout ratio.
4. The firm's return on equity (ROE) is 20%.
5. The existing 12% irredeemable debenture is currently trading at Sh. 112 cum-interest.
6. The 14% preference shares are currently trading at Sh.33.50 cum-dividend at the securities exchange.
7. The corporate tax rate applicable is 30%.

Required:

- i. The cost of ordinary share capital. (3 marks)
- ii. The cost of 12% irredeemable debenture capital. (2 marks)
- iii. The cost of 14% irredeemable preference share capital (2 marks)
- iv. The firm's weighted average cost of capital (WACC). (5 marks)

ANSWER

i) Cost of ordinary share capital

$$Ex - \text{div price} = 45 - 5 = 40$$

$$K_e = \frac{D_o (1+g)}{P_o} + g \times 100$$

$$g = ROE \times (1 - DPR)$$

Where: Return on equity (ROE) – 20%

Dividend payout ratio (DPR) – 60%

$$g = 20\% (1 - 0.6)$$

$$= 8\%$$

$$K_e = \frac{D_o (1+g)}{P_o} + g \times 100$$

$$D_o = 5, g = 8\% \text{ \& } P_o = 40\%$$

$$= \frac{5(1.08)}{40} + 0.08 \times 100 = 21.5\%$$

ii. Cost of 12% irredeemable debenture capital

$$\text{ex-interest price} = 112 - \frac{12}{100} \times 20$$

$$= 112 - 2.4 = \text{Sh.}109.6$$

$$K_i = \frac{I(1-t)}{P_o} \times 100\% = \frac{12}{100} \times 20 = 2.4\%$$

$$T = \text{tax rate } 30\% \quad P_o = 109.60$$

$$K_i = \frac{2.4(1-0.3)}{109.6} \times 100\% = 1.53\%$$

iii. Cost of 14% irredeemable preference share capital

$$\text{Dividend per share (D}_p) = \frac{14}{100} \times 25 = \text{Sh.}3.5$$

$$\text{Ex-dividend price} = 33.5 - 3.5 = \text{Sh. } 30$$

$$K_p = \frac{D_p}{P_o} \times 100\% = \frac{3.5}{30} \times 100\% = 11.67\%$$

Cost of retained earnings

$$K_r = \frac{D_o(1+g)}{P_o} + g \times 100\% \quad \text{where; } D_o = 5, g = 8\% \text{ and } P_o = 40$$

$$K_r = \frac{5(1+0.08)}{40} + 0.08 \times 100\% = \frac{5.4}{40} + 0.08 \times 100\%$$

$$(0.135 + 0.08)100\% = 21.5\%$$

iv. WACC

Number of ordinary shares = ordinary share capital ÷ par value

$$= 50,000,000 \div 20$$

$$= 2,500,000$$

$$\begin{aligned}
 \text{Number of debentures} &= \text{debenture capital} \div \text{par value} \\
 &= 25,000,000 \div 20 \\
 &= 1,250,000
 \end{aligned}$$

$$\begin{aligned}
 \text{Number of preference shares} &= \text{preference share capital} \div \text{par value} \\
 &= 15,000,000 \div 25 \\
 &= 600,000
 \end{aligned}$$

Source	Amount	Weight	Cost of source	Weighted cost
Ordinary shares	$2,500 \times 40 = 100,000,000$	0.35	21.5%	7.53%
Retained earnings	$30,000,000 = 30,000,000$	0.11	21.5%	2.37%
Preference shares	$30 \times 600,000 = 18,000,000$	0.6	11.67%	0.70%
Debentures	$1,250,000 \times 109.6$	0.48	1.53%	0.73%
				11.33%

$$\text{W.A.C.C} = 11.33\%$$

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QUESTION 16

November 2018 Question One B (i)

Differentiate between the following terms as applied in finance:

- (i) —Operating leverage” and "financial leverage". (2 marks)

ANSWER

i) Operating leverage refers to the use of fixed operating cost in order to increase the profitability of the firm. It is measured using the degree of operating leverage (D.O.L)

$$D.O.L = \frac{\% \text{ change in EBIT}}{\% \text{ change in sales}} \text{ or } \frac{\text{Contribution}}{\text{Contribution} - \text{Fixed cost}}$$

Financial leverage refers to the use of fixed financing cost in order to improve earnings per share. It is measured using the Degree of Financial leverage (D.F.L)

$$D.F.L = \frac{Q(SP - VC) - FC}{Q(SP - VC) - FC - I - \frac{P_d}{1-T}}$$

Where;

Q = units sold

SP = Selling price per unit

VC = Variable cost

FC = Fixed cost

I = interest

P_d = Preference dividends

T = corporate tax rate

QUESTION 17

November 2018 Question One C

The fixed operating cost for Gahaleni Pharmaceutical Ltd. are Sh.5.8 million per annum and the variable cost ratio is 0.20.

Additional information:

1. The company has Sh.2 million in bonds outstanding with an annual coupon interest rate of 8 per cent.
2. The company has 30,000 preference shares outstanding which pay Sh.2 dividend per share each year.
3. The company has 100,000 ordinary shares outstanding.
4. Revenues of the company are Sh.8 million per annum.
5. The company is in the 30% corporate tax bracket.

Required:

- | | |
|--|-----------|
| (i) The degree of operating leverage. | (4 marks) |
| (ii) The degree of financial leverage. | (3 marks) |
| (iii) The degree of combined leverage. | (3 marks) |

ANSWER

i) Degree of operating leverage

$$\text{Degree of operating leverage} = \frac{\text{Contribution}}{\text{Contribution} - \text{Fixed operating costs}}$$

$$\begin{aligned}\text{Contribution} &= (1 - 0.2) \times 8,000,000 \\ &= 6,400,000\end{aligned}$$

$$\begin{aligned}\text{D.O.L} &= \frac{6,400,000}{6,400,000 - 5,800,000} \\ &= 10.67\end{aligned}$$

ii) Degree of financial leverage

$$\text{Degree of financial leverage}$$

$$= \frac{\text{Contribution}}{\text{Contribution} - \text{Fixed operating costs} - \text{interest paid} - \frac{\text{Preference divided}}{1 - \text{tax}}}$$

$$= \frac{6,400,000}{6,400,000 - 5,800,000 - \left[\frac{8}{100} \times 2,000,000 \right] - \frac{30,000 \times 2}{1 - 0.3}}$$

$$= \frac{6,400,000}{6,400,000 - 5,800,000 - 160,000 - 85,714.29}$$

$$= \frac{6,400,000}{354,285.71} = 18.06$$

iii) Degree of combined leverage

Degree of combined leverage

$$\begin{aligned} D.C.L &= \text{Degree of operating leverage} \times \text{Degree of financial leverage} \\ &= 10.67 \times 18.06 \\ &= 192.70 \end{aligned}$$

QUESTION 18**November 2018 Question Four B**

NewWays Ltd. intends to raise new capital to expand its production level.

The company plans to undertake the following financial decisions:

1. Issue 200,000 ordinary shares which have a par value of Sh.10 at Sh.16 per share.
The floatation cost per share is Sh. 1.
2. Issue 75,000, 12% preference shares which have a par value of Sh.20 at Sh.18 per share. The total floatation cost is Sh. 150,000.
3. Issue 50,000, 18% debentures which have a par value of Sh. 100 at Sh.80 per debenture.
4. Borrow Sh. 5,000.000. 18% long-term loan. The total floatation cost is Sh.200,000.

Additional information:

1. The company paid 28% ordinary dividends which is expected to grow at the rate of 4% per annum.
2. The corporate tax rate is 30%.

Required:

- (i) The total capital to be raised net of floatation costs. (2 marks)
- (ii) The weighted marginal cost of capital (WMCC) for the company. (6 marks)

(Total: 20 marks)

ANSWER

Source	Market Value	Floatation	Net floatation
Ordinary shares	$200,000 \times 16 = 320,000$	$200,000 \times 1 = 200,000$	3,000,000
12% preference shares	$750,000 \times 18 = 1,350,000$	$7,500,000 \times 2 = 150,000$	1,200,000
18% debentures	$50,000 \times 80 = 4,000,000$	0	4,000,000
Loan	5,000,000	200,000	4,800,000
	13,550,000	550,000	13,000,000

Cost of ordinary shares

$$K_e = \frac{d_0(1 + g)}{P_0 - f} + g \times 100\%$$

Where;

$$d_0 = \frac{28}{100 \times 10} = 2.8$$

$$g = 4\%$$

$$P_0 = 16$$

$$f = 1$$

$$= \left[\frac{2.8(1 + 0.04)}{16 - 1} + 0.04 \right] \times 100$$

$$= \left[\frac{2.912}{15} + 0.04 \right] \times 100$$

$$= [0.1941 + 0.04] \times 100$$

$$= 23.41\%$$

Cost of preference shares

$$K_p = \frac{D_p}{P_0 - f} \times 100$$

Where;

$$D_p = \text{the dividends paid} = \frac{12}{100} \times 20 = 2.4$$

$$P_0 = \text{is the market price per share} = 18$$

$$f \text{ is the floatation cost} = \frac{150,000}{75,000} = 2$$

$$K_p = \frac{2.4}{18 - 2} \times 100 = 15\%$$

Cost of Debentures (Kd)

$$K_d = \frac{I(1-t)}{P_0 - f} \times 100$$

Where;-

$$I = \frac{18}{100} \times 100 = 18$$

$$t = 30\%$$

$$P_0 = 80$$

$$f = 0$$

$$K_d = \frac{18(1-0.3)}{80 - 0} \times 100$$

$$= 15.75\%$$

Cost of long term loan

$$K_L = \frac{I(1-t)}{M_v} \times 100$$

$$I = \frac{18}{100} \times 5,000,000 = 900,000$$

$$K_l = \frac{9,000,000(1-0.3)}{4,800,000} \times 100\%$$

$$= 13.125\%$$

Source	Amount	Weight	Cost of capital	Weighted cost
Ordinary shares	3,000,000	$3,000 \div 13,000 = 0.23$	23.41%	5.38%
12% preference shares	1,200,000	$1,200 \div 13,000 = 0.09$	15%	1.35%
18% debentures	4,000,000	$4,000 \div 13,000 = 0.31$	15.75%	4.88%
18% Loan	4,800,000	$4,800 \div 13,000 = 0.37$	13.125%	4.86%
	13,000,000			16.7%

$$WMC = 16.47\%$$

QUESTION 19

May 2018 Question Three A and B

a) Argue three cases for and three cases against the use of market values for various components of cost of capital in determining the weighted average cost of capital (WACC) of a firm. (6 marks)

(b) Akiba Limited has the following capital structure:

	Sh. "000"
3,000,000 fully paid ordinary shares	30,000
Retained earnings	20,000
1,000,000 10% preference shares	20,000
6% debentures (Sh. 150 par value)	30,000

Additional information:

1. The current market price per share (MRS) is Sh.30,
2. The expected dividend per share in the following year is Sh. 1.20.
3. The average growth rate in both earnings and dividends has been maintained at 10% over the last 10 years.
4. The trend is expected to remain the same into the foreseeable future.

5. The debentures are trading at Sh. 110 at the securities market.
6. The debentures mature in 100 years period.
7. The preference shares were issued 4 years ago and they are still trading at face value.
8. The corporate tax rate is 30%.

Required:

Weighted average cost of capital (WACC) for the company. (8 marks)

ANSWER

a) Argument for use of market values in WACC computation

- i. They reflect current market conditions in which new funds may be acquired
- ii. Assets are valued at market
- iii. By treating the entire balance sheet as market value, capital strength is shown as liquidated value
- iv. Investors also demand market required rate of return on the market value of the capital and not on the book value of the capital
- v. Weightings based on book values are unreliable

Argument against use of market values in WACC computation

- Hard to obtain market value of equity and debt especially for unlisted company.

b) Weighted average cost of capital (WACC) for the company

Cost of ordinary shares

$$K_e = \frac{D_1}{P_0} + g \times 100\%$$

$$= \frac{1.2}{30} + 0.1 \times 100\%$$

$$= 14\%$$

Cost of retained earnings

$$K_r = \frac{D_1}{P_0} + g \times 100\%$$

$$= \frac{1.2}{30} + 0.1 \times 100\%$$

$$= 14\%$$

Cost of Debentures (Kd)

$$K_d = \left[\frac{I + \left(\frac{MV - MPS}{n} \right)}{\frac{MV + MPS}{2}} \right] (1 - t) \times 100\%$$

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Where;-

I = interest paid $0.06 \times 150 = 9$

t = tax rate 30% or 0.3

MPS = Market price per debenture

Maturity value

n = number of years – 100 years

$$K_d = K_d = \left[\frac{I + \left(\frac{MV - MPS}{n} \right)}{\frac{MV + MPS}{2}} \right] (1 - t) \times 100\%$$

$$K_d = K_d = \left[\frac{9 + \left(\frac{100 - 110}{100} \right)}{\frac{100 + 110}{2}} \right] (1 - 0.3) \times 100\%$$

$$= \frac{8.9}{105} \times 0.7 \times 100\%$$

$$= 5.93\%$$

Cost of preference shares

$$K_P = \frac{D_p}{P_0} \times 100\%$$

Where;

$$D_p = \frac{10}{100} \times 20 = \text{sh. } 2$$

$$P_0 = \frac{20,000,000}{1,000,000} = 20$$

$$K_P = \frac{2}{20} \times 100 = 10\%$$

Source	Amount (000)	Weight	Cost of capital	WACC
Ordinary shares	3,000x30=90,000	$\frac{90,000}{152,000} = 0.59$	14%	8.26%
Retained earnings	20,000	$\frac{20,000}{152,000} = 0.13$	14%	1.82%
10% preference shares	20,000	$\frac{20,000}{152,000} = 0.13$	10%	1.3%
6% debentures	$\frac{30,000}{150} \times 110 = 22,000$	$\frac{22,000}{152,000} = 0.14$	5.93%	0.83%

QUESTION 20**November 2017 Question Three B**

Ray Properties Ltd. is planning to build a business mall. The project will cost Sh. 180 million. The firm's current optimal capital structure is as follows:

	Sh."000"
Ordinary shares (Sh. 10 par value)	480.000
10% debt (Sh.100 par value)	384.000
Retained profit	<u>96.000</u>
	<u>960.000</u>

Additional information:

1. The firm will issue a new 15% debenture at Sh.120 each with a floatation cost of Sh.10 per unit. The par value of each debenture is Sh.100.
2. New ordinary shares will be issued at the current market price of Sh.30 each with a floatation cost of Sh.5 per share.
3. The most recent dividend paid by the company was Sh.5 per share.
4. The dividend is expected to grow at the rate of 5% per annum in perpetuity.
5. The firm expects to retain Sh. 18 million to finance this investment.
6. The corporate tax rate is 30%.

Required:

- (i) The amount to be raised from equity capital, if the capital structure is to remain unchanged. (3 marks)
- (ii) The number of ordinary shares the company should issue to raise the desired external equity capital. (3 marks)
- (iii) The firm's weighted marginal cost of capital (WMCC). (6 marks)

ANSWER

Ray properties Ltd

i. Amount to be raised

Source	Amount -000"	Weight
--------	--------------	--------

Ordinary shares	480,000	$480,000 \div 960,000 = 0.5$
10% debt	384,000	$384,000 \div 960,000 = 0.4$
Retained profit	<u>96,000</u>	$96,000 \div 960,000 = \underline{0.1}$
	960,000	1.0

Amount to be raised from equity capital

$$\begin{aligned}
 &= (\text{weight ordinary shares} + \text{weight retained earnings}) \times \text{Project cost} \\
 &= (0.5 + 0.1) \times 180,000,000 \\
 &= \text{Sh. } 108,000,000
 \end{aligned}$$

ii. Number of ordinary shares

$$\begin{aligned}
 &= \frac{\text{amount to be raised from ordinary share issue}}{\text{market price per each ordinary share} - \text{floatation cost per share}} \\
 &= \frac{0.5 \times 180,000,000}{30 - 5} \\
 &= 3,600,000 \text{ shares}
 \end{aligned}$$

iii. Weighted marginal cost of capital

Cost of equity

$$K_e = \frac{D_o(1 - g)}{P_o - f} + g \times 100\%$$

Where:

D_o is dividend per share = Sh. 5

g is growth rate in dividends = 5% or 0.05

P_o is ex dividend market price per share = Sh. 30

F if floatation cost per share = 5

$$K_e = \frac{5(1 + 0.05)}{30 - 5} + 0.05 \times 100\%$$

$$K_e = \frac{5(1.05)}{30 - 5} + 0.05 \times 100\%$$

$$= \frac{5.25}{25} + 0.05 \times 100\%$$

$$= (0.21 + 0.05) \times 100\%$$

$$= \mathbf{26\%}$$

Cost of retained earnings (K_r)

$$K_r = \frac{D_o(1-g)}{P_o} + g \times 100\% = \frac{5(1.05)}{30} + 0.05 \times 100\%$$

$$\frac{5.25}{30} + 0.05 \times 100 = (0.175 + 0.05) \times 100\% = \mathbf{22.5\%}$$

Cost of debenture

$$K_d = \frac{I(1-T)}{P_o - f} \times 100\% \quad I \text{ is interest paid } \frac{15}{100} \times 100 = \text{Sh. } 15$$

T is tax rate = 30% or 0.3

P_o is market value of each debenture = Sh. 120

F is floatation cost per each debenture issued = Sh. 10

$$K_d = \frac{15(1 - 0.3)}{120} \times 100\%$$

$$= \mathbf{8.75\%}$$

Weighted marginal cost of capital (WMCC)

Source	Amount “000”	Weight	Cost	WMCC
Ordinary shares	90,000	0.5	26%	13%
Retained earnings	18,000	0.1	22.5%	2.25%
15% debenture	72,000	0.4	8.75%	3.5%
	180,000			18.75%

QUESTION 21

May 2017 Question One B and C

(b) Discuss the relevance of cost of capital to a business enterprise. (6 marks)

(c) Upendo Ltd.’s existing capital structure is given as follows:

	Sh. “000”
Ordinary share capital (Sh.20 par)	20,000
Reserves	5,000
10% Debenture (Sh.100 par)	10,000
8% Preference shares (Sh.20 par)	<u>15,000</u>
Additional information:	<u>50,000</u>

Additional information:

1. The most recent earnings per share (EPS) of the company is Sh.5.
2. The firm adopts 40% pay-out ratio as its dividend policy.
3. Ordinary shares of the company are currently selling for Sh.50 each.
4. The existing 10% debenture is currently trading at 110% of par at the securities exchange.
5. Existing 8% preference shares are currently trading at Sh.25 each.
6. Corporate tax rate applicable is 30%.

Required:

- (i) The annual dividend growth rate using Gordon’s growth model. (2 marks)
- (ii) Cost of ordinary share capital. (2 marks)
- (iii) Cost of 10% debenture capital. (1 mark)

- (iv) Cost of 8% preference share capital. (1 mark)
- (v) The weighted average cost of capital (WACC) of the firm. (4 marks)

ANSWER

b) Relevance of cost of capital to a business enterprise

1. Used to evaluate the performance of management in raising funds efficiently and in optimally investing the funds
2. The cost of capital is used as a discount rate that is used in evaluation of investment proposals
3. Used in lease or buy decisions
4. It has an effect in capital structure decisions i.e in designing a firm's debt policy, it helps in the choice of the combination of debt and equity
5. In firm's value maximization - in order to maximize the value of the firm, all input costs must be minimized especially cost of capital

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c) Upendo's Ltd annual dividend growth rate using Gordon's growth model

i. Annual dividend using Gordon's growth model

Sustainable growth rate (g) = retention ratio × return on equity

Number of ordinary shares

Ordinary share capital ÷ par value (20,000 ÷ 20) = 1,000

Total equity = ordinary share capital + reserves

$$20,000 + 5,000 = 25,000$$

Book value per share (NAV) = total equity ÷ number of ordinary shares

$$25,000 \div 1,000 = \text{Sh. } 25$$

$$\text{Return on equity} = \frac{\text{EPS}}{\text{NAV}} \times 100\% \quad \frac{5}{25} \times 100\% = 20\%$$

$$\text{Sustainable growth rate (g)} = 0.4 \times 20\% = 12\%$$

ii. Cost of ordinary share capital

$$K_e = \frac{D_o(1 + g)}{P_o} + g \times 100\%$$

Where

$$D_o = \text{dividend per share} \frac{40}{100} \times 5 = \text{Sh. } 2$$

$$g = \text{growth rate} = 12\%$$

$$K_e = \frac{2(1.12)}{50} + 0.12 \times 100\% = 16.48\%$$

iii. Cost of 10% debenture capital (K_d)

$$K_d = \frac{I(1-T)}{P_o} \times 100\%$$

Where

$$I \text{ is interest paid} = \frac{10}{100} \times 100 = \text{Sh. } 10$$

$$T \text{ is tax rate} = 30\% \text{ or } 0.3$$

$$P_o = \text{market price per debenture} = \frac{110}{100} \times 100 = \text{Sh. } 110$$

$$K_d = \frac{10(1 - 0.3)}{110} \times 100\% = 6.36\%$$

iv. **Cost of 8% preference share capital (K_p)**

$$K_p = \frac{D_p}{P_0} \times 100\%$$

Where:

$$D_p = \text{dividend per share} = \frac{8}{100} \times 20 = 1.60$$

$$P_0 = \text{market value} = \text{Sh. } 25$$

$$K_p = \frac{1.6}{25} \times 100\% = 6.4\%$$

v. **Weighted average cost of capital (WACC) of the firm**

Source	Market value(000)	weight	Cost of source	WACC
Ordinary share capital	$1000 \times 50 = 50,000$	0.63	16.48%	10.38%
10% debentures	$100 \times 110 = 11,000$	0.14	6.36%	0.89
8% preference shares	$750 \times 25 = 18,750$	0.24	6.4%	1.54%
	79,750	1.00		12.81%

Number of ordinary share = ordinary share capital ÷ par value

$$20,000 \div 20 = 1,000$$

Number of debentures = debenture capital ÷ par value

$$= 10,000 \div 100 = 100$$

Number of preference shares = preference share capital ÷ par value capital

$$= 15,000 \div 20 = 750$$

QUESTION 22

November 2016 Question Two B

Sandy Ltd presented the following extracts of the statement of financial position as at 31 October 2016:

	Sh."000"	Sh."000"
Equity		
Ordinary shares (Sh5 nominal value)	800,000	
Reserves	3,600,000	4,400,000
Long term liabilities		
4% preference shares (Sh. 1 nominal value)	600,000	
7% bonds (redeemable after 6 years)	600,000	
Long term bank loan	200,000	1,400,000

Additional information:

- Ordinary shares of Sandy Ltd ex-div market value of sh.47.00 per share and an ordinary dividend of Sh.3.63 per share has just been paid.
- The following dividends have been paid over the past four years:

Year	2013	2014	2015	2016
Dividend per share (Sh.)	3.09	3.22	3.36	3.50

- The preference shares are not redeemable and have an ex-div market value of 40 cents per share.
- The 7% bond is redeemable at 5% premium to their nominal value of sh. 100 per bond and have an ex-interest market value of Sh. 104.50.
- The bank loan has a variable interest rate that has averaged 4% per year in recent years.
- The corporate tax rate is 30%

Required:

- The weighted average cost of capital (WACC). (4 marks)
- Explain four reasons why the cost of equity could be greater than the cost of debt. (4 marks)

ANSWER**i)The weighted average cost of capital (WACC).****Growth model**

$$K_e = \left(\frac{D_n}{D_0} - 1 \right) \times 100\%$$

$$= \left[\left(\frac{3.63}{3.09} \right)^{\frac{1}{4}} - 1 \right] \times 100\%$$

$$= (1.04109 - 1) \times 100\%$$

$$= 4.109\%$$

Cost of ordinary shares(K_e)

$$K_e = \frac{D_0(1 + g)}{P_0 - f} + g \times 100\%$$

$$= \left[\frac{3.63(1.04109)}{47} \right] + 0.05 \times 100\%$$

$$= 0.1215 \text{ or } 12.15\%$$

Cost of preference shares(K_p)

$$K_p = \frac{D_p}{P_0} \times 100$$

$$D_p = \text{Dividends per preference share} = \frac{4}{100} \times 1 = 0.04$$

$$P_0 = \text{Market price per share} = \text{sh. } 0.4$$

$$\frac{0.04}{0.4} \times 100\% = 10\%$$

Cost of bonds(K_d)

$$K_d = \left[\frac{I + \frac{MV - MPS}{n}}{\frac{MV + MPS}{2}} \right] (1 - T) \times 100\%$$

$$I = \text{interest paid} = \frac{7}{100} \times 100 = \text{sh.}7$$

$$MV = \text{Maturity value} = \frac{105}{100} \times 100 = \text{sh.}105$$

$$MPS = \text{Market price per share} = \text{sh.} 104$$

$$T = 0.3 = 30\%$$

$$N = 6 \text{ years}$$

$$K_d = \left[\frac{7 + \frac{105 - 104.5}{6}}{\frac{105 + 104.50}{2}} \right] (1 - 0.3) \times 100\%$$

$$= 4.73\%$$

Cost of loan(K_L)

$$K_L = I(1 - T) \times 100\%$$

$$= 4(1 - 0.3) \times 100\%$$

$$= 2.8\%$$

$$\text{Number of ordinary shares} = 800,000 \div 5 = 160,000$$

$$\text{Number of preference shares} = 600,000 \div 1 = 600,000$$

$$\text{Number of bonds} = 600,000 \div 100 = 6,000$$

$$\text{Market value ordinary shares} = \frac{800,000}{5} \times 47 = 7,500,000$$

$$\text{Market value preference share} = 600,000 \times 0.4 = 240,000$$

$$\text{Market value bonds} = 104.5 \times 6000 = 627,000$$

	Market value	Weighs	Cost of source	Weighted costs.
Equity	7,520,000	0.88	12.15%	10.69%
Preference shares	240,000	0.03	10%	3%
Redeemable bonds	627,000	0.07	4.73%	0.33%
Bank loan	<u>200,000</u>	0.02	2.80%	<u>0.06%</u>
	<u>8,587,000</u>			<u>14.08%</u>

$$WACC = 14.08\%$$

ii) Reasons why the cost of equity could be greater than the cost of debt

- Ordinary and preference dividends are not allowable tax deductions. Thus there is no tax shield benefit
- Equity holders are always subordinates to debt holders and do not receive a contractual obligation to be repaid their capital. Thus they require a return of investment higher than cost of debt for risk assumed by them.
- Greater uncertainty of receiving dividends and repayment of capital at end.
- Debt is secured against securities resulting to less risk of loss. Less risk brings less returns, hence lesser cost

QUESTION 23

November 2016 Question Five C

The following data was extracted from Mwakuhenga Limited's financial statements for the year ended 30 June 2016:

	Sh.
Total sales	3,000,000
Variable costs	<u>(900,000)</u>
Contribution	2,100,000
Fixed costs	<u>(1,500,000)</u>
Earnings before interest and tax (EBIT)	600,000

Interest	<u>(75,000)</u>
Profit before tax	<u>525,000</u>

Required:

Using the concept of leverage, determine:

- (i) The percentage taxable income if EBIT increases b (3 marks)
- (ii) The percentage EBIT if there is a 10% increase in sales. (3 marks)
- (iii) The percentage taxable income if sales increase by 8%. (4 marks)

ANSWER

Using the concept of leverage to determine:

(i) Degree of financial leverage

$$DFL = \frac{\text{Operating profit (EBIT)}}{\text{profit before tax}} = \frac{600,000}{525,000} = 1.15$$

Percentage change in taxable income if operating profit increases by 6%
 $= 6\% \times 1.15 = 6.9\%$

(ii) Degree of operating leverage

$$DOL = \frac{\text{Contribution}}{\text{Operating profit}} = \frac{2,100,000}{600,000} = 3.50$$

If sales increase by 10%, operating profit increases by $10\% \times 3.5 = 35\%$

(iii) Degree of combined leverage

$$DCL = \frac{\text{Contribution}}{\text{Profit before tax}} = \frac{2,100,000}{525,000} = 4$$

If sales increase by 8%, profit before tax increases by $8 \times 4 = 32\%$

Or

Degree of combined leverage = Degree of financial leverage $\times$ Degree of operating leverage

$$= 1.15 \times 3.5 = 4.025$$

Percentage change in taxable income if sales increase by 8%

$$= 4.025 \times 8$$

$$= 32.2\%$$

$$= 32\%$$

QUESTION 24

May 2016 Question Two A

The existing capital structure of Mwarakaya Limited is given as follows:

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	Sh.000
Ordinary share capital (Sh. 100 par value)	40,000
Reserves	15,000
12% debentures (Sh. 100 par value)	25,000
10% preference share capital (Sh. 20 par value)	<u>20,000</u>
	<u>100,000</u>

Additional information:

1. Ordinary shares of Mwarakaya Limited are currently selling at Sh.80 each.
2. The 12% debentures and 10% preference shares are currently selling at Sh.90 and Sh.30 respectively.
3. The most recent ordinary dividend paid by the company is Sh.2.00. This is expected to grow at the rate of 10% each year in perpetuity.
4. The corporate tax rate is 30%.

Required:

The weighted average cost of capital (WACC).

(6 marks)

ANSWER**Weighted average cost of capital****i) Cost of equity(K_e)**

$$K_e = \frac{D_0(1 + g)}{P_0} + g \times 100$$
$$= \frac{D_1}{P_0} + g$$

Where K_e = Cost of equity D_0 = Dividend per share = sh.2 P_0 = Current share market price = sh.80 g = growth rate = 10%.

$$K_e = \left[\frac{2.0(1 + \frac{10}{100})}{80} + 0.1 \right] \times 100 = 12.75\%$$

ii) Cost of debentures

$$K_d = \frac{I(1 - T)}{MVD} \times 100 = 9.33\%$$

Where I = interest rate T = tax rate = 30% or 0.3 Mvd = market price of debenture.

$$K_d = \frac{\left(\frac{12}{100} \times 100\right) (1 - 0.3)}{90} \times 100 = 9.33\%$$

iii) Cost of 10% preferred shares (K_p)

$$K_p = \frac{D_p}{P_0} \times 100\%$$

Where;-

K_p =cost of preference stock

D_p = Dividend of preferred share

P_0 = market price of preference share=sh. 30.

$$K_p = \frac{\left(\frac{10}{100} \times 20\right)}{30} \times 100 = 6.67\%$$

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Weighted average cost of capital (WACC)

SOURCE	Market value Sh.000	Market value weight	Cost of each item	Weighted cost
Ordinary share capital	$\frac{40,000}{10} \times 80 = 32,000$	0.38	12.75%	4.85%
12% debenture capital	$\frac{25,000}{100} \times 900 = 22,500$	0.27	9.337%	2.52%
10% preference share capital	$\frac{20,000}{10} \times 30 = 30,000$	0.36	6.67%	2.40%
	84,800		WACC	9.77%

QUESTION 25

November 2015 Question Two B

Wendy Limited has the following capital structure:

Debt	35%
Equity	50%
Preference shares	15%

The management of the company has provided the data below:

Bond yield to maturity	9%
Corporate tax rate	30%
Growth rate of ordinary dividends	9%
Market price of one ordinary share	Sh.30
Dividend for one ordinary share	Sh. 1.20
Market price of one preference share	Sh. 100
Floatation cost of one preference share	Sh.2.00
Dividend for one preference share	Sh.8.50

Required:

The company's weighted average cost of capital (WACC). (6 marks)

ANSWER

A. WACC for Wendy Ltd

Cost of preference shares K_p

$$K_p = \frac{D_p}{P_o} \times 100\%$$

$$D_p = \text{Dividend per share} = \text{Sh. } 8.5$$

$$P_o = \text{Market price per share} = \text{Sh. } 100$$

$$K_p = \frac{8.5}{100} \times 100\% = 8.5\%$$

Cost of preference shares

$$K_p = \frac{D_o(1+g)}{P_o} + g \times 100$$

D_o = Dividend per share = 100

g = growth rate = 9%

P_o = Market price per ordinary share = 30

$$\frac{12(1.09)}{30} + 0.09 \times 100 = 13.36\%$$

$$WACC = \left(\frac{35}{100} \times 9 \right) + \left(\frac{50}{100} \times 13.36 \right) + \left(\frac{15}{100} \times 8.5\% \right)$$

$$3.15 + 6.68 + 1.28 = 11.11\%$$

QUESTION 26

November 2015 Question Four A

Fila Ltd. intends to raise finance as follows:

Debenture: Raise Sh.100 million through a debenture issue. Each debenture will have a face value of Sh.1,000 and will be issued at 2% floatation cost and a discount of Sh.60. The coupon rate will be 10% with a maturity period of 10 years.

Equity: The firm will raise Sh.100 million from ordinary shares. The current level of dividend is Sh.5 per share and this has been growing at 10% per annum. The current market price per share is Sh.40 and floatation cost will be 5% of the market price.

Long term debt: Raise Sh.20 million long-term debt at par with an interest rate of 10% per annum.

Corporate tax rate is 30%.

Required:

The marginal cost of capital (MCC) of Fila Ltd.

(8 marks)

ANSWER**A. MCC of Fila Limited**

Cost of debenture (K_d)

$$K_d = \frac{I + \frac{M_v - P_o}{n}}{\frac{M_v + P_o}{2}} \times (1 - T) \times 100\%$$

I is interest paid = $\frac{10}{100} \times 1,000 = \text{Sh.}100$

M_v is maturity value = Sh. 1,000

P_o is market price per debenture

$$1,000 - 60 - \left(\frac{2}{100} \times 1,000\right) = \text{Sh.} 920$$

T is corporate tax = 30%

$$K_d = \left[\frac{100 + \frac{1000 - 920}{10}}{\frac{1000 + 920}{2}} \right] \times (1 - 0.3) \times 100\% = \frac{100 + 8}{960} \times 0.7 \times 100 = 7.88\%$$

Cost of long term debt

$$K_L = I (1 - T)$$

$$= 10\% (1 - 0.3) = 7\%$$

Cost of equity (K_e)

D_o is dividend per share = Sh. 5

g is growth rate = 10%

P_o is current market price per share = Sh.40

f is floatation cost per share = $0.05 \times 40 = \text{Sh.} 2$

$$K_e = \frac{D_o(1.1)}{P_o - f} + g \times 100\%$$

$$= \frac{5(1.1)}{40-2} + 0.1 \times 100\% = 24.47\%$$

Source	Amount	Weight	Cost of capital	Weighted debt
Debenture	100 million	0.45	7.88%	3.55%
Equity	100 million	0.45	24.47%	11.01%
Long term debt	20 million	0.1	7.00%	0.70%
	220 million		WCC	15.26%

QUESTION 27

November 2015 Question Five A

Distinguish between "required rate of return" and "expected rate of return".

(4 marks)

ANSWER

B. Distinctions between required rate of return and expected rate of return

Required rate of return is the absolute minimum return on investment you would accept for that investment to be worthwhile. The expected rate of return is what you actually make in an investment

QUESTION 28

September 2015 Pilot Paper Question Three B

Ruiru Tanners Ltd. has a total of Sh.100 million invested in net assets as at the end of December 2014. The firm intends to increase its production capacity during the year 2015 by Sh.100 million. The company utilises debt, preferred stock and equity capital within its capital structure. Several alternative financing arrangements are available, namely;

- The company can issue 9% debentures with a par value of Sh.100 each at an issue price of Sh.90 each (market price). Maximum amount available is Sh.20,000,000. Any extra debt finance will be raised through the issue of 12% debentures at Sh.960 each. The par value of this debenture is Sh. 1,000 each.

- The company can issue additional 15% preference shares with a par-value of Sh.50 at Sh.75 each
- The company can issue new ordinary shares at the current market price of Sh.88 per share. Floatation cost equal to Sh.8 per share sold. The company's ordinary shareholders have consistently enjoyed a dividend whose annual growth rate on average has been 10% and this is expected to continue into the foreseeable future. The company's earnings per share this year is Sh. 10 and adopts a constant dividend payout ratio of 40% each year.
- The company can generate Sh.10 million from the internal sources to finance this expansion programme.

Additional information:

1. The company pays corporation tax at the rate of 30%.
2. The firm's existing capital structure which is considered to be optimal is given below:

Debt capital:	Sh.,,,"000"	Sh. "000
6% debenture capital	10,000	
8% term loan	<u>20,000</u>	30,000
Preference shares (Sh.50 par value)		30,000
Ordinary shares (Sh.5 par value)	15,000	
Retained earnings	<u>25,000</u>	<u>40,000</u>
		<u>100,000</u>

Required:

- (i) The amount of funds to be raised from each source during the year 2015 so as to maintain the firm's existing optimal capital structure. (3 marks)
- (ii) The number of ordinary shares to be issued to raise desired external equity.
The levels of financing at which marginal cost of capital changes (Hint: break points in weighted marginal cost of capital curve). (2 marks)
- (iii) The firm's weighted marginal cost of capital if it were to raise only Sh.20 million. (3 marks)

- (iv) The firm's weighted marginal cost of capital for the funds to be raised during the year 2015 for the three levels of financing. (4 marks)

ANSWER

Ruiru tanners Ltd

i. Amount of funds to be raised

Current capital structure

Source	Amount	Weight
6% debenture capital	10,000	0.10
8% term loan	20,000	0.20
Preference shares	30,000	0.30
Ordinary shares	15,000	0.15
Retained earnings	25,000	0.25
	100,000	1.00

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Amount to be raised from each source

$$9\% \text{ debenture capital} = 0.1 \times 100,000,000 = 10,000,000$$

$$15\% \text{ preference shares} = 0.3 \times 100,000,000 = 30,000,000$$

$$\text{Ordinary shares} = 0.15 \times 100,000,000 = 15,000,000$$

$$\text{Retained earnings} = 0.25 \times 100,000,000 = 25,000,000$$

$$8\% \text{ term loan} = 0.20 \times 100,000,000 = 20,000,000$$

ii. The number of ordinary shares

$$= \frac{15,000,000 + 15,000,000}{80} = 375,000 \text{ ordinary shares}$$

iii. The levels of financing

$$\text{Breakpoint} = \frac{\text{amount}}{\text{proportion in capital structure}}$$

$$\text{Breakpoint ordinary shares} = \frac{\infty}{0.15} = \infty$$

$$\text{Breakpoint retained earnings} = \frac{10,000,000}{0.25} = 40,000,000$$

$$\text{Breakpoint 9\% debentures} = \frac{20,000,000}{0.1} = 200,000,000$$

$$\text{Breakpoint 12\% debentures} = \frac{>20,000,000}{0.1} = \infty$$

$$\text{Breakpoint 15\% preference shares} = \frac{\infty}{0.30} = \infty$$

iv. Firm's weighted marginal cost of capital

Cost of retained earnings (K_r)

$$K_r = \frac{D_o(1+g)}{P_o} + g \times 100\%$$

Where: D_o = dividend per share = Sh. $\frac{40}{100} \times 10 = 4$

g = growth rate = 10%

P_o = current market price per share = Sh. 88

$$K_r = \frac{4(1.1)}{88} + 0.1 \times 100\% = 15\%$$

Cost of ordinary shares (K_e)

$$K_e = \frac{D_o(1+g)}{P_o - F} + g \times 100\%$$

$$= \frac{4(1.1)}{88-8} + 0.1 \times 100\% = 15.5\%$$

Cost of 9% debenture (K_d)

$$I = \frac{9}{100} \times 100 = 9$$

$$K_d = \frac{I(1-T)}{P_o} \times 100\% = 7\%$$

Cost of 12% debenture K_d

$$I = \frac{12}{100} \times 1,000 = 120$$

$$K_d = \frac{120(1-0.3)}{960} \times 100\% = 8.75\%$$

Cost of 15% preference shares

$$D_p = \frac{15}{100} \times 50 = 7.5$$

$$K_p = \frac{D_p}{P_o} \times 100\% = \frac{7.5}{75} \times 100\% = 10\%$$

v. Firms weighted marginal cost of capital

Therefore, weighted marginal cost of capital if firm plans to raise 20 million

$$WACC = K_e W_e + K_d W_d + K_p W_p$$

$$(0.4 \times 15.5) + (0.3 \times 8.75) + (0.3 \times 10) = 6 + 2.6225 + 3 = 11.825\%$$

QUESTION 29

September 2015 Pilot Paper Question Four A

A Ltd. is considering taking over B Ltd. The forecasted annual net operating cash flows to be generated by the target firm are given as follows:

Year	Net cash flow (NCF) Sh."million"
1	5
2	8
3-7	10
8-10	15
11 - α	12

The firm's minimum required rate of return is 5% above the risk free rate of return. The risk free rate of return 15%.

Required:

The maximum price payable by A Ltd. to acquire B Ltd. (6 marks)

ANSWER

A. A limited

Required rate of return is $5\% + 15\% = 20\%$

Year	Cash flows(million)	D.F 20%	P.V (million)
1	5	0.8333	4.1665
2	8	0.6944	5.5552
3-7	10	$3.6046 - 1.5278 = 2.0768$	20.768
8-10	15	$4.1925 - 3.6046 = 0.5879$	8.8185
11-∞	12	0.1615	1.938
			41.2462

A limited should pay a maximum of Sh. 41.2462 million to acquire B

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TOPIC 7

INTRODUCTION TO CAPITAL BUDGETING DECISIONS

QUESTION 1

December 2023 Question One B

Mapema Ltd. is considering investing in one of the following two mutually exclusive projects. The relevant cash flows of each of the projects are as shown in the table below:

Annual cash flows

Initial investment	Project X Sh.“000”	Project Y Sh.“000”
	(57,750)	(55,500)
Cash flows		
Year:1	37,500	45,000
2	(16,500)	12,000
3	30,000	(6,000)
4	22,500	33,000
5	9,000	(15,000)
6	7,500	22,500

Additional information:

1. The firm's cost of capital is 15%.
2. Cash flows accrue at the end of the year.

Required:

Compute the following for each project:

- (i) Discounted payback period. (5 marks)
- (ii) Modified internal rate of return (MIRR). (5 marks)
- (iii) Profitability index (PI). (2 marks)

ANSWER

(i) Discounted payback period.

Project X				
Year	Cashflows Sh. "000"	D.F@15%	PV Sh. "000"	Cumulative Cashflows Sh. "000"
1	37,500	0.8696	32,610.00	32,610.00
2	-16,500	0.7561	-12,475.65	20,134.35
3	30,000	0.6575	19,725.00	39,859.35
4	22,500	0.5718	12,865.50	52,724.85
5	9,000	0.4972	4,474.80	57,199.65
6	7,500	0.4323	3,242.25	60,441.90

$$\begin{aligned}
 \text{D.P.P} &= 5 + \frac{57,750 - 57,199.25}{3,242.25} \\
 &= 5 + 0.17 = 5.17 \text{ Years}
 \end{aligned}$$

Project Y				
Year	Cashflows Sh. "000"	D.F@15%	PV Sh. "000"	Cumulative Cashflows Sh. "000"
1	45,000	0.8696	39,132.00	39,132.00
2	12,000	0.7561	9,073.20	48,205.20
3	-6,000	0.6575	-3,945.00	44,260.20
4	33,000	0.5718	18,869.40	63,129.60
5	-15,000	0.4972	-7,458.00	55,671.60
6	22,500	0.4323	9,726.75	65,398.35

$$\begin{aligned}
 \text{D.P.P} &= 3 + \frac{55,500 - 44,260.20}{18,869.40} \\
 &= 3 + 0.60 \\
 &= 3.60 \text{ Years}
 \end{aligned}$$

(ii) **Modified internal rate of return (MIRR).**

Project X				
Year	Cashflows Sh."000""	D.F@15%	PV Sh."000""	Cumulative Cashflows Sh."000""
1	37,500	0.8696	32,610.00	32,610.00
2	-16,500	0.7561	-12,475.65	20,134.35
3	30,000	0.6575	19,725.00	39,859.35
4	22,500	0.5718	12,865.50	52,724.85
5	9,000	0.4972	4,474.80	57,199.65
6	7,500	0.4323	3,242.25	60,441.90
		PVCIF	60,441.90	
		Less: PVCOF	<u>57,750.00</u>	
		NPV	2,691.90	

Year	Cashflows Sh."000""	D.F@20%	PV Sh."000""	Cumulative Cashflows Sh."000""
1	37,500	0.8333	31,250.00	31,250.00
2	-16,500	0.6944	-11,458.33	19,791.67
3	30,000	0.5787	17,361.11	37,152.78
4	22,500	0.4823	10,850.69	48,003.47
5	9,000	0.4019	3,616.90	51,620.37
6	7,500	0.3349	2,511.73	54,132.11
		PVCIF	54,132.11	
		Less: PVCOF	<u>57,750.00</u>	
		NPV	-3,617.89	

$$\begin{aligned}
 \text{MIRR} &= 15 + \left(\frac{2691.90}{2691.90 - -3617.89} \times (20 - 15) \right) \\
 &= 15 + \left(\frac{2691.90}{6309.79} \times 5 \right) \\
 &= 15 + 2.13
 \end{aligned}$$

=17.13%

Modified internal rate of return Project Y

Project Y				
Year	Cashflows Sh."000"	D.F@15%	PV Sh."000"	Cumulative Cashflows Sh."000"
1	45,000	0.8696	39,132.00	39,132.00
2	12,000	0.7561	9,073.20	48,205.20
3	-6,000	0.6575	-3,945.00	44,260.20
4	33,000	0.5718	18,869.40	63,129.60
5	-15,000	0.4972	-7,458.00	55,671.60
6	22,500	0.4323	9,726.75	65,398.35
		PVCIF	65,398.35	
		Less: PVCOF	<u>55,500.00</u>	
		NPV	9,898.35	

Year	Cashflows Sh."000"	D.F@30%	PV Sh."000"	Cumulative Cashflows Sh."000"
1	45,000	0.7692	34,615.38	34,615.38
2	12,000	0.5917	7,100.59	41,715.98
3	-6,000	0.4552	-2,731.00	38,984.98
4	33,000	0.3501	11,554.22	50,539.20
5	-15,000	0.2693	-4,039.94	46,499.26
6	22,500	0.2072	4,661.46	51,160.73
		PVCIF	51,160.73	
		Less: PVCOF	<u>55,500.00</u>	
		NPV	-4,339.27	

$$\text{MIRR} = 15 + \left(\frac{9898.35}{9898.35 - -4339.27} \times (30 - 15) \right)$$

$$\begin{aligned}
 &= 15 + \frac{9898.35}{14237} \times 15 \\
 &= 15 + 10.43 \\
 &= 25.43\%
 \end{aligned}$$

(iii) **Profitability index (PI).**

Profitability index of Project X

PI = Present Value of Cash Inflows / Initial Investment

PVCIF	60,441.90
PVCOF	57,750.00
Profitability Index	1.05

Profitability index of Project Y

PI = Present Value of Cash Inflows / Initial Investment

PVCIF	65,398.35
PVCOF	55,500.00
Profitability Index	1.18

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QUESTION 2

December 2023 Question Two C

Tabaka Fabricators Enterprise purchases one of its raw materials branded —Zed” externally. The annual demand for material “Zed” is 400,000 units. The company is planning to change its purchasing system to a just in time (JIT) system to improve efficiency.

The following information is provided:

	Current System	Proposed system (JIT)
Purchase cost per unit (Sh.)	25	25
Ordering cost per order (Sh.)	10,000	2,500
Inventory holding cost	20%	20%

Additional information:

1. To implement the new system, a one time cost of Sh.140,000 will be incurred.
2. The new system is expected to have a lifespan of 8 years.
3. The required rate of return for the firm is 18%.
4. The corporate tax rate is 30%.

Required:

Advise the management of Tabaka Fabricators Enterprise on whether to implement the proposed system. (8 marks)

ANSWER**Economic Order Quantity**

$$EOQ = \sqrt{\frac{2 \times D \times S}{H}}$$

D = Annual demand in units

S = Cost per order

C = Cost per unit

I = Holding cost (%)

H = Holding cost (1 × C)

$$EOQ = \sqrt{\frac{2 \times 400,000 \times 10,000}{0.20 \times 25}}$$

=

40,000 Units under old system

$$\begin{aligned} \text{Total cost under old system} &= \frac{400,000}{40,000} \times 10,000 + \left(\frac{40,000}{2} \times 5 \right) \\ &= 100,000 + 100,000 \\ &= \text{Sh. } 200,000 \end{aligned}$$

$$EOQ = \sqrt{\frac{2 \times 400,000 \times 2,500}{0.20 \times 25}}$$

= 20,000 Units under JIT system

$$\begin{aligned}\text{Total cost under JIT system} &= \frac{400,000}{20,000} \times 2500 + \left(\frac{20,000}{2} \times 5 \right) \\ &= 50,000 + 50,000 \\ &= \text{Sh. } 100,000\end{aligned}$$

$$\begin{aligned}\text{Annual cost savings pretax} &= 200,000 - 100,000 \\ &= \text{Sh. } 100,000\end{aligned}$$

Tabaka Fabricators Enterprise should implement the proposed system.

QUESTION 3

August 2023 Question Two B

Fila Ltd. is considering to purchase a new machine so as to improve its production process which is currently being undertaken manually. The machine costs Sh.13,950,000. The firm will incur installation cost of Sh.450,000. The machine will have an economic life of 6 years but will require an overhaul at the end of the fourth year. The overhaul will cost Sh.1,125,000. After six years, the machine could be disposed of for Sh.900,000.

The company estimates that it will cost Sh.2,100,000 per year to operate the new machine. The current manual production method costs Sh.5,250,000 per annum. In addition to reducing annual operating costs, the new machine will allow the company to increase production capacity by 120,000 units per annum. The company realises a contribution margin of Sh.45 per unit.

Additional information:

1. The company applies straight-line method of depreciation.
2. Corporate tax rate is 30%.
3. Fila Ltd. requires 20% return on all investment.

Required:

Using the Net Present Value (NPV) project evaluation method, advise Fila Ltd. on whether or not to purchase the machine. (12 marks)

ANSWER

Effective cost of new machine

Purchase cost 13,950,000

Add: Installing cost 450,000

14,400,000

Depreciation new machine

$$= \frac{14,400,000 - 900,000}{6}$$

Sh. 2,250,000 PA

Depreciation overhaul cost PA

$$\frac{1,125,000}{2} = \text{Sh.}562,500 \text{ PA}$$

Operating cash inflows

Saving cost = Sh. 5,250,000 – 2,100,000

Sh. 3,150,000

Year	1	2	3	4	5	6
Increase Contribution	5400000	5400000	5400000	5400000	5400000	5400000
Annual Cost Saving	3150000	3150000	3150000	3150000	3150000	3150000
Less: Depreciation	2,250,000	2,250,000	2,250,000	2,250,000	2,250,000	2,250,000
Less: Depreciation Overhaul Cost					562,500	562,500
Before tax cash flows	6,300,000	6,300,000	6,300,000	6,300,000	5,737,500	5,737,500
Less: Tax 30%	1890000	1890000	1890000	1890000	1721250	1721250
After tax cash flows	4,410,000	4,410,000	4,410,000	4,410,000	4,016,250	4,016,250
Add: Depreciation	2,250,000	2,250,000	2,250,000	2,250,000	2,812,500	2,812,500
Less: Overhaul cost				1,125,000		
Add: Salvage value						900,000
Cash flows	6,660,000	6,660,000	6,660,000	5,535,000	6,828,750	7,728,750

Discount factor 20%	0.8333	0.6944	0.5787	0.4823	0.4019	0.3349
PVCIF	5549778	4624704	3854142	2669531	2744475	2588358.4
Cumulative PVCIF	5549778	10174482	14028624	16698155	19442629	22030988
Less: Initial Investment						14,400,000
NPV						7,630,988

Advice:

Fila Limited should purchase the machine since it has a positive net present value.

QUESTION 4

April 2023 Question One B

Cipo Ltd. is evaluating an investment project which requires the importation of a new machine at a cost of Sh.3,700,000. The machine has a useful life of six years and a salvage value of sh.1,000,000.

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Additional information:

- The following additional costs would be incurred in relation to the machine:

Sh.

Modification cost	1,000,000
Import duty	900,000
Installation cost	375,000
Freight charges	225,000

- The machine is expected to increase the company's annual cash flow (before tax) as shown below:

Year	1	2	3	4	5	6
	“000”	“000”	“000”	“000”	“000”	“000”
Increase in cash flow	1,760	1,360	1,050	900	840	750

3. The machine is to be fully depreciated over its useful life using the straight-line method.
4. The corporate rate of tax is 30% while the cost of capital is 10%.
5. The maximum acceptable payback period for the company for all capital project is four years.

Required:

- (i) Total initial cost. (2 marks)
- (ii) Annual net cash flow. (3 marks)
- (iii) Payback period of the machine. (3 marks)
- (iv) Net present value (NPV) of the machine. (3 marks)
- (v) Advise the company's management on whether to import the machine based on your results in (b) (iii) and (b) (iv) above. (1 mark)

ANSWER

(i) Total initial cost.

Purchase cost	3,700,000
Modification cost	1,000,000
Import duty	900,000
Installation cost	375,000
Freight charges	225,000
	6,200,000

(ii) Annual net cash flow.

$$\text{Depreciation annually} = \frac{6,200,000 - 1,000,000}{6}$$

$$= \text{Sh. } 866,667$$

Year	1	2	3	4	5	6
Increase CFS	1,760,000	1,360,000	1,050,000	900,000	840,000	750,000
Less: Dep	866,667	866,667	866,667	866,667	866,667	866,667
PBT	893,333	493,333	183,333	33,333	-26,667	-116,667
Less: 30% tax	<u>268,000</u>	<u>148,000</u>	<u>55,000</u>	<u>10,000</u>	<u>-8,000</u>	<u>-35,000</u>
PAT	625,333	345,000	128,333	23,333	-18,667	-81,667
Add: Dep	<u>866,667</u>	<u>866,667</u>	<u>866,667</u>	<u>866,667</u>	<u>866,667</u>	<u>866,667</u>
Annual Net CF	1,472,000	1,212,000	995,000	890,000	848,000	785,000
Salvage value	-	-	-	-	-	1,000,000
Total net CF	<u>1,492,000</u>	<u>1,212,000</u>	<u>995,000</u>	<u>890,000</u>	<u>848,000</u>	<u>1,785,000</u>

(iii) Payback period of the machine.

Year	Cash flows	Cumulative cash flows
1	1,492,000	1,492,000
2	1,212,000	2,704,000
3	995,000	3,699,000
4	890,000	4,589,000
5	848,000	5,437,000
6	1,785,000	7,222,000

$$\text{Payback period} = 5 + \frac{6,200,000 - 5,437,000}{1,785,000}$$

$$5 + 0.43$$

$$= 5.43 \text{ years}$$

(iv) Net present value (NPV) of the machine.

Year	Cash Flows	PVIF@10%	PV
1	1,492,000	0.9091	1,356,377.20
2	1,212,000	0.8264	1,001,596.80

3	995,000	0.7513	747,543.50
4	890,000	0.6830	607,870.00
5	848,000	0.6209	526,523.20
6	1,785,000	0.5645	1,007,632.50
	PVCIF		5,247,543.20
	PVCOF		(6,200,000)
	NPV		-952,456.80

(v) Advice

The company shouldn't import the machine since it has a negative net present value

QUESTION 5

December 2022 Question One C

Your client, Alfred Otieno, wishes to invest Sh.500,000 for two years (with interest compounded) but with the right to withdraw at a moment's notice. The following investment options are available:

Option I: Invest with Nyumba Building Society currently offering an interest rate of 14% per annum after tax with interest paid half yearly.

Option II: Invest with Kijiji Bank Ltd. at an interest rate of 17% per annum with interest paid annually.

Option III: Invest with Faida Bank Ltd. which is offering 16% per annum, interest paid every three months.

Required:

- (i) Using suitable computations, advise your client on the best investment option he should consider. (6 marks)
- (ii) Calculate the effective rate of interest for option I. (2 marks)
- (iii) Identify TWO situations that could affect your recommendation in (c) (i) above. (2 marks)

ANSWER**Alfred Otieno****(i) Advise**

$$\text{FV Option I} = 500,000 (1.07)^4$$

$$= 500,000 \times 1.3108$$

$$= \text{Sh. } 655,400.00$$

Future value option II

$$= 500,000 (1.17)^2$$

$$= 500,000 \times 1.3689$$

$$= \text{Sh. } 684,450$$

Future value option III

$$= 500\,000 (1.04)^8$$

$$= 500\,000 \times 1.3686$$

$$= \text{Sh. } 684\,300$$

Advice:

Best option is option II because it gives the highest future value.

(ii) Effective rate of interest for option I.

$$r = \left(1 + \frac{0.14}{2}\right)^2 - 1$$

$$= 1.07^2 - 1$$

$$= 0.1449 \text{ or } 14.49\%$$

(iii) Situations that could affect your recommendation in (c) (i) above.

- Increase in interest rate for other options.
- Increase in number of compounding intervals for other options.

QUESTION 6**December 2022 Question Three C**

Jawabu Ltd. is evaluating a project with an expected useful life of 6 years and the following characteristics:

1. Fixed capital investment of Sh.4,000,000.
2. The initial investment in net worth working capital is Sh.400,000.

At the end of each year, net working capital must be increased so that the cumulative investment in net working capital is one-sixth of the next year projected sales.
3. The fixed capital is depreciated on cost at the following rates: 30% in year 1, 35% in year 2, 20% in year 3, 10% in year 4, 5% in year 5 and 0% in year 6.
4. Sales are Sh.2,400,000 in year 1, they grow at 25% annual rate for the next two years and then grow at 10% annual rate for the last three years.
5. Fixed cash operating expenses are Sh.300,000 for year 1-3 and Sh.260,000 for year 4-6.
6. Variable cash operating expenses are 40% of sales in year 1, 39% of sales in year 2 and 38% of sales in year 3 - 6.
7. The corporate tax rate is 30%. If taxable income on the project is negative in any year, the loss will offset gains elsewhere in the corporation, resulting in a tax savings.
8. Fixed capital investment will be sold for Sh.300,000 when the project is complete and recapture its cumulative investment in networking capital. Income taxes will be paid on any gains on disposal.
9. The project required rate of return is 12%.

Required:

Determine the suitability of the project using the Net Present Value (NPV) method.

(8 marks)

ANSWER**Jawabu Ltd.**

The suitability of the project using the Net Present Value (NPV) method.

Annual depreciation expense

$$yr_1 = \frac{30}{100} \times 4,000,000 = \text{Sh.}1,200,000$$

$$yr_2 = \frac{35}{100} \times 4,000,000 = \text{Sh.}1,400,000$$

$$yr_3 = \frac{20}{100} \times 4,000,000 = \text{Sh.}800,000$$

$$yr_4 = \frac{10}{100} \times 4,000,000 = \text{Sh.}400,000$$

$$yr_5 = \frac{5}{100} \times 4,000,000 = \text{Sh.}200,000$$

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Sales

$$yr_1 = 2,400,000$$

$$yr_2 = 2,400,000 \times 1.25 = 3,000,000$$

$$yr_3 = 2,400,000 \times 1.25^2 = 3,750,000$$

$$yr_4 = 3,750,000 \times 1.1 = 4,125,000$$

$$yr_5 = 3,750,000 \times 1.1^2 = 4,537,500$$

$$yr_6 = 3,750,000 \times 1.1^3 = 4,991,250$$

Net working capital

$$yr_0 = 400,000$$

$$yr_1 = 500,000 - 400,000 = 100,000$$

$$yr_2 = \left(\frac{1}{6} \times 3,750,000\right) - 500,000 = 125,000$$

$$yr_3 = \left(\frac{1}{6} \times 4,125,000\right) - 625,000 = 62,500$$

$$yr_4 = \left(\frac{1}{6} \times 4,537,500\right) - 687,500 = 68,750$$

$$yr_5 = \left(\frac{1}{6} \times 4,991,250\right) - 756,250 = 831,875 - 756,250 = \text{Sh.}75,625$$

$$yr_6 = \text{Recovery of Sh.}831,875$$

	1	2	3	4	5	6
Sales	2,400,000	3,000,000	3,750,000	4,125,000	4,537,500	4,991,250
Less:						
Fixed cash	300,000	300,000	300,000	260,000	260,000	260,000
Operating expenses						
Depreciation	1,200,000	1,400,000	800,000	400,000	200,000	0
Profit before tax	900,000	1,300,000	2,650,000	3,465,000	4,077,500	4,731,250
Less: tax	270,090	390,000	795,000	1,039,500	1,223,250	1,419,375
Net income	630,000	910,000	1,855,000	2,425,500	2,854,250	3,311,875
Add:						
Depreciation	1,200,000	1,400,000	800,000	2,425,500	200,000	0
Less:						
(increase)						
decrease						
networking capital	(100,000)	(125,000)	(62,500)	(68,750)	(75,625)	831,875
Cash flows	1,730,000	2,185,000	2,592,500	2,756,750	2,978,625	4,143,750
D.F 12 %	0.8929	0.7972	0.7118	0.6355	0.5674	0.5066
PV	1,544,717	1,741,882	1,845,341.5	1,751,914.63	1,690,071.83	2,099,233.75

$$\begin{aligned}
 NPV &= (1,544,717 + 1,741,882 + 1,845,341.50 + 1,751,914.63 + 1,690,071.83 + \\
 &\quad 2,099,223.75 - 4,400,000) \\
 &= 10,673,150.71 - 4,400,000 \\
 &= \text{Sh.}6,273,150.71
 \end{aligned}$$

QUESTION 7

August 2022 Question One B

Dima Ltd. has developed a new product and is considering whether to put it into production. The following information is available:

1. Development costs will be Sh.4.8 million.
2. Production will require purchase of new machinery at a cost of Sh.2.4 million payable immediately. The machinery has a production life of four years and a production capacity of 30,000 units per annum.

3. Production costs per unit: Sh.

- Variable material cost 8.00
- Variable labour cost 12.00
- Variable overheads 12.00

Fixed production costs including straight line depreciation on plant and machinery will amount to Sh.200,000 per annum.

4. Selling price is Sh.80.00 per unit. Demand is projected at 25,000 units per annum.
5. The retail price index is expected to increase at a rate of 5% per annum over the period and selling price will increase at the same rate. Annual inflation rates on production costs are as follows:

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	(%)
Variable material cost	4
Variable labour cost	10
Variable overheads	4
Fixed costs	5

6. The weighted average cost of capital (WACC) in nominal terms is 15%.

Required:

Advise the firm whether to undertake the production using the net present value (NPV) approach. (8 marks)

ANSWER

Dima Ltd.

Advise on whether to undertake the production using the net present value (NPV) approach.

	Year 1	Year 2	Year 3	Year 4
Selling price per unit	80	84	88.2	92.61
Material cost	8	8.32	8.65	9.00
Labour cost	12	13.2	14.52	15.97
Variable cost	<u>12</u>	<u>12.48</u>	<u>12.98</u>	<u>13.50</u>
Contribution per unit	48	50	52.05	54.14
No. of units	<u>25,000</u>	<u>25,000</u>	<u>25,000</u>	<u>25,000</u>
Total contribution	1,200,000	1,250,000	1,301,250	1,353,500
Less: Fixed costs	200,000	210,000	220,500	231,525
Cash flows	1,000,000	1,040,000	1,080,750	1,121,975

Year	Cash flows	PVIF15%	PV
1	1,000,000	0.8696	869,600
2	1,040,000	0.7561	786,344
3	1,080,750	0.6575	710,593.125
4	1,121,975	0.5718	<u>641,545.305</u>
		PVCIF	3,008,082.43
		Less: PVCOF	<u>(7,200,000)</u>
		NPV	- <u>4191917.57</u>

Advice: Firm should not undertake production

QUESTION 8

August 2022 Question Five A

Hamsa Manufacturing Ltd. is considering two alternative investment proposals. The first proposal requires a major renovation of the company's manufacturing facility. The second proposal involves replacing a few obsolete items of equipment in the manufacturing facility. The company is only able to select one of the two proposals.

The cash flows associated with each proposal are shown below:

Year	Proposal I (Renovation)	Proposal II (replacement of some items)
	Sh."000"	Sh."000"
0	-9,000	-1,000
1	3,500	600
2	3,000	500
3	3,000	400
4	2,800	300
5	2,500	200

The firm discounts cash flows at the rate of 15%.

Required:

- (i) Rank the two investment proposals using the net present value (NPV) approach.

(4 marks)

(ii) Rank the two investment proposals using the internal rate of return (IRR) approach.

(4 marks)

(iii) Compare the rankings under the NPV and IRR approaches and comment on any differences.

(3 marks)

ANSWER

Hamsa Manufacturing Ltd.

(i) Ranking the two investment proposals using the net present value (NPV) approach.

Proposal 1

Year	Cash flows (000)	PVIF @ 15%	PV(000)
1	3,500	0.8696	3,043.60
2	3,000	0.7561	2,268.30
3	3,000	0.6575	1,972.50
4	2,800	0.5718	1,601.04
5	2,500	0.4972	<u>1,243</u>
		PVCIF	10,128.44
		Less: 10	<u>9,000</u>
		NPV	<u><u>1,128.44</u></u>

Proposal 11

Year	Cash flows (000)	PVIF @ 15%	PV(000)
1	600	0.8696	521.76
2	500	0.7561	378.05

3	400	0.6575	263
4	300	0.5718	171.54
5	200	0.4972	<u>99.44</u>
		PVCIF	1,433.79
		Less: PVCOF	<u>1,000</u>
		NPV	<u>433.79</u>

Proposal NPV (000) Rank

Proposal	NPV (000)	Rank
I	1,128.44	1
II	433.79	2

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- (ii) Ranking the two investment proposals using the internal rate of return (IRR) approach.

IRR for proposal I

Year	Cash flows (000)	D.F 20%	PV(000)
1	3,500	0.8333	2,916.55
2	3,000	0.6944	2,083.20
3	3,000	0.5787	1,736.10
4	2,800	0.4823	1,350.44
5	2,500	0.4019	<u>1,004.75</u>

PVCIF	9,091.04
LESS: PVCOF	<u>9,000.00</u>
NPV	<u>91.04</u>

Year	Cash flows (000)	DF 22%	PV(000)
1	3500	0.8197	2,868.95
2	3000	0.6719	2,015.70
3	3000	0.5507	1,652.10
4	2800	0.4514	1,263.92
5	2500	0.3700	<u>925</u>
		PVCIF	8,725.67
		LESS: PVCOF	<u>9,000.00</u>
		NPV	<u>- 274.33</u>

IRR Proposal I

$$\begin{aligned}
 &= 20 + \left[\frac{91.04}{91.04 + 274.33} \times (22 - 20) \right] \\
 &= 20 + \frac{91.04}{365.37} \times 2 \\
 &= 20 + 0.50 = 20.50\%
 \end{aligned}$$

IRR proposal II

Assume discount rate is 22%

Year	Cash flows (000)	D.F 22%	PV(000)
------	------------------	---------	---------

1	600	0.8197	491.82
2	500	0.6719	335.95
3	400	0.5507	220.28
4	300	0.4514	135.42
5	200	0.3700	<u>74</u>
		PVCIF	1,257.77
		Less: PVCOF	1000.00
		NPV	<u>257.47</u>

IRR proposal II

Assume discount rate is 28%

Year	Cash flows (000)	D.F 28%	PV(000)
1	600	0.7813	468.78
2	500	0.6104	305.20
3	400	0.4768	190.72
4	300	0.3725	111.75
5	200	0.2910	<u>58.20</u>
		PVCIF	1,134.65
		Less: PVCOF	<u>1,000.00</u>
		NPV	<u>134.65</u>

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Assume discount rate is 36%

Year	Cash flows (000)	D.F 36%	PV(000)
1	600	0.7353	441.18
2	500	0.5407	270.35
3	400	0.3975	159
4	300	0.2923	87.69
5	200	0.2149	<u>42.98</u>
		PVCIF	1001.2
		Less: PVCOF	<u>1000</u>

NPV 1.20

Assume discount rate is 38%

Year	Cash flows (000)	D.F 36%	PV(000)
1	600	0.7246	434.76
2	500	0.5251	262.55
3	400	0.3805	152.20
4	300	0.2757	82.71
5	200	0.1998	<u>39.82</u>
		PVCIF	972.04
		Less: PVCOF	1,000
		NPV	<u>- 27.96</u>

$$\begin{aligned}
 \text{IRR} &= 36 + \left[\frac{1.2}{1.2 + 27.96} \times (38 - 36) \right] \\
 &= 36 + \left(\frac{1.2}{29.16} \times 2 \right) \\
 &= 36 + 0.08 \\
 &= 36.08\%
 \end{aligned}$$

Proposal internal rate of return rank

Proposal	Internal rate of return	Rank
I	20.50%	2
II	36.08%	1

(iii) Comparing the rankings under the NPV and IRR approaches

Under NPV proposal I is taken because it has highest NPV. Under IRR Project II is taken because it has highest internal of return

QUESTION 9

April 2022 Question Two C

Tuli Ltd. intends to invest in one of two machines; A or B.

The following data is available:

Year	Machine A	Machine B
	Cash inflows "Sh.000"	Cash inflows "Sh.000"
0	(30,000)	(30,000)
1	8,400	8,400
2	9,600	9,000
3	14,000	8,000
4	16,000	10,000
5	4,000	20,000

Additional information:

- Both machines have a useful life of 5 years.
- Depreciation is on a straight line basis.
- The corporate tax rate is 30%.

Required:

Using pay back period as the criterion for project selection, advise the management of

Tuli Ltd. on which machine to invest in. (8 marks)

ANSWER

Tuli Ltd

Advice to the management on which machine to invest in.

Cash flows after tax machine A

Depreciation —00”

$$= \frac{30,000}{5}$$

$$= 6,000$$

Year

$$1 = (8,400 \times 0.7) + (0.3 \times 6,000) = 5,880 + 1,800 = \text{Sh. } 7,680$$

$$2 = (9,600 \times 0.7) + 1,800 = 8,520$$

$$3 = (14,000 \times 0.7) + 1,800 = \text{Sh. } 11,600$$

$$4 = (16,000 \times 0.7) + 1800 = \text{Sh. } 13,000$$

$$5 = (4,000 \times 0.7) + 1800 = 4,600$$

Cashflows machine B

$$1 = (8,400 \times 0.7) + 1800 = \text{Sh. } 7,680$$

$$2 = (9,000 \times 0.7) + 1800 = \text{Sh. } 8,100$$

$$3 = (8,000 \times 0.7) + 1800 = \text{Sh. } 7,400$$

$$4 = (10,000 \times 0.7) + 1800 = \text{Sh. } 8,800$$

$$5 = (20,000 \times 0.7) + 1800 = \text{Sh. } 15,800$$

Payback period project A

Year	Cashflows “000”	Cumulative cashflows “000”
1	7,680	7,680
2	8,520	16,200
3	11,600	27,800
4	13,000	40,000
5	4,600	45,400

$$\text{PBP} = 3 + \frac{30,000 - 27,800}{13,000}$$

$$= 3 + 0.17$$

$$= 3.17 \text{ years}$$

Payback period project B

Year	Cashflows “000”	Cumulative cashflows “000”
1	7,680	7,680
2	8,100	15,780
3	7,400	23,180

4	8,800	31,980
5	15,800	47,780

$$\begin{aligned}\text{PBP} &= 3 + \frac{30,000 - 23,180}{8,800} \\ &= 3 + 0.78 \\ &= 3.78 \text{ years}\end{aligned}$$

Invest in project A because it has lowest PBP

QUESTION 10

December 2021 Question Five A

Bob Ltd. is contemplating replacing an existing machine which was purchased 5 years ago at a cost of Sh.825,000.

The machine was expected to have a useful life of 10 years and an estimated salvage value of Sh.40,000 at the end of the 10th year. However, if replaced the existing machine can be sold in the market today at Sh.300,000.

The new machine will have an initial cost of Sh.1,200,000 and will have a useful life of 5 years. As a result of increased efficiency that will arise due to replacement of the existing machine, the firm will save Sh.168,000 per annum in production costs throughout the new machine's life. In addition, there will be annual savings in reduction of wastages estimated at Sh.80,000.

The salvage value of the new machine is estimated at Sh.100,000 at the end of its useful life.

Additional information:

1. The firm provides for depreciation on a straight-line basis.
2. The cost of capital is 14%.

3. The corporation tax rate applicable is 30%.

Required:

- (i) Using the net present value (NPV) approach, advise the firm on the replacement decision. (8 marks)
- (ii) State any two assumptions you made in your calculations in (a) (i) above. (2 marks)

ANSWER

Bob Ltd.

(i) Replacement decision using the net present value (NPV) approach.

(a) Depreciation of old machine

$$\begin{aligned} &= (825,000 - 40,000) \div 10 \\ &= \text{Sh. } 78,500 \end{aligned}$$

(b) Book value of old machine

$$\begin{aligned} &= 825,000 - (78,500 \times 5) \\ &= 825,000 - 392,500 \\ &= \text{Sh. } 432,500 \end{aligned}$$

(c) initial cash outlay

Purchase cost of new machine	1,200,000
Less: Disposal value of old machine	(300,000)
Less: Tax shield 30% (300,000 – 432,500)	(39,750)
	860,250

(d) Depreciation incremental

Depreciation of new machine

$$\begin{aligned} &= (1,200,000 - 100,000) \div 5 \\ &= 220,000 \end{aligned}$$

Incremental depreciation

$$= 220,000 - 78,500 = 141,500$$

(e) Incremental salvage value

$$= 100,000 - 40,000 = \text{Sh. } 60,000$$

(f) incremental operating cash flows

Savings production costs	168,000
Wastage reduction	<u>80,000</u>
	<u>248,000</u>
Less: Incremental depreciation	<u>(141,500)</u>
Profit before tax	106,500
Less tax 30%	<u>(31,950)</u>
Profit after tax	74,550
Add: Depreciation	<u>141,500</u>
Cash flows	216,050

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NPV

Year	Cash flows	PVIF 14%	PV
1 – 5	216,050	3.4331	741,721.26
5	60,000	0.5194	31,164.00
PV of cash inflows			772,885.26
Less: initial outlay			<u>(860,250)</u>
NPV			<u>(87,364.74)</u>

Advice

Do not replace old machine because it has a negative NPV

(ii) Assumptions in (a) (i) above.

- Capital gains or losses on disposal are subjected to taxation
- Cost of capital of 14% does not change from year to year.

QUESTION 11

September 2021 Question Two B

Jahazi Limited is considering investing in the purchase of a machine for its manufacturing process at a cost of Sh.5,000,000. Installation cost of the machine is expected to be Sh.500,000. The machine is expected to have a useful life of five years, at the end of which, salvage value is estimated at Sh.800,000. This investment shall lead to increase in sales. In order to support increased sales, the firm requires an extra investment in working capital at the start of the machine's useful life. Inventory balances will increase by Sh.1,200,000, debtors balances will rise by Sh.1,500,000 and creditors balance will increase by Sh.1,700,000. The additional investment in working capital will be recovered at the end of the machine's useful life.

The quantity of the product to be manufactured and sold in each year are estimated as follows:

Year	Quantity (Units)
1	20,000
2	25,000
3	30,000
4	35,000
5	40,000

Additional information:

1. The unit selling price and unit variable costs incurred are estimated at Sh.45 and Sh.15 respectively. These are expected to remain constant each year.
2. The firm's estimated fixed operating costs excluding depreciation are Sh.100,000 per annum.

3. The machine will require an overhaul at the end of the second year. This overhaul cost will amount to Sh.240,000.
4. The overhaul cost will be amortised separately on a straight line basis over the remaining useful life of the asset.
5. The firm provides for depreciation on a reducing balance basis at the rate of 32% per annum.
6. The cost of capital is 13%.
7. The corporation tax rate is 30%.

Required:

Using the net present value (NPV) technique, advise on the suitability or otherwise of the project. (12 marks)

ANSWER

Jahazi Limited

Suitability or otherwise of the project.

Initial outlay		
Purchase cost machine	5,000,000	
Installation cost	<u>500,000</u>	
Effective cost	5,500,000	
Add: Increase inventory	1,200,000	
Add: Increase debtors	1,500,000	
Less: Increase creditors	(1,700,000)	
Add: Overhaul cost ($240,000 \times 1.13^{-2}$)	<u>187,944</u>	
	<u>6,687,944</u>	
	45 – 15	Contribution per unit =
	= Ksh. 30	selling unit per unit – variable cost per unit

Year Contributions

1 $20,000 \times 30 =$ 600,000

2	$25,000 \times 30 =$	750,000
3	$30,000 \times 30 =$	900,000
4	$35,000 \times 30 =$	1,050,000
5	$40,000 \times 30 =$	1,200,000

Depreciation annually

$$\text{Year 1} = \frac{32}{100} \times 5,500,000 = 1,760,000$$

$$\text{Year 2} = \frac{32}{100} \times (5,500,000 - 1,760,000) = 1,196,800$$

Depreciation annually 1

$$\text{Year 1} = \frac{32}{100} \times 5,500,000 = 1,760,000$$

$$\text{Year 2} = \frac{32}{100} \times 3,740,000 = 1,196,800$$

$$\text{Year 3} = \frac{32}{100} \times 2,543,200 = 813,824$$

$$\text{Year 4} = \frac{32}{100} \times 1,729,376 = 553,400$$

$$\text{Year 5} = \frac{32}{100} \times 1,175,976 = \underline{376,312}$$

$$\underline{\underline{4,700,336}}$$

Net book value year 5

$$= 5,500,000 - 4,700,336$$

$$= 799,664 \approx \text{Sh } 800,000$$

Working capital

Increase inventory + increase debtors – increase creditors

$$= 1,200,000 + 1,500,000 - 1,700,000$$

$$= 2,700,000 - 1,700,000$$

Initial outlay

Purchase cost 5,000,000

Add: Installation cost 500,000

Effective cost 5,500,000

Increase working capital	1,000,000
Overhaul cost ($240,000 \times 0.7831$)	<u>187,944</u>
	<u>6,687,944</u>

Amortization of overhaul cost P.A

$240,000 \div 3$
 = Ksh 80,000

Contribution per unit = selling price per unit – variable cost per unit

45 – 15

Ksh 30 per unit

Contribution per year

Year	Annual contribution
1	$30 \times 20,000 = 600,000$
2	$30 \times 25,000 = 750,000$
3	$30 \times 30,000 = 900,000$
4	$30 \times 35,000 = 1,050,000$
5	$30 \times 40,000 = 1,200,000$

Year	1	2	3	4	5
Contribution	600,000	750,000	900,000	1,050,000	1,200,000
Less: Fixed costs	100,000	100,000	100,000	100,000	100,000
Less: Depreciation	1,760,000	1,196,800	813,824	553,400	376,312
Less: Amortization	0	0	80,000	80,000	80,000
Profit before tax	(1,260,000)	(546,800)	(93,824)	316,600	643,688
Tax 30%	<u>(378,000)</u>	<u>(164,040)</u>	<u>(28,147)</u>	<u>(94,980)</u>	<u>(193,106)</u>
Net income	(882,000)	(382,760)	(65,677)	221,620	450,580
Add: Depreciation	1,760,000	1,196,800	813,824	553,400	376,312
Add: Amortization	0	0	80,000	80,000	80,000
Add: Resale value					800,000

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	<u>878,000</u>	<u>818,040</u>	<u>828,147</u>	<u>855,020</u>	<u>1,706,892</u>
Recovery working capital					1,000,000
Cash flow	<u>878,000</u>	<u>814,040</u>	<u>828,147</u>	<u>855,020</u>	<u>2,706,892</u>

Year	cashflows	DF 13%	PV
1	878,000	0.8850	777,030
2	814,040	0.7831	637,474.72
3	828,147	0.6931	573,988.69
4	855,020	0.6133	524,383.77
5	2,706,892	0.5428	1,469,300.98
PV of cash inflows			3,982,178.16
Less: Initial outlay			6,687,944
NPV			(2,705,765.84)

Advice:

You should not purchase new asset because it will give a negative figure

QUESTION 12

September 2021 Question Five A

Differentiate between the following terms as used in capital budgeting decisions:

- (i) "Hard capital" and "Soft capital" (2 marks)
- (ii) "Sunk costs" and "Opportunity costs". (2 marks)
- (iii) "Independent projects" and "Mutually exclusive projects". (2 marks)

ANSWER

Differentiation of capital budgeting decisions term

(i) "Hard capital" and "Soft capital"

Hard capital rationing represent rationing that is being imposed on a company by circumstances beyond its control.

Soft capital rationing refers to a situation where a company has freely chosen to impose some restrictions on its capital expenditure even though it may have the ability to make much higher capital investments than it chooses to

The company may choose from any of a number of methods for imposing investment restrictions on itself.

(ii) "Sunk costs" and "Opportunity costs".

Opportunity cost is the cost of a missed opportunity i.e the profit / gain forgone when choosing one business alternative over another. Sunk costs represents past costs that have already been incurred and cannot be recovered.

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(iii) "Independent projects" and "Mutually exclusive projects".

The mutually exclusive projects refer to a set of projects out of which only one project can be selected. A decision to undertake one project from mutually exclusive projects excludes all other projects from consideration.

An independent project is a project whose acceptance or rejection is independent of the acceptance or rejection of the other projects

QUESTION 13

May 2021 Question One A and B

- (a) Explain four categories of capital investment projects. (4 marks)

- (b) Nyakati Limited intends to invest in a four-year mini project whose initial outlay is Sh.32,000,000. The project is expected to generate the following cash flows at the end of each year:

Year	1	2	3	4
Cash flows (Sh. "000")	12,000	15,000	9,000	6,000

The cost of capital is 12%.

Ignore taxation.

Required:

Advise the management on whether to undertake the project using the internal rate of return (IRR) method. (4 marks)

ANSWER

a) Categories of capital investment projects

1. **Mutually exclusive projects** - projects that cannot be done concurrently since they serve same purpose.
2. **Independent projects** – projects that are stand alone and do not affect other projects.
3. **Dependent projects** – projects that rely on one another and the investment in one forces one to invest in another.
4. **Divisible projects** - are those that start generating revenue before they are complete
5. **Indivisible projects** - are those that cannot start generating revenue before they are complete.

(b) Nyakati Limited

Advice on whether to undertake the project using the internal rate of return (IRR) method.

Year	Cash flows (000)	D.F 12%	PV (000)
0	-32,000	1	(32,000)
1	12,000	0.8929	10,714.8
2	15,000	0.7972	11,958.0
3	9,000	0.7118	6,406.2
4	6,000	0.6355	3,813.0
		NPV	892

Year	Cash flows (000)	D.F 16%	PV (000)
0	(32,000)	1.0000	(32,000)
1	12,000	0.8921	10,345.2
2	15,000	0.7432	11,148.0
3	9,000	0.6407	5,766.3
4	6,000	0.5523	3,313.8
		NPV	-1,426.7

$$\begin{aligned}
 \text{IRR} &= \text{LDR} + \frac{\text{NPVLDR}}{\text{NPVLDR} - \text{NPVHDR}} \times (\text{HDR} - \text{LDR}) \\
 &= 12 + \left[\frac{892}{892 - (-1426.7)} \times (16 - 12) \right] \\
 &= 12 + 1.54 \\
 &= 13.54\%
 \end{aligned}$$

The management should undertake the project

QUESTION 14**May 2021 Question Four B**

Alpha Ltd. intends to introduce a new product, branded "Q" into the market. This will require an initial investment in machinery costing Sh.4,800,000. The machinery will be installed at a cost of Sh.200,000 and is estimated to have a useful life of four years and a salvage value of Sh.800,000.

Additional information:

1. Capital allowance will be provided on the machinery on a reducing balance basis.
2. Annual profits from the sale of Product "Q" will amount to Sh.1,920,000 before deducting depreciation on machinery.
3. An investment in working capital amounting to Sh.340,000 will be required on commencement of the project.
4. The firm pays corporation tax at the rate of 30%.
5. Cost of capital is 15% per annum.

Required:

- i. The annual depreciation rate. (3 marks)
- ii. The total initial cash outlay. (2 marks)
- iii. The total terminal cash flows. (2 marks)
- iv. The annual net operating cash flows. (5 marks)
- v. Using the net present value approach, advise the management of this company on the suitability or otherwise of the project. (4 marks)

ANSWER**Alpha Ltd.**

- i. **The annual depreciation rate.**

Total initial outlay

Purchase cost	4,800,000
Add: installation cost	<u>200,000</u>
	<u>5,000,000</u>

Annual depreciation rate (r)

$$R = 1 - \left(\frac{s}{c}\right)^{\frac{1}{n}}$$

Where: r - rate of depreciation

S - Scrap value = 800,000

c - Cost of asset = 5,000,000

n - Useful life

$$r = 1 - \left(\frac{800,000}{5,000,000}\right)^{\frac{1}{4}}$$

$$r = 36.75\%$$

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ii. The total initial cash outlay.

Purchase cost of machine	4,800,000
Add: installation cost	200,000
	5,000,000

iii. The total terminal cash flows.

Salvage value machine	800,000
Recovery working capital	340,000
	1,140,000

iv. The annual net operating cash flows.

Year	Depreciation	Net book value
1	$\frac{36.75}{100} \times (5,000,000 - 800,000)$ $= 1,543,500$	$5,000,000 - 800,000 = 1,543,500$ $= 3,456,500$
2	$\frac{36.75}{100} \times (3,456,500) = 1,270,263.75$ $= 2,186,236.25$	$3,456,500 - 1,270,263.75$ $= 2,186,236.25$
3	$0.3675 \times 2,186,236.25 = 803,441.8219$	$2,186,236.25 - 803,441.8219 =$ $1,382,794.43$
4	$0.3675 \times 1,382,794.43 = 508,176.95$	$1,382,794.43 - 508,176.95 =$ $874,617.48$

Annual net operating cash flows

Year 1

$$\begin{aligned}
 &= (1,920,000 \times 0.7) + (1,543,500 \times 0.3) \\
 &= 1,344,000 + 463,050 \\
 &= \text{sh. } 1,807,050.00
 \end{aligned}$$

Year 2

$$\begin{aligned}
 &= 1,344,000 + (1,270,263 \times 0.3) \\
 &= 1,344,000 + 381,079.13 \\
 &= 1,725,079.13
 \end{aligned}$$

Year 3

$$\begin{aligned}
 &= 1,344,000 + (803,441.82 \times 0.3) \\
 &= 1,344,000 + 241,032.55 \\
 &= 1,585,032.55
 \end{aligned}$$

Year 4

$$= 1,344,000 + (508,176.95 \times 0.3)$$

$$= 1,344,000 + 152,453.09$$

$$= 1,496,453.09$$

v. Advice to the management

Year	Cashflow	D.F 15%	Pv	
0	(5,000,000)	1.0000	- 5,000,000	Management should invest in the project since it has a positive NPV
1	1,807,050	0.8696	1,571,410.68	
2	1,725,079.13	0.7561	1,304,332.33	
3	1,583,032.55	0.6575	1,042,158.90	
4	2,636,453.09	0.5718	1,507,523.88	
		NPV	425,425.79	

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QUESTION 15

November 2020 Question Four A

Ulanda Engineering Works Ltd. is contemplating the purchase of a new machine to replace the existing one. The existing machine was purchased two years ago at an installed cost of Sh.500,000. The machine was estimated to have an economic life of 5 years with nil salvage value but a critical analysis of its performance now shows that it is usable for the next five years with a resale value of Sh.100,000. The current disposal value of existing machine is Sh.200,000.

The new machine would cost Sh.600,000 and require Sh.50,000 in installation cost. Since the machine is not locally available, the company plans to import it and will pay import duty and freight charges of Sh.150,000 and Sh.100,000 respectively. The new machine shall require an overhaul at the end of third year which is expected to cost Sh.100,000.

The overhaul cost is to be amortised on a straight line basis over the remaining useful life of the machine.

To support the increased business resulting from purchase of the new machine, accounts receivable would increase by Sh.250,000, inventories and accounts payable shall increase by Sh.200,000 and Sh.300,000 respectively.

At the end of five years, the new machine would be sold for Sh.250,000.

The estimated profit before depreciation and taxes over the next five years period for both machines are given as follows:

	Existing machine	New machine
Year	Sh."000"	Sh."000"
1	120	260
2	150	280
3	130	250
4	145	240
5	135	270

Additional information:

1. The corporation tax rate is 30%.
2. The company uses the straight line method of depreciation.
3. The cost of capital is 13%.

4. Capital gains are tax exempt.

Required:

- (i) The incremental initial cash outlay. (4 marks)
- (ii) The incremental net operating cash flows associated with the proposed machine replacement. (6 marks)
- (iii) Should the existing machine be replaced? Justify your answer. (4 marks)

ANSWER

(i) The incremental initial cash outlay.

Book value of old machine before critical analysis

$$= 500,000 - \left(\frac{500,000}{5} \times 2 \right)$$

$$= 500,000 - 200,000 = \text{Sh } 300,000$$

Depreciation for the next 5 years

$$= \frac{300,000 - 100,000}{5} = \text{Sh } 40,000 \text{ per year}$$

Incremental initial outlay

	Sh	Sh
Cost of new machine		600,000
Add:		
Installation costs	50,000	
Import duty	150,000	
Freight charges	100,000	<u>300,000</u>
Effective cost new machine		900,000
<u>Tax effect on disposal</u>		

Disposal value of old machine	200,000	
Book value	<u>(300,000)</u>	
Trading balance charge	<u>(100,000)</u>	
Tax rate 30%	30%	<u>0</u>
		900,000
Less: Disposal value of old machine	<u>(200,000)</u>	
		700,000
Add: investment working capital	<u>150,000</u>	
(250,000 + 200,000 – 300,000)		
Incremental initial outlay		<u>850,000</u>

(ii) The incremental net operating cash flows associated with the proposed machine replacement.

1) Incremental depreciation

$$\text{Depreciation of new machine} = \frac{900,000 - 250,000}{5}$$

= Sh 130,000 per year

Depreciation of old machine

$$\frac{300,000 - 100,000}{5} = \text{Sh. 40,000 per annum}$$

Incremental depreciation

$$\text{Sh 130,000} - 40,000$$

$$\text{Sh. 90,000}$$

2) Incremental salvage value

Salvage value of new machine	Sh 250,000
Less: Salvage value of old machine	<u>Sh 100,000</u>

Sh 150,000

3) Incremental terminal cash flows

Incremental salvage value	Sh 150,000
Recovery working capital (250,000 + 200,000 – 300,000)	<u>Sh 150,000</u>
	<u>Sh. 300,000</u>

4) Amortization of overhaul cost

$$= \frac{100,000}{2} = \text{Sh } 50,000$$

5) Incremental profit before depreciation and tax (PBDT)

Year	New machine	Old machine	Incremental PBDT
1	260,000	120,000	140,000
2	280,000	150,000	130,000
3	250,000	130,000	120,000
4	240,000	145,000	95,000
5	270,000	135,000	135,000

Year	1	2	3	4	5
Incremental PBDT	140,000	130,000	120,000	95,000	135,000
Less: Depreciation	90,000	90,000	90,000	90,000	90,000
Profit before tax	50,000	40,000	30,000	5,000	45,000
Less: Amortization	<u>0</u>	<u>0</u>	<u>0</u>	<u>-50,000</u>	<u>-50,000</u>
Profit before tax	50,000	40,000	30,000	(45,000)	(5,000)
Less: Tax 30%	<u>(15,000)</u>	<u>(12,000)</u>	<u>(9,000)</u>	<u>(13,500)</u>	<u>(1,500)</u>
Profit after tax	35,000	28,000	21,000	(31,500)	(3,500)
Terminal cash flow	0	0	0	0	300,000
Incremental net operating Cash flows	35,000	28,000	21,000	(31,500)	296,500

(iii) Should the existing machine be replaced? Justify your answer.

Year	Cash flows	D.F 13%	PV
1	35,000	0.8850	30,975
2	28,000	0.7831	21,926.8
3	21,000	0.6931	14,555.1
4	-31,500	0.6133	-19,318.95
5	296,500	0.5428	160,940.2
PV of cash inflows			209,078.15
Less: initial outlay			<u>(850,000)</u>
NPV			<u>(640,921.85)</u>

Advice:

Should not replace the machine since NPV is negative

QUESTION 16

May 2019 Question Four B

Juhudi Industries intends to replace an existing machine with a new one which is more efficient. The existing machine was acquired 2 years ago at a cost of Sh.4 million. The useful life of this machine was originally expected to be 5 years with no salvage value. However, the valuer has now estimated that the machine shall have an economic useful life of 10 more years and a salvage value of Sh.500.000.

The new machine is estimated to cost Sh. 8 million. An additional installation cost of Sh.400.000 shall be incurred. The new machine has a useful economic life of 10 years. The financial analyst of the company estimates that the existing machine could be sold for Sh.2.5 million at the current prevailing market price.

The new machine is expected to increase sales whereby debtors would increase by Sh.320.000. inventory by Sh, 140.000 while creditors would increase by Sh.300,000. The profit before depreciation and tax over the next 10 years for the two machines is given as follows:

Year	New machine Sh.“000”	Existing machine Sh.“000”
1	350	280
2	400	300
3	420	320
4	410	340
5	410	340
6	380	320
7	380	310
8	350	280
9	300	260
10	280	240

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Additional information:

1. The company's required cost of capital is 10%.
2. Corporate tax rate is at 30%.
3. The company uses a straight-line method of depreciation.

Required:

Using the net present value (NPV) method, advise the management of Juhudi Industries on whether to replace the existing machine with the new one. (12 marks)

ANSWER

A. Use of NPV method

$$\text{Depreciation of old machine} = \frac{\text{original cost} - \text{salvage value}}{\text{useful life}}$$

$$= \frac{4,000,000 - 0}{5} = 800,000$$

Book value of old machine today

Cost - accumulated depreciation

$$4,200,000 - (800,000 \times 2) = \text{Sh. } 2,400,000$$

Gain on disposable old machine

$$\text{Sh. } (2,500,000 - 2,400,000) = \text{Sh. } 100,000$$

Step 1: incremental outlay

Cost of new machine	8,000,000
Add: installation cost	<u>400,000</u>
Effective cost of new machine	<u>8,400,000</u>
Investment working capital	
(320,000 + 140,000 - 300,000)	160,000
Capital gain tax disposal gains	
(0.3 × 100,000)	30,000
Less: disposal value of old machine	<u>(2,500,000)</u>
Incremental initial outlay	<u>6,090,000</u>

Step 2: incremental depreciation

Depreciation of new machine

$$(8,400,000 - 0) \div 10 = 840,000 \text{ per annum}$$

Less depreciation recalculated old machine

$$(2,400,000 - 500,000) \div 10 = 190,000 \text{ per annum}$$

Incremental depreciation

$$840,000 - 190,000 = \text{Sh. } 650,000 \text{ per annum}$$

Step 3: incremental salvage value

Salvage value new machine Sh. 0

Less: salvage value old machine Sh. 500,000

Sh. (500,000)

Step 4 : incremental terminal benefits

Recovery working capital (step 1) Sh. 160,000

Incremental salvage value (step 3) Sh. (500,000)

Sh. (340,000)

Step 5: incremental operating cash inflows

1. $(350,000 - 280,000) \times 0.7 + 0.3 \times 650,000 = 244,000$
2. $(400,000 - 300,000) \times 0.7 + 0.3 \times 650,000 = 265,000$
3. $(420,000 - 320,000) \times 0.7 + 0.3 \times 650,000 = 265,000$
4. $(410,000 - 340,000) \times 0.7 + 0.3 \times 650,000 = 244,000$
5. $(410,000 - 340,000) \times 0.7 + 0.3 \times 650,000 = 244,000$
6. $(380,000 - 320,000) \times 0.7 + 0.3 \times 650,000 = 237,000$
7. $(380,000 - 310,000) \times 0.7 + 0.3 \times 650,000 = 244,000$
8. $(350,000 - 280,000) \times 0.7 + 0.3 \times 650,000 = 244,000$
9. $(300,000 - 260,000) \times 0.7 + 0.3 \times 650,000 = 223,000$
10. $(280,000 - 240,000) \times 0.7 + 0.3 \times 650,000 = 223,000$

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Step 6: determination of net profit value

Year	Cash flows	Discount factor 10%	PV
1	244,000	0.9091	221,820.4
2	265,000	0.8264	218,996.0
3	265,000	0.7513	199,094.5
4	244,000	0.6830	166,652.0
5	244,000	0.6209	151,499.6
6	237,000	0.5645	133,786.5
7	244,000	0.5132	125,220.8
8	244,000	0.4665	113,826.0
9	223,000	0.4241	94,574.3

10	117,000	0.3855	-45,103.5
		PV of cash inflows	1,380,366.6
		Less: initial outlay	(6,090,000)
		NPV	-4,709,000

You should not replace old machine

QUESTION 17

May 2019 Question Five C and D

- (c) i) Explain the term “abandonment” as used in capital budgeting decisions. (2 marks)
- ii) Palakumi Agribusiness Ltd. is analysing a youth empowerment project. The following information is provided:

Year	Cash flow (Sh. “million”)	Abandonment value (Sh. “million”)
0	(16)	-
1	8	12
7	6	8
3	5	6
4	4	-

The company’s cost of capital is 10%.

Required:

Advise the management of Palakumi Agribusiness Ltd. on the optimal time to abandon the project. (5 marks)

- (d) Bidii Enterprises is a small medium enterprise (SME) in floriculture industry. The company intends to invest Sh.300.000 in a project that has a useful economic life of 4 years.

The following are the expected cash flows:

Year	1	2	3	4
Cash flows (Sh.)	140,000	120,000	80,000	60,000

The company's required rate of return is 14%.

Required:

The modified internal rate of return (MIRR) of the project. (3 marks)

ANSWER

C) Capital budgeting

- I.** Abandonment means discounting a project even when it's profitable to continue operating it. NPV of abandonment should exceed NPV without abandonment for this to be viable

II. Abandon year 1

Year	Cash flows	D.F 10%	P.V
1	20,000,000	0.9091	18,182,000
		Less: initial outlay	16,000,000
		Net present value	2,182,000

Abandon year 2

Year 1	Cash flows	D.F 10%	P.V
1	8,000,000	0.9091	7,272,800
2	14,000,000	0.8264	11,569,600
		P.V of cash inflows	18,842,400
		Less: initial outlay	16,000,000
		Net present value	2,842,400

Abandon year 3

Year	Cash flows	D.F 10%	P.V
1	8,000,000	0.9091	7,272,800

2	6,000,000	0.8264	4,958,400
3	11,000,000	0.7513	8,262,300
	P.V of cash inflows		20,495,500
	Less: initial outlay		16,000,000
	Net present values		4,494,500

Do not abandon option

Year	Cash flows	D.F 10%	P.V
1	8,000,000	0.9091	7,272,800
2	6,000,000	0.8264	4,958,400
3	5,000,000	0.7513	3,756,500
4	4,000,000	0.6830	<u>2,732,000</u>
	P.V of cash inflows		<u>18,719,700</u>
	Less: initial outlay		(16,000,000)
	Net present value		2,719,700

Advice: The management of Palakumi agribusiness Ltd Should abandon the project in year 3 because that is where the project net present value is highest

D) Modified internal rate of return

Year	Total cash flow
1	$140,000 \times 1.14^3$ 207,416.16
2	$120,000 \times 1.14^2$ 155,952.00
3	$80,000 \times 1.14^2$ 91,200.00
4	$60,000 \times 1.14^0$ 60,000.00
	514,568.16

$$\text{MIRR} = \left[\left(\frac{514,568.16}{300,000} \right)^{1/4} - 1 \right] \times 100\% = 14.44\%$$

QUESTION 18

November 2018 Question Two A and B

(a) Describe four limitations of the net present value (NPV) method of investment appraisal. (4 marks)

(b) The management of Bundacho Limited is in the process of evaluating two projects, namely Alpha and Beta.

The estimated pre-tax cash flows of each of the projects are as follows:

Year	Project Alpha	Project Beta
	Pre-tax cash flows	Pre-tax cash flows
	Sh. "000"	Sh. "000"
1	2,590	4,300
2	2,880	3,290
3	3,050	3,200
4	2,950	3,700
5	-	4,850
6	-	4,420

Additional information:

1. Project Alpha costs Sh.3.8 million and has an estimated lifespan of 4 years.
2. Project Beta costs Sh.8 million with an estimated lifespan of 6 years.
3. Both projects have a zero salvage value.
4. An investment in working capital of Sh.825,000 will be required irrespective of the project to be undertaken.
5. The cost of capital for the company is 12%.
6. The corporate tax rate is 30%.

Required:

Using the discounted payback period method, recommend to the management of Bundacho Limited on which project to undertake. (8 marks)

ANSWER

a) Limitations of NPV

1. Cannot be used to compare projects of different sizes
2. Requires some guesswork on a project cost of capital
3. Requires some guesswork on a project future cash flows
4. Ignores sunk costs e.g research and development costs incurred before project start.

b) Using the discounted payback period method to recommend to the management of Bundacho Limited on which project to undertake

$$\text{Depreciation per year alpha} = \frac{3,800,000 - 0}{4} = \text{sh. } 950,000$$

$$\text{Depreciation per year Beta} = \frac{800,000 - 0}{6} = \text{sh. } 1,333,333.33$$

For working capital it will be utilized in first year and recovered at end of project life.

PROJECT ALPHA				
Year	1	2	3	4
Pretax cash flows (000)	2,590	2,880	3,050	2,950
Less: Depreciation (000)	(950)	(950)	(950)	(950)
Profit before tax (000)	1,640	1,930	2,100	2,000
Tax rate at 30%	(492)	(579)	(630)	(600)
Profit after tax	1,148	1,351	1,470	1,400
Add: Depreciation	950	950	950	950
Add: Recovery working capital	0	0	0	825
Cash flows	2,098	2,301	2,420	3,175

Discounted payback period Alpha

Year	Cashflows (000)	D.F 12%	Discounted Cashflows (000)	Cumulative Cashflows(000)
1	2,098	0.8929	1873.30	1873.30
2	2,301	0.7972	1834.36	3707.66
3	2,420	0.7118	1722.56	5430.22
4	3,175	0.6355	2017.71	7447.93

Initial outlay project Alpha

$$= 3,800,000 + 825,000 = 4,625,000$$

Discounted Payback period

$$= 2\text{years} + \frac{4625 - 3707.66}{1722.56} \times 12\text{months}$$

$$= 2\text{years } 6.39 \text{ months}$$

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PROJECT BETA

Year	1	2	3	4	5	6
	(000)	(000)	(000)	(000)	(000)	(000)
PBDT	4,300	3,290	3,200	3,700	4,850	4,420
Less:	(1,333.333)	(1,333.333)	(1,333.333)	(1,333.333)	(1,333.333)	(1,333.333)
Depreciation						
PBT	2966.667	1956.667	1866.667	2366.667	3561.667	3086.667
Less tax	(890.000)	(587.000)	(560.000)	(710.000)	(1055.000)	(926.000)
PAT	2076.667					
Add:	1,333.333	1,333.333	1,333.333	1,333.333	1,333.333	1,333.333
Depreciation						
Add:	0	0	0	825		
Recovery WC						
Cashflows	3,409.333	2,703	2,640	2,990	3,840	4,319

Year	Cashflows (000)	D.F 12%	PV
1	3,409,333	0.8929	3,044,193.44
2	2,703,000	0.7972	2,154,831.60
3	2,640,000	0.7118	1,879,152.00
4	2,990,000	0.6355	1,900,145.00
5	3,840,000	0.5674	2,178,816.00
6	4,319,000	0.5066	2,188,000.40

Project Beta Discounted payback period

Year	Discount Cashflows	Cumulative Cashflows (000)
1	3,044,193.44	3,044,193.44
2	2,154,831.60	5,199,025.04
3	1,879,152.00	7,078,177.04
4	1,900,145.00	8,978,322.04
5	2,178,816.00	11,157,138.04
6	2,188,005.40	13,345,143.44

Discounted Payback period

$$\begin{aligned}
 &= 3 \text{ years} + \frac{8,825,000 - 7,078,177.04}{1,900,145} \\
 &= 3.92 \text{ years}
 \end{aligned}$$

Advise:

Management of Bundaaho limited should undertake project Alpha because it has the lowest discounted payback period

QUESTION 19

November 2018 Question Three A

Propose four factors that might lead to soft capital rationing in a limited company.

(4 marks)

ANSWER

Factors that might lead to soft capital rationing in a limited company

1. Policy of reliance on internally generated funds
2. Dilution effect : To prevent outsiders from gaining control of the firm
3. Capital rationing for units of the organization: Fixed allocation from the headquarters to departments
4. Higher cut off ration: setting the firm at rate for investments higher than the firms cost of money.
5. When desirable investment is larger than the size of the firm.

QUESTION 20

May 2018 Question Two B

Kubusa Ltd. is contemplating the acquisition of a new machine to replace the one currently being used in production process. The existing machine was acquired 2 years ago at a cost of Sh.8 million. The existing machine was estimated to have a useful life of 5 years with no salvage value. However, a critical analysis of the machine now shows that the machine is usable for the next 5 years with a salvage value of Sh. 1.5 million. The existing machine can be disposed of now at Sh.4 million.

The new machine is expected to cost Sh. 12.56 million with a salvage value of Sh.4 million at the end of its useful life of 5 years. The new machine will also require an additional investment in working capital of Sh.2.6 million at the start of its useful life which will however be recovered at the end of its useful life.

The following information relates to the estimated earnings before depreciation and tax (EBDT) over the coming live year period for the two machines:

Year	New machine (Sh. “000”)	Existing machine (Sh. “000”)
1	5,400	3,200
2	5,400	2,800
3	5,400	3,000
4	5,400	2,400
5	5,400	2,000

Kubusa Ltd.’s cost of capital is 13%. The company applies straight-line method of depreciation.

The corporate tax rate is 30%.

Required:

Using the net present value (NPV) technique, advise the management of Kubusa Ltd. on whether to replace the existing machine with the new machine. (14 marks)

ANSWER

Using the net present value (NPV) technique to advise the management of Kubusa Ltd. on whether to replace the existing machine with the new machine.

Depreciation of old machine pre – critical analysis

$$= \frac{8,000,000 - 0}{5}$$

$$= sh. 1,600,000$$

Current book value of existing machine before contemplation of disposal

$$= 8,000,000 - (1,600,000 \times 2)$$

$$= shs. 4,800,000$$

Depreciation of existing machine post critical analysis

$$= \frac{4,800,000 - 1,500,000}{5}$$

$$= \text{sh. } 660,000 \text{ per year}$$

Step 1: Incremental outlay

Purchase cost new machine	12,560,000
Add: investment working capital	2,600,000
Less: Disposal value new machine	(4,000,000)
Less: Tax shield benefit disposal loss (0.3x800,000)	(240,000)
	<u>10,920,000</u>

Step 2: Incremental depreciation

Depreciation new machine ($\frac{12,560,000 - 4,000,000}{5}$)	1,712,000
Less: Depreciation old machine	(660,000)
	<u>1,052,000</u>

Step 3: Incremental salvage value

Salvage value new machine	4,000,000
Less: Salvage value old machine	(1,500,000)
	<u>2,500,000</u>

Step 4: Incremental terminal benefit

Recovery working capital (Step 1)	2,600,000
Add: Incremental Salvage value(step 3)	<u>2,500,000</u>
	<u>5,100,000</u>

Step 5: Incremental operating cash in flows

Year	EBIT sh.000 New machine	EBIT sh.000 Old machine	Change EBIT sh.000
1	5,400	3,200	2,200
2	5,400	2,800	2,600

3	5,400	3,000	2,400
4	5,400	2,400	3,000
5	5,400	2,000	3,400

Cash flows per year

$$\text{Cashflows} = \text{EBDT}(1 - T) + Tx\text{Depreciation}$$

$$\text{Year 1} = (2,200 \times 0.7) + 0.3 \times 1052 = 1855.60$$

$$\text{Year 2} = (2,600 \times 0.7) + 0.3 \times 1052 = 2135.60$$

$$\text{Year 3} = (2,400 \times 0.7) + 0.3 \times 1052 = 1995.60$$

$$\text{Year 4} = (3,000 \times 0.7) + 0.3 \times 1052 = 2415.60$$

$$\text{Year 5} = (3,400 \times 0.7) + 0.3 \times 1052 = 2695.60$$

Year	Cashflows (000)	D.F 13%	PV (000)
1	1855.60	0.8850	1642.206
2	2135.60	0.7831	1672.388
3	1995.60	0.6931	1383.150
4	2415.60	0.6133	1481.487
5	7795.60	0.5428	<u>4231.452</u>
		PV of cash inflows	10,410.683
		Less: initial outlay (Step 1)	<u>(10,920.000)</u>
	NPV		<u>-509.317</u>

Advice:

Management should not replace existing machine with new machine since doing so will result to a negative NPV thus destruction of shareholders wealth.

QUESTION 21

May 2018 Question Four C

- (i) Describe two types of capital rationing in capital budgeting. (1 marks)
- (ii) Amani Contractors Ltd. is intending to invest in four independent projects. The following information relates to the four projects:

Project	A	B	C	D
Present values of cash inflows (Sh. —million’')	50	60	100	70
Initial outlay (Sh. —million’')	<u>(30)</u>	<u>(45)</u>	<u>(60)</u>	<u>(40)</u>
Net present values (Sh. —million’')	<u>20</u>	<u>15</u>	<u>40</u>	<u>30</u>

Additional information:

1. The company has a capital limitation of Sh.90 million.
2. The company’s required rate of return is 10%.
3. Any surplus funds can be re-invested to generate a return of net cash flow of 14% in perpetuity.
4. The projects are indivisible.
5. The projects can be combined to achieve a higher return subject to the company’s capital limitation.

Required:

Advise on the optimal project combination.

(5 marks)

ANSWER

i) Two types of capital rationing in capital budgeting

- Externally generated capital rationing also called hard capital rationing scarcity of funds to invest in viable projects which emanate from factors external to the organization
- Internally generated capital rationing also called soft capital rationing. Scarcity of funds that arise from factors inside the company e.g. reliance on internally generated funds to finance projects.

ii) Advice on the optimal project combination

The optimal project combination

Project	Initially outlay (M)	Net present Value (M)
A	30	20
C	60	40
Surplus funds	0	$\frac{0.14 \times 0}{0.1} = 0$
		60

Optimal project is to invest in A&C and invest surplus funds at rate of 14%

QUESTION 22

November 2017 Question Four C

Karem Bottling Company is considering replacing one of the bottling machines with a more efficient one.

The old machine has a current net book value of Sh.2,400,000 with a remaining useful life of five years. The old machine has an estimated re-sale value of Sh.200,000 at the end of its useful life.

The existing machine's current disposal value is estimated to be Sh. 1,060,000.

The new machine has a purchase price of Sh.4,700,000 and an estimated useful life of 5 years. The machine is expected to have an estimated market value of Sh.600,000 at the end of the five years.

The machine is expected to economise on electric power usage and repair costs which will save the company Sh.920,000 each year. In addition, the new machine is expected to reduce the number of defective bottles which will save an additional amount of Sh. 120,000 annually.

The company's corporate tax rate is 30% with a required rate of return of 12%.

The company provides for depreciation on a straight line basis.

Assume capital gains are taxable.

Required:

- (i) The initial net cash outlay. (3 marks)
- (ii) The incremental net operating cash flows for years 1 through year 5. (4 marks)
- (iii) The total terminal cash flows. (2 marks)
- (iv) Using net present value (NPV) criteria, advise the management of Karem Bottling Company whether or not to purchase the new machine. (4 marks)

ANSWER

Karem bottling company

i. Initial net cash outlay

Purchase cost of new machine	4,700,000
Less: disposal proceeds of old machine	1,060,000
Less: tax shield benefit disposal loss	
30%(2,400,000 – 1,060,000)	<u>(402,000)</u>
	<u>3,238,000</u>

ii. The incremental net operating cash flows for years 1 through year 5

Incremental depreciation

Depreciation of new machine = $4,700,000 - 600,000 \div 5 = \text{Sh. } 820,000$

Depreciation of old machine = $(2,400,000 - 200,000) \div 5 = \text{Sh. } 440,000$

Incremental depreciation = $820,000 - 440,000 = \text{Sh. } 380,000$

iii. Incremental salvage value

Salvage value of new machine Sh. 600,000

Salvage value of old machine Sh. (200,000)

400,000

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iv. Total terminal cash flows

Recovery working capital	0
Add: incremental salvage value	<u>400,000</u>
	<u>Sh. 400,000</u>

Incremental net operating cash inflows

Savings on electric power usage and repair costs	Sh. 920,000
Add: savings in defective bottles annually	Sh. 120,000
Less: incremental depreciation	<u>Sh. (380,000)</u>
Profit before tax	660,000
Less: tax 30%	<u>(198,000)</u>
Profit after tax	462,000
Add: incremental depreciation	<u>380,000</u>
Operating cash inflows	<u>842,000</u>

v. Whether to purchase new machine

Year	Cash flows	D.F 12%	PV
1-5	842,000	3.6048	3,035,241.60
5	400,000	0.5674	226,960
	PV cash inflows		3,262,201.60
	Less: initial outlay		3,238,000.00
	NPV		24,201.60

Advice

Karem bottling company should replace existing machine since it has a positive NPV

QUESTION 23**May 2017 Question Three B**

Roka Limited has two mutually exclusive projects namely; project A and project B with initial cash outlay of Sh.50,000 each. The projects have a useful life of 5 years. The company's cost of capital is 12% with a corporate tax rate of 30%.

The expected cash flows for the projects before depreciation and tax are given below:

Year	Project A Sh.“000”	Project B Sh.“000”
1	42	62
2	42	32
3	42	22
4	42	52
5	42	52

The company uses straight line method of depreciation.

Required:

Using the profitability index approach, advise the management of Roka Limited on the project to consider. (10 marks)

ANSWER

Using the profitability index approach, advise the management of Roka Limited on the project to consider

$$= \frac{\text{annual cash flows project A}}{\text{depreciation per annum}}$$

$$= \frac{50,000}{10,000}$$

$$= 5 \text{ per annum}$$

Annual cash flows

$$= PBDT (1 - T) + \text{depreciation} \times T$$

$$= 42,000 \times (1 - 0.3) + 10,000 \times 0.3$$

$$= \text{Sh. } 32,400$$

Project B

Year	1	2	3	4	5
Profit before depreciation and tax(PBDT	62,000	32,000	22,000	52,000	52,000
Less: depreciation	10,000	10,000	10,000	10,000	10,000
Profit before tax	52,000	22,000	12,000	42,000	42,000
Less: tax 30%	(15,600)	(6,600)	(3,600)	(12,600)	(12,600)
Profit after tax	36,400	15,400	8,400	29,400	29,400
Add: depreciation	10,000	10,000	10,000	10,000	10,000
Cash flows	46,400	25,400	18,400	39,400	39,400

PV of cash flows and profitability

Project A

Year	Cash flows	D.F 12%	PV
1-5	32,400	3.6048	116,795.52

$$\text{Profitability index (PI)} = \frac{\text{PV of cash inflows}}{\text{initial outlay}} = \frac{116,795.52}{50,000} = 2.34$$

Every one shilling invested in profit A generates Sh. 2.34

Project B

Year	Cash flows	D.F 12%	PV
1	46,400	0.8929	41430.56
2	25,400	0.7972	20248.88
3	18,400	0.7118	13097.12
4	39,400	0.6355	25038.70

5	39,400	0.5674	22,355.56
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PV cash inflows 122,170.82

$$\text{Profitability index} = \frac{122,170.82}{50,000} = 2.44$$

Every one shilling invested in project B generates Sh. 2.44

Advise:

Management of Roka limited should undertake project B because it has the highest profitability index

QUESTION 24

November 2016 Question Four C

Mwarakaya Ltd. is considering the acquisition of a new machine to replace the existing machine currently being used in production processes. The existing machine was acquired 2 years ago at a cost of Sh.2,000,000. It was originally estimated to have a useful life of 5 years with no salvage value.

A critical evaluation of the machine now shows that the machine is usable for another 5 years with a salvage value of Sh.250,000 at the end of this period. The disposal value of the existing machine is currently estimated at Sh. 1,250,000.

The new machine is estimated to cost Sh.3,140,000 and its estimated salvage value is Sh. 1,000,000 at the end of its useful life of 5 years. The new machine will also require an additional investment in working capital of Sh.650,000 at the start of the asset's useful life.

The investment in working capital will however be recovered at the end of the 5 years useful life.

The following information relates to the estimated earnings before depreciation and tax (EBDT) over the coming five-year period for the two machines.

Year	New machine Sh.	Existing machine Sh.
1	1,400,000	800,000
2	1,350,000	700,000
3	1,300,000	750,000
4	1,450,000	650,000
5	1,200,000	600,000

The cost of capital is 10% and the firm applies the straight line method of depreciation.

The corporate tax rate is 30%.

Required:

Using the net present value (NPV) technique, advise the company's management on whether to replace the existing machine. (10 marks)

ANSWER

Net present value (NPV) of the replacement decision:

Depreciation of existing machine before critical evaluation:

$$= \frac{2,000,000}{5} = sh. 400,000 \text{ per annum}$$

Book value of existing machine

$$2,000,000 - (400,000 \times 2) = sh. 1,200,000$$

Gain if existing machine sold now

$$= 1,250,000 - 1,200,000 = sh. 50,000$$

Step 1: Incremental initial outlay

Purchase cost of new machine	3,140,000
Add: Additional investment working capital	650,000
Less: Disposal value existing machine	(1,250,000)
Add: Tax on disposal gains (0.3x50,000)	<u>15,000</u>
<i>Initial outlay</i>	<u>2,555,000</u>

Step 2: Incremental Depreciation

Depreciation new machine:

$$= \frac{3,140,000 - 1,000,000}{5} = sh. 428,000$$

Depreciation new machine after critical evaluation

$$= \frac{1,200,000 - 250,000}{5} = sh. 190,000$$

Incremental depreciation = 428,000 - 190,000 = **sh. 238,000**

Step 3: Incremental salvage value

Salvage value new machine	1,000,000
Less: salvage value of old machine	<u>(250,000)</u>
	<u>750,000</u>

Step 4: Incremental terminal benefits

Recovery of working capital	650,000
Add: Incremental salvage value	<u>750,000</u>
	<u>1,400,000</u>

Step 5: Incremental operating cash inflows

Year	New machine	Old machine	Change	Cash inflows
1	1,400,000	800,000	600,000	$(600,000 \times 0.7) + (0.3 \times 238,000) = 491,400$
2	1,350,000	700,000	650,000	$(650,000 \times 0.7) + (0.3 \times 238,000) = 526,400$
3	1,300,000	750,000	550,000	$(550,000 \times 0.7) + (0.3 \times 238,000) = 456,400$
4	1,450,000	650,000	800,000	$(800,000 \times 0.7) + (0.3 \times 238,000) = 631,400$
5	1,200,000	600,000	600,000	$(600,000 \times 0.7) + (0.3 \times 238,000) = 491,400$
				<u>1,400,000</u>

NPV of the project

YEAR	Cashflows Sh."000"	FVIF _{n,10%}	Present value
1	491,400	0.9091	446,732.74
2	526,400	0.8264	435,016.96
3	456,400	0.7513	342,893.32
4	631,400	0.6830	431,246.20
5 (490,400 + 1,400,000)	1,891,400	0.6209	1,174,370.26
Sum discounted cash flow			2,830,265
Incremental initial cost N.P.V			(2,555,000)
			275,258.48

Advice:

Management of Mwarakaya limited company should replace existing machine due to positive NPV

QUESTION 25

May 2016 Question One A and B

(a) Explain four principles of capital budgeting

(b) A firm is considering the following investment projects:

Project	Cash flows (Sh.)			
	Year 0	Year 1	Year 2	Year 3
A	1,000,000	500,000	500,000	-
B	1,000,000	-	650,000	850,000
C	1,000,000	300,000	500,000	1,000,000
D	1,000,000	800,000	400,000	400,000

The firm's opportunity cost of capital is 15%.

Required:

- (i) Rank the projects using payback period method. (3 marks)
- (ii) Rank the projects using net present value (NPV) method. (3 marks)

ANSWER

a) Principles of capital budgeting

1. The capital budgeting cash flows are not the same as accounting net income.
2. The timing of Cashflows is critical
3. Post tax cash flows are used in analysis. Taxes have to be fully incorporated in capital budgeting decisions
4. Decision are based on Cashflows and not an accounting concepts such as net income
5. The financing costs are ignored. Financing costs are already reflected in the required rate of return and therefore including them again in the cashflow and in the discounting rate could lead to double counting
6. Cashflows are based on opportunity costs with comparison made between the incremental cashflow that occur with and without investment

b) i) Ranking the projects using payback period method

Payback period refers to time taken to recover money initially invested in the project

Payback period for project A

Year	Cashflows	Cumulative cashflows
1	500,000	500,000
2	500,000	1,000,000

Payback period Project A = 2 years

Payback period for project B

Year	Cashflows	Cumulative cashflows
1	0	0
2	650,000	650,000
0	850,000	1,500,000

Payback period Project B = 2 years + $\frac{1,000,000}{850,000} = 2.41 \text{ years}$

Payback period for project C

Year	Cashflows	Cumulative cashflows
1	300,000	300,000
2	500,000	800,000
0	1,000,000	1,800,000

Payback period Project C = 2 years + $\frac{1,000,000 - 800,000}{1,000,000} = 2.2 \text{ years}$

Payback period for project D

Year	Cashflows	Cumulative cashflows
1	800,000	800,000
2	400,000	1,200,000
0	400,000	1,600,000

$$\text{Payback period Project C} = 1 \text{ years} + \frac{1,000,000 - 800,000}{400,000} = 1.5 \text{ years}$$

Project	Payback period	Rank
A	2 years	2
B	2.41 years	4
C	2.2 years	3
D	1.5 years	1

ii) Ranking the projects using net present value (NPV) method

Net present value is the difference between the present value of cash inflows and the initial outlay

NPV A

Year	Cashflows	D.F 15%	PV
1-2	500,000	1.6257	812,850
			(1,000,000)

NPV B

Year	Cashflows	D.F 15%	PV
2	650,000	0.7561	491,465
3	850,000	0.6575	558,875
	PV of cash inflows		1,050,340
	Less: Initial outlay		(1,000,000)
	NPV		<u>50,340</u>

NPV C

Year	Cashflows	D.F 15%	PV
1	300,000	0.8696	260,880
2	500,000	0.7561	378,050
3	1,000,000	0.6575	675,500
	PV of cash inflows		1,296,430
	Less: Initial outlay		(1,000,000)
	NPV		<u>296,430</u>

NPV D

Year	Cashflows	D.F 15%	PV
1	800,000	0.8696	695,680
2-3	400,000	1.4136	565,440
	PV of cash inflows		1,261,120
	Less: Initial outlay		(1,000,000)
	NPV		<u>261,120</u>

Project	Payback period	Rank
A	-187,150	4
B	50,340	3
C	296,430	1
D	261,120	2

QUESTION 26

May 2016 Question Four D

Laika Ltd. has identified five investment projects with the following details:

Investment project	Initial outlay (Sh. “millions”)	Net present value of investment (Sh. “million”s)
A	120	24.0
B	160	43.2
C	100	17.0
D	90	21.6
E	110	19.8

Additional information:

1. None of the investment projects could be delayed.
2. Amount available for investment is limited to Sh.300 million, therefore, the company cannot undertake all the investment projects.
3. All the five projects are divisible.

Required:

Advise the management of Laika Ltd. on the most appropriate investment projects to undertake. (5 marks)

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ANSWER

Advising the management of Laika Ltd. on the most appropriate investment projects to undertake

We have to calculate the profitability index and rank projects according to that

$$\text{Profitability index} = \frac{NPV}{\text{Initial outlay}}$$

Project	Profitability index	Rank
A	$24 \div 120 = 0.20$	3
B	$43.2 \div 160 = 1 \text{ } 0.27$	1
C	$17 \div 100 = 0.17$	5
D	$21.6 \div 90 = 0.24$	2
E	$19.8 \div 110 = 0.18$	4

Capital limit of investment=Sh.300 millions

Investment projects to undertake based on sh.300 millions.

Investment	Initial outlay	Net present value
B	160	43.2
D	90	21.6
A(balance)	50	$\frac{50}{120} \times 24 = 10$
	300	74.8

Advice:

Laika ltd should undertake project B,D and A

QUESTION 27

November 2015 Question Three B and C

b) The following details relate to a capital project in XYZ Limited:

Project cost	Sh.65,000,000
Annual cash flows (after tax)	Sh.21,000,000
Project economic life	5 years
Required rate of return	12%

Required:

Assess the suitability of the capital project using the following methods:

- (i) Internal rate of return (IRR). (5 marks)
- (ii) Profitability index (PI). (3 marks)

(c) Nile group of hotels is considering the acquisition of Victoria hotel at a cost of Sh.200 million. The group of hotels cost of capital is currently 16% due to its high gearing level. Victoria hotel has no debt.

As a result of acquisition, the cost of capital for Nile group of hotels will drop to 12%. Total cash flows will also increase by 25 million per annum in perpetuity.

Required:

- (i) Using the net present value (NPV) approach, advise the management of Nile group of hotels on the acquisition of Victoria hotel. (3 marks)
- (ii) If the acquisition was funded by borrowing so that there is no impact on gearing after acquisition and the cost of capital was not reduced, advise the management of Nile group of hotels whether to proceed with the acquisition of Victoria hotel. (3 marks)

ANSWER

b) XYZ limited

i. Internal rate of return of return (IRR)

$$\text{Target discount rate} = \frac{65,000,000}{21,000,000} = 3.0952$$

Assume discount rate of 18% then,

$$\begin{aligned} \text{NPV} &= (3.1272 \times 21,000,000) - 65,000,000 \\ &= 65,671,200 - 65,000,000 \\ &= 671,200 \end{aligned}$$

Assume a discount rate of 20% then,

$$\begin{aligned} \text{NPV} &= (2.9906 \times 21,000,000) - 65,000,000 \\ &= (2,197,400) \end{aligned}$$

$$\text{IRR} = \text{LDR} + \frac{\text{NPV}_{\text{LDR}}}{\text{NPV}_{\text{LDR}} - \text{NPV}_{\text{HDR}}} \times (\text{HDR} - \text{LDR})$$

$$18 + \frac{671,200}{671,200 - (2,197,400)} \times (20 - 18)$$

$$18 + 0.47 = 18.47\%$$

ii. Profitability index (PI)

$$PI = \frac{\text{present value of cash inflows}}{\text{Project cost}} = \frac{21,000,000 \times 3.6048}{65,000,000}$$

$$= \frac{75,700,800}{65,000,000} = 1.16$$

For every Sh.1 invested in project XYZ generates Sh. 1.16

c) Nile group

i. Advice to management of Nile group

$$NPV = \frac{\text{Cash flows in perpetuity}}{\text{Cost of capital}} - \text{initial outlay}$$

$$\frac{25,000,000}{0.12} - 200,000,000$$

$$208,333,333.3 - 200,000,000 = \text{Sh. } 8,333,333.3$$

Nile group should proceed with the acquisition

ii. Whether to proceed with the acquisition of Victoria hotel.

$$NPV = \frac{25,000,000}{0.16} - 200,000,000$$

$$= 156,250,000 - 200,000,000$$

$$NPV = \text{Sh. } -43,750,000$$

Nile group should not proceed with the purchase of Victoria Hotel due to negative NPV

QUESTION 28

September 2015 Pilot paper Question Two B and C

The management of Swara Ltd. is considering replacing an existing machine which was bought 3 years ago at a cost of Sh.20 million. The machine was expected to have a useful life of 5 years with no resale value at the end of this period. A critical evaluation of this asset shows that the existing machine is usable for another five years at the end of which resale value is estimated at Sh.2 million. The current disposal value of the existing machine is estimated at Sh.10 million.

The new machine is not locally available. The management expect to import this machine at a cost of Sh.40 million. Installation cost of this machine is estimated at Sh.500,000.

Import duty payable and freight charges are estimated at Sh.300,00 and Sh.200,000 respectively. This machine is expected to have a useful life of five years, at the end of which resale value is estimated at Sh.5 million.

This investment is expected to lead to increased sales. To support increase in sales, the firm will require an extra investment in working capital at the beginning of the new machine's useful life. Inventory balance is expected to increase by Sh.800,000, debtors balance will increase by Sh.700,000 and creditors balance will increase by Sh. 1,000,000. However, the firm will require an extra investment in working capital at the end of the second year of Sh.250.000. The total investment in working capital will be recovered at the end of the machine's useful life.

The earnings before depreciation and tax to be generated by each asset during each year are given as follows:

Earning before depreciation and tax (EBDT)

Year	New machine Sh.“000”	Existing machine Sh.“000”
1	70,000	50,000
2	75,000	55,000
3	85,000	60,000
4	80,000	55,000
5	70,000	65,000

Additional information:

1. The new machine shall require an overhaul at the end of third year. The overhaul cost is estimated at Sh.2 million. The cost will be amortised separately on a straight line basis.
2. The firm provides for depreciation on all their non-current assets on a straight line basis.
3. The firm pays corporation tax at the rate of 30%.
4. The firm's capital structure which is optimal comprises of 70% equity and 30% debt. The cost of equity is 10% and before tax cost of debt is 8%.

Required:

Using the net present value technique, advise on whether the firm should replace the existing machine. (15 marks)

(c) State two limitations of the net present value method. (2 marks)

ANSWER

b) Swara Ltd

Incremental cash outlay

Step 1

Import cost of new machine	40,000,000
Add: installation cost of new machine	500,000
Add: import duty payable	300,000
Add: freight charges	<u>200,000</u>
Effective cost of new machine	<u>41,000,000</u>
Working capital investment (800,000 + 700,000 – 1,000,000)	500,000
Tax recapture on disposal gains 0.3(10,000,000 - 8,000,000)	600,000
Less: disposal proceeds old machine	(10,000,000)
Incremental initial outlay	32,100,000

Step 2: depreciation incremental

Depreciation new machine = $\frac{41,000,000 - 5,000,000}{5}$	7,200,000
Depreciation old machine $\frac{8,000,000 - 2,000,000}{5}$	1,200,000
Incremental depreciation (7,200,000 – 1,200,000)	6,000,000

Step 3: incremental salvage value

Salvage value new machine	5,000,000
Less: salvage value of old machine	(2,000,000)
Incremental salvage value	3,000,000

Step 4 : incremental terminal benefit

Incremental salvage value	3,000,000
Recovery of working capital (year 1):	500,000
Recovery of working capital (year 2):	250,000
Incremental terminal benefits	3,750,000

Step 5: incremental operating cash inflows

Year	1 (000)	2 (000)	3(000)	4(000)	5(000)
EBDT	20,000	20,000	25,000	25,000	5,000
Less: depreciation	(6,000)	(6,000)	(6,000)	(6,000)	(6,000)
Less: amortization overhaul cost	<u>0</u>	<u>0</u>	<u>0</u>	<u>(1,000)</u>	<u>(1,000)</u>
EBT	14,000	14,000	19,000	18,000	(2,000)
Less: tax	<u>(4,200)</u>	<u>(4,200)</u>	<u>(5,700)</u>	<u>(5,400)</u>	<u>600</u>
Profit after tax	9,800	9,800	13,300	12,600	(1,400)
Add: depreciation	6,000	6,000	6,000	6,000	6,000
Add: amortization	-	-	-	1,000	1,000
Less: working capital	-	(500)	-	-	-
Less: overhaul cost	-	-	(3,000)	-	-
Add: terminal benefits					
Cash flows	15,800	15,300	16,300	19,600	5,600

$$WACC = \left(\frac{70}{100} \times 10\% \right) + \left(\frac{30}{100} \times 8\% \times 0.7 \right) = 8.68\%$$

Year	Cash flows	D.F 8.68%	P.V (000)
1	15,800	0.9201	14,537.58
2	15,300	0.8466	12,952.98
3	16,300	0.7790	12,697.70
4	19,600	0.7168	14,049.28
5	5,600	0.6596	<u>3,693.76</u>
PV of cash inflows			57,931.30
Less: initial outlay			(32,100.00)
NPV			25,831.30

Swara Ltd should replace the existing machine because of the positive net present value

Workings

1. Depreciation old machine = $\frac{20,000,000-0}{5} = 4,000,000$
2. Book value of old machine at contemplation of replacement date $20,000,000 - (4,000,000 \times 3) = 8,000,000$
3. Amortization of overhaul cost = $2,000,000 \div 2 = \text{Sh. } 1,000,000$

c) Limitations of NPV

1. It requires guesswork about the future cash flows
2. It requires guesswork about the discount rate
3. Can be misleading when comparing projects of different sizes

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TOPIC 8

FINANCIAL ANALYSIS AND FORECASTING

QUESTION 1

December 2023 Question Four A

In relation to measuring business performance in a company:

- (i) Examine **FOUR** applications of financial ratio analysis by the management of the company. (4 marks)
- (ii) Describe **FOUR** weaknesses associated with financial ratio analysis. (4 marks)

ANSWER

- (i) **Applications of financial ratio analysis by the management of the company.**

Financial ratio analysis is a powerful tool for company management, offering valuable insights into various aspects of a company's financial health and performance. Here are some key applications:

Assessing Financial Performance:

- **Profitability:** Ratios like Profit Margin, Return on Assets (ROA), and Return on Equity (ROE) help evaluate how efficiently the company generates profits from its resources and investments.
- **Liquidity:** Ratios like Current Ratio and Quick Ratio assess the company's ability to meet short-term debt obligations and its overall financial flexibility.
- **Solvency:** Ratios like Debt-to-Equity Ratio and Debt Coverage Ratio measure the company's ability to meet long-term debt obligations and financial stability.
- **Efficiency:** Ratios like Inventory Turnover Ratio and Receivables Turnover Ratio evaluate how effectively the company manages its assets and resources.

Strategic Decision Making:

- **Investment decisions:** Ratios can help assess the feasibility and potential returns of new investments by comparing the company's performance to industry benchmarks.
- **Budgeting and forecasting:** Ratios can be used to set realistic financial goals and budgets based on historical trends and industry norms.
- **Performance management:** Tracking changes in key ratios over time helps identify areas for improvement and measure the effectiveness of implemented strategies.
- **Mergers and acquisitions:** Ratios can be used to evaluate the financial strength and compatibility of potential acquisition targets.

Benchmarking and Competitor Analysis:

- **Comparing performance:** Ratios allow comparison of the company's financial performance to competitors within the industry, identifying areas of strength and weakness.
- **Identifying industry trends:** Tracking changes in industry-wide ratios can help the company anticipate market shifts and adjust its strategies accordingly.
- **Competitive advantage:** Understanding competitors' financial strengths and weaknesses can inform strategies for gaining a competitive edge.

Risk Management:

- **Identifying financial risks:** Early warning signs of potential financial trouble can be detected through changes in key ratios, prompting preventative measures.
- **Monitoring debt levels:** Ratios help track debt levels and ensure they remain within manageable limits.
- **Evaluating profitability trends:** Declining profitability ratios can indicate potential risks that need to be addressed.

Communication and Reporting:

- **Communicating financial performance:** Financial ratios simplify complex financial data into easily understandable metrics for stakeholders like investors, creditors, and boards of directors.
- **Supporting loan applications:** Strong financial ratios can enhance the company's creditworthiness and improve the chances of securing loans.
- **Meeting regulatory requirements:** Certain industries and regulations might require companies to report specific financial ratios.

(ii) Weaknesses associated with financial ratio analysis.

Financial ratio analysis is a valuable tool for assessing a company's financial health, but it's important to understand its limitations and weaknesses to avoid misinterpretations or misleading conclusions. Here are some key weaknesses to consider:

Overreliance on Historical Data:

- Ratios reflect past performance, which may not always be indicative of future results. External factors like economic changes, competitor actions, or technological advancements can significantly impact future performance.
- Trending analysis helps mitigate this limitation, but unforeseen events can still disrupt projected outcomes.

Industry Dependence and Benchmarking:

- Comparing ratios across different industries can be misleading due to differing business models, accounting practices, and regulatory environments.
- Choosing appropriate industry benchmarks and adjusting for relevant factors is crucial for meaningful comparisons.

Potential for Manipulation:

- Creative accounting practices or selective data presentation can distort ratios, making them unreliable indicators of true financial health.

- Careful analysis beyond just the numbers, considering qualitative factors and management practices, is necessary to detect potential manipulation.

Limited Scope and Oversimplification:

- Ratios provide a snapshot of specific financial aspects, potentially neglecting other crucial factors like brand value, intellectual property, or talent pool.
- Combining ratio analysis with other financial tools and qualitative assessments offers a more comprehensive understanding.

Focus on Quantitative Measures:

- Ratios often overlook qualitative factors like management competence, employee morale, or customer satisfaction, which can significantly impact performance.
- Combining quantitative and qualitative analysis ensures a more holistic view of the company's health.

Subjectivity in Interpretation:

- Different analysts might interpret the same ratios differently, leading to varying conclusions.
- Establishing clear benchmarks, considering industry context, and using multiple ratios for triangulation can reduce subjectivity.

Inability to Predict Future Performance:

- While ratios can identify potential risks or strengths, they are not crystal balls and cannot predict the future with certainty.
- Scenario planning and considering various external factors remain essential for strategic decision-making.

Focus on Short-term Analysis:

- Emphasis on ratios like liquidity and current ratios can be detrimental to long-term investments or strategies requiring initial capital expenditure.
- Balancing short-term and long-term perspectives through comprehensive financial analysis is crucial.

QUESTION 2**August 2023 Question Three B**

Utamu Enterprises operates a fruit juice processing business. The financial manager has realised a consistent relationship among the following items as a percentage of sales:

- | | | |
|----|---------------------------|-----|
| 1. | Current assets | 65% |
| 2. | Accounts payable | 10% |
| 3. | Non-current assets | 25% |
| 4. | Other current liabilities | 12% |

The statement of financial position for the year ended 31 December 2022 was as follows:

Utamu Enterprises
Statement of financial position as at 31 December 2022

	Sh.	Sh.
Non-current assets		2,500,000
Current assets		6,500,000
Total assets		<u>9,000,000</u>
Current liabilities:		
Accounts payable	1,000,000	
Notes payable	1,200,000	
Other current liabilities	<u>1,200,000</u>	3,400,000
Equity and non-current obligations:		
Ordinary share capital	2,000,000	
Retained earnings	2,600,000	
Long term loan	<u>1,000,000</u>	<u>5,600,000</u>
Total equity and non-current obligations		<u>9,000,000</u>

Additional information:

1. The profit after tax margin is 5% and the company sales for the year ended 31 December 2022 were Sh.10 million.
2. Sales are expected to grow by 10% every year for the next 5 years.
3. The notes payable were paid off.

Required:

Using the percentage of sales method of financial forecasting:

- (i) Determine the external financial requirement for the business for the year ending 31 December 2027. (8 marks)
- (ii) Prepare a program statement of financial position for the year ending 31 December 2027. (8 marks)

ANSWER

i) External Financing Requirement

	1	2	3	4	5
Increase in current assets	650,000.00	715,000.00	786,500.00	865,150.00	951,665.00
Increase in non current assets	250,000.00	275,000.00	302,500.00	332,750.00	366,025.00
Total Increase in assets	900,000.00	990,000.00	1,089,000.00	1,197,900.00	1,317,690.00
Less: Increase in current liabilities	-980,000.00	132,000.00	145,200.00	159,720.00	175,692.00
Less: Profit after tax	550,000.00	605,000.00	665,500.00	732,050.00	805,255.00
External financing requirement	1,330,000.00	253,000.00	278,300.00	306,130.00	336,743.00
Sales	11,000,000.00	12,100,000.00	13,310,000.00	14,641,000.00	16,105,100.00
Profit after tax	550,000.00	605,000.00	665,500.00	732,050.00	805,255.00

ii) Program Statement of Financial Position

Year	0	1	2	3	4	5
Current Assets	6,500,000.00	7,150,000.00	7,865,000.00	8,651,500.00	9,516,650.00	10,468,315.00
Non-current Assets	2,500,000.00	2,750,000.00	3,025,000.00	3,327,500.00	3,660,250.00	4,026,275.00
Total Assets	9,000,000.00	9,900,000.00	10,890,000.00	11,979,000.00	13,176,900.00	14,494,590.00
Account Payable	1,000,000.00	1,100,000.00	1,210,000.00	1,331,000.00	1,464,100.00	1,610,510.00
Note Payable	1,200,000.00	0.00	0.00	0.00	0.00	0.00
Other Current Liabilities	1,200,000.00	1,320,000.00	1,452,000.00	1,597,200.00	1,756,920.00	1,932,612.00
Total Current Liabilities	3,400,000.00	2,420,000.00	2,662,000.00	2,928,200.00	3,221,020.00	3,543,122.00
Ordinary Share Capital	2,000,000	2,000,000	2,000,000	2,000,000	2,000,000	2,000,000
Retained earnings	2,600,000	3,150,000.00	3,755,000.00	4,420,500.00	5,152,550.00	5,957,805.00
Long term loans	1,000,000	2,330,000.00	2,473,000.00	2,630,300.00	2,803,330.00	2,993,663.00
Total Equity and Liabilities	9,000,000.00	9,900,000.00	10,890,000.00	11,979,000.00	13,176,900.00	14,494,590.00

QUESTION 3

April 2023 Question Four C

The following statement of profit or loss relates to Memuko Ltd. for the year ended 31 December 2022 and 31 December 2021:

Memuko Ltd

Statement of profit or loss for the years ended 31 December:

	2022	2021
	Sh.“000”	Sh.“000”
Net sales	52,678.5	46,485
Cost of goods sold	(19,039.5)	(16,632)
Gross margin	33,639	29,853
Selling and administrative expenses	(19,737)	(17,037)

Other operating expenses	<u>(1,228.5)</u>	<u>(469.5)</u>
Operating income	12,673.5	12,346.5
Interest expense	<u>(1,099.5)</u>	<u>(532.5)</u>
Other income (net expenses)	<u>9,715.5</u>	<u>1,482</u>
Income before taxes	21,289.5	13,296
Income tax expense	<u>(3,576)</u>	<u>(3,060)</u>
Net income	<u>17,713.5</u>	<u>10,236</u>

Required:

- (i) Present the common size analysis for Memuko Ltd. income statement. (8 marks)
- (ii) Analyse the changes in cost of goods sold, selling and administrative expenses, operating income and income before taxes for the year ended 31 December 2022 compared to 31 December 2021. (4 marks)

ANSWER

(i) Common size analysis

	2022	2021
Net sales	$\frac{52,678.5}{52,678.5} \times 100\%$ $= 100\%$	$\frac{46,485}{46,485} \times 100\%$ $= 100\%$
Cost of goods sold	$\frac{19,039}{52,678.5} \times 100$ $= 36.14\%$	$\frac{16,632}{46,485} \times 100$ $= 35.78\%$
Gross margin	$\frac{33,639}{52,678.5} \times 100 = 63.86\%$	$\frac{29,853}{46,485} \times 100\%$ $= 64.22\%$
Selling & administrative expenses	$\frac{19,737}{52,678.5} \times 100 = 37.47\%$	$\frac{17,037}{46,485} \times 100$ $= 36.65\%$
Other operating expenses	$\frac{1,228.5}{52,675.5} \times 100$ $= 2.33\%$	$\frac{469.5}{46,485} \times 100 = 1.01\%$

Operating income	$\frac{12673.5}{52678.5} \times 100$ = 24.06%	$\frac{12346.5}{46485} \times 100$ = 26.56%
Interest expense	$\frac{1099.5}{52678.5} \times 100$ = 2.09%	$\frac{532.5}{46485} \times 100 = 1.15\%$
Other income (net expenses)	$\frac{9715.5}{52678.5} \times 100$ = 18.44%	$\frac{1482}{46485} \times 100$ = 3.19%
Income before taxes	$\frac{21289.5}{52678.5} \times 100$ = 40.41%	$\frac{13296}{46485} \times 100$ = 28.60%
Income tax expense	$\frac{3576}{52678.5} \times 100 = 33.63\%$	$\frac{10236}{46485} \times 100$ = 22.02%

Memuko Ltd

Common Size income statement

	2022	2021
Net sales	100.00%	100.00%
Cost of goods sold	36.14%	35.78%
Gross margin	63.86%	64.22%
Selling and admin expenses	37.47%	36.65%
Other operating expenses	2.33%	1.01%
Operating income	24.06%	26.56%
Interest income	2.09%	1.15%
Other income (net expenses)	18.44%	3.19%
Income before taxes	40.41%	28.60%
Income tax expenses	6.79%	6.58%
Net income	33.63%	22.02%

(ii) Changes.

- **Cost of goods sold** as a percent of sales increased from 35.78% in 2021 to 36.14% in 2022. This may be attributable to increase in input prices for purchases made by Memuko Ltd or a decrease in selling prices of goods sold by Memuko Ltd.
- **Selling and administrative** costs as a percent of sales increased from 36.65% in 2021 to 37.47 in 2022. This may be attributable to pay rise made by company for existing staff and new employees hired by company.
- **Operating income** as a percent of sales declined from 26.56% in 2021 to 24.06% in 2022. This is attributable to faster growth in sales as compared to operating income.
- **Income before tax** as a percent of sales increased from 28.60% in 2021 to 40.41 in 2022. This is attributable to faster growth of profit before tax as compared to sales growth.

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QUESTION 4**December 2022 Question Two B**

The following information relates to security X and security Y returns for the year 2021:

Probability	Returns %	
	R _x	R _y = 6 + 0.2 R _x
0.10	30	= 6 + 0.2 × 30 = 12
0.20	20	10
0.40	15	9
0.20	10	?
0.10	-50	?

Required:

- (i) Determine the expected returns for security X and security Y. (2 marks)

- (ii) Determine the standard deviation for each security. (4 marks)
- (iii) Determine the covariance between security X and security Y. (4 marks)
- (iv) Determine the correlation coefficient between security X and security Y. (2 marks)

ANSWER

- (i) The expected returns for security X and security Y.

$$R_x = (0.1 \times 30) + (0.2 \times 20) + (0.40 \times 15) + (0.2 \times 10) + (0.1 \times -50)$$

$$= 3 + 4 + 6 + 2 - 5$$

$$= 10\%$$

Return % Ry

Probability Returns

0.10 12%

0.20 10%

0.40 9%

0.20 = $6 + 0.2 \times 10 = 8\%$

0.10 = $6 + 0.2 \times -50 = -4\%$

$$= (0.1 \times 12) + (0.2 \times 10) + (0.4 \times 9) + (0.2 \times 8) + (0.1 \times -4)$$

$$= 1.2 + 2 + 3.6 + 1.6 - 0.4$$

$$= 8\%$$

- (ii) The standard deviation for each security.

Standard deviation

$$\begin{aligned}\delta_x &= \sqrt{0.1(30^2) + 0.2(20^2) + 0.4(15^2) + 0.2(10^2) + 0.5(-50^2) - 10^2} \\ &= \sqrt{1530 - 100} \\ &= 37.82\end{aligned}$$

$$\begin{aligned}\delta_y &= \sqrt{0.1(12^2) + 0.2(10^2) + 0.4(9^2) + 0.2(8^2) + 0.1(-4^2) - 8^2} \\ &= \sqrt{81.2 - 64} \\ &= 4.15\end{aligned}$$

(iii) The covariance between security X and security Y.

Probability	X	X - $\bar{x}$	Y	Y - $\bar{y}$	P (X- $\bar{x}$)(Y - $\bar{y}$)
0.10	30	20	12	4	8
0.20	20	10	10	2	4
0.40	15	5	9	1	2
0.20	10	0	8	0	0
0.10	-50	-60	-4	-12	<u>72</u>
					<u>86</u>

Covariance security X and Y = 86

(iv) The correlation coefficient between security X and security Y.

$$COV_{x,y} = corr_{x,y} \times \sigma_x \times \sigma_y$$

$$\begin{aligned} \text{corr}_{x,y} &= \frac{\text{cov}_{x,y}}{\sigma_x \sigma_y} = \frac{86}{37.82 \times 4.15} \\ &= 0.55 \end{aligned}$$

QUESTION 5**August 2022 Question Three A**

Examine four shortcomings of the percentage of sales method of forecasting. (4 marks)

ANSWER**Shortcomings of the percentage of sales method of forecasting**

- **Relies on past behavior:** This forecasting model uses past sales behavior to predict future trends. However, there's no guarantee that this behavior will stay the same.
- **Short term focus:** The percentage of sales method focuses on short term sales. This can present a challenge when there's potential long term consequence the business should be aware of.
- **Missed variables:** Business variables that don't have an immediate effect on sales may be ignored by companies relying on the percentage of sales method.
- Assumes the firm is operating at 100% capacity which is unrealistic
- Assumes the net profit margin will remain constant which is unrealistic as net profit margin keeps on fluctuating
- Assumes no issue of new ordinary shares or preference shares. This is unrealistic given a company may raise funds for acquisition of new fixed assets by issuing new ordinary shares.

QUESTION 6**April 2022 Question Five C**

The following information relates to Jolop Ltd. for the financial year ended 31 December 2021:

Jolop Ltd.**Statement of comprehensive income for the year ended 31 December 2021**

	Sh."000"	Sh."000"
Sales:	200,000	
Less: Sales returns	<u>5,000</u>	195,000
Cost of sales:		
Opening stock	3,000	
Purchases	127,000	
Closing stock	<u>(5,000)</u>	<u>(125,000)</u>
Gross profit		70,000
Other income (loss):		
Loss on sale of plant		(5,000)
Dividend received		10,000
Total revenue		75,000
Less expenses:		
Administrative expenses	15,000	
Distribution expenses	<u>20,000</u>	<u>(35,000)</u>
Net income/profit		<u>40,000</u>

Jolop Ltd.**Statement of financial position as at 31 December 2021**

Assets:	Sh."000"
Non-current assets:	
Land	150,000
Building	200,000
Plant and equipment	<u>200,000</u>

Total non-current assets	<u>550,000</u>
Current assets:	
Inventory	80,000
Accounts receivable	50,000
Bank balances	<u>20,000</u>
Total current assets	<u>150,000</u>
Total assets	<u>700,000</u>
Equity and liabilities:	
Equity:	
Equity share capital (50 million shares of Sh.10 each)	500,000
General reserve	50,000
Retained earnings	<u>70,000</u>
Total equity	<u>620,000</u>
Current liabilities:	
Accounts payable	<u>80,000</u>
Total equity and liabilities	<u>700,000</u>

Required:

Compute and interpret the following for Jolop Ltd.:

- | | |
|--|-----------|
| (i) Operating profit ratio. | (2 marks) |
| (ii) Net profit to capital employed ratio. | (2 marks) |
| (iii) Stock turnover ratio. | (2 marks) |
| (iv) Debt collection period. | (2 marks) |
| (v) Gross profit ratio. | (2 marks) |
| (vi) Quick/acid test ratio. | (2 marks) |

ANSWER

Jolop Ltd.**(i) Operating profit ratio.**

$$\begin{aligned}\text{Operating profit ratio} &= \frac{\text{Operating profit}}{\text{Sales}} \times 100\% \\ &= \frac{40,000}{195,000} \times 100\% \\ &= 20.51\%\end{aligned}$$

The company generated an operating profit of Sh. 20.51 for every Sh. 100 of sales in 2021

(ii) Net profit to capital employed ratio.

$$\begin{aligned}\text{Capital employed} &= \text{Total assets} - \text{current liabilities} \\ &= 700,000 - 80,000 \\ &= \text{Sh. } 620,000 \\ &= \frac{40,000}{620,000} \times 100\% = 6.45\%\end{aligned}$$

The company made a net profit of Sh. 6.45 for every Sh. 100 of capital employed in 2021

(iii) Stock turnover ratio.

$$\frac{\text{Cost of sales}}{\text{Inventory average}} = \frac{125,000}{80,000} = 1.56 \text{ times}$$

The company sold its stock an average of 1.56 times in 2021

(iv) Debt collection period.

$$\begin{aligned}&\frac{\text{Average account receivable}}{\text{Credit sales}} \times 365 \\ &= \frac{50,000}{195,000} \times 365 \\ &= 93.59 \text{ days}\end{aligned}$$

It took an average of 93.59 days to collect amounts owed by debtors to the company

(v) Gross profit ratio.

$$\begin{aligned}\frac{\text{Gross profit}}{\text{Sales}} &\times 100\% \\ &= \frac{70,000}{195,000} \times 100\% \\ &= 35.90\%\end{aligned}$$

(vi) Quick/acid test ratio.

$$\begin{aligned}\frac{\text{C.A} - \text{Stock}}{\text{C.L}} &= \frac{150,000 - 80,000}{80,000} \\ &= \frac{70,000}{80,000} \\ &= 0.875 \text{ times}\end{aligned}$$

QUESTION 7**December 2021 Question Two B**

QRS Limited is planning to expand its production capacity as per its marketing strategy as a result of access to new markets. The company requires additional capital of Sh.95,000,000 which it intends to raise through a rights issue. The company's financial year ends on 30 June.

Additional information:

1. The company's pre-tax profits for the year ended 30 June 2021 was Sh.177,500,000. This represents a 30% increase over the profits of the previous year.
2. The current market price per share is Sh.12.
3. The rights issue will comprise one share for every four shares held at a rights issue price of Sh.8 per share.
4. The dividend per share for the year ended 30 June 2021 was Sh.0.70. The company expects to pay a dividend per share of Sh.0.90 for the year ending 30 June 2022.
5. Corporate tax rate is 30%.

Required:

- (i) The existing number of ordinary shares before the rights issue. (1 mark)

- (ii) The earnings per share (EPS) and price-earnings (P/E) ratio for the year ended 30 June 2021. (4 marks)
- (iii) The ex-rights price per share. (3 marks)

ANSWER

QRS Limited

- (i) The existing number of ordinary shares before the rights issue.

$$\begin{aligned}\text{New shares issued} &= \frac{\text{Amount raised}}{\text{Rights issue price}} \\ &= 95,000,000/8 \\ &= 11,875,000 \text{ Shares}\end{aligned}$$

Original Shares in issue

$$\begin{aligned}&= 4/1 \times 11,875,000 \\ &= 47,500,000 \text{ shares}\end{aligned}$$

- (ii) The earnings per share (EPS) and price-earnings (P/E) ratio for the year ended 30 June 2021.

$$\begin{aligned}\text{New shares issued} &= \frac{100-30}{100} \times 177,500,000 \\ &= \text{Sh. } 124,250,000\end{aligned}$$

Earnings per share 2021

$$\begin{aligned}&= \frac{124,250,000}{47,500,000} \\ &= \text{Sh. } 2.62\end{aligned}$$

PIE ratio 2021 = $12 \div 2.62$

= 4.58

(iii) The ex-rights price per share.

$$\frac{(4 \times 12) + (1 \times 8)}{4 + 1}$$

= Sh. 11.20

QUESTION 8

December 2021 Question Four D

The following information relates to Kabaka Ltd. for the year ended 31 March 2021:

	Sh."000"
Profit after tax	12,000
10% Debenture	12,000
Ordinary share capital (Sh.20 par value)	80,000

Additional information:

1. The market price per share is Sh.72.
2. The corporate tax rate is 30%.

Required:

Calculate the following ratios:

- (i) Times interest earned ratio. (2 marks)
- (ii) Price earnings ratio. (2 marks)

ANSWER

Kabaka Ltd

(i) Times interest earned ratio.

	Sh. "000"
Profit after tax	12000
Add: Tax (30/70 x 12000)	5143
Profit before tax	17143
Add: Debenture interest 10/100 x 12000	1200
Operating profit	18343

Times interest earned ratio

$$= \text{Operating profit} \div \text{interest expt.}$$

$$= 18343000 \div 1200000$$

$$= 15.29$$

(ii) Price earnings ratio.

$$\text{Earnings per share (EPS)} = \frac{\text{Profit after tax}}{\text{Number of shares}}$$

$$\text{Number of shares} = 80,000,000 \div 20$$

$$= 4000000$$

$$\text{EPS} = 12000000 \div 4000000$$

$$= 3$$

$$\text{P/E Ratio} = 72 \div 3$$

$$= 24$$

QUESTION 9

May 2019 Question Three B

The following is a summary of financial data for Hakika Ltd. for the financial year ended 31 December 2017 and 31 December 2018:

	2018	2017
	Sh.“000”	Sh.“000”
Income statement:		
Earnings before interest and tax (EBIT)	29,498	27,012
Interest	(3,106)	(3,726)
Tax	(8,694)	(7,452)
Earnings after interest and tax (EAIT)	17,698	15,834
Dividend payable	9,600	6,200
Statement of financial position:		
Shareholders' funds	79,800	70,174
Long-term debt	28,000	35,000
	107,800	105,174
Additional information:		
	2018	2017
1. The number of outstanding shares ("000")	28,000	28,000
2. Price-earnings (P/E) ratio : Hakika Ltd.	14.00	13.00
Industry	15.20	15.00

Required:

Calculate the following ratios for years 2017 and 2018:

- (i) Return on capital employed (ROCE). (2 marks)
- (ii) Interest coverage ratio. (2 marks)
- (iii) Earnings per Share (EPS). (2 marks)
- (iv) Dividend yield. (2 marks)

ANSWER

i. Return on capital (ROCE)

$$\frac{\text{operating profit}}{\text{capital employed}} \times 100\% \quad 2018 = \frac{29,498}{107,800} \times 100\% = 27.36\%$$

$$2017 = \frac{27,012}{105,174} \times 100\% = 25.68\%$$

In 2018 for every Sh. 100 of capital employed generated Shs.27.36 as operating profit. In 2017 every Sh 100 of capital employed generated Shs 25.68 as operating profit.

ii. Interest coverage ratio

$$\frac{\text{operating profit}}{\text{interest paid}}$$

$$2018 = 29,498,000 \div 3,106,000 = 9.50$$

$$2017 = 27,012,000 \div 3,726,000 = 7.25$$

In 2017, for every Sh.100 of interest to be paid there was Sh 725 as operating profit to cater for it.

In 2018, for every Sh.100 of interest to be paid, there was Sh. 950 as operating profit to cater for it.

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iii. Earnings per share (EPS)

$$\frac{\text{earnings after tax}}{\text{number of outstanding shares}}$$

$$2018 = 17,698,000 \div 28,000,000 = 0.63$$

$$2017 = 15,834,000 \div 28,000,000 = 0.57$$

iv. Dividend yield (DY)

$$\text{P/E ratio} = \frac{\text{Market price per share}}{\text{Earnings per share}}$$

Market price per share = Earnings per share $\times$ P/E ratio

2018 market price per share = $14 \times 0.63 = \text{Sh.}8.82$

2017 market price per share = $13 \times 0.57 = \text{Sh. } 7.41$

Dividend per share = $\frac{\text{dividend payable}}{\text{number of ordinary shares}}$

2018 dividend per share = $9,600,000 \div 28,000,000 = 0.34$

2017 dividend per share = $6,200,000 \div 28,000,000 = 0.22$

Dividend yield = $\frac{\text{dividend per share}}{\text{market price per share}} \times 100\%$

2018 dividend per share = $\frac{0.34}{8.82} \times 100\% = 3.85\%$

2017 dividend yield = $\frac{0.22}{7.41} \times 100\% = 2.97\%$

QUESTION 10

November 2018 Question Four A

The following are the financial statements for ABC Ltd. and X YZ Ltd. for the year ended 30 September 2018:

Income statement for the year ended 30 September 2018:

	ABC Ltd.	XYZ Ltd.
	Sh.“million	Sh.“million'
	"	
Revenues	4,000	6,000
Cost of sales	(3,000)	(4,800)
Gross profit	1,000	1,200
Expenses:		

Distribution costs	200	150	
Administration expenses	290	250	
Finance costs	<u>10</u>	<u>(500)</u>	<u>400</u>
Profit before tax		500	400
Tax paid		<u>(120)</u>	<u>(90)</u>
Profit after tax		380	310
Dividends paid		(150)	(100)
Retained profits for the year		230	210
Retained profit brought forward		<u>220</u>	<u>2,480</u>
Retained profit carried forward		450	2,690

Statement of financial position as at 30 September 2018:

	ABC Ltd.	XYZ Ltd.
	Sh. “million”	Sh. “million”
Non-current assets:		
Land and buildings	1,200	5,000
Furniture and motor vehicles	<u>600</u>	<u>1,000</u>
	1,800	6,000
Current assets:		
Inventories	400	800
Trade receivables	850	750
Financial assets	100	230
Cash at bank	=	<u>100</u>
	<u>1,350</u>	<u>1,880</u>
	<u>3,150</u>	<u>7,880</u>
Financed by:		
Equity and liabilities:		
Ordinary share capital	1,000	1,600
Retained profits	<u>450</u>	<u>2,690</u>
	1,450	4,290

Non-current liabilities:

Bank loan	500	3,000
-----------	-----	-------

Current liabilities:

Trade payables	1,080	590
Bank overdraft	<u>120</u>	<u>1,200</u> = <u>590</u>
	<u>3,150</u>	<u>7,880</u>

Required:

- (i) Vertical common size statements of income for the year ended 30 September 2018.
(6 marks)
- (ii) Vertical common size statements of financial position as at 30 September 2018.
(6 marks)

ANSWER

- i) For common size income statement every item should be expressed as a percent of sales.

ABC LTD

Common size income statement

For year ended 30 September 2018

	Sh "Million"	As % sales
Revenue	4,000	$\frac{4,000}{4,000} \times 100\% = 100\%$
Cost of sales	(3,000)	$\frac{3,000}{4,000} \times 100\% = 75\%$
Gross profit	1,000	$\frac{1,000}{4,000} \times 100\% = 25\%$
Expenses:		
Distribution costs	200	$\frac{200}{4,000} \times 100\% = 5\%$
Administrative expenses	290	$\frac{290}{4,000} \times 100\% = 7.25\%$
Finance costs	10	$\frac{10}{4,000} \times 100\% = 0.25\%$

Profit before tax	500	$\frac{500}{4,000} \times 100\% = 12.5\%$
Tax paid	120	$\frac{120}{4,000} \times 100\% = 3\%$
Profit after tax	380	$\frac{380}{4,000} \times 100\% = 9.5\%$
Dividend paid	150	$\frac{150}{4,000} \times 100\% = 3.75\%$
Retained profit for the year	230	$\frac{230}{4,000} \times 100\% = 5.75\%$
Retained profit brought forward	220	$\frac{220}{4,000} \times 100\% = 5.5\%$
Retained profit carried forward	450	$\frac{450}{4,000} \times 100\% = 11.25\%$

XYZ LTD

Common size income statement

For year ended 30 September 2018

	Sh "Million"	As % sales
Revenue	6,000	$\frac{6,000}{6,000} \times 100\% = 100\%$
Cost of sales	4,800	$\frac{4,800}{6,000} \times 100\% = 80\%$
Gross profit	1,200	$\frac{1,200}{6,000} \times 100\% = 20\%$
Expenses:		
Distribution costs	150	$\frac{150}{6,000} \times 100\% = 2.5\%$
Administrative expenses	250	$\frac{250}{6,000} \times 100\% = 4.17\%$
Finance costs	400	$\frac{400}{6,000} \times 100\% = 6.67\%$
Profit before tax	400	$\frac{400}{6,000} \times 100\% = 6.67\%$
Tax paid	90	$\frac{90}{6,000} \times 100\% = 1.5\%$
Profit after tax	310	$\frac{310}{6,000} \times 100\% = 5.17\%$
Dividend paid	100	$\frac{100}{6,000} \times 100\% = 1.67\%$

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Retained profit for the year	210	$\frac{210}{6,000} \times 100\% = 3.5\%$
Retained profit brought forward	2,480	$\frac{2,480}{6,000} \times 100\% = 41.33\%$
Retained profit carried forward	2,690	$\frac{2,690}{6,000} \times 100\% = 44.83\%$

Common size income statement

For year ended 30 September 2018

	ABC LTD	XYZ LTD
Revenue	100%	100%
Cost of sales	75%	80%
Gross profit	25%	2.5%
Expenses:		4.17%
Distribution costs	5%	6.67%
Administrative expenses	7.25%	6.67%
Finance costs	0.25%	1.5%
Profit before tax	12.50%	5.17%
Tax paid	3%	1.5%
Profit after tax	9.5%	5.17%
Dividend paid	3.75%	1.67%
Retained profit for the year	5.75%	3.5%
Retained profit brought forward	5.5%	41.33%
Retained profit carried forward	11.25%	44.83%

Common size Balance sheet

For year ended 30 September 2018

	ABC Ltd Sh. “million”	%Total Assets	XYZ Ltd Sh. “million”	%Total Assets
Non-current Assets:				
Land and buildings	1,200	$\frac{1,200}{4,9500} \times 100\%$ = 24.24%	5,000	$\frac{5,000}{4,9500} \times 100\% = 36.02\%$
Furniture and fittings	600	$\frac{600}{4,9500} \times 100\%$ = 12.12%	1,000	$\frac{1,000}{4,9500} \times 100\% = 7.20\%$
Total non-current assets	1,800	$\frac{1,800}{4,9500} \times 100\%$ = 36.6%	6,000	$\frac{6,000}{4,9500} \times 100\% = 43.23\%$
Current assets:				
Inventories	400	$\frac{400}{4,9500} \times 100\%$ = 8.08%	800	$\frac{800}{4,9500} \times 100\% = 5.76\%$
Trade receivables	850	$\frac{850}{4,9500} \times 100\%$ = 7.17%	750	$\frac{750}{4,9500} \times 100\% = 5.40\%$
Financial assets	100	$\frac{100}{4,9500} \times 100\%$ = 2.02%	230	$\frac{230}{4,9500} \times 100\% = 1.66\%$
Cash at Bank	0	$\frac{0}{4,9500} \times 100\% = 0\%$	100	$\frac{100}{4,9500} \times 100\% = 0.72\%$
Total current assets	1,350	$\frac{1,350}{4,9500} \times 100\%$ = 27.27%	1,880	$\frac{1,880}{4,9500} \times 100\% = 13.54\%$
Financed by:				
Equity and liabilities:				
Ordinary share capital	1,000	$\frac{1,000}{4,9500} \times 100\%$ = 20.20%	1,600	$\frac{1,600}{4,9500} \times 100\%$ = 11.53%
Retained profits	450	$\frac{450}{4,9500} \times 100\%$ = 9.09%	2,690	$\frac{2,690}{4,9500} \times 100\%$ = 19.38%

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Total equity	1,450	$\frac{1,450}{4,9500} \times 100\%$ = 29.29%	4,290	$\frac{4,290}{4,9500} \times 100\%$ = 30.91%
Non-Current liabilities:				
Bank loan	500	$\frac{500}{4,9500} \times 100\%$ = 10.10%	3,000	$\frac{3,000}{4,9500} \times 100\%$ = 21.61%
Current Liabilities:				
Trade payables	1,080	$\frac{1,080}{4,9500} \times 100\%$ = 21.82%	590	$\frac{590}{4,9500} \times 100\%$ = 4.25%
Bank overdraft	120	$\frac{120}{4,9500} \times 100\%$ = 2.42%	-	—
Total current liabilities	200	$\frac{200}{4,9500} \times 100\%$ = 24.24%	590	$\frac{590}{4,9500} \times 100\%$ = 4.25%

Vertical Common size Balance sheet

For year ended 30 September 2018

	ABC Ltd	XYZ Ltd
Non-current Assets:		
Land and buildings	24.24%	36.02%
Furniture and fittings	12.12%	7.20%
Total non-current assets	36.6%	43.23%
Current assets:		
Inventories	8.08%	5.76%
Trade receivables	7.17%	5.40%
Financial assets	2.02%	1.66%
Cash at Bank	0%	0.72%
Total current assets	27.27%	13.54%

Financed by:		
Equity and liabilities:		
Ordinary share capital	20.20%	11.53%
Retained profits	9.09%	19.38%
Total equity	29.29%	30.91%
Non-Current liabilities:		
Bank loan	10.10%	21.61%
Current Liabilities:		
Trade payables	21.82%	4.25%
Bank overdraft	2.42%	—
Total current liabilities	24.24%	4.25%

QUESTION 11

May 2018 Question Five A and C

- (a) Distinguish between “financial planning” and “financial forecasting”. (2 marks)
- (c) Furunzi Express Ltd.’s records got lost in a fire that burnt down the accounts office. The following information was however obtained from the laptop of (he accountant for the year ended 31 December 2017:

Opening stock	Sh.450,000
Stock turnover ratio	10 times
Net profit margin	15%
Gross profit margin	20%
Current ratio	4:1
Long-term loan	Sh. 1,000,000
Depreciation of fixed assets (10%)	Sh. 100,000
Closing stock	Sh.510,000

Additional information:

- Credit period allowed by suppliers is one month.
- Average debt collection period is 2 months.
- On 31 December 2017, current assets consisted of stock, debtors and cash only.

4. There was no bank overdraft.
5. All purchases are made on credit.
6. Cash sales were 1/4 of total sales.

Required:

- | | |
|------------------------|-----------|
| (i) Cost of sales. | (2 Marks) |
| (ii) Cross profit. | (2 Marks) |
| (iii) Total sales. | (2 Marks) |
| (iv) Total purchases. | (2 Marks) |
| (v) Net profit. | (2 Marks) |
| (vi) Debtors value. | (2 Marks) |
| (vii) Creditors value. | (2 Marks) |

ANSWER

a) Distinguish between “financial planning” and “financial forecasting”

Financial planning refers to the task of determining how a business or individual will afford to achieve its strategic goals and objectives. **Financial forecasting** refers to predicting future performance of a company e.g. by use of percentage of sales method.

c) Computing

i) Cost of sales

Gross profit margin

$$\frac{20}{100} = \frac{1}{5}$$

Mark up

$$\frac{1}{5 - 1} \text{ or } 25\%$$

$$\text{Stock turnover} = \frac{\text{Cost of sales}}{\text{Average stock}}$$

$$\text{Cost of sales} = \text{Stock turnover} \times \text{average stock}$$

$$\text{Average stock} = \frac{\text{Opening stock} + \text{closing stock}}{2}$$

$$\frac{450,000 + 510,000}{2} = 480,000$$

$$\text{Cost of sales} = 480,000 \times 10$$

$$= \text{shs. } 4,800,000$$

ii) Gross profit

$$\text{Mark up} = \frac{\text{Gross profit}}{\text{Cost of sales}} \times 100\%$$

$$\text{Gross profit} = \frac{\text{Mark up}}{100} \times \text{cost of sales}$$

$$= \frac{25}{100} \times 4,800,000$$

$$= \text{sh. } 1,200,000$$

iii) Total sales

$$\text{Gross profit margin} = \frac{\text{Gross profit}}{\text{sales}} \times 100\%$$

$$\text{Sales} = \frac{\text{Gross profit}}{\text{Gross profit margin}} \times 100\%$$

$$= \frac{1,200,000}{20} \times 100\%$$

$$= \text{sh. } 6,000,000$$

iv) Total Purchases

$$\text{Cost of sales} = \text{Opening stock} + \text{closing stock} - \text{Closing stock}$$

$$\text{Purchases} = \text{Cost of sales} + \text{closing stock} - \text{Opening stock}$$

$$= 4,800,000 + 510,000 - 450,000$$

$$= \text{Sh. } 4,860,000$$

v) Net profit

$$\text{Net profit} = \frac{\text{Net profit margin}}{100} \times \text{sales}$$

$$= \frac{15}{100} \times 6,000,000$$

$$= \text{shs. } 900,000$$

vi) Debtors' value

$$\text{Debtors value} = \frac{\text{Average debt collection period}}{12 \text{ months}} \times \text{Credit sales}$$

$$\text{Credit sales} = \frac{3}{4} \times 6,000,000$$

$$= sh. 4,500,000$$

$$Debtors\ value = \frac{2}{12} \times 4,500,000 = Sh. 750,000$$

vii) Credit value

$$Credit\ value = \frac{Current\ period}{12\ months} \times Credit\ Purchases$$

$$= \frac{1}{12} \times 4,860,000$$

$$= sh. 405,000$$

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QUESTION 12

November 2017 Question One C

Illustrate how the problem of window dressing manifests itself in measuring business performance using financial ratio analysis. (4 marks)

ANSWER

How the problem of window dressing manifests itself in measuring business performance using financial ratio analysis

Window dressing entails carrying some transactions to provide a misleading impression of the company's performance. For example, analysts always consider it praise worthy if a company has no bank overdraft at year end so companies will borrow overdrafts and ensure it is paid before financial end. This means the bank overdraft then will not feature in the financial statements. After this company will then borrow an overdraft in the next period thus endless cycle continues forever.

QUESTION 13

May 2017 Question Three A

The following information relates to Tsuma Enterprises Ltd. for the four months given below:

Sales	Sh. "Million"
September	60
October	60
November	70
December	90

All sales will be made on credit.

Half of the debtors are expected to pay within the month of sale and are also expected to claim a 2% cash discount. The remaining debtors are expected to pay by the beginning of the following month.

Raw materials purchases	Sh. "Million"
September	20
October	40
November	40
December	30

The firm plans to pay its creditors in full in the month following that of purchase.

Wages and salaries:	Sh. "Million"
September	12
October	15
November	17
December	13

Additional information:

1. All employees are paid in the month in which the wage or salary is earned.
2. Rent of Sh.10 million for each quarter is paid in March, June, September and December.
3. Other cash overheads of Sh.2 million per month are payable.
4. A new plant due for delivery in September will be paid in November at a cost of Sh.25 million.
5. On 1 October, the firm plans to have Sh.10 million in the bank.

Required:

A cash budget for the three months ending in December. (10 marks)

ANSWER

Tsuma enterprises Ltd cash budget for the three months ending in December

Debtor's collection schedule

Month	50% less 2% discount	50% full amount	Total
September	29.4	-	29.4
October	29.4	30	59.4
November	34.3	30	64.3
December	44.1	35	79.1

Creditor's payment schedule

Month	Raw materials "M"	Repayment "m"
September	20	-
October	40	20
November	40	40
December	30	40

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**TUMA ENTERPRISES LTD CASH BUDGET FOR PERIOD ENDED 31ST
DECEMBER**

Details	October Sh “million”	November Sh “million”	December Sh. “million”
Bal B/F	10	32.4	12.7
Sales receipts	<u>59.4</u>	<u>64.3</u>	<u>79.1</u>
	<u>69.4</u>	<u>96.7</u>	<u>91.8</u>
Payments			
Raw materials purchases	20	40	40
Wages and salaries	15	17	13
Rent	-	-	10
Other overheads	2	2	2
Plant payment	-	25	-
	37	84	65
Bal C/F	32.4	12.7	26.8

QUESTION 14

May 2017 Question Four A

The following are the summarised financial statements for Bokasa Limited. Bokasa Limited statement of financial position as at 31 December:

Bokasa Limited statement of financial position as at 31 December:

	2015		2016	
	Sh. “000”	Sh. “000”	Sh. “000”	Sh. “000”
Non-current assets		4,995		12,700
Current assets:				
Inventory	40,145		50,455	
Accounts receivable	40,210		43,370	
Cash at bank	<u>12,092</u>		<u>5,790</u>	
		<u>92,447</u>		<u>99,615</u>

Total assets	97,442	I 12,315
Current liabilities:		
Accounts payable	34,389	39,215
Taxation	<u>2,473</u>	<u>3,260</u>
	36,862	42,475
Long-term liabilities:		
10% loan notes	<u>19,840</u>	<u>19,480</u>
Total liabilities	<u>(56,702)</u>	<u>(62,315)</u>
Net assets	<u>40,740</u>	<u>50,000</u>
Equity:		
Called-up share capital Sh.0.25 per share	9,920	9,920
Retained earnings	<u>30,820</u>	<u>40,080</u>
Shareholders' funds	<u>40,740</u>	<u>50,000</u>

Bokasa Limited income statement for the year ended 31 December

	2015	2016
	Sh. "000"	Sh. "000"
Revenue	486,300	583,900
Operating profit	17,238	20,670
Interest payable	(1,984)	(1,984)
Profit before taxation	15,254	18,686
Taxation	<u>(5,734)</u>	<u>(7,026)</u>
Profit for the year	<u>9,520</u>	<u>11,660</u>
Notes:	31 December 2015	31 December 2016
	Sh. "000"	Sh. "000"
1. Retained profit brought forward	23,540	30,820
2. Dividends paid during the year	2,240	2,400

Required:

For each of the two years, calculate:

- | | |
|--|-----------|
| (i) Earnings per share (EPS). | (2 marks) |
| (ii) Dividend cover. | (2 marks) |
| (iii) Current ratio. | (2 marks) |
| (iv) Acid test ratio. | (2 marks) |
| (v) Return on capital employed (ROCE). | (2 marks) |

ANSWER

B. Calculating Bokasa Ltd

i. Earnings per share (EPS)

EPS = profit for the year ÷ outstanding shares

Number of shares = ordinary shares capital ÷ par value

$$9,920 \div 0.25 = 39,680$$

$$2015 \text{ EPS} = 9,520 \div 39,680 = \text{Sh. } 0.24$$

$$2016 \text{ EPS} = 11,660 \div 39,680 = 0.29$$

ii. Dividend cover

Dividends per share (DPS) = dividends paid ÷ number of shares

$$2015 \text{ DPS} = 2,240 \div 39,680 = \text{Sh. } 0.06$$

$$2016 \text{ DPS} = 2,400 \div 39,680 = \text{Sh. } 0.06$$

iii. Current ratio (CR)

$$C.R = \frac{\text{current assets}}{\text{current liabilities}}$$

$$2015 \text{ C.R} = \frac{92,447}{36,862} = 2.51$$

$$2016 \text{ C.R} = \frac{99,615}{42,475} = 2.35$$

iv. Acid test ratio

$$\text{Acid test ratio} = \frac{\text{current assets} - \text{stock}}{\text{current liabilities}}$$

$$2015 \text{ acid test ratio} = \frac{92,447 - 40,145}{36,862} = 1.42$$

$$2016 \text{ acid test ratio} = \frac{99,615 - 50,455}{42,475} = 1.16$$

v. Return on capital employed (ROCE)

$$\text{ROCE} = \frac{\text{Operating profit}}{\text{Capital employed}} \times 100\%$$

Capital employed = fixed assets + current assets – current liabilities

2015 capital employed = 92,447 + 4,995 – 36,862 = Sh. 60,580

2016 capital employed = 112,315 – 42,475 = Sh. 69,840

$$R.O.C.E \text{ 2015} = \frac{17,238}{60,580} \times 100\% = 28.45\%$$

$$R.O.C.E \text{ 2016} = \frac{20,670}{69,840} \times 100\% = 29.60\%$$

QUESTION 15

November 2016 Question One C

The following data was extracted from the financial statements of XYZ Limited for the year ended 30 September 2016.

Total assets	Sh.7,000,000
Total liabilities	Sh.4,000,000
Preference share capital	Sh.500,000
Earnings per share (EPS)	Sh. 1.10
Price-earnings (P/E) ratio	15
Outstanding number of ordinary shares	400,000

Required:

- (i) Book value per share. (2 marks)
- (ii) Market price per share. (2 marks)
- (iii) Market value to book value ratio. (2 marks)

ANSWER

Calculating

i) Book value per share

$$\begin{aligned}
 \text{Total equity} &= \text{total assets} - \text{Total liabilities} \\
 &= 7,000,000 - 4,000,000 - 500,000 \\
 &= \text{sh. } 2,500,000
 \end{aligned}$$

$$\begin{aligned}
 \text{Book value per share} &= \frac{\text{Total equity}}{\text{No. of shares}} \\
 &= \frac{2,500,000}{400,000}
 \end{aligned}$$

$$= \text{sh. } 6.25$$

ii) Market price per share

$$\text{Market price per share (MPS)} = \text{P/E ratio} \times \text{E.P.S}$$

$$= 1.10 \times 15 = \text{Sh.} 16.50$$

(iii) Market value to book value ratio

Market value to book value ratio

$$= \frac{\text{Market price per share}}{\text{Book value per share}}$$

$$= \frac{16.50}{6.25}$$

$$= 2.64$$

QUESTION 16

November 2016 Question Three A

The management of Georgina Ltd wishes to establish the amount of external financial needs for the year ending 31 December 2016. The statement of financial position of the company as at 31 December 2015 was as follows:

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	Sh."000"
Plant and machinery	31,200
Furniture and fittings	18,720
Motor Vehicles	12,480
Inventory	19,200
Account receivables	14,400
Cash and bank	<u>3,600</u>
	<u>99,600</u>
Financed by:	
Ordinary share capital	42,000
Retained profit	17,600
14% debenture capital	10,000

Account payables	18,000
Accrued expenses	<u>12,000</u>
	<u>99,600</u>

Additional information:

1. The sales for the year ended 31 December 2015 amounted to Sh. 120, 000,000.
2. The company forecasts that sales will increase by 10% for the year ending 31 December 2016.
3. For the year ended 31 December 2015, the after tax profit of the company amounted to sh. 18,000,000
4. The company adopts 80% payout ratio as its dividend policy. The payout ratio is expected to remain constant each year in perpetuity.
5. The after-tax profit margin is also expected to remain constant each year.
6. Assets are expected to vary directly with sales while account payables and accrued expenses form the spontaneous sources of financing.
7. Any external financing will be affected through long term debt financing.

Required:

- i) The amount of external 12% long term debt financing that would be required for the year ending 31 December 2016. (4 marks)
- ii) A forecast statement of financial position as at 31 December 2016 (6 marks)
- iii) Comment on two weakness of the method of forecasting applied in (a) (i) and (a)(ii) above. (2 marks)

ANSWER

(i) Amount of external financing required for the year ended 21 December 2016:

$$\text{Profit margin (2015)} = \frac{\text{Profit}}{\text{Sales}}$$

$$\frac{18,000,000}{120,000,000} \times 100\% = 15\%$$

Incremental sales:

$$\text{Sales (2016)} = \frac{110}{100} \times 120,000,000$$

$$= \text{sh. } 132,000,000$$

Profit after tax

$$= \frac{15}{100} \times 132,000,000 = \text{sh. } 19,800,000$$

Retained Earnings 2016:

$$= \frac{20}{100} \times 19,800,000 = \text{sh. } 3,960,000$$

<u>% of sales method</u>		
Plant and machinery	$\left[\frac{\text{Plant and machinery}}{\text{sales}} \times 100 \right]$ $= \left[\frac{31,200}{120,000} \times 100 \right]$	26%
Furniture and fittings	$\left[\frac{\text{Furniture and fittings}}{\text{sales}} \times 100 \right]$ $= \left[\frac{18,720}{120,000} \times 100 \right]$	15.6%
Motor vehicles	$\left[\frac{\text{Motor vehicles}}{\text{sales}} \times 100 \right]$ $= \left[\frac{12,480}{120,000} \times 100 \right]$	10.4%
Inventory	$\left[\frac{\text{Inventory}}{\text{sales}} \times 100 \right]$ $= \left[\frac{19,200}{120,000} \times 100 \right]$	16%

Accrued Receivables	$\left[\frac{\text{Accrued Receivables}}{\text{sales}} \times 100 \right]$ $= \left[\frac{14,400}{120,000} \times 100 \right]$	12%
Cash and Bank	$\left[\frac{\text{Cash and Bank}}{\text{sales}} \times 100 \right]$ $= \left[\frac{3,600}{120,000} \times 100 \right]$	3%
Accounts payable	$\left[\frac{\text{Accounts payable}}{\text{sales}} \times 100 \right]$ $= \left[\frac{18,000}{120,000} \times 100 \right]$	15%
Accrued expenses	$\left[\frac{\text{Accrued expenses}}{\text{sales}} \times 100 \right]$ $= \left[\frac{12,000}{120,000} \times 100 \right]$	10%

Incremental in assets

Plant and machinery	0.26x12,000,000	3,120,000
Furniture and fittings	0.156x12,000,000	1,872,000
Motor vehicles	0.104x12,000,000	1,248,000
Inventory	0.16x12,000,000	1,920,000
Account Receivables	0.12x12,000,000	1,440,000
Cash and Bank	0.03x12,000,000	360,000

Increase in total assets **9,960,000**

Less: Accounts payable	0.15x12,000,000	(1,800,000)
Less: Accrued expenses	0.1x12,000,000	(1,200,000)
Less: Retained earnings for the year		(3,960,000)

External 12% long-term debt finance **3,000,000**

(ii) **Forecasted statement of financial position:**

Georgina Ltd.

Forecasted statement of financial position as at 31 December

2016:

	Sh. “000”	Sh. “000”
Plant and machinery	31,200+3,120	34,320
Furniture and fittings	18,720+1,872	20,592
Motor vehicles	12,480+1,248	<u>13,728</u>
Total fixed assets:		68,640
Current assets:		
Inventory	19,200+1,920	21,120
Accounts receivable	14,400+1,440	15,840
Cash and bank	3,600+360	3,960
Total current assets		<u>40,920</u>
Total assets		<u>109,560</u>
Financed by:		
Ordinary share capital		42,000
Retained profit	17,600 + 3,960	21,560
Shareholder’s funds		63,560
14% debenture capital		10,000
12% long term debt		<u>3,000</u>
Total long term capital		76,560
Current liabilities		
Accounts payable	15/100 x 132000	19,800
Accrued expenses	10/100 x 132000	13,200
Current liabilities		<u>33,000</u>
		<u>109,560</u>

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iii) Limitations of percentage of sales method

1. Assumes constant net profit margin which is unrealistic as it will keep changing from one year to another

2. Assumes that dividend can only be paid in cash which is unrealistic as there are other forms of dividends e.g. bonus issues
3. Unrealistic assumption that value of money remains unchanged
4. Assumes a direct between sales and balance sheet items which is imaginary

QUESTION 17

May 2016 Question Three A

The following data was extracted from the financial statements of Jaribuni Limited for the year ended 31 December 2015:

	Sh. "millions"
Cash and cash equivalents	200
Fixed assets	567
Sales(credit)	2,000
Net income	100
Current liabilities	211
Notes payable to bank	40
Current ratio	3:1
Debtors collection period	40.55 days
Return on equity	12%

Assume 365 days in a year.

Required:

- | | |
|-------------------------------|-----------|
| (i) Accounts receivable. | (2 marks) |
| (ii) Current assets. | (2 marks) |
| (iii) Return on total assets. | (2 marks) |
| (iv) Equity. | (2 marks) |
| (v) Quick ratio. | (2 marks) |

ANSWER

(i) Accounts receivable

$$\text{Debtors period} = \frac{\text{Debtors}}{\text{credit sales}} \times 365 \text{ days}$$

$$\text{Debtors (Accounts receivable)} = \frac{\text{Debtors period} \times \text{credit sales}}{365}$$

$$= \frac{40.55 \times 2,000}{365} = 222.19178082$$

$$= \text{sh. } 222,191,780.82$$

(ii) Current assets

$$\text{current ratio} = \frac{\text{current assets}}{\text{current liabilities}}$$

$$\begin{aligned} \text{Total assets, Current assets} &= \text{Fixed assets} \times \text{Current ratio} \\ &= 211,000,000 \times 3 \\ &= \text{Sh. } 633,000,000 \end{aligned}$$

(iii) Return on total assets

$$\begin{aligned} \text{Total assets} &= \text{Current assets} + \text{Fixed} \\ &= 211 \times 3 = \text{Sh. } 633 \text{ millions} \end{aligned}$$

(iii) Return on total assets

$$\begin{aligned} \text{Total assets, Current assets} &= \text{Fixed assets} \times \text{Current ratio} \\ &= 633,000,000 + 567,000,000 \\ &= \text{Sh. } 1,200,000 \end{aligned}$$

$$\text{Return on total Assets} = \frac{\text{Net income}}{\text{Total assets}} \times 100$$

$$= \frac{1,000,000}{1,200,000} \times 100 = 8.33\%$$

v) **Equity**

$$\text{Return on Equity (ROE)} = \frac{\text{Net Income}}{\text{Equity}} \times 100$$

$$\text{Equity} = \frac{\text{Net Income}}{\text{ROE}} \times 100 = \frac{1,000,000}{8.33} \times 100 = \text{sh. } 833,000,000$$

v) **Quick Ratio**

$$\text{Current assets} = \text{Cash} + \text{Debtors} + \text{Inventory}$$

$$\text{Inventory} = \text{Current assets} - \text{cash} - \text{debtors}$$

$$633,000,000 - 200,000,000 - 222,191,780.82$$

$$= 210,808,219.18$$

$$\text{Quick Ratio} = \frac{\text{Current assets} - \text{Inventory}}{\text{Current Liabilities}}$$

$$= \frac{633,000,000 - 210,808,219.18}{211,000,000}$$

$$= 2:1$$

QUESTION 18

November 2015 Question Two D

The following data was extracted from the financial statements of Kapecha Limited as at 30 September 2015:

	Sh. “million”
10% preference shares (Sh.10 par value)	16
Ordinary shares (Sh.10 par value)	<u>16</u>
	32
Retained earnings	<u>28</u>
	60
15% debentures	<u>48</u>
	<u>108</u>

The company's net profit before interest was Sh.80 million. The company's dividend pay-out ratio was 50%. Corporate tax rate is 30%.

Required:

Dividend per share (DPS).

(4 marks)

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ANSWER

DPS for Kapecha Limited

$$\begin{aligned}
 &\text{Number of ordinary shares} \\
 &= \text{ordinary share capital} \div \text{par value} \\
 &= 16,000,000 \div 10 = 1,600,000
 \end{aligned}$$

$$\begin{aligned}
 &\text{Debenture interest annually} \\
 &= \frac{15}{100} \times 48,000,000 = \text{Sh. } 7,200,000
 \end{aligned}$$

	Sh.
EBIT	80,000,000
Less interest	7,200,000
Profit before tax	72,800,000

Less: tax 30%	21,840,000
Profit after tax	50,960,000
Dividends paid	$0.5 \times 50,960,000$
	= 25,480,000

Dividends per share = dividend paid ÷ number of shares

$$25,480,000 \div 1,600,000$$

$$\text{DPS} = \text{Sh. } 15.925$$

QUESTION 19

November 2015 Question Three A

The following information relates to Mongwe Limited for the year ended 31 October 2015:

Earnings yield	25%
Dividend for the year	10% of share nominal value
Nominal value per share	Sh.40
Market price per share	Sh.150

Required:

- (i) Earnings per share (EPS). (2 marks)
- (ii) Dividend cover. (2 marks)
- (iii) Price-earnings (P/E) ratio. (2 marks)

ANSWER

A. Mongwe Limited

i. Earnings per share (EPS)

$$\text{Earning yield} = \frac{\text{Earnings per share}}{\text{Market price per share}} \times 100\%$$

$$\text{EPS} = \frac{\text{Earnings yield}}{100\%} \times \text{market price per share}$$

$$= \frac{25}{100} \times 150 = \text{Sh. } 37.50$$

ii. Dividend cover

$$\frac{\text{Earnings per share}}{\text{Dividend per share}} = \frac{37.50}{0.1 \times 40} = 9.375$$

For every Sh. 1 bob of dividends to be paid, the company has Sh. 9.375 of earnings to cater for the same

iii. Price-earnings (P/E) ratio

$$\frac{\text{Market price per share}}{\text{Earnings per share}} = \frac{150}{37.50} = 4$$

For every Sh. 1 of earnings of Mongwe Limited an investor pays Sh. 4

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QUESTION 20

November 2015 Question Four B

The following information was extracted from the financial statements of Tana Enterprises Ltd. for the year ended 31 December 2013 and 31 December 2014:

Statement of financial position

	2014	2013
	Sh. „million“	Sh.„million“
Assets:		
Non-current assets	1,850	1,650
Depreciation	(350)	(225)
Net non-current assets	1,500	1,425
Intangible assets	150	150
Current assets:		
Inventory	330	230
Accounts receivable	220	170

Cash	100	90
Total current assets	<u>650</u>	<u>490</u>
Total assets	<u>2,300</u>	<u>2,065</u>
Equity and liabilities:		
Ordinary share capital (Sh.2 par value) 100 million shares issued)	200	200
Additional paid in ordinary share capital	325	325
Retained earnings	<u>550</u>	<u>470</u>
Ordinary shareholders' equity	<u>1,075</u>	<u>995</u>
Preference share capital (10%, Sh.100 par value)	150	150
Long-term liabilities:		
Long-term debt	625	540
Deferred tax	<u>100</u>	<u>80</u>
Total long-term liabilities	<u>725</u>	<u>620</u>
Current liabilities:		
Accounts payable	85	105
Accruals	65	85
Current portion of long-term debt	75	-
Short-term bank notes	<u>125</u>	<u>110</u>
Total current liabilities	<u>350</u>	<u>300</u>
Total equity and liabilities	<u>2,300</u>	<u>2,065</u>

Statement of comprehensive income

	2014	2013
	Sh.	Sh. “million”
	“million”	
Net sales	3,500	2,990
Cost of goods sold	2,135	1,823
Selling, general and administrative expenses	<u>1,107</u>	<u>974</u>
Operating profit	258	193
Net interest expense	<u>74</u>	<u>64</u>
Income from operations	184	129
Income taxes	<u>55</u>	<u>38</u>
Net income	129	91
Preference dividends	<u>15</u>	<u>15</u>
Net income available for ordinary shareholders	114	76
Dividends declared	40	30

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Assume that a year has 365 days.

Required:

Compute and interpret the following ratios for the year ended 31 December 2014:

- (i) Cash conversion cycle. (6 marks)
- (ii) Equity turnover. (2 marks)
- (iii) Fixed charge cover. (2 marks)
- (iv) Return on capital. (2 marks)

ANSWER**Tana enterprises****I. Cash conversion cycle**

Inventory period + Debtors collection – Creditors period

$$\text{Inventory period} = \frac{\text{Average inventory}}{\text{Cost of sales}} \times 365$$

$$\frac{(330+230) \div 2}{2135} \times 365 = 47.88 \text{ days}$$

$$\text{Debtors collection period} = \frac{\text{Average debtors}}{\text{Credit sales}} \times 365$$

$$\frac{(220+170) \div 2}{3,500} \times 365 = 20.34 \text{ days}$$

$$\text{Payables period} = \frac{\text{Average creditors}}{\text{Credit purchases}} \times 365$$

Credit purchases = cost of sales + closing stock – opening stock
 $2135 + 330 - 230 = \text{Sh. } 2235 \text{ million}$

$$\text{Payable period} = \frac{(85 + 105) \div 2}{2235} \times 365 = 15.51 \text{ days}$$

Thus conversion cycle is $47.88 + 20.34 - 15.51 = 52.71 \text{ days}$

II. Equity turnover

$$\text{Equity turnover} = \frac{\text{Sales}}{\text{Average equity}} = \frac{3,500}{(1075+995) \div 2} = 3.38$$

This implies every Sh. 1 of equity in Tana Enterprises Ltd generates Sh. 3.38 of revenue

III. Fixed charge cover

$$\frac{\text{EBIT} + \text{fixed charges}}{\text{interest payments} + \text{fixed charges}} = \frac{258}{74} = 3.49$$

This implies the company has Sh. 3.49 of earnings for every Sh. 1 of fixed charges it has to pay

IV. Return on capital

$$\frac{NOPAT}{Equity + interest\ bearing\ debt - cash} \times 100\%$$

$$\frac{258(1-0.3)}{1075+625-100+150} \times 100\% = \frac{180.6}{1750} \times 100\% = 10.32\%$$

Every Sh. 100 of invested capital in the company generates Sh. 10.32 net operating profit after tax

QUESTION 21

Pilot paper 2015 Question Five A

The most recent statement of financial position for Upendo Ltd. is presented below:

Upendo Ltd.

Statement of financial position As at 30th November 2014

	Sh. “000”		Sh. “000”
Inventory	2,000	Accrued expenses	2,200
Debtors	3,000	Long-term debt	8,800
Cash at bank	3,800	Ordinary shares	2,200
Fixed assets (NBV)	13,200	Retained profit	6,600
	22,000		22,000

The company is about to embark on an advertising campaign which is expected to raise sales from their present level of Sh.27.5 million to Sh.38.5 million by the end of the next financial year ended 30 November 2015.

The firm is presently operating at full capacity and therefore will have to increase its investment in both current and fixed assets to support the projected level of sales. It is estimated that both categories of assets will rise in direct proportion to the projected increase in sales.

For the year just ended, the firm's after tax profit margin was 6% but is expected to rise to 7% of projected sales. The firm adopts a stable predictable dividend policy. The ordinary dividend payable for the year ended 30 November 2015 is expected to increase by 10% from the last year's dividend of Sh.1 million.

Upendo Ltd's trade creditors and accrued expenses are expected to vary directly with sales. In addition long term debt financing will be used to finance next year's operations that are not forthcoming from other sources.

Required:

- (i) Estimate the amount of additional funds to be raised through long term debt financing.
(4 marks)
- (ii) Prepare a forecast statement of financial position as at 30 November 2015.
(6 marks)
- (iii) Using the results obtained in (a) (i) and (ii) above, compute and interpret the following financial ratios for the year ended 30 November 2015:
 - a) Return on equity (2 marks)
 - b) Total assets turnover (2 marks)
 - c) Capital gearing ratio (2 marks)

ANSWER

Upendo limited

i. Estimated amount of additional funds

$$\text{Profit after tax 2015} = \frac{7}{100} \times 38,500,000 = \text{Sh. } 2,695,000$$

$$\text{Total dividend share 2015} = \frac{110}{100} \times 1,000,000 = \text{Sh. } 1,100,000$$

$$\text{Retained earnings 2015} = \text{Sh. } 2,695,000 - 1,100,000 = \text{Sh. } 1,595,000$$

Additional funds required

$$\text{Total assets as percentage of sales} = \frac{22,000}{27,500} \times 100\% = 80\%$$

$$\begin{aligned} \text{Current liabilities (creditors and accrued expenses) as percentage of sales} \\ = \frac{2200+2200}{27,500} \quad 16\% \end{aligned}$$

$$\text{Increase in sales (000)} = 38,500 - 27,500 = \text{Sh. } 11,000$$

Additional funds required

$$\text{Increase in total assets} \quad \frac{80}{100} \times 11,000 = \quad \quad \quad 8,800$$

$$\text{Less: increase in current liabilities} \quad \frac{16}{100} \times 11,000 = (1,760)$$

$$\text{Less: retained earnings} \quad \quad \quad \underline{(1,595)}$$

$$\text{Long term debt issued} \quad \quad \quad \underline{\underline{5,445}}$$

ii. Forecast statement of financial position

Upendo Ltd statement of financial position 2014

Percentage of sales

$$\text{Inventory} = \frac{2,000}{27,500} \times 100\% = 7.27\%$$

$$\text{Increase in inventory 2015} = \frac{7.27}{100} \times 11,000 = 800$$

$$\text{Debtors} = \frac{3,000}{27,500} \times 100\% = 10.91\%$$

$$\text{Increase debtors 2015} = \frac{10.91}{100} \times 11,000 = 1,200$$

$$\text{Cash at bank} = \frac{3,800}{27,500} \times 100\% = 13.82$$

$$\text{Increase cash 2015} = \frac{13.82}{100} \times 11,000 = \text{Sh. } 1,520$$

$$\text{Fixed assets} = \frac{13,200}{27,500} \times 100\% = 48\%$$

$$\text{Increase in fixed assets} = \frac{48}{100} \times 11,000 = \text{Sh. } 5,280$$

$$\text{Trade creditors} = \frac{2,200}{27,500} \times 100\% = 8\%$$

$$\text{Increase in trade creditors 2015} = \frac{8}{100} \times 11,000 = \text{Sh. } 880$$

$$\text{Accrued expenses} = \frac{2,200}{27,500} \times 100\% = 8\%$$

$$\text{Increase in accruals 2015} = \frac{8}{100} \times 11,000 = \text{Sh. } 880$$

UPENDO LTD

STATEMENT OF FINANCIAL POSITION AS AT 30TH NOVEMBER 2015

	Sh. (000)		Sh. (000)
Inventory	2,800	Trade creditors	3,080
Debtors	4,200	Accrued expenses	3,080
Cash at bank	5,320	Long term debt	14,245
Fixed assets (NBV)	18,480	Ordinary shares	2,200
		Retained profit	8,195
	<u>30,800</u>		<u>30,800</u>

iii. Computed ratios

a) Return on equity

$$= \frac{\text{Profit after tax}}{\text{Equity}} \times 100$$

$$= \frac{2,695}{10,395} \times 100\% = 25.93\%$$

Every Sh. 100 of equity in Upendo Ltd generated Sh. 25.93 as profit after tax in 2015

b) Total assets turnover

$$= \frac{\text{Total sales}}{\text{Total assets}}$$

$$= \frac{38,500}{30,800} = 1.25$$

Every Sh. 100 invested in total assets in 2015 generated Sh. 125 as sales

c) Capital gearing ratio

$$\frac{\text{Fixed charge capital}}{\text{fixed charge capital} + \text{equity}} \times 100\%$$

$$= \frac{14,245}{14,245 + 2,200 + 8195} \times 100\%$$

$$= 57.81\%$$

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TOPIC 9

WORKING CAPITAL MANAGEMENT

QUESTION 1

December 2023 Question Five C

Merip Ltd. expects to make payments of Sh.9,000,000 in the coming year. The firm's investment in marketable securities generates an annual return of 20%. The minimum cash balance maintained by the firm at all times is Sh.300,000. The firm incurs a cost of Sh.30 per transaction when converting marketable securities into cash.

Assume a year has 360 days.

Required:

Using Baumol's model of cash management, determine:

- (i) The optimal cash balance. (2 marks)
- (ii) Total relevant cost incurred in each year. (2 marks)
- (iii) The average cash balance. (2 marks)

ANSWER

(i) The optimal cash balance.

The optimal cash balance (C*) can be calculated using the following formula:

$$C = \sqrt{2 \times \text{Total cash payments} \times \frac{\text{Ordering cost}}{\text{Holding cost rate}}}$$

$$\sqrt{2 \times 9,000,000 \times \frac{30}{0.20}}$$

$$= \text{Sh. } 51,961.52$$

(ii) **Total relevant cost incurred in each year.**

$$= \left(\frac{9,000,000}{51,961.52} \times 30 \right) + \left(\frac{51,961.52}{2} \times 0.20 \right)$$

$$[9,000,000/51,961.52 \times 30] + [51,961.52/2 \times 0.20]$$

$$= 5,196.153 + 5,196.152 = \text{Sh. } 10,392.306$$

(iii) **The average cash balance.**

$$\text{Average cash balance} = \frac{1}{2} \times [C + M]$$

$$= \frac{1}{2} \times [51,961.52 + 300,000] = \text{Sh. } 175,980.76$$

QUESTION 2

August 2023 Question Two A

- (a) (i) Distinguish between “temporary working capital” and “permanent working capital” in relation to working capital management. (2 marks)
- (ii) The monthly working capital requirement for Mbuni Ltd. are given as follows:

Month	Total working capital requirement Sh. “000”
January	3,000
February	2,500
March	2,000
April	1,500
May	1,500
June	1,700
July	1,800
August	2,800
September	3,200
October	3,500
November	3,600
December	3,800

The short term cost of financing working capital is 15% per annum whereas the long-term financing cost is 20% per annum.

The firm adopts an aggressive policy in financing its working capital needs. 80% of the firm's permanent working capital are financed using short term funds and the balance is financed using long term funds.

Required:

Determine the total cost of financing the working capital needs of the firm. (6 marks)

ANSWER

(i) Difference between „temporary working capital“ and „permanent working capital.“

Feature	Temporary Working Capital	Permanent Working Capital
Definition	The amount of working capital that is required to meet short-term fluctuations in the business.	The amount of working capital that is required to meet the on-going day-to-day operations of the business.
Time period	Required for a short period of time, such as seasonal variations in demand or temporary increases in inventory.	Required for a long period of time, such as the ongoing day-to-day operations of the business.
Financing	Financed through short-term sources, such as trade credit and bank overdrafts.	Financed through long-term sources, such as long-term debt and equity.
Management	Managed by closely monitoring the levels of inventory and accounts receivable.	Managed by ensuring that the business has sufficient cash flow to meet its short-term obligations.

(ii) Total cost of financing the working capital needs of the firm

Month	Total Working Capital	Permanent Working Capital	Temporary working capital
January	3,000,000	1,500,000	1,500,000
February	2,500,000	1,500,000	1,000,000
March	2,000,000	1,500,000	500,000
April	1,500,000	1,500,000	0
May	1,500,000	1,500,000	0
June	1,700,000	1,500,000	200,000
July	1,800,000	1,500,000	300,000
August	2,800,000	1,500,000	1,300,000
September	3,200,000	1,500,000	1,700,000
October	3,500,000	1,500,000	2,000,000
November	3,600,000	1,500,000	2,100,000
December	3,800,000	1,500,000	2,300,000
Total	30,900,000	18,000,000	12,900,000

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Cost of financing working capital needs:

$$\begin{aligned}
 &= \{80/100 \times 18,000,000 \times 15/100\} + (20/100 \times 18,000,000 \times 20/100) + (15/100 \times 12,900,000) \\
 &= \text{Sh. } 2,160,000 + 720,000 + 1,935,000 \\
 &= \text{Sh. } 4,815,000
 \end{aligned}$$

QUESTION 3

April 2023 Question Five C

The following financial statement data relates to Bamaco Ltd. for the year ended 31 December 2022:

	Sh. "000"
Inventory - Opening balance	4,000
- Ending balance	4,600
Trade receivables	5,000

Trade payables	3,400
Credit sales	50,000
Cost of goods sold	40,000

Number of days in a year is 365 days.

Required:

Compute the net operating cycle of Bamaco Ltd.

(6 marks)

ANSWER

Inventory holding period

$$= \frac{\text{Average inventory}}{\text{Cost of goods sold}} \times 365$$

$$= \frac{(4000+4600) \div 2}{40,000} \times 365 = 39.24 \text{ days}$$

Debt collection period

$$\frac{\text{Average debtors}}{\text{Credit sales}} \times 365$$

$$= \frac{5000}{50000} \times 365 = 36.50\%$$

Purchases = cost of sales – Opening inventory + Closing Inventory

$$= 40,000 + 4600 - 4000$$

Ksh. 40,600

Creditors' period

$$\frac{\text{Average creditors}}{\text{Credit purchases}} \times 365$$

$$\frac{3400}{40600} \times 365$$

$$= 30.57$$

Net operating cycle = inventory holding period + debtors collection period – creditors period

$$= 39.24 + 36.50 - 30.57$$

$$= 45.17 \text{ days}$$

QUESTION 4

December 2022 Question Three B

Baraka Traders Ltd. has a minimum cash balance limit of Sh.100,000. The standard deviation of the daily cash flows is Sh.25,000. The interest rate of the marketable securities is 9.2% per annum.

The transactional cost for each sale or purchase of security is Sh.200.

Assume a 365-day year.

Required:

Using Miller-Orr cash management model, determine:

- (i) Target cash level. (2 marks)
- (ii) Upper cash limit. (2 marks)
- (iii) Average cash balance. (2 marks)
- (iv) The spread. (1 mark)

ANSWER

Baraka Traders Ltd.

- (i) Target cash level.

$$Z = \left(\frac{3b\sigma^2}{4i} \right)^{1/3} + L$$

Where:

b = Transaction cost = sh. 200

δ = Volatility of daily cash = sh. 25,000

i = Daily interest rate = 9.2% ÷ 365

L = Minimum cash balance = sh. 100,000

$$\begin{aligned} Z &= \left(\frac{3 \times 200 \times 25000^2}{4 \times \frac{0.092}{365}} \right)^{1/3} + 100,000 \\ &= 71,915.99 + 100,000 \\ &= \text{Sh. } 171,915.99 \end{aligned}$$

(ii) Upper cash limit.

$$\begin{aligned} H &= 3Z - 2L \\ &= (3 \times 171,915.99) - (2 \times 100,000) \\ &= 515,747.97 - 200,000 \\ &= 315,747.97 \end{aligned}$$

(iii) Average cash balance.

$$\begin{aligned} &= \frac{4Z - L}{3} \\ &= \left((4 \times 171,915.99) - 100,000 \right) \div 3 \\ &= (687,663.96 - 100,000) \div 3 \\ &= \text{Sh. } 195,887.99 \end{aligned}$$

(iv) The spread.

$$\begin{aligned}
 S &= H - L \\
 &= 315,747.97 - 100,000 \\
 &= \text{Sh. } 215,747.97
 \end{aligned}$$

QUESTION 5

December 2022 Question Four C

The following are the summary statement of profit and loss and statement of financial position for Miranda Ltd. for the year ended 31 October 2022.

Summary Statement of profit and loss

	Sh.“000”
Sales (150,000 units)	15,000
Variables costs	(13,500)
Fixed costs	<u>(600)</u>
Profit	<u>900</u>

Statement of financial position

	Sh.“000”	Sh.“000”
Non-current assets		22,500
Current assets:		
Trade receivables	3,000	
Inventory	1,200	
Bank	360	
Trade payables	<u>(900)</u>	<u>3,660</u>
		<u>26,160</u>
Financed by:		
Equity capital		15,000

12% long- term debt	3,840
6% preference share capital	<u>7,320</u>
	<u>26,160</u>

Additional information:

1. Return on investment (ROI) is 4.8%.
2. All sales are on credit and the company operates a very strict credit control system.
3. A suggestion has been made that a relaxation of credit policy would increase sales by 40%, if the company were to introduce a 2% discount (at present no discount is given) on accounts paid within 10 days.
4. It is envisioned that 70% of the customers would take advantage of the discount and the average collection period of the remainder would be half of what it is at present.
5. Bad debts would remain at 2% of firm's credit sales.
6. Investors required rate of return is 10%.

Assume 360 days in a year.

Required:

- (i) The current average collection period. (1 mark)
- (ii) The new level of profits after change in credit policy. (6 marks)
- (iii) Explain the effect of the new level of investment in account receivable. (2 marks)

ANSWER

Miranda Ltd.

- (i) **The current average collection period.**

$$= \frac{3,000}{15,000} \times 360 = 72 \text{ days}$$

(ii) **The new level of profits after change in credit policy.**

New sales

$$= \frac{140}{100} \times 15\,000\,000$$

$$= \text{Sh.}21,000,000$$

Customer sales subject to discount

$$\frac{70}{100} \times \text{Sh.}21,000,000$$

$$= \text{Sh.}14,700,000$$

Customers who won't take advantage of discount

$$= 21,000,000 - 14,700,000$$

$$= \text{Sh.}6,300,000$$

Discount allowed

$$= \frac{2}{100} \times 14,700,000$$

$$= \text{Sh.}294,000$$

Bad debts

$$\text{Current policy} = \frac{2}{100} \times 1,500,000 = 300,000$$

$$\text{Proposed policy} = \frac{2}{100} \times 21,000,000 = 420,000$$

Debtors proposed policy

$$\begin{aligned}
 &= \left(\frac{10}{360} \times 14,700,000\right) + \left(\frac{36}{360} \times 6,300,000\right) \\
 &= 408,333 + 630,000 \\
 &= \text{Sh.1,038,333}
 \end{aligned}$$

Opportunity cost debtors

$$\begin{aligned}
 \text{Current policy} &= \frac{4.8}{100} \times 3,000,000 \\
 &= \text{Sh.144,000}
 \end{aligned}$$

$$\begin{aligned}
 \text{Proposed policy} &= \frac{4.8}{100} \times 1,038,333 \\
 &= \text{Sh.49,840}
 \end{aligned}$$

(iii) Effect of the new level of investment in account receivable.

Details	New policy	Current policy	Change
Sales	21,000,000	15,000,000	6,000,000
Less: variable costs	18,900,000	13,500,000	5,400,000
Contribution	2,100,000	1,500,000	600,000
Less: Dad debts	(420,000)	(300,000)	(120,000)
Less: Discount	(294,000)	0	(294,000)
Less: Opportunity cost debtors	(49,840)	(144,000)	94,160
Less: Fixed costs	<u>(600 000)</u>	<u>(600 000)</u>	<u>0</u>
Profit	736,160	456,000	280,160

The change in credit policy will lead to a decline in the account receivable balance

QUESTION 6

August 2022 Question Two B (iii)

Distinguish between the following terms as used in finance:

- (iii) —liberal credit policy” and “conservative credit policy”. (2 marks)

ANSWER

Distinction between “Liberal credit policy” and “conservative credit policy” as used in finance:

For liberal credit policy, selling on credit to as many customers as possible without even considering credit worthiness while a conservative credit policy entails selling goods on credit on highly selective basis to only credit worthy customers

QUESTION 7

August 2022 Question Four B

The following information relates to inventory relationship of QZ Ltd.:

1. Annual purchases of Sh.2,160,000.
2. Purchase price per unit is Sh.60.
3. Carrying cost is 15% of the purchase price per unit.
4. Cost per order placed is Sh.240.
5. Desired stock levels is 300 units. This stock level was in hand initially.
6. Lead time is 7 days.

Assume a 365-day year.

Required:

- (i) The economic order quantity (EOQ) for the company. (3 marks)
- (ii) The optimal number of orders to be placed in a year. (1 mark)

- (iii) The re-order level. (2 marks)
- (iv) Assuming that for any orders of at least 2,000 units, the firm will get 5% discount on the purchase price.

Analyse whether the company should take advantage of the discount or not (4 marks)

ANSWER

QZ Ltd.

- (i) The economic order quantity (EOQ) for the company.

$$q = \sqrt{\frac{2dCo}{Ch}}$$

Where D - Annual demand in units = 36,000

Co - Ordering cost per order = 240

Ch - Carrying cost per unit

$$= \frac{15}{100} \times 60 = \text{Sh } 9$$

$$Q = \sqrt{\frac{2 \times 36,000 \times 240}{9}}$$

$$= \sqrt{1,920,000}$$

$$= 1,385.64 \approx 1386 \text{ units}$$

- (ii) The optimal number of orders to be placed in a year.

$$O = \frac{d}{Q}$$

$$= 3600 \div 1386 = 25.97 \approx 26 \text{ orders}$$

- (iii) The re-order level.

$$RI = \text{Safety stock} + \text{average usage} \times \text{lead time}$$

$$= 300 + \frac{36,000}{365} \times 7$$

$$= 300 + 690 = 990 \text{ units}$$

(iv) Whether the company should take advantage of the discount or not.

No discount

$$TC = Pc + \frac{D}{q}Co + \frac{q}{2}Ch$$

$$= 2,160,000 + \left(\frac{36,000}{1,386} \times 240 \right) + \left(\frac{1,386}{2} \times 9 \right)$$

$$= 2,160,000 + 6,233.77 + 6,237$$

$$= \text{Sh } 2,172,470.77$$

Discount offer

$$\text{New purchase price} = 0.95 \times 60 = \text{Sh } 57$$

$$\text{Carrying cost per unit} = \frac{15}{100} \times 57 = 8.55$$

$$Tc = (36,000 \times 57) + \left(\frac{36,000}{2,000} \times 240 \right) + \left(\frac{2,000}{2} \times 8.55 \right)$$

$$= 2,052,000 + 4,320 + 8,550$$

$$= \text{Sh.}2,064,870.00$$

Advice

Should take advantage of discount offer since it will reduce total costs.

4C

$$V = \text{Annuity} \left[\frac{(1+r)^n - 1}{r} \right]$$

$$\text{Annuity} = \text{FV} \div \left[\frac{(1+r)^n - 1}{r} \right]$$

$$= 30,000,000 \div \left(\frac{1.04^8 - 1}{0.04} \right)$$

$$= 30,000,000 \div 9.2142$$

$$= \text{Sh.}3,255,844.24$$

QUESTION 8

April 2022 Question Two B

The following information relates to Triplex Ltd:

1. The company has annual sales revenue of Sh.6,000,000 and all sales are on 30 days credit, although customers on average take 10 days more than the recommended 30 days to pay.
2. Contribution represents 60% of sales and the company currently has no bad debts. Accounts receivable are financed by an overdraft at an annual interest rate of 7%.
3. Triplex Ltd. plans to offer an early settlements discount of 1.5% for payment within 15 days and to extend the maximum credit offered to 60 days.
4. The company expects that these changes will increase annual credit sales by 5% while also leading to additional incremental costs equal to 0.5% of sales revenue.
5. The discount is expected to be taken by 30% of customers, while the remaining customers would take an average of 60 days to pay.

Required:

Evaluate whether the proposed changes in credit policy will increase the profitability of Triplex Ltd. (8 marks)

ANSWER

Whether the proposed changes in credit policy will increase the profitability of Triplex Ltd.

New sales

$$= \frac{105}{100} \times 6,000,000$$

$$= 6,300,000$$

Contribution current policy

$$= \frac{60}{100} \times 6,000,000$$

$$= 3,600,000$$

Proposed policy

$$= \frac{60}{100} \times 6,300,000$$

$$= \text{Sh. } 3,780,000$$

Debtors:

$$\text{Debtors} = \frac{\text{Debtors period}}{365 \text{ days}} \times \text{credit sales}$$

$$\text{Current policy} = \frac{40}{365} \times 6,000,000$$

$$= \text{Sh. } 657,534$$

Proposed policy

$$\text{Discount offer} = \frac{15}{365} \times \frac{30}{100} \times 6,300,000$$

$$= \text{Sh. } 77,671$$

$$\text{No discount} = \frac{60}{365} \times \frac{70}{100} \times 6,300,000$$

$$= \text{Sh. } 724,932$$

Total debtors proposed policy

$$= 77,671 + 724,932$$

$$= \text{Sh. } 802,603$$

Opportunity cost debtors

$$\text{Current policy} = \frac{7}{100} \times 657,534$$

$$= \text{Sh. } 46,027$$

$$\text{Proposed policy} = \frac{7}{100} \times 802,603$$

$$= \text{Sh. } 56,182$$

Incremental costs

$$= \frac{0.5}{100} \times 6,300,000 = \text{Sh. } 31,500$$

Discount allowed

$$= \frac{1.5}{100} \times \frac{30}{100} \times 6,300,000$$

$$= \text{Sh. } 28,350$$

Details	Proposed	Current	Change
Sales	6,300,000	6,000,000	300,000
Less: Variable costs	<u>2,520,000</u>	<u>2,400,000</u>	<u>120,000</u>
Contribution	3,780,000	3,600,000	180,000
Expenses:			
Opportunity cost: Debtors	56,182	46,027	10,155
Incremental costs	31,500	0	31,500
Discount allowed	<u>28,350</u>	<u>0</u>	<u>28,350</u>
Profit	<u>3,663,968</u>	<u>3,553,973</u>	<u>109,995</u>

Advice:

Triplex Ltd should change the credit policy since it will lead to an increase in profitability.

QUESTION 9

December 2021 Question Five B and C

- (b) Analyse three motives of holding cash. (6 marks)
- (c) Jasmine Limited generates Sh.100,000 per month excess cash which it intends to invest in short-term securities. The interest rate it can expect to earn on its investment is 10% per annum. The transaction costs associated with each separate investment of funds is constant at Sh.500.

Required:

- (i) The optimal amount of cash to be invested in each transaction. (1 mark)
- (ii) The number of transactions per annum. (1 mark)
- (iii) The total cost of making the transactions in (c) (ii) above. (1 mark)

(iv) The opportunity cost of holding cash per annum.

(1 mark)

ANSWER

(b) Motives of holding cash.

- **Transaction motive:** a firm holds cash to be deployed in payment of operational costs like salaries
- **Speculative motive:** a firm sets aside some cash to be utilized for taking advantage of profitable opportunities that might arise.
- **Precautionary motive:** a business sets aside some cash too cater for emergencies that might arise.

(c) Jasmine Limited

(i) The optimal amount

$$C = \sqrt{\left(\frac{2bT}{i}\right)}$$

Where: $b = 500$

$$T = 12 \times 100000 = 1,200,000$$

$I = 10\%$ or 0.1

$$C = \sqrt{\frac{2 \times 500 \times 1,200,000}{0.1}}$$

$$= \text{Ksh. } 109,544.5115$$

(ii) The number of transactions per annum.

$$N = T/C = 1200000 \div 109,544.5115$$

$$= 10.95 = 11$$

(iii) The total cost of making the transactions in (c) (ii) above.

$$= \frac{1,200,000}{109,544.5115} \times 500$$

$$= \text{Sh. } 5477.23$$

(iv) The opportunity cost of holding cash per annum.

$$= \frac{109,544.5115}{2} \times 0.1$$

$$= \text{Sh. } 5,477.23$$

QUESTION 10

September 2021 Question Four C

Upesi Wholesalers Ltd. deals in sale of foodstuffs to retailers. Owing to economic depression, the firm intends to relax its credit policy to boost productivity and sales. The firm's current credit policy is "net 30" and the average debt collection period is 45 days.

The current annual credit in amount to Sh.60 million. The firm intends to change to "net 60" where sales are expected to increase by 25%. Credit and debt analysis costs will increase from the current 2% to 2.5% of credit sales. Bad debts will also increase from the current 1.5% to 2% of credit sales, variable costs account for 75% of sales and return on assets is 12%. Assume a 360 day year.

Required:

Advise the company on whether to adopt the new credit policy. (6 marks)

ANSWER

Upesi Wholesalers Ltd.

Whether to adopt the new credit policy.

$$\text{New sales} = \frac{125}{100} \times 60,000,000$$

$$= \text{Sh } 75,000,000$$

$$\text{Current debtors} = \frac{45}{360} \times 60,000,000$$

$$= \text{Sh. } 7,500,000$$

$$\text{Proposed debtors} = \frac{60}{360} \times 75,000,000$$

$$= \text{Sh } 12,500,000$$

$$\text{Variable cost old policy} = \frac{75}{100} \times 60,000,000$$

$$= \text{Sh } 45,000,000$$

$$\begin{aligned} \text{Variable cost proposed policy} &= \frac{75}{100} \times 75,000,000 \\ &= \text{Sh. } 56,250,000 \end{aligned}$$

Credit and debt analysis costs

$$\begin{aligned} \text{Current policy} &= \frac{2}{100} \times 60,000,000 \\ &= \text{Sh. } 1,200,000 \end{aligned}$$

$$\begin{aligned} \text{Proposed policy} &= \frac{2.5}{100} \times 75,000,000 \\ &= \text{Sh. } 1,875,000 \end{aligned}$$

Bad debts

$$\begin{aligned} \text{Current policy} &= \frac{1.5}{100} \times 60,000,000 \\ &= \text{Sh. } 900,000 \end{aligned}$$

$$\begin{aligned} \text{Proposed policy} &= \frac{2}{100} \times 75,000,000 \\ &= \text{Sh } 1,500,000 \end{aligned}$$

Opportunity cost

$$\begin{aligned} \text{Current policy} &= \frac{12}{100} \times 7,500,000 \\ &= \text{Sh. } 900,000 \end{aligned}$$

$$\begin{aligned} \text{Proposed policy} &= \frac{12}{100} \times 12,500,000 \\ &= \text{Sh. } 1,500,000 \end{aligned}$$

Details	Proposed	Current	Change
Sales	75,000,000	60,000,000	15,000,000
Less: variable costs	56,250,000	45,000,000	11,250,000
Contribution	18,750,000	15,000,000	3,750,000
Credit and debit analysis	(1,875,000)	(1,200,000)	(675,000)
Bad debts	(1,500,000)	(900,000)	(600,000)
Opportunity cost	<u>(1,500,000)</u>	<u>(900,000)</u>	<u>(600,000)</u>
Profit before tax	13,875,000	12,000,000	1,875,000

Less tax 30%	4,162,500	3,600,000	562,500
PAT	9,712,500	8,400,000	1,312,500

Advice

The company should change the current policy because it will lead to increase in profitability.

QUESTION 11

May 2021 Question Three C

ABC Limited's current annual sales are Sh.1.8 million with a cost of sales of 80% and bad debts average 1% of total sales.

The current debt collection period is one month and the management considers that if credit terms were eased (Option A), the effects would be as follows:

	Present Policy	Option A
Additional sales (%)	-	25%
Average collection period	1 month	2 months
Bad debts (% of sales)	1%	3%

Additional information:

1. The company requires a 20% return on its investments.
2. The cost of sales are 75% variable and 25% fixed.

Required:

Advise the management on whether or not to ease the credit terms. (4 marks)

ANSWER

ABC Limited

Advise on whether or not to ease the credit terms.

$$\text{Debtors collection period} = \frac{\text{average debtors}}{\text{sales}} \times 12$$

$$\text{Average debtors} = \frac{\text{debtors collection period}}{12} \times \text{sales}$$

$$\text{Debtors present policy} = \frac{1}{12} \times 1,800,000 = 150,000$$

$$\text{Debtors policy} = \frac{2}{12} \times \frac{125}{100} 1,800,000 = 375,000$$

$$\text{Bad debts present policy} = \frac{1}{100} 1,800,000 = 18,000$$

$$\begin{aligned} \text{Bad debts option A} &= \frac{3}{100} \times \frac{125}{100} \times 1,800,000 \\ &= \frac{3}{100} \times 2,250,000 = \text{sh. } 67,500 \end{aligned}$$

$$\text{Opportunity cost present policy} = \frac{20}{100} \times 150,000 = 30,000$$

$$\text{Opportunity cost option A} = \frac{20}{100} \times 375,000 = 75,000$$

$$\text{Fixed cost of present policy} = \frac{25}{100} \times \frac{80}{100} \times 1,800,000 = 360,000$$

$$\text{Fixed cost of option A} = \frac{25}{100} \times \frac{125}{100} \times 1,800,000 \times \frac{80}{100} = \text{sh. } 450,000$$

$$\text{New sales} = \frac{125}{100} \times 1,800,000 = \text{sh. } 2,250,000$$

Details	Option A	Current	Change
Sales	2,250,00	1,800,000	450,000
Less :fixed cost of sales	450,000	360,000	90,000
Less :variable sales cost	1,350,000	1,080,000	270,000
Gross contribution	450,000	360,000	90,000
Less: bad debts	67,500	18,000	49,500
Less :opportunity cost	75,000	30,000	45,000
	307,500	312,000	(4,500)

Advice:

They should not change current policy as it will result to a decline in profitability.

QUESTION 12

May 2021 Question Five A, B and C

(a) Discuss four factors that a firm should consider in formulating a working capital policy on the management of trade receivables. (4 marks)

(b) Phoenix Ltd. is considering amendments to its current inventory management policy.

The following information relates to the proposed ordering policy:

1. The current policy is to order 200,000 units when the inventory level falls to 70,000 units.
2. Forecast demand to meet production requirements during the next year is 1,250,000 units.
3. The cost of placing and processing an order is Sh.500, while the cost of holding a unit is Sh.1 per unit per year. Both costs are expected to be constant during the next year.
4. Orders are received two weeks after being placed with the supplier.
5. Assume one year has 50 weeks.

Required:

- (i) The cost of the current ordering policy. (3 marks)
- (ii) Determine the savings that could be made by using the economic order quantity (EOQ) model. (3 marks)
- (c) The following data was extracted from the financial statements of Mbuni Ltd. for the year ended 31 December 2020:

Statement of financial position as at 31 December 2020:

Sh."000"		Sh."000"	
Cash	?	Notes payable	100,000
Accounts receivable	?	Long term debt	?
Plant and equipment	?	Ordinary shares	100,000

	—	Retained earnings	100,000
Total Assets	<u>?</u>	Total Liabilities	?

Additional information:

Long term debt to equity ratio	0.5 to 1.0
Total assets turnover	2.5 times
Average collection period (Assume 360 days in a year and that all sales are on credit)	18 days
Inventory turnover	9 times
Acid test ratio	1 to 1
Gross profit margin	0.1

Required:

Determine the following:

- (i) Long term debt. (1 mark)
- (ii) Total liabilities and shareholders' equity. (1 mark)
- (iii) Cost of sales. (1 mark)
- (iv) Inventory. (1 mark)
- (v) Accounts receivable. (1 mark)
- (vi) Cash. (1 mark)
- (vii) Complete the statement of financial position of Mbuni Ltd. for the year ended 31 December 2020 using the figures obtained in (c) (i) to (c) (vi) above.

(2 marks)

ANSWER

a) Factors that a firm should consider in formulating a working capital policy on the management of trade receivables

- **Competitors:** the kind of credit terms they are offering
- **Credit period:** it should be in line with what similar competing firms are offering otherwise there is a possibility of loss of clients to competing firms
- **Factoring arrangement:** factoring entails the sale of debtors to a third party. An effective credit policy should in corporate factoring.

- **Credit standards:** This refers to minimum condition customers must meet to qualify for credit. credit worthiness can be on the basis of conditions, capital capacity and character

(b) Phoenix Ltd.

(i) The cost of the current ordering policy.

$$\begin{aligned} TC &= \left(\frac{1,250,000}{200,000} \times 500 \right) + \left(\frac{200,000}{2} \times 1 \right) \\ &= 3125 + 100,000 = 103,125 \end{aligned}$$

(ii) Savings that could be made by using the economic order quantity (EOQ) model.

$$\begin{aligned} Q &= \sqrt{\frac{2DCO}{Ch}} \\ &= \sqrt{\frac{2 \times 1,250,000 \times 500}{1}} \\ &= \sqrt{1,250,000,000} \\ &= 35,355.34 = 35,355 \text{ units} \\ T.C &= \frac{D}{Q} Co + \frac{Q}{2} Ch \\ &= \left(\frac{1,250,000}{35,355} \times 500 \right) + \frac{35,355}{2} \times 1 \\ &= 17,677.84 + 17,677.50 = 35,355 \\ &\text{Cost savings} \\ &= 103,125 - 35,355.34 = \text{sh. } 67,769.66 \end{aligned}$$

(c) Mbuni Ltd.

(i) Long term debt.

$$\begin{aligned} \text{Long term debt} &= \text{long term debt to equity ratio} \times \text{total equity} \\ &= 0.5 \times (100,000 + 100,000) \\ &= 0.5 \times 200,000 \\ &= \text{Sh. } 100,000 \end{aligned}$$

(ii) Total liabilities and shareholders' equity.

Total liabilities = note payable + long term debt + ordinary shares + retained earnings

$$= 100,000 \times 4$$

$$= \text{sh. } 400,000$$

Shareholders equity = ordinary shares + retained earnings

$$= 100,000 + 100,000 = \text{sh. } 200,000$$

(iii) Cost of sales.

Total asset = total asset turnover $\times$ total asset

$$= 2.5 \times 400,000$$

$$= \text{sh. } 1,000,000$$

$$\text{Gross profit} = 0.1 \times 1,000,000$$

$$= \text{sh. } 100,000$$

Cost of sales = sales – gross profit

$$= 1,000,000 - 100,000$$

$$= 900,000$$

(iv) Inventory.

Inventory = cost of sales $\div$ inventory turnover

$$= 900,000 \div 9$$

$$= \text{sh. } 100,000$$

(v) Accounts receivable.

$$\text{Debtors collection period} = \frac{\text{average debtors}}{\text{sales}} \times 365$$

$$\begin{aligned} \text{debtors} &= \frac{18}{360} \times 1,000,000 \\ &= \text{sh. } 50,000 \end{aligned}$$

(vi) **Cash.**

$$\begin{aligned} \text{Cash} &= 400,000 - 100,000 - 50,000 \\ &= \text{sh. } 250,000 \end{aligned}$$

(vii) **Complete the statement of financial position of Mbuni Ltd. for the year ended 31 December 2020 using the figures obtained in (c) (i) to (c) (vi) above.**

Statement of financial position As at 31st December 2020

	Sh''000''		Sh —00''
Cash	250,000	Notes payable	100,000
Accounts receivable	50,000	Long term debt	100,000
Plant and equipment	0	Ordinary shares	100,000
Inventory	100,000	Retained earnings	100,000
Total assets	400,000	Total liabilities	400,000

QUESTION 13

November 2020 Two B

Nderu Suppliers Ltd. is reviewing its working capital commitments for enhanced efficiency.

The following information relating to the period ended 31 March 2020 is provided:

Turnover for the year	Sh.15,000,000
Costs as percentages of sales	(%)
Direct materials	30
Direct labour	25
Variable overheads	10
Fixed overheads	15
Selling and distribution	5

Additional information:

1. On average:

- Account receivables take two and a half months before payment.
- Raw materials are in inventory for three months.
- Work in progress represents two months' worth of half produced goods.
- Finished goods represent one month's production.

2. Credit is taken as follows:

- Direct materials 2 months
- Direct labour 1 week
- Variable overheads 1 month
- Fixed overheads 1 month
- Selling and distribution Half a month

3. Work in progress and finished goods are valued at material, labour and variable expenses cost.

4. Labour force is paid for 50 working weeks a year.

Required:

Assess the working capital requirements for the company.

(8 marks)

ANSWER**(a) Working capital requirements for Nderu Suppliers Ltd.**

$$\text{Direct material cost} = \frac{30}{100} \times 15,000,000$$

$$\text{Sh } 4,500,000$$

$$\text{Direct labour cost} = \frac{25}{100} \times 15,000,000$$

$$= \text{Sh } 3,750,000$$

$$\text{Variable overheads} = \frac{10}{100} \times 15,000,000$$

$$= \text{Sh } 1,500,000$$

$$\text{Fixed overheads} = \frac{15}{100} \times 15,000,000$$

$$= \text{Sh } 2,250,000$$

$$\text{Selling and distribution overheads} = \frac{5}{100} \times 15,000,000$$

$$= 750,000$$

$$\text{Account receivables} = \frac{\text{Debtors collection period}}{12} \times \text{credit sales}$$

$$\frac{2.5}{12} \times 15,000,000$$

$$= \text{Sh } 3,125,000$$

$$\text{Raw materials} = \frac{\text{Raw materials holding period}}{12} \times \text{direct material cost}$$

$$\frac{3}{12} \times 4,500,000$$

$$= 1,125,000$$

Cost of production

Material + Labour + Variable overheads

$$= 4,500,000 + 3,750,000 + 1,500,000$$

$$= 9,750,000$$

Cost of work in progress

$$= \frac{1}{12} \times 9,750,000$$

$$= \text{Sh } 812,500$$

Value of work in progress inventory

$$= \frac{2}{12} \times 4,875,000$$

$$= \text{Sh } 812,500$$

Value of finished goods inventory

$$\frac{1}{12} \times 9,750,000$$

$$\text{Sh } 812,500$$

Direct material accruals

$$\frac{2}{12} \times 4,500,000$$

$$= 750,000$$

Direct labour accruals

$$= \frac{1}{50} \times 3,750,000$$

Sh 75,000

Variable overhead accruals

$$\frac{1}{12} \times 1,500,000$$

= Sh 125, 000

Fixed overhead accruals

$$= \frac{1}{12} \times 2,250,000$$

= Sh 187, 500

Selling and distribution overheads accruals

$$= \frac{0.5}{12} \times 750,000$$

Sh. 312, 500

Nderu suppliers Ltd

Statement of working capital requirement

For the period ended 31 March 2020

Current assets:	Sh	Sh
Account receivables		3,125,000
Raw materials		1,125,000
Work in progress		812,500
Finished goods		<u>812,500</u>

Total current assets:	<u>5,875,000</u>
Current liabilities:	
Direct material accruals	750,000
Direct labour accruals	75,000
Variable overhead accruals	125,000
Fixed overhead accruals	187,500
Selling and distribution overhead accruals	<u>312,500</u>
Total current liabilities	<u>1,450,000</u>
Working capital requirement	<u>4, 425,000</u>

QUESTION 14

November 2020 Question Three A

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The following information relates to Bawabu Traders:

1. The minimum cash balance is Sh.8,000.
2. The variance of daily cash flows is Sh.4,000,000, equivalent to a standard deviation of Sh.2,000 per day.
3. The transaction cost for buying or selling securities is Sh.50.
4. The interest rate is 0.025 per cent per day.

Required:

Using Miller-Orr Model of managing cash, determine the following:

- (i) The spread. (2 marks)
- (ii) Upper cash limit. (2 marks)

- (iii) Return point. (2 marks)
- (iv) Propose a decision rule for cash management to the company based on your calculations in (a) (i) to (a) (iii) above. (2 marks)

ANSWER

Miller-Orr Model of managing cash

(i) The spread (S)

$$\begin{aligned} S &= H - L \\ &= 13,451.36 - 8,000 \\ &= 5,451.36 \end{aligned}$$

(ii) Upper cash limit (H)

$$\begin{aligned} H &= 3Z - 2L \\ &= (3 \times 9,817.12) - (2 \times 8,000) \\ &= 29,451.36 - 16,000 \\ &= \text{Sh } 13,451.36 \end{aligned}$$

(iii) Return point.

$$Z = \sqrt[3]{\frac{3b\delta^2}{4 \times i/365}} + L$$

Where:

b = transaction cost = Sh 50

δ^2 = variance of daily cash flows = Sh 4,000,000

i = interest rate daily = 0.025

L = minimum cash balance = 8,000

$$\begin{aligned} Z &= \left(\frac{3 \times 50 \times 4,000,000}{4 \times 0.025} \right)^{1/3} + 8,000 \\ &= 6,000,000,000^{1/3} + 8,000 \\ &= \text{Sh. } 9,817.12 \end{aligned}$$

(iv) Decision rule for cash management to the company based on calculations in (a) (i) to (a) (iii) above.

Optimal cash to hold is Sh 9,817.12, when cash rises to Sh 13,451.36, Bawabu traders should deploy Sh. 3,634.24 (Sh 13,451.36 – 9817.12) to buy marketable securities.

When cash reaches minimum balance of Sh 8,000, Bawabu traders should sell securities worth Sh 1,817.12 to restore cash to target level of sh 9,817.12

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QUESTION 15

November 2019 Question One B

Sasumua Ltd. is considering a review of its credit policy as a way of enhancing its working capital management:

The following information relates to the company:

- Annual sales amount to Sh.6,000,000.
- Credit sales are 80% of all sales.
- Bad debts average 3% of all credit sales.
- Average collection period for debtors is 45 days.

- Gross profit on sales is 75%.
- The company's cost of capital is 12% per annum.
- Terms of credit sales are net 60 days.

The company's credit manager is recommending a review of the credit policy of the company. The expected outcome of this action will be:

- Increase in total sales by 40%.
- Credit sales will be 60% of all sales.
- Average collection period will decrease to 30 days.
- Bad debts will increase to 5% of credit sales.
- An additional part-time credit consultant will be hired at Sh.500,000 per annum.
- Gross profit margin will increase to 80%.
- Terms of credit sales will be 5/15 net 45. All credit customers will enjoy the 5% cash discount subject to the terms.
- No change is expected in the firm's cost of capital.
- The tax rate is 30%.

Required:

Advise the company on whether to adopt the revised credit policy. (7 marks)

ANSWER

Advise to the company on whether to adopt the revised credit policy

$$\text{Total new sales} = \frac{140}{100} \times 6000\ 000 = \text{Sh. } 8,400,000$$

$$\text{New credit sales} = \frac{60}{100} \times 8,400\ 000 = \text{Sh. } 5,040,000$$

$$\text{Old credit sales} = \frac{80}{100} \times 6000\ 000 = \text{Sh. } 4,800,000$$

$$\text{New bad debts} = \frac{5}{100} \times 5040\ 000 = \text{Sh. } 252000$$

$$\text{Old bad debts} = \frac{3}{100} \times 4,800\ 000 = \text{Sh. } 144,000$$

$$\text{Discount allowed} = \frac{5}{100} \times 5040000 = \text{sh. } 252,000$$

$$\text{Old debtors} = \frac{45}{365} \times 4,800,000 = \text{sh. } 591,781$$

$$\text{New debtors} = \frac{15}{365} \times 5040\ 000 = \text{sh. } 207,123$$

	New	Old	Change
Sales	8,400,000	6,000,000	2,400,000
Less: Variable costs	<u>(1,680,000)</u>	<u>(1,500,000)</u>	<u>(180,000)</u>
Gross profit	6720 000	4,500,000	2,220,000
Bad debts	(252 000)	(144,000)	(108,000)
Discount allowed	(252 000)	-	(252 000)
Consultant pay	(500 000)	-	(500 000)
Opportunity cost debtors	<u>(24,855)</u>	<u>71,014)</u>	<u>46,159</u>
Profit before tax	5,691,145	4,284,986	1,406,159
Less: Tax 30%	<u>(1,707,344)</u>	<u>(1,285,500)</u>	<u>(421,848)</u>
Profit after tax	<u>3,987,802</u>	<u>2,999,486</u>	<u>954,311</u>

Advice: Company should adopt new policy

QUESTION 16

November 2019 Question Two B

Maua Horticultural Ltd. runs a flower export business. The company has two sources of funds at different interest rates. The finance cost for short-term funds is 20% while the cost of long-term funds is 25%. These costs are expected to remain constant in the next two years.

The following are the projected monthly working capital requirements of the company for the year ending 31 December 2020:

Month	Working capital required (Sh."000")
January	35,000
February	35,000
March	52,500
April	70,000
May	105,000
June	157,500
July	210,000
August	242,500
September	157,500
October	87,500
November	70,000
December	52,500

Required:

- (i) The average monthly permanent and seasonal working capital requirements for the company. (6 marks)

- (ii) Total cost of working capital finance for the company under an aggressive financing policy, conservative financing policy and matching financing policy. (6 marks)

- (iii) Advise the company on the appropriate working capital financing policy to adopt. (2 marks)

ANSWER

i) The average monthly permanent and seasonal working capital requirements for the company.

Month	Working capital (000)	Permanent Working capital (000)	Seasonal working capital (000)
January	35,000	35,000	0
February	35,000	35,000	0
March	52,500	35,000	17,500
April	70,000	35,000	35,000
May	105,000	35,000	70,000
June	157,500	35,000	122,500
July	210,000	35,000	175,000
August	242,500	35,000	207,500
September	70,000	35,000	122,500
October	87,500	35,000	52,500
November	70,000	35,000	35,000
December	<u>52,500</u>	<u>35,000</u>	<u>17,500</u>
	<u>1,275,000</u>	<u>429,000</u>	<u>855,000</u>

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Average permanent working capital = $420,000,000 \div 12 = Sh. 35,000,000$

Average seasonal working capital = $855,000,000 \div 12 = Sh. 71,250,000$

ii) Total cost of working capital finance for the company under an aggressive financing policy, conservative financing policy and matching financing policy

Cost of aggressive financing policy

$$= \frac{20}{100} \times (71,250,000 + 35,000,000)$$

$$= 21,250,000$$

Cost of conservative financing policy

$$= \frac{25}{100} \times (71,250,000 + 35,000,000)$$

$$= 26,562,500$$

Cost of marching policy

$$\left(\frac{25}{100} \times 71,250,000\right) + \left(\frac{25}{100} \times 35,000,000\right)$$

$$= 14,250,000 + 8,750,000$$

$$= \text{Sh. } 23,000,000$$

iii) Advise to the company on the appropriate working capital financing policy to adopt

Best policy to adopt is the aggressive financing policy because it costs less.

QUESTION 17

May 2019 Question Two C

In an effort to lower its accounts receivable balances, Chizingo Manufacturing Limited is considering to switch from its existing no discount policy to a 2% cash discount for payment done by 15th day after sale.

It is estimated that 60% of the customers would take the discount and the average collection period is expected to decline from 60 days to 45 days.

The company's management projects an increase of 20,000 units in annual sales to 220,000 units at the existing price of Sh.2,500 per unit.

Additional information:

1. The variable cost per unit is Sh.2,100 and the average cost per unit is Sh.2,300.
2. The company requires a 15% return on investment (ROI).
3. The corporate tax rate is 30%.

4. All sales are on credit basis.

Assume 365 days in a year.

Required:

Advise the management of Chizingo Manufacturing Limited on whether to offer the cash discount to customers. (6 marks)

ANSWER

Chizigo manufacturing limited

$$\text{Old sales} = 200,000 \times 2,500 = \text{Sh. } 500,000,000$$

$$\text{New sales} = 220,000 \times 2,500 = \text{Sh. } 550,000,000$$

$$\text{Average debtors} = \frac{\text{average collection period}}{365} \times \text{credit sales}$$

$$\begin{aligned}\text{Average debtors old policy} &= \frac{60}{365} \times 500,000,000 \\ &= \text{Shs. } 82,191,780.82\end{aligned}$$

Average debtors under new policy will be categorized under:

- i. Discount after debtors
- ii. Non-discount after debtors

$$\begin{aligned}\text{Average debtors discount offer} &= \frac{15}{365} \times 0.6 \times 550,000,000 \\ &= 13,561,643.84\end{aligned}$$

Average debtors non-discount offer

$$= \frac{15}{365} \times 0.4 \times 550,000,000 = \text{Shs. } 9,041,095.89$$

Total average debtors new policy

$$13,561,643.84 + 9,041,095.89 = 22,602,739.73$$

Opportunity cost

$$\text{Old policy} = \frac{15}{100} \times 82,191,780.82$$

$$= \text{Shs. } 12,328,767.12$$

$$\text{New policy} = \frac{15}{100} \times 22,602,739.73$$

$$= \text{Shs. } 3,390,410.96$$

$$\text{Discount allowed new policy} = \frac{2}{100} \times \frac{60}{100} \times 550,000,000 = 6,600,000$$

Details	New policy	Old policy	Change
Rates	550,000,000	500,000,000	50,000,000
Less: variable costs	(462,000,000)	(420,000,000)	(42,000,000)
Contribution	88,000,000	80,000,000	8,000,000
Less: other costs	(44,000,000)	(40,000,000)	(4,000,000)
Less: discount allowed	(6,600,000)	0	(6,600,000)
Less: opportunity cost debtors	<u>(3,390,410.96)</u>	<u>(12,328,767.12)</u>	<u>(8,938,356.16)</u>
Profit before tax	34,009,589.04	27,671,232.88	6,338,356.16
Less: tax 30%	(10,202,876.71)	(8,301,369.864)	(1,901,506.848)
Profit after	<u>23,806,712.33</u>	<u>19,369,863.02</u>	<u>4,436,849.312</u>

The management of Chizingo manufacturing Limited should change credit policy since it's profitable to do so.

QUESTION 18**November 2018 Three D**

Bahati Enterprises is considering amendments to its current credit policy. The firm's current annual credit sales amount to Sh.6,000,000.

The current credit terms are net 30 with an average debtors collection period of 45 days.

The following information relates to the proposed credit policy:

1. The credit period to be extended to net 60.
2. Annual sales are expected to increase by 20%.
3. Bad debts are expected to increase from 2% to 2.5% of the annual credit sales.
4. Credit analysis and debt collection costs are expected to increase by Sh.84,000 per annum.
5. The return on investment in debtors is 12%.
6. For every Sh.100 of sales, Sh.75 is the variable cost.
7. Assume one year has 360 days.

Required:

Advise the management of Bahati Enterprises on whether to adopt the proposed credit policy.

(8 marks)

ANSWER

Advising the management of Bahati Enterprises on whether to adopt the proposed credit policy

$$\text{New sales} = \frac{120}{100} \times 6,000,000$$

$$= \text{sh. } 7,200,000$$

Average debtors:

$$\text{Average debtors} = \frac{\text{Average collection period}}{360} \times \text{credit sales}$$

$$\text{Average debtors new policy} = \frac{60}{360} \times 7,200,000 = \text{sh. } 1,200,000$$

$$\text{Average debtors old policy} = \frac{45}{360} \times 6,000,000 = \text{sh. } 750,000$$

Opportunity cost:

$$\text{Debtors new policy} = \frac{12}{100} \times 1,200,000 = \text{sh. } 144,000$$

$$\text{Debtors old policy} = \frac{12}{100} \times 750,000 = \text{sh. } 90,000$$

Bad debts:

$$\text{New policy} = \frac{2.5}{100} \times 7,200,000 = \text{sh. } 180,000$$

$$\text{Old policy} = \frac{2}{100} \times 6,000,000 = \text{sh. } 120,000$$

Variable cost:

$$\text{New policy} = \frac{75}{100} \times 7,200,000 = \text{sh. } 5,400,000$$

$$\text{Old policy} = \frac{75}{100} \times 6,000,000 = \text{sh. } 4,500,000$$

Details	New policy	Old Policy	Change
Sales	7,200,000	6,000,000	1,200,000
Less: variable costs	(5,400,000)	(4,500,000)	(900,000)
Contribution	1,800,000	1,500,000	300,000
Less: Bad debts	180,000	(120,000)	(60,000)
Less: opportunity cost	(144,000)	(90,000)	(54,000)

Less: collection cost	(84,000)	0	(84,000)
Profit before tax	1,392,000	1,290,000	102,000
Less: tax	(417,600)	(387,000)	(30,000)
Profit after tax	974,400	903,000	71,400

The management of Bahati enterprises should adopt new policy

QUESTION 19

May 2018 Question One A

Explain three working capital financing policies and their implications in an organisation.
(6 marks)

ANSWER

Working capital financing policies refers to approach adopted by the firm to finance its working capital needs.

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i) Aggressive financing policy

- Refers to the use of short term funds to finance the permanent and seasonal working capital requirements of the company
- It hurts the profitability of company and results to the deterioration in the liquidity position of the company.

ii) Conservative financing policy

- Refers to the use of long term funds to finance both permanent and seasonal working capital requirements of the company
- It's a cheaper option since long term funds are paid over a long period of time

iii) Matching policy

Refers to the sue of longtime funds to finance permanent working capital requirement of the company and short term funds to finance seasonal working capital requirements of the company

QUESTION 20**May 2018 Question One C**

Maandani Enterprises maintains a minimum cash balance of Sh. 10,000. The standard deviation of the daily cash flows is Sh. 2,500. The transaction cost of each marketable security is Sh.20.

The interest rate of a marketable security is 9.2% per annum,
Assume 365 days in a year.

Required:

Using the Milier-Orr model of cash management, determine:

- (i) The optimal cash balance. (3marks)
- (ii) The upper cash limit. (2 marks)
- (iii) The average cash balance. (2 marks)
- (iv) The spread. (2 marks)

ANSWER

i)

$$\sqrt[3]{\frac{3C\delta^2}{4i}} + L$$

Where:

$$C = \text{Transaction cost} = 20$$

$$\delta^2 = \text{Variance of daily cash flows} = 2500^2$$

$$i = \text{Daily interest rate} = \frac{0.092}{365}$$

$$L = \text{Minimum cash balance sh. 10,000}$$

$$\sqrt[3]{\frac{3 \times 20 \times 2500^2}{4 \times \frac{0.092}{365}}} + 10,000$$

$$= 7191.60 + 10,000$$

$$= shs. 17191.60$$

ii) Upper cash limit (H)

$$H = 3Z - 2L$$

$$= (3 \times 1719.60) - (2 \times 10,000)$$

$$= 51,574.80 - 20,000$$

$$= shs. 31,574.80$$

iii) Average Balance

$$= \frac{4Z - L}{3}$$

$$= \frac{(4 \times 17191.60) - 10,000}{3}$$

$$= \frac{(68,766.60 - 10,000)}{3}$$

$$= sh. 19,588.80$$

iii) The spread

$$Spread = H - L$$

$$= 31,574.80 - 10,000$$

$$= shs. 21,574.80$$

QUESTION 21

November 2017 Question One A

Explain four factors that might be considered when establishing an effective credit policy in an organisation. (4 marks)

ANSWER

Factors to be considered when establishing an effective credit policy in an organization

1. How sensitive are the goods and sensitive you're offering
2. Competitors – the kind of credit terms they are offering
3. Factoring arrangement – factoring entails the sale of debtors to a third party. An effective credit policy should incorporate factoring
4. Credit period – it should be in line with what similar competing firms are offering, otherwise there is possibility of loss of clients to competing firms
5. Credit standards – these refers to minimum conditions customers must meet to qualify for credit. A credit worthiness can be on the basis of conditions, capital, capacity and character.

QUESTION 22

November 2017 Question One D

The management of Gumbo Ltd. intends to change the company's credit policy from net 30 to 3/10 net 60. If this change is effected, annual sales are expected to increase by 25% from the current level of Sh. 12 million. The proportion of bad debts is also expected to increase from 10% to 15% of the credit sales.

A new credit assistant officer will also have to be employed at a salary of Sh.500,000 per annum. If there is a change in the firm's credit policy, it is expected that 60% of the credit customers will benefit from the cash discount offer.

The inventory level and variable costs are however expected to remain constant at 20% and 40% of the annual sales respectively. The firm's rate of return on investment is 14% per annum.

The corporate tax rate is 30%.

All sales are on credit.

Assume a 360-day financial year.

Required:

Advise the management of Gumbo Ltd. on whether to adopt the new credit policy

(8 marks)

ANSWER

Credit policy

$$\text{New sales} = \frac{125}{100} \times 12,000,000 = \text{Sh. } 15,000,000$$

Bad debts

$$\text{New policy} \frac{15}{100} \times 15,000,000 = \text{Sh. } 2,250,000$$

$$\text{Bad policy} \frac{10}{100} \times 15,000,000 = \text{Sh. } 1,500,000$$

Customers who will benefit from discount offer

$$= \frac{60}{100} \times 15,000,000 = \text{Sh. } 9,000,000$$

Customers who'll not benefit from discount offer

$$= \frac{40}{100} \times 15,000,000 = \text{Sh. } 6,000,000$$

$$\text{Average debtors} = \frac{\text{credit collection period}}{360} \times \text{credit sales}$$

Average debtors new policy

$$\text{Discount offer} = \frac{10}{360} \times 9,000,000 = \text{Sh. } 250,000$$

$$\text{No discount} = \frac{60}{360} \times 6,000,000 = \text{Sh. } \underline{1,000,000}$$

$$\text{Total average debtors new policy} = \underline{\underline{1,250,000}}$$

Average debtors old policy

$$= \frac{30}{360} \times 12,000,000 = \text{Sh. } 1,000,000$$

Discount allowed new policy

$$= \frac{3}{360} \times 9,000,000 = \text{Sh. } 270,000$$

Inventory

$$\text{New policy} = \frac{20}{100} \times 15,000,000 = \text{Sh. } 3,000,000$$

$$\text{Old policy} = \frac{20}{100} \times 12,000,000 = \text{Sh. } 2,400,000$$

Opportunity cost

$$\text{Debtors new policy} = \frac{14}{100} \times 1,250,000 = \text{Sh. } 175,000$$

$$\text{Debtors old policy} = \frac{14}{100} \times 1,000,000 = \text{Sh. } 140,000$$

$$\text{Inventory new policy} = \frac{14}{100} \times 3,000,000 = \text{Sh. } 420,000$$

$$\text{Inventory old policy} = \frac{14}{100} \times 2,400,000 = \text{Sh. } 336,000$$

Details	New policy	Old policy	Change
Sales	15,000,000	12,000,000	3,000,000
Less: variable cost 40%	(6,000,000)	(4,800,000)	(1,200,000)
Contribution	9,000,000	7,200,000	1,800,000
Less: bad debts	(2,250,000)	(1,200,000)	(1,050,000)
Less: salary new credit assistant	(500,000)	0	(500,000)
Less: discount allowed	(270,000)	0	(280,000)
Less: opportunity cost inventory debtors	(420,000)	(336,000)	(84,000)
Debtors	<u>(175,000)</u>	<u>(140,000)</u>	<u>(35,000)</u>
Profit before tax	5,385,000	5,524,000	(139,000)
Less : 30%	(1,615,500)	(1,657,200)	41,700
Profit after tax	<u>3,769,500</u>	<u>3,866,800</u>	<u>97,300</u>

Advice

Management of Gumbo Ltd should not change credit policy

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QUESTION 23

November 2017 Question Five D

Chitsaka Limited estimates that it requires Sh. 12,000.000 for its operations during the following year.

The company will sell marketable securities and deposits into a cost-free no-interest bank account.

The marketable securities currently provide an interest yield of 5% per year.

The cost of selling marketable securities is Sh.60 per transaction regardless of the size of the transaction.

Assume a 365-day financial year.

Required:

Using the Baumol cash management model, determine:

- (i) The optimal size of transaction for selling the marketable securities. (4 marks)
 (ii) The frequency with which the securities should be sold. (2 marks)

ANSWER

Using the Baumol cash management model to determine the optimal size of transaction for selling the marketable securities

i. Optimal size of transaction

$$C = \sqrt{\frac{2FT}{I}}$$

Where;

F is annual cash requirement = Sh.12,000,000

T = is transaction cost = Sh. 60

I is interest annually = 5% or 0.05

$$\begin{aligned} C &= \sqrt{\frac{2 \times 12,000,000 \times 60}{0.05}} \\ &= \sqrt{28,800,000,000} \\ &= Sh. 169,705.63 \end{aligned}$$

ii. Using the Baumol cash management model to determine the Frequency with which the securities should be sold

$$\text{Frequency} = \frac{F}{C}$$

$$= \frac{12,000,000}{169,705.63}$$

$$= 70.71 \text{ times}$$

$$\text{Frequency in days} = 365 \div 70.71 = 5.16 \text{ days}$$

QUESTION 24**November 2016 Question Three B**

The following information was extracted from the financial statements of a manufacturing company:

	Sh.
Average total debtors outstanding	48,000
Raw materials consumption	440,000
Total production cost	1,000,000
Total cost of sales	1,050,000
Sales for the year	1,600,000
Value of average stock maintained:	
Raw material	32,000
Work in progress	35,000
Finished goods	26,000
Number of days in a year	365
Average period of credit allowed to suppliers	16 days

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Required:

- (i) The operating cycle in days (6 marks)
- (ii) The amount of working capital required. (2 marks)

ANSWER**The operating cycle in days**

$$\text{Debtors period} = \frac{\text{Average debtors}}{\text{credit sales}} \times 365$$

$$= \frac{48,000}{1,600,000} \times 365 \text{ days} = 10.95 \text{ days}$$

$$\text{Raw material period} = \frac{\text{Average raw material}}{\text{Raw material consumption}} \times 365 \text{ days}$$

$$= \frac{32,000}{440,000} \times 365 = 26.55 \text{ days}$$

$$\text{WIP conversion period} = \frac{\text{Average work in progress}}{\text{Production cost}} \times 365 \text{ days}$$

$$= \frac{35,000}{1,000,000} \times 365 = 12.78 \text{ days}$$

$$\text{Finished goods holding period} = \frac{\text{Average finished goods}}{\text{Cost of sales}} \times 365 \text{ days}$$

$$= \frac{26,000}{1,050,000} \times 365 = 9.04 \text{ days}$$

$$\text{Creditors period} = \frac{\text{Average Creditors}}{\text{Purchases}} \times 365$$

$$\text{Creditors period} = 16 \text{ days}$$

$$16 = \frac{\text{Average Creditors}}{(440,000 + 32,000)} \times 365$$

$$16 = \frac{365 \times \text{Average Creditors}}{472,000}$$

$$\text{Average Creditors} = \frac{16 \times 472,000}{365}$$

$$\begin{aligned}\text{Average Creditors} &= \frac{7,552,000}{365} \\ &= 20,690\end{aligned}$$

Length of operating cycle

Raw materials holding period	26.55
WIP conversion period	12.78
Debtors period	10.95
Finished goods holding period	9.04
	59.32

Operating cycle in days = 59.32

b ii) The amount of working capital required

Average Raw materials	32,000
Average Work In Progress	35,000
Average Finished goods	26,000
Average Debtors	<u>48,000</u>
	141,000
Less: Average creditors	<u>(20,690)</u>
Amount of working capital required	<u>120,310</u>

QUESTION 25**May 2016 Question Two D**

Mwatata Ltd. currently operates with terms of net 80 days. The firm's average investment in accounts receivable amount to Sh.4,400,000 per annum. Eighty per cent of the firm's sales are always on credit. The company is considering introducing terms of 2/20 net 90 days.

The relaxation of terms of sale will increase the firm's total sales by 60%. All cash customers and 40% of the credit customers will take advantage of the cash discount. The average collection period will increase to 80 days up from the current average collection period of 72 days. Bad debts are expected to remain at 3% of credit sales.

Inventory levels are estimated to be 5% of the firm's turnover and creditors will increase by Sh.1,000,000.

Gross margin on sales is 40%. The cost of capital is 16%. Corporate tax rate is 30%.

Assume 360 days in a year.

Required:

Advise the management of Mwatata Ltd. on whether to switch to the new credit policy.

(8 marks)

ANSWER

$$\text{credit sales} = \frac{\text{Average debtors}}{\text{Average collection period}} \times 360$$

$$72 = \frac{4,400,000}{\text{credit sales}} \times 360$$

$$\text{Credit sales} = \frac{360}{72} \times 4,400,000 = 22,000,000$$

Remember 80% of sales are always on credit

$$80\% = 22,000,000$$

$$\text{Therefore } 100\% = \frac{100}{80} \times 22,000,000 = 27,500,000$$

$$\text{New sales amount} = \frac{160}{100} \times 27,500,000 = \text{sh. } 44,000,000$$

$$\text{New Credit sales} = \frac{80}{100} \times 44,000,000 = \text{sh. } 35,200,000$$

$$\text{Discount offer Credit sales} = \frac{40}{100} \times 35,200,000 = \text{sh. } 14,080,000$$

Credit customers who will not take advantage of discount

$$= \frac{60}{100} \times 35,200,000 = \text{sh. } 21,080,000$$

Discount allowed

$$= \frac{2}{100} \times [14,080,000 + (44,000,000 - 35,200,000)] = \text{sh. } 457,600$$

Average debtors

i. Current policy = sh. 4,400,000

ii. New policy

a) Average debtors discount offer = $\frac{20}{100} \times 14,080,000 = \text{sh. } 782,222.22$

b) Average debtors no discount = $\frac{80}{100} \times 21,120,000 = \text{sh. } 4,693,333.33$

Total average debtors new policy

$$= 782,222.22 + 4,693,333.33 = \text{sh. } 5,475,555.55$$

Inventory

i. Current policy = $\frac{5}{100} \times 27,500,000 = \text{sh. } 1,375,000$

ii. New policy = $\frac{5}{100} \times 44,000,000 = \text{sh. } 2,200,000$

Bad debts

i. Current policy = $\frac{3}{100} \times 22,000,000 = \text{sh. } 660,000$

ii. New policy = $\frac{3}{100} \times 35,200,000 = \text{sh. } 1,056,000$

Contribution

i. Current policy = $\frac{40}{100} \times 27,500,000 = sh. 11,000,000$

ii. New policy = $\frac{40}{100} \times 44,000,000 = sh. 17,600,000$

Opportunity cost

Debtors current policy = $\frac{16}{100} \times 4,400,000 = sh. 704,000$

Debtors new policy = $\frac{16}{100} \times 5,475,555.55 = sh. 876,088.89$

Inventory Current policy = $\frac{16}{100} \times 1,375,000 = sh. 220,000$

Inventory new policy = $\frac{16}{100} \times 2,200,000 = sh. 352,000$

Creditors new policy = $\frac{16}{100} \times 1,000,000 = sh. 160,000$

Details	New policy	Current policy	Change
Contribution	17,600,000	11,000,000	6,600,000
Less: Discount	(457,600)	0	(457,600)
Less: Bad debts	(1,056,000)	(660,000)	(396,000)
Opportunity costs			
Less: Debtors	(876,088.89)	(704,000)	(172,088.89)
Less: Inventory	(352,000)	(220,000)	(132,000.00)
Less: Creditors	(160,000)	0	(160,000.00)
Profit before tax	14,698,311.11	9,416,000	5,282,311.11

Less tax 30%	(4,409,493.33)	(2,824,800)	(1,584,693.33)
Profit after tax	10,288,817.78	6,591,200	3,697,617.78

Advise: Switch to the new credit policy

QUESTION 26

May 2016 Question Three B

Manjewa Limited maintains a minimum cash balance of Sh.2.000.000. The standard deviation of its daily net cash flow is estimated at Sh.22.000. The transaction cost of buying and selling of marketable securities is Sh.60 per transaction. The rate of interest for the marketable securities is 5% per annum.

Assume 365 days in a year.

Required:

Using the Miller-Orr cash management model, determine:

- (i) The spread. (5 marks)
- (ii) The upper cash limit. (2 marks)
- (iii) The return point. (3 marks)

ANSWER

(i) The spread

$$\text{Standard deviation } (\sigma) = \text{Sh. } 22,000$$

$$\text{Variance } (\sigma^2) = (22,000)^2 = \text{Sh. } 484,000,000$$

$$\text{Daily interest rate } (r) = \frac{5\%}{365 \text{ days}} = 0.0137\%$$

$$Spread = 3x \left[\frac{3/4 \times C \times \sigma^2}{r} \right]^{\frac{1}{2}}$$

Where:

C = Cost per transaction of marketable securities

r = daily interest rate

σ^2 = Variance of daily net cash flow

$$Spread = 3x \left[\frac{3/4 \times 60 \times 484,000,000}{0.000137} \right]^{\frac{1}{3}} = sh. 162,517.58$$

(ii) The upper cash limit

$$Upper\ cash\ limit = Minimum\ balance + Spread$$

$$= 2,000,000 + 162,517.58$$

$$= Sh. 2,162,517.58$$

(iii) The return point

$$The\ return\ point = Minimum\ balance + \left(\frac{1}{3} \times spread \right)$$

$$= 2,000,000 + (1/3 \times 162,517.58)$$

$$= 2,000,000 + 54,172.53$$

$$= 2,054,172.53$$

QUESTION 27

November 2015 Question Two C

Cindy Ltd. currently gives credit terms of net 30 days. The company's average annual sales amount to Sh. 120 million. The average collection period is 45 days. The management intends to increase the credit period to net 60 days. This plan is expected to increase sales by 15 per cent. After the change in credit terms, the average collection period is expected to be 75 days. Variable costs are 80% of sales. The company's required rate of return on receivables is 20%.

Corporate tax rate is 30%.

Assume a 360 days year.

Required:

Advise the management of Cindy Ltd. on whether to relax its credit terms. (6 marks)

ANSWER

B. Cindy Limited

$$\text{New sales} = \frac{115}{100} \times 120 \text{ million} = \text{Sh. 138 million}$$

$$\text{Average debtors} = \frac{\text{average collection period}}{360 \text{ days}} \times \text{credit sales}$$

$$\text{Current policy} = \frac{45}{360} \times 120 \text{ million} = 5 \text{ million}$$

$$\text{Proposed policy} = \frac{75}{360} \times 138 \text{ million} = 28.75\%$$

Opportunity cost

$$\text{Current policy} = \frac{20}{100} \times 15,000,000 = \text{Sh. 3,000,000}$$

$$\text{Proposed policy} = \frac{20}{100} \times 28,750,000 = \text{Sh. 5,750,000}$$

Variable costs

$$\text{Current policy} = \frac{80}{100} \times 120,000,000 = 96,000,000$$

$$\text{Proposed policy} = \frac{80}{100} \times 138,000,000 = 110,400,000$$

	Proposed policy Sh.	Current policy Sh.	Change Sh.
Sales	138,000,000	120,000,000	18,000,000
Less: variable costs	(110,400,000)	(96,000,000)	(14,400,000)
contribution	27,600,000	24,000,000	3,600,000
Less: opportunity cost	(5,750,000)	(3,000,000)	(2,750,000)
Profit before tax	21,850,000	21,000,000	850,000
Less tax	(6,555,000)	(6,300,000)	(255,000)
Profit after tax	15,295,000	14,700,000	595,000

Management of Cindy Limited should relax its credit terms

QUESTION 28

September 2015 Pilot Paper Question Four B

Shafana Ltd. currently operates with terms of net 72 days. The firm's average investment in accounts receivable is Sh.2,400,000 per year. Eighty percent of the firm's sales are always on credit. The company is considering introducing terms of 2/20 net 90 days.

The firm's total sales per annum will increase by 50%. All cash customers and 40% of credit customers will take advantage of the cash discount.

Average collection period will increase to 80 days. Gross margin on sales is 40% while the cost of capital is 16%.

Required:

Advise the company on whether to switch to the new credit policy (Assume a year has 360 days). (6 marks)

ANSWER**B. Shafana Ltd**

$$\text{Average collection period} = \frac{\text{average debtors}}{\text{credit sales}} \times 360$$

$$\text{Credit sales} = \frac{\text{average debtors}}{\text{credit collection period}} \times 360$$

$$= \frac{2,400,000 \times 360}{72} = 12,000,000$$

$$\text{If } 80\% = 12,000,000$$

$$100\% = ?$$

$$\frac{100}{80} \times 12,000,000 = \text{Sh. } 15,000,000$$

$$\text{New sales} = \frac{150}{100} \times 15,000,000 = \text{Sh. } 22,500,000$$

$$\text{New credit sales} = \frac{80}{100} \times 22,500,000 = \text{Sh. } 18,000,000$$

Discount offer credit sales

$$\frac{40}{100} \times 18,000,000 = 7,200,000$$

Non discount offer credit sales

$$\frac{60}{100} \times 18,000,000 = \text{Sh. } 10,800,000$$

Total discount offer sales

$$7,200,000 + \frac{20}{100} \times 22,500,000 = 11,700,000$$

Discount allowed to customers

$$\frac{2}{100} \times 11,700,000 = \text{Sh. } 234,000$$

Average debtors proposed policy

$$\frac{\text{average collection period}}{360} \times \text{credit sales}$$

$$\text{Average debtors discount offer credit sales} = \frac{20}{360} \times 7,200,000 = \text{Sh. } 400,000$$

Average debtors no discount credit sales

$$\frac{80}{360} \times 10,800,000 = \text{Sh. } 2,400,000$$

Total average debtors new policy

$$2,400,000 + 400,000 = \text{Sh. } 2,800,000$$

Opportunity cost

$$\text{Current policy} = \frac{16}{100} \times 2,400,000 = \text{Sh. } 384,000$$

$$\text{Proposed policy} = \frac{16}{100} \times 2,800,000 = \text{Sh. } 448,000$$

Contribution

$$\text{Current policy} = \frac{40}{100} \times 15,000,000 = 6,000,000$$

$$\text{Proposed policy} = \frac{40}{100} \times 22,500,000 = 9,000,000$$

Details	Proposed policy	New policy	Change
Contribution	9,000,000	6,000,000	3,000,000
Less: discount allowed	(234,000)	0	(234,000)
Less: opportunity cost debtors	<u>(448,000)</u>	<u>(384,000)</u>	=
Profit before tax	8,318,000	5,616,000	2,766,000
Less: tax 30%	(2,495,400)	(1,684,800)	(829,800)
Profit after tax	5,822,600	3,931,200	1,936,200

Shafana Ltd should switch to the new credit policy

TOPIC 10

DIVIDEND DECISION

QUESTION 1

August 2023 Question Four B

Describe **FOUR** forms of dividend payments that a company could use while making dividend decisions. (4 marks)

ANSWER

Forms of dividend payment

There are several forms of dividend payment that a company can choose from. The most common forms are:

- **Cash dividends:** This is the most common form of dividend payment. The company simply transfers cash to the shareholders' accounts.
- **Stock dividends:** This type of dividend gives shareholders additional shares of the company's stock. This can be a good way for companies to conserve cash, as they do not have to pay out any cash. However, it can also dilute the value of existing shares.
- **Property dividends:** This type of dividend gives shareholders physical assets, such as land or buildings. This is a rare form of dividend payment.
- **Scrip dividends:** This type of dividend gives shareholders the option to receive cash or additional shares of stock. This is a way for companies to give shareholders more flexibility.
- **Liquidating dividends:** This type of dividend is paid out when a company is dissolving. It is a way for the company to distribute its remaining assets to its shareholders.

QUESTION 2

April 2023 Question Three B

Bright Moon Ltd. currently owns 100,000 outstanding ordinary shares with a market price of Sh.10 per share. The firm has Sh.1 million earnings after tax and intends to invest Sh.2 million during the year 2023. The firm is also considering declaring a dividend of Sh.5 per share at the end of the year. The firm's opportunity cost of capital is 10%.

Required:

- (i) The price of the share at the end of the year 2023 assuming dividend is not declared. (2 marks)
- (ii) The price of the share at the end of the year 2023 assuming dividend is declared. (2 marks)
- (iii) The number of new shares to be issued in (b) (i) and (b) (ii) above. (6 marks)

ANSWER

(i) Dividend is not declared

$$\begin{aligned}
 P_1 &= P_0 (1 + K) - D_1 \\
 &= 10 (1 + 0.1) - 0 \\
 &= (10 \times 1.1) \\
 &= \text{Sh } 11
 \end{aligned}$$

(ii) Dividend is declared

$$\begin{aligned}
 P_1 &= 10(1 + 0.1) - 5 \\
 &= 11 - 5 \\
 &= \text{Sh. } 6
 \end{aligned}$$

(iii) The number of new shares to be issued in (b) (i)

$$M = \frac{i - x + ND1}{P_1}$$

Where: i = investment

X = Earnings

N = Old shares already issued

D1 = Dividend per share

$$\frac{2,000,000 - 1,000,000 + (100,000 \times 0)}{11}$$

$$\frac{1,000,000}{11} = 90,909 \text{ shares}$$

Number of shares in b (ii) above

$$\frac{2000000 - 1000000 + (100000 \times)}{6}$$

$$\frac{1,500,000}{6}$$

$$= 250,000 \text{ shares}$$

QUESTION 3

December 2022 Question Four A

Explain THREE factors that influence the dividend policy of a firm (6 marks)

ANSWER

Factors that influence the dividend policy of a firm

- **Growth:** Companies with higher potential for growth may choose may choose to reinvest their earnings back to the business, which will decrease the amount of dividends paid out
- **Cash flow:** Companies need to have enough cash on hand to pay dividends and meet other financial obligations. If a company's cash flow is not sufficient, they may need to reduce or eliminate dividends.
- **Debt:** Companies with high levels of debt may be hesitant to pay out dividends, as they need to use their cash to cover interest expenses and loan payments.

- **Investor preference:** Companies will often tailor their dividend policy to meet the preference of their investors. Some investors may prefer higher growth and lower dividends.
- **Industry:** Different industries may have different dividend policies for example, companies in the technology sector may not pay out dividends at all, while companies in more mature industries may have more consistent dividend payouts
- **Legal requirements:** Dividends cannot be paid out of share capital of company and are only paid out of earnings made but company

QUESTION 4

August 2022 Question Three D

Explain the following dividend theories:

- | | |
|-------------------------------------|-----------|
| (i) Bird-in-hand theory. | (2 marks) |
| (ii) Clientele effect theory. | (2 marks) |
| (iii) Information signaling theory. | (2 marks) |

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ANSWER

(i) Bird-in-hand theory.

Investors prefer sure dividends to capital gains which are uncertain

(ii) Clientele effect theory.

It states that the type of dividend policy adopted by a company will depend on nature of shareholders in company. Wealthy shareholders will focus more on capital gains and will prefer a company that pays low or zero dividends to reduce their tax liability. Shareholders from humble backgrounds will prefer companies that pay dividends to be utilized to meet their day to day needs.

(iii) Information signaling theory.

Argues that, in an efficient market, managers can use dividend policy to signal (send out) important information to the investors which is only known to them.

QUESTION 5

April 2022 Question One B

State five arguments against Modigliani and Miller's (MM's) view that dividend policy is irrelevant as a means of affecting shareholders' wealth. (5 marks)

ANSWER

Arguments against Modigliani and Miller's (MM's)

- The assumption of no uncertainty is unrealistic. Dividends are relevant under certain conditions as well
- Perfect markets do not exist. Taxes are present in the market
- Shareholder's wealth is not affected by dividends as per MM. However, there are transaction costs associated with the selling of shares to make cash inflows. This makes the investors prefer dividends.
- According to this theory, there is no difference between internal and external financing. However, on considering the floatation costs of new issues, it is false.
- Information is readily available to everyone, but the tools and sophistication with which institutional investors analyse securities is far better than what a retail investor might use. The information that a company's manager might have is still superior than what an institutional investor might have in spite of the sophisticated tools that they possess.

QUESTION 6

December 2021 Question Four C

Arctic Developers Ltd. deals in the construction and sale of residential homes. The company is evaluating different forms of paying dividends to its shareholders.

In relation to the above statement, advise the company on four forms of paying dividends that it could adopt. (4 marks)

ANSWER

Arctic Developers Ltd.

Forms of paying dividends

- **Cash dividends:** Hereby company pays dividends in cash.
- **Bonus share:** also called the stock dividend. Hereby free shares are issued to shareholders out of reserves of company.
- **Property dividend:** the company makes the payment in the form of assets under the property dividend.
- **Liquidating dividend:** when the company returns the original capital contributed by the shareholders as a dividend, it is called liquidating dividend. It is often seen as a sign of closing down the company.

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QUESTION 7

September 2021 Question Two A

Maridadi Ltd. is a firm that operates in the textile industry. The firm's operating profit, that is, earnings before interest and tax (EBIT) over the next five years are forecasted as follows:

Year	2021	2022	2023	2024	2025
Operating profit (EBIT) Sh. "000"	10,000	12,000	15,000	16,000	18,000

Additional information:

1. The firm expects to incur a fixed financing cost of Sh.20,000,000 in the year 2021.
This is expected to rise at a constant rate of 10% each year.
2. The firm's acceptable investments to be financed each year are given as follows:

Year	2021	2022	2023	2024	2025
Acceptable projects Sh. "000"	3,000	4,500	5,000	6,500	7,000

3. Corporation tax rate is 30%.

4. The number of issued ordinary shares are 10,000,000. These are expected to remain constant each year.

Required:

(i) Dividend per share (DPS) payable in each year assuming the firm adopts 40% payout ratio as its dividend policy. (4 marks)

(ii) Dividend per share (DPS) payable in each year assuming the firm adopts a residual dividend policy. (4 marks)

ANSWER

Maridadi Ltd.

(i) **Dividend per share (DPS) payable assuming the firm adopts 40% payout ratio as its dividend policy.**

Year	1 "000"	2 "000"	3 "000"	4 "000"	5 "000"
Operating profit	10,000	12,000	15,000	16,000	18,000
Interest	(20,000)	$1.1 \times 20,000$	(24,200)	(26,620)	(29,282)
		= 22,000			
Profit before tax	(10,000)	(10,000)	(9,200)	(10,620)	(11,282)
Tax 30%	(3,000)	(3,000)	(2,760)	(3,186)	(3,384.6)
Profit after tax	(7,000)	(7,000)	(6,440)	(7,434)	(7,897.4)

Dividend per share will be zero since net income is negative

(ii) **Dividend per share (DPS) payable assuming the firm adopts a residual dividend policy.**

Dividend per share will be zero since net income is negative

QUESTION 8

May 2021 Question Two B

Describe four factors that could affect a company's dividend policy. (4 marks)

ANSWER

Factors that could affect a company's dividend policy

- **Taxation policy** - high taxes may reduce the earning of a company thus reducing the rate of dividends.
- **Age of corporation** – well established company may have a stable dividend policy unlike young companies.
- **Repayment of loans** - A company having loan obligation may find it difficult to have a stable dividend policy
- **Government policies** – government may restrict payment /distribution of dividend beyond a certain percentage in a particular industry e.g.: banking sector.
- **Contractual restrictions** – may be restrictions put in place by lenders on the dividend payment so as to protect their interest.
- **Legal restrictions** – a company can only pay dividend out of current/ past profits of a company and only if a company is insolvent.
- **State of capital market** - if the market conditions are favorable, the company may pay high dividends and vice versa

QUESTION 9

November 2020 Question B (i)

Walter's model on dividend policy believes in the relevance concept of a dividend. According to this concept, a dividend decision of the company affects its valuation.

Required:

Discuss four assumptions of Walter's model.

(8 marks)

ANSWER

Assumptions of Walter's model

- The firm has a perpetual life
- The earnings per share and dividend per share remains constant. They should not be varying from year to year rather they should be stable
- The rate of return (r) and the cost of capital (k) remain constant irrespective of any changes in the investments
- All the earnings are either retained or distributed completely among the shareholders
- All the financing is done through retained earnings. No external financing is used

QUESTION 10

May 2019 Question One A

Explain the following dividend theories:

- i. Information signalling theory. (2 marks)
- ii. Tax differential theory. (2 marks)
- iii. Bird-in-hand theory. (2 marks)
- iv. Agency theory. (2 marks)

ANSWER

A. Explanation of theories.

1. **Information signaling theory** – in this theory management decision to increase, maintain or reduce dividend sends a signal of their expectations about future business performance. If management increases dividends to be paid, then management is indicating they believe future business prospects are good. If they

reduce dividends, then the management expects hard times ahead for the business.

2. Tax differential theory – people prefer capital gains to dividends and income.

The reason for this is dividend income is taxed at higher rate as compared to capital gains. Therefore, investors opt for capital gains to minimize their tax liability

3. Bird-in-hand theory – dividend are preferable to capital gains. The reason for this is dividends are sure and investors can say for certain when and how much they will receive as dividends. For capital gains there is an element of uncertainty thus investors are unsure of how much they can make as capital gains

4. Agency theory – is a theory that tries to explain the conflict that may exist between principal e.g shareholder and the agent e.g management who may not work for the principal in best interest. Some of the ways that can be used to reduce this agency conflict include:

- i. Hiring of auditors
- ii. Firing of non-performing directors at annual general meeting
- iii. Threat of takeover
- iv. Share options

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QUESTION 11

November 2018 Question One A

Examine four assumptions of the Modigliani and Miller (MM) dividend irrelevance Theory. (4 marks)

ANSWER

Assumptions of Modigliani and miller dividend theory

1. Existence of perfect capital market
2. No taxes
3. The company does not change its existing investment policy
4. All the investors are certain about the future market prices and the dividends

5. Investors are indifferent between dividend income and capital gain income
6. There are neither floatation nor transaction costs
7. The dividend policy does not affect capital budgeting
8. The cost of equity is constant at any dividend payout rate

QUESTION 12

May 2017 Question Five A

Highlight four factors that might influence a company when establishing a dividend policy. (4 marks)

ANSWER

Factors that might influence a company when establishing a dividend policy

1. State of the economy
2. Contractual restrictions
3. Legal restrictions
4. Government policies
5. Taxation policy
6. Age of corporation
7. Stability of earnings
8. Liquidity of funds
9. Past dividend rates

QUESTION 13

May 2016 Question Two B

The following information relates to the dividend per share (DPS) for Zomollo Ltd.:

Earnings per share (EPS) for year 2016 Sh.6.00

Dividend per share (DPS) for year 2015 Sh.2.40

Target payout ratio	0.60
Adjustment rate	0.70

Required:

Using the Lintner model, predict the dividend per share for the year ended 31

December 2016. (3 marks)

ANSWER

Lintner model is a basic model that incorporate dividend decisions

$$D_1 = D_0 + af[(E_1 \times t_p) - D_0]$$

Where:

D_1 = Dividend per share in year 1

af = Adjustment factor

t_p = Target pay-out ratio

E_1 = Earnings per share for year 1

D_0 = Dividend per share for year 0

$$D_1 = 2.40 + 0.7[(6 \times 0.6) - 2.4]$$

$$= 2.40 + 0.7[3.6 - 2.4]$$

$$= 2.40 + 0.84$$

$$= sh. 3.24$$

QUESTION 14

May 2016 Question Five B

Highlight four factors that could be taken into account when making dividend decisions

(4 marks)

ANSWER

Factors that could be taken into account when making dividend decision

- Availability of projects to be financed.
- Profit stability of the firm.
- Existence of Bond covenants.
- Investor's expectations.
- Ability to raise money from other alternative sources.
- Legal requirements.
- Capital impairments rule.
- The growth stage of the firm.

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TOPIC 11

INTRODUCTION TO PORTFOLIO ANALYSIS

QUESTION 1

December 2023 Question Two B

Job Nyangaya plans to invest in two securities; security A and security B.

The returns on each security is dependent on the state of the economy as shown below:

State of economy	Probability	Return on security A	Return on security B
Boom	0.40	27%	36%
Average	0.50	21%	33%
Recession	0.10	18%	31.5%

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Required:

- The standard deviation for security A and security B. (4 marks)
- The covariance between security A and security B. (2 marks)
- The correlation of coefficient between security A and security B. (2 marks)

ANSWER

- The standard deviation for security A and security B.

State	Prob-ability	Return A	Expected Return A	A- $\bar{A}$	P(A- $\bar{A}$) ²	Return B	Expected Return B	B - $\bar{B}$	P(B- $\bar{B}$) ²
Boom	0.4	27%	10.80%	3.90%	6.08%	36%	14.40%	1.950%	1.52%
Average	0.5	21%	10.50%	-2.10%	2.21%	33%	16.50%	-1.05%	0.55%
Recession	0.1	18%	1.80%	-5.10%	2.60%	32%	3.15%	-2.55%	0.65%
			23.10%		10.89%		34.05%		2.72%

Standard deviation for Security A = $\sqrt{10.89} = 3.30\%$

Standard deviation for Security B = $\sqrt{2.72} = 1.65\%$

(ii) **The covariance between security A and security B.**

State	Probability	Return A	A - $\bar{A}$	Return B	B - $\bar{B}$	P(A - $\bar{A}$)(B - $\bar{B}$)
Boom	0.4	27%	3.900%	36%	1.950%	3.04%
Average	0.5	21%	-2.100%	33%	-1.050%	1.10%
Recession	0.1	18%	-5.100%	32%	-2.550%	1.30%
						5.44%

Covariance security A and B = 5.44%

(iii) **The correlation of coefficient between security A and security B.**

$$\rho(A, B) = \frac{\text{Cov}(A, B)}{\sigma_A \cdot \sigma_B}$$

$$= \frac{2.72}{3.30 \times 1.65}$$

$$= 0.50$$

QUESTION 2

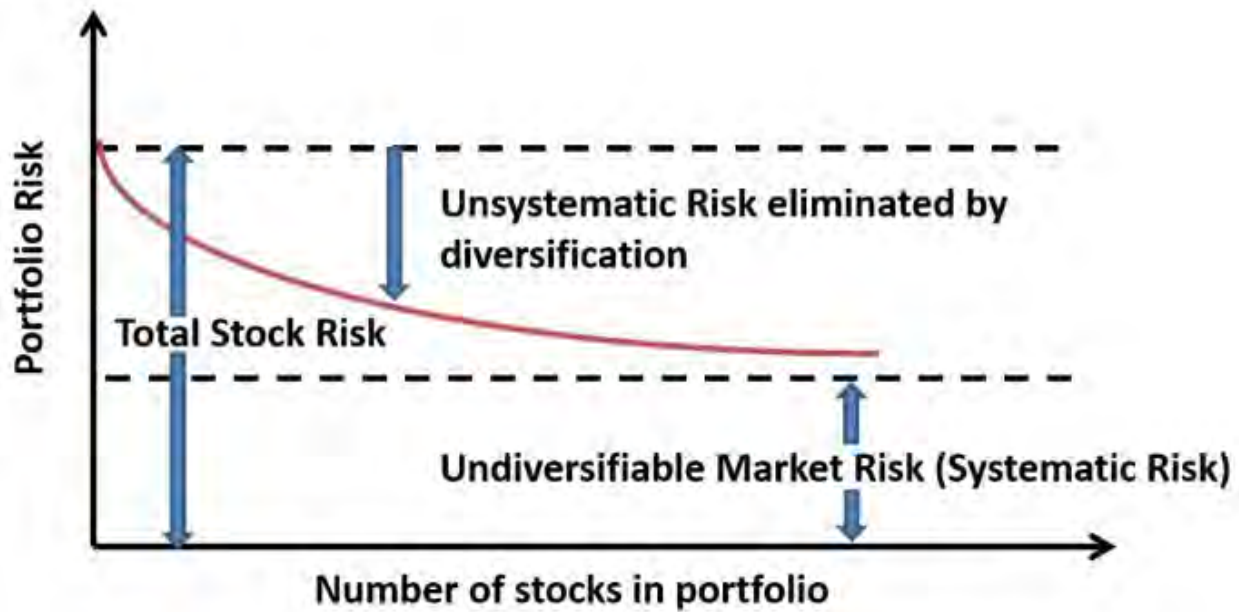
August 2023 Question Three A

By aid of a diagram, differentiate between “systematic risk” and “unsystematic risk” in relation to portfolio analysis. (4 marks)

ANSWER

Unsystematic risk refers to uncertainty inherent in a company or industry investment e.g business risk; financial risk and operational risk.

Systematic risk is risk that is non-diversifiable e.g interest risk and terrorism



QUESTION 3

April 2023 Question Three C

The Maji Group portfolio comprises of Maji A shares with an expected return of 10% and a standard deviation of 20% and Maji B shares with an expected return of 16% and a standard deviation of 40%. The correlation between Maji A shares and Maji B shares is 0.4. The portfolio is comprised of 30% Maji A and 70% Maji B.

Required:

- (i) The expected return of the portfolio. (2 marks)
- (ii) The standard deviation of the portfolio. (3 marks)

ANSWER

(i) The expected return of the portfolio.

$$\begin{aligned}
 & (0.3 \times 10\%) + (0.7 \times 16\%) + (0.7 \times 16\%) \\
 & = 7\% + 11.2\% \\
 & = 18.2\%
 \end{aligned}$$

(ii) The standard deviation of the portfolio.

$$\begin{aligned}
 \delta_p &= \sqrt{W_A^2 \delta_A^2 + W_B^2 \delta_B^2 + 2W_A W_B \delta_A \delta_B \text{Corr}_{AB}} \\
 &= \sqrt{0.3^2 \times 20^2 + 0.7^2 \times 40^2 + 2 \times 0.3 \times 0.7 \times 20 \times 40 \times 0.4}
 \end{aligned}$$

$$\begin{aligned} & \sqrt{36 + 784 + 134.4} \\ &= \sqrt{954.4} \\ & 30.89\% \end{aligned}$$

QUESTION 4**August 2022 Question Five B**

A financial expert has provided you with the following data regarding returns on Mebco Ltd.'s shares for the years 2017 – 2021:

Year	Return on Mebco Ltd.'s shares (%)
2017	18
2018	16
2019	10
2020	6
2021	8

Required:

The risk in Mebco Ltd.'s shares return as measured by the standard deviation. (4 marks)

ANSWER

The risk in Mebco Ltd.'s shares return as measured by the standard deviation.

Standard deviation

Year	Return %	$X - \bar{X}$	$(X - \bar{X})^2$
2017	18	6.4	40.96
2018	16	4.4	19.36
2019	10	-1.6	2.56
2020	6	-5.6	31.36

2021	<u>8</u>	-3.6	12.96
	58		107.20

$$\text{Mean} = 58 \div 5 = 11.6\%$$

$$\delta = \sqrt{\frac{107.20}{5-1}} = \sqrt{26.8}$$

$$= 5.18\%$$

QUESTION 5

April 2022 Question Four C

Fredrick Onyango, a prospective investor, is considering buying shares of Company A and Company B which are currently selling at Sh.40 and Sh.50 respectively at the securities exchange. He wishes to invest Sh.1,000,000 in both companies' shares in the ratio of 6:4 for share A and share B respectively.

The forecasted end of year market prices and the probabilities of their occurrence in different economic conditions are given as follows:

Economic condition	Probability	End of year market price (Sh.)	
		Share A	Share B
Best	0.40	50	60
Fair	0.30	42	50
Poor	0.30	35	40

Required:

- Expected rate of return for each investment. (3 marks)
- Portfolio expected return. (2 marks)
- Assuming that the returns of share A and share B are perfectly positively correlated, compute the portfolio risk. (6 marks)

ANSWER**Fredrick Onyango.**

- (i) Expected rate of return for each investment.**

Security A**Condition Return**

$$\text{Best} \quad \frac{50 - 40}{40} \times 100\% = 25\%$$

$$\text{Fair} \quad \frac{42 - 40}{40} \times 100\% = 5\%$$

$$\text{Poor} \quad \frac{35 - 40}{40} \times 100\% = -12.5\%$$

$$\begin{aligned}\bar{A} &= (0.4 \times 25) + (0.3 \times 5) + (0.3 \times -12.5) \\ &= 10 + 1.5 - 3.75 \\ &= 7.75\%\end{aligned}$$

Security B**Probability condition Returns**

$$0.40 \quad \frac{60 - 50}{50} \times 100\% = 20\%$$

$$0.30 \quad \frac{50 - 50}{50} \times 100\% = 0\%$$

$$0.30 \quad \frac{40 - 50}{50} \times 100\% = -20\%$$

$$\begin{aligned}\bar{B} &= (0.4 \times 20) + (0.3 \times 0) + (0.3 \times -20) \\ &= 8 + 0 - 6 \\ &= 2\%\end{aligned}$$

- (ii) Portfolio expected return.**

$$\begin{aligned}\bar{P} &= \left(\frac{6}{10} \times 7.75\right) + \left(\frac{4}{10} \times 2\right) \\ &= 4.65 + 0.80 \\ &= 5.45\%\end{aligned}$$

- (iii) The portfolio risk.**

$$\delta_A = \sqrt{0.4(25)^2 + 0.3(5)^2 + 0.3(-12.5)^2 - 7.75^2}$$

$$\begin{aligned}
 &= \sqrt{250 + 7.5 + 46.875 - 60.0625} \\
 &= \sqrt{244.3125} \\
 &= 15.63
 \end{aligned}$$

$$\begin{aligned}
 \delta_B &= \sqrt{0.4(20)^2 + 0.3(0)^0 + 0.3(-20)^2 - 2^2} \\
 &= \sqrt{160 + 0 + 120 - 4} \\
 &= \sqrt{276} \\
 &= 16.61
 \end{aligned}$$

Portfolio risk δ_p

$$\begin{aligned}
 \delta_p &= \sqrt{\delta_A^2 W_A^2 + \delta_B^2 W_B^2 + 2W_A W_B \text{corr}_{A,B} \delta_A \delta_B} \\
 &= \left(\sqrt{(0.6^2 \times 15.63^2) + (0.4^2 \times 16.61^2) + (2 \times 0.4 \times 0.6 \times +1 \times 15.63 \times 16.61)} \right) \\
 &= \sqrt{87.946884 + 44.1427336 + 124.614864} \\
 &= \sqrt{256.704484} \\
 &= 16.022\%
 \end{aligned}$$

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QUESTION 6

September 2021 Question Three C

Theophilus Akumu has a capital of Sh.1,000,000 which she intends to invest in two securities; namely Security A and Security B.

She plans to invest Sh.200,000 in Security A, and Sh.800,000 in Security B.

The returns of the two securities have the following characteristics depending on the state of the economy:

State of the economy		Returns (%)	
	Probability	Security	
		A	B
Recession	0.40	18	24
Stable	0.50	14	22
Expansion	0.10	12	21

Required:

- (i) Expected return for the portfolio comprising of Security A and Security B. (3 marks)
- (ii) Correlation coefficient between Securities A and B. (3 marks)

ANSWER

Theophilus Akumu

(i) Expected return for the portfolio comprising of Security A and Security B.

Expected return

$$\begin{aligned}\bar{A} &= (0.4 \times 18) + (0.5 \times 14) + (0.1 \times 12) \\ &= 7.2 + 7.0 + 12 \\ &= 15.4\%\end{aligned}$$

$$\begin{aligned}\bar{B} &= (0.4 \times 24) + (0.5 \times 22) + (0.1 \times 21) \\ &= 9.6 + 11.0 + 2.1 \\ &= 22.7\%\end{aligned}$$

Portfolio return

$$\begin{aligned}&= \left(\frac{200,000}{1,000,000} \times 15.4 \right) + \left(\frac{800,000}{1,000,000} \times 22.7 \right) \\ &= 3.08 + 18.16 \\ &= 21.24\%\end{aligned}$$

(ii) Correlation coefficient between Securities A and B.

P	A - $\bar{A}$	$P(A - \bar{A})^2$	B - $\bar{B}$	$P(B - \bar{B})^2$	P (A - $\bar{A}$) (B - $\bar{B}$)
0.4	18-15.4= 2.6	2.704	24 - 22.7 = 1.3	0.676	2.6 × 1.3 × 0.4 = 1.352
0.5	14-15.4= -1.4	0.98	22 - 22.7 = -0.7	0.245	-1.4 × -0.7 × 0.5 = 0.490
0.1	12- 15.4= - 3.4	1.156	21 - 22.7 = - 1.7	0.289	-3.4 × -1.7 × 0.5 = 0.578
		$\delta_A^2 = 4.84$		$\delta_B^2 = 1.2$	2.42
		$\delta_A = 2.20$		$\delta_B = 1.1$	

$$\begin{aligned}\text{Correlation coefficient} &= \frac{\text{Covariance A and B}}{\delta_A \delta_B} \\ &= \frac{2.42}{2.20 \times 1.1} \\ &= +1\end{aligned}$$

QUESTION 7**May 2021 Question Two D**

Eliud Mwaniki is considering investing in Security A and Security B in equal proportions. The following forecasts have been provided:

State	Probability	Returns (%)	
		Security A	Security B
Recession	0.30	12	6
Stable	0.40	15	7.5
Expansion	0.30	10	5

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Required:

- (i) Expected return for the portfolio. (4 marks)
- (ii) Standard deviation for security A. (3 marks)

ANSWER**Eliud Mwaniki**

- (i) Expected return for the portfolio.

$$\begin{aligned}x_A &= (0.3 \times 12) + (0.4 \times 15) + (0.3 \times 10) \\ &= 3.6 + 6.0 + 3.0 = 12.6\% \\ x_B &= (0.3 \times 6) + (0.4 \times 7.5) + (0.3 \times 5) \\ &= 1.8 + 3.0 + 1.5 \\ &= 6.3\%\end{aligned}$$

(ii) Standard deviation for security A.

$$\begin{aligned}
 \sigma_A &= \sqrt{0.3(12 - 12.6)^2 + 0.4(15 - 12.6)^2 + 0.3(10 - 12.6)^2} \\
 &= \sqrt{0.108 + 2.304 + 2.028} \\
 &= \sqrt{4.44} \\
 &= 2.11\%
 \end{aligned}$$

Or

$$\begin{aligned}
 \sigma_A &= \sqrt{(0.3 \times 12^2) + (0.4 \times 15^2) + (0.3 \times 10^2) - 12.6^2} \\
 &= \sqrt{43.2 + 90 + 30 - 158.76} \\
 &= \sqrt{163.2 - 158.76} \\
 &= \sqrt{4.44} \\
 &= 2.11\%
 \end{aligned}$$

QUESTION 8**November 2020 Question Five B (ii)**

Explain the risk-return trade off in the context of investments.

(4 marks)

ANSWER**Risk-return trade off in the context of investments.**

Asserts that for a rational investor, the higher the risk, the higher the return demanded to compensate for additional risk assumed. The lower the risk, the lower the return demanded from the investment



QUESTION 9**November 2019 Question Three A**

Using a well labeled diagram, distinguish between "systematic risk" and "unsystematic risk". (4 marks)

ANSWER**Differences between a systematic risk and unsystematic risk**

COMPARISON BASIS	SYSTEMATIC RISK	UNSYSTEMATIC RISK
Meaning	Refers to the hazard which is associated with the market or market segment	Refers to the risk associated with a particular security company or industry
Protection types	Asset allocation interest risk, Market risk and purchasing power risk	Portfolio diversification Business risk and financial risk
Nature factors affect	Uncontrollable external factors large number of securities in the market	Controllable internal factors only a particular company



QUESTION 10**November 2019 Question Three (B)**

An investor has presented the following information relating to forecasted returns of two securities, Y and Z together with their respective probabilities:

Probabilities (Pi)	Forecasted returns (%)	
0.10	10	8
0.20	12	10
0.35	8	7
0.05	15	12
0.15	14	11
0.15	9	8

Required:

- (i) The standard deviation of security Y and security Z returns. (6 marks)
- (ii) The relative risk of security Y and security Z. (2 marks)
- (iii) Advise the investor on which of the two securities to invest in. (1 mark)

ANSWER**i) The standard deviation of security Y and security Z returns**

$$\begin{aligned} \text{Mean } Y &= 0.1 \times 10 + 0.2 \times 12 + 0.35 \times 8 + 0.05 \times 15 + 0.15 \times 14 + 0.15 \times 9 \\ &= 10.4\% \end{aligned}$$

Standard deviation of security Y

$$\begin{aligned} \delta_y &= \sqrt{0.1(10)^2 + 0.2(12)^2 + 0.35(8)^2 + 0.05(15)^2 + 0.15(14)^2 + 0.15(9)^2 - 10.4^2} \\ &= \sqrt{10 + 28.8 + 22.4 + 11.25 + 29.44 + 12.15 - 108.6} \\ &= \sqrt{114 - 108.6} \end{aligned}$$

$$= \sqrt{5.4} = 2.32\%$$

Mean Security z

$$\begin{aligned} & 0.10 \times 8 + 0.2 \times 10 + 0.35 \times 7 + 0.05 \times 12 + 0.15 \times 11 + 0.15 \times 8 \\ &= 0.8 + 2 + 2.45 + 0.6 + 1.65 + 1.2 = 8.7\% \end{aligned}$$

Standard deviation z

$$\begin{aligned} \delta_z &= \sqrt{0.1(8)^2 + 0.2(10)^2 + 0.35(7)^2 + 0.05(12)^2 + 0.15(11)^2 + 0.15(8)^2 - 8.7^2} \\ &= \sqrt{78.5 - 75.69} = \sqrt{2.81} \\ &= 1.68\% \end{aligned}$$

ii) The relative risk of security Y and security Z

$$\text{Coefficient of variation} = \frac{\text{Standard Deviation}}{\text{Mean}} \times 100\%$$

$$\text{Coefficient of variation} = \frac{2.32}{10.4} \times 100\% = 22.31\%$$

$$\text{Coefficient of variation} = \frac{1.68}{8.7} \times 100\% = 19.31\%$$

iii) Advise to the investor on which of the two securities to invest in

Invest in security z because it has the least risk.

QUESTION 11**November 2019 Question Four (C)**

A financial analyst has predicted the following returns on the securities of two companies, Coral Ltd. and Reef Ltd., operating in the same industry, during the financial year ending 31 December 2019 under different states of the economy.

State of economy	Probability	Forecasted return (%)	
		Coral Ltd.	Reef Ltd.
Boom	0.20	16	14
Normal	0.60	12	10
Recession	0.20	8	6

A prospective investor is considering investing Sh.500,000 in the shares of both firms. He wishes to invest Sh.300,000 in shares of Coral Ltd. and the balance in the shares of Reef Ltd. The prospective investor feels that his 2 — asset portfolio will not only guarantee him his required return but will assist him to eliminate diversifiable risks.

Required:

On the basis of 2 - asset portfolio, determine:

- (i) Portfolio's expected rate of return. (3 marks)
- (ii) Portfolio's actual risk using the mathematical model. (5 marks)

ANSWER

i) Portfolio's expected rate of return

Expected return Coral Ltd

$$\begin{aligned}
 &= 0.2 \times 16 + 0.6 \times 12 + 0.2 \times 8 \\
 &= 3.2 + 7.2 + 1.6 = 12\%
 \end{aligned}$$

Expected return reef Ltd

$$\begin{aligned}
 &= 0.2 \times 14 + 0.6 \times 10 + 0.2 \times 6 \\
 &= 2.8 + 6 + 1.2 = 10\%
 \end{aligned}$$

Portfolio return

$$\begin{aligned}
 &= \left(\frac{300\,000}{500\,000} \times 12\% \right) + \left(\frac{200\,000}{500\,000} \times 10\% \right) \\
 &= 7.2\% + 4.0\% = 11.2\%
 \end{aligned}$$

ii) Portfolio's actual risk using the mathematical model.

State	Probability	A	$A - \bar{A}$	$P_i(A - \bar{A})^2$	B	$B - \bar{B}$	$P_i(B - \bar{B})^2$	$P_i(A - \bar{A})(B - \bar{B})$
Boom	0.20	16	4	3.2	14	2.8	1.568	2.24
Normal	0.60	12	0	0	10	-12	1.568	0.00
Recession	0.20	8	-4	<u>3.2</u>	6	-5.2	<u>0.864</u>	<u>4.16</u>
				<u>6.4</u>			<u>7.84</u>	<u>6.4</u>

$$\text{Standard deviation reef Ltd B} = \sqrt{7.84} = 2.8$$

$$\text{Standard deviation coral ltd A} = \sqrt{6.4} = 2.53$$

$$\text{Weight securtity A } (W_A) = 300,000 \div 500\,000 = 0.6$$

$$\text{Weight Security B } (W_B) = 1 - 0.6 = 0.4$$

$$\text{Actual portfolio risk} = \sqrt{W_A^2 \delta_A^2 + W_B^2 \delta_B^2 + 2W_A W_B \text{COV}_{A\&B}}$$

$$= \sqrt{0.6^2 \times 6.4 + 0.4^2 \times 7.84 + 2 \times 0.4 \times 0.6 \times 6.4}$$

$$= \sqrt{2.304 + 1.2544 + 3.072}$$

$$= 2.57\%$$

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QUESTION 12

November 2018 Question Five C

Daima Investment Bank has provided the following information relating to two of its securities namely; A and B:

State of economy	Probability (P;)	Security returns (%)	
		A	B
Stable	0.30	12	6
Expansion	0.40	15	7.5
Recession	0.30	10	5

Required:

- (i) The expected return for each security. (2 marks)
- (ii) The standard deviation for each security. (2 marks)
- (iii) The correlation coefficient between the two securities' returns. (3 marks)
- (iv) Determine the expected return of a portfolio constituting 60% of Security A and 40% of Security B. (2 marks)
- (v) Compute the risk of the portfolio in (c) (iv) above. (3 marks)

ANSWER**i)**

$$\text{Expected return } A = (\bar{A})$$

$$\begin{aligned}\bar{A} &= 0.3 \times 12 + 0.4 \times 15 + 0.3 \times 10 \\ &= 3.6 + 6 + 3 \\ &= 12.6\%\end{aligned}$$

$$\text{Expected return } B = (\bar{B})$$

$$\begin{aligned}\bar{B} &= 0.3 \times 6 + 0.4 \times 7.5 + 0.3 \times 1.5 \\ &= 1.8 + 3 + 1.5 \\ &= 6.3\%\end{aligned}$$

ii) Standard deviation of A

$$\begin{aligned}\delta_A &= \sqrt{(12^2 \times 0.3 + 15^2 \times 0.4 + 10^2 \times 0.3) - 12.6^2} \\ &= \sqrt{(43.2 + 90 + 30) - 158.76} \\ &= \sqrt{4.44} \\ &= 2.11\end{aligned}$$

Standard deviation of B

$$\begin{aligned}\delta_B &= \sqrt{(6^2 \times 0.3 + 7.5^2 \times 0.4 + 5^2 \times 0.3) - 6.3^2} \\ &= \sqrt{(10.8 + 22.5 + 7.5) - 39.69} \\ &= \sqrt{1.11} \\ &= 1.05\end{aligned}$$

iii) Coefficient of correlation

P	A	(A - $\bar{A}$)	B	(A - $\bar{A}$)(B - $\bar{B}$)
0.30	12	-0.6	6	0.18
0.40	15	2.4	7.5	2.88
0.30	10	-2.6	5	3.38
				6.44

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$$r_{A\&B} = \frac{COV_{A\&B}}{\delta_A \delta_B} = \frac{6.44}{2.11 \times 1.5}$$

$$= 2.9068$$

iv) Portfolio return

$$\begin{aligned}&= 0.6 \times 12.6 + 0.4 \times 6.3 \\ &= 7.56 + 2.52 \\ &= 10.08\%\end{aligned}$$

v) Portfolio risk

$$\begin{aligned}\delta_p &= \sqrt{w_A^2 \delta_A^2 + w_B^2 \delta_B^2 + 2(w_A w_B COV_{A\&B})} \\ &= \sqrt{0.6^2 \times 4.44 + 0.4^2 \times 1.11 + 2[0.6 \times 0.4 \times 6.44]} \\ &= \sqrt{4.8672} \\ &= 2.2061\end{aligned}$$

QUESTION 13**November 2017 Question Five C**

A prospective investor is intending to buy ordinary shares of a firm listed at the securities exchange whose market price per share is Sh.30.

The forecasted market price per share for the following five months is estimated as follows:

Month	Forecasted market price per share (Sh.)	Probability
1	33	0.2
2	30	0.1
3	27	0.3
4	36	0.15
5	39	0.25

Required:

The expected return from the investment.

(4 marks)

ANSWER

The expected return from the investment

Month	Return	Probability	Expected
1	$\frac{33 - 30}{30} \times 100\% = 10\%$	0.2	2%

2	$\frac{30-30}{30} \times 100\% = 0\%$	0.1	0%
3	$\frac{27-30}{30} \times 100\% = -10\%$	0.3	-3%
4	$\frac{36-30}{30} \times 100\% = 20\%$	0.15%	3.0%
5	$\frac{39-30}{30} \times 100\% = 10\%$		7.5%
			9.5%

QUESTION 14

November 2015 Question Five D

Makata Limited intends to invest its surplus funds in shares with the following return expectations:

Economic condition	Probability	Share returns
Boom	0.20	40%
Average	0.60	15%
Recession	0.20	-10%

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Required:

Using the coefficient of variation, assess the risk level associated with the investment.

(4 marks)

ANSWER

Makata limited

$$\begin{aligned}\text{Expected return} &= (0.2 \times 40) + (0.6 \times 15) - (0.2 \times 10) \\ &= 8 + 9 - 2 = 15\%\end{aligned}$$

Standard deviation

$$\delta = \sqrt{(0.2 \times 40^2) + (0.6 \times 15^2) + (0.2 \times 10^2) - 15^2}$$

$$\sqrt{320 + 135 + 20 - 225} = 15.81\%$$

$$\text{Coefficient of variation} = \frac{\text{standard deviation}}{\text{mean}} \times 100\%$$

$$\frac{15.81}{15} \times 100\% = 105.41\%$$

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TOPIC 12

ISLAMIC FINANCE

QUESTION 1

December 2013 Question Five A

Describe THREE contracts applied in Islamic Finance.

(6 marks)

ANSWER

- **Murabaha:** A cost-plus financing contract where the bank purchases goods on behalf of the customer and resells them with a markup to cover its profit.
- **Musharakah:** A partnership contract where both parties contribute capital and share profits and losses based on their pre-agreed ratio.
- **Mudarabah:** An investment contract where one party (mudarib) manages the capital of another (rab al-mal) and shares the profits based on a pre-agreed ratio.
- **Ijara:** A leasing contract where the owner of an asset grants its use to another party for a fixed period in exchange for rental payments.
- **Salam:** A forward purchase contract where a customer pays the full price of goods in advance, and the seller agrees to deliver them at a future date.
- **Istisna:** A manufacturing contract where a customer orders the production of specific goods and pays the cost of materials and labor in advance.

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QUESTION 2

April 2013 Question Three A

Highlight FIVE reasons for the increased popularity of Islamic Finance in the recent past.

(5 marks)

ANSWER

- Economic growth in middle East and North Africa region

- Growing oil wealth, with demand for suitable investments soaring in the gulf region
- Competitiveness of Islamic finance products, attracting muslim and non muslim investors
- Strong demand from a large number of immigrant and nonimmigrant muslims for shariah compliant financial services and transactions.
- Ethical credibility is helping to fuel growth
- Protection of economic stability
- Market growth
- Encouragement of social responsibility

QUESTION 3

December 2022 Question Five B

Discuss THREE activities that are prohibited under Islamic Finance. (6 marks)

ANSWER

Activities that are prohibited under Islamic Finance

Islamic finance prohibits certain activities that are considered to be harmful or unethical, such as:

- Riba (usury or interest)
- Gharar (excessive uncertainty or risk)
- Maysir (gambling or speculation)
- Haram (forbidden) activities such as investing in tobacco, alcohol, or pork-related products
- Investing in businesses that are considered to be harmful to society, such as weapons manufacturing or pornography.

QUESTION 4

August 2022 question Three C

Explain four sources of finance as used in Islamic Financing. (4 marks)

ANSWER

Sources of finance as used in Islamic Financing

- **Mudarabah:** Profit and loss sharing contract is a special kind of partnership where one partner gives money to another for investing in a commercial enterprise.
- **Murabahah:** Sale of goods where the purchase and selling price, other costs and the profit margin are clearly stated at the time of sale agreement
- **Bai Salam:** A contract in which advance payment is made for goods to be delivered later on.
- **Hibah:** This is a token given voluntarily by a debtor in return for a loan
- **Istisna:** Is a process where payments are made in stages to facilitate step wise progress in the manufacturing / processs construction / works
- **Sukuk:** Issuing an Islamic bond
- **Ijarah:** Means lease, rent, or wage

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QUESTION 5

April 2022 Question Five B

Explain two differences between "conventional finance" and "islamic finance". (4 marks)

ANSWER

Differences between "conventional finance" and "islamic finance"

- **Musharakah financing:** The financier and client are considered partners under non conventional institutions it's a lender and borrower.

- **Murabaha:** parties are seen as financier (seller) and the buyer (client) form an arrangement. In non-islamic financial institutions parties involved are borrower and lender.
- **Savings account:** for Islamic finance institutions only profit sharing is permissible for non-Islamic financial institutions interests is paid to depositors.
- For **Sukuk(Islamic bond)** the issuer of a sukuk sales it to investors and issues a certificates and later rents it for a fee. Issuer also undertakes to buy back the certificate in the future at the par value of the instrument: under conventional banks, bonds pay interest.
- **Ijarah:** an asset owned by a client could be sold and leased back to seller. Here the seller gets proceeds on sales and have the right to continue the use of such asset at a rental fee, not interest predetermined under non – Islamic financial institutions rental fee is interest predetermined
- **Mudaraba:** is an instrument which allows bank to purchase assets, machinery or domestic appliances. The customer agrees to take up asset and pays back at a mutually agreed mark – up on the cost acquired by bank hire purchase interest charged under conventional finance.

QUESTION 6

December 2021 Question Four B

In relation to Islamic Finance, explain the following terms:

- | | |
|-------------------|-----------|
| (i) Sukuk. | (2 marks) |
| (ii) Gharar. | (2 marks) |
| (iii) Musharakah. | (2 marks) |

ANSWER

Islamic Finance

(i) Sukuk.

This refers to an Islamic bond that complies with Islamic religious law commonly known as Sharia since the traditional western interest-paying bond structure is not permissible, the issuer of a Sukuk essentially sells an investor group a certificate and then user then

proceeds to purchase an asset that the investor group has direct partial ownership interest in.

The issuer must also make a contract promise to buy back the bond at a future date at par value.

(ii) Gharar.

It means uncertainty and is associated with deception involving risk and uncertainty e.g. derivative contracts, gambling (maysir) and selling something of uncertain quality.

(iii) Musharakah.

Refers to a joint partnership where two or more persons combine either their capital or labour, forming a business in which all partners share the profit according to a specific ratio, while the loss is shared according to the ratio of contribution.

QUESTION 7

September 2021 Question Four B

Examine four challenges of Islamic banking in your country.

(4 marks)

ANSWER

Challenges of Islamic banking in your country

- Different applications of similar contracts
- Different Quran interpretations
- Different spelling of the Arabic terms used in Islamic financing between two functions such as Shia and Sunni with both groups interpreting aspects of Quran differently and having different spellings of the Arabic terms.
- Different practices allowed, that is, the opening of “Islamic windows” within conventional banks which is acceptable in many countries outside middle Eastern countries but not acceptable in some middle Eastern countries
- Competition has been caused due to changing of environment which has led to innovation. One such innovation is concept of interest free banking. Acceptability by general public on the grounds of religion and its positive results has shifted the

trend in its spread. Therefore, to be competent, some of the conventional banks had to introduce a dual banking system or else change into fully fledged Islamic banks.

QUESTION 8

May 2021 Question Three A

Explain the following concepts in the context of Islamic Finance:

- (i) Takaful. (2 marks)
- (ii) Riba. (2 marks)
- (iii) Mudarabah financing. (2 marks)
- (iv) Murabahah financing. (2 marks)

ANSWER

Islamic Finance concepts

i) Takaful.

Is an alternative form of cover that a Muslim can avail himself against the risk of loss due to misfortunes. Takaful is based on the idea that what is uncertain with respect to an individual may cease to be uncertain with respect to a very large number of similar individuals. Insurance by combining the risk of many people enables each individual to enjoy the advantage provided by the law of large numbers.

ii) Riba.

Riba or the usury is the predetermined interest collected by a lender, which the lender receives over and above the principal amount it has lent out.

iii) Mudarabah financing.

The mudarabah (profit sharing) is a contract with one party providing 100% of the capital and other party providing its specialized knowledge to invest the capital and manage the investment project.

iv) Murabahah financing.

This concept refers to the sale of goods such as real estate commodities or a vehicle where the purchase and selling price, other costs and the profit margin are clearly stated at the time of sale agreement.

QUESTION 9

November 2020 Question One A

Islamic banking is grounded on Sharia Law. To earn money, Islamic banks use equity participation system.

Required:

With reference to the above statement:

- (i) Explain the term "equity participation system". (2 marks)
- (ii) Discuss three principles of Islamic finance. (6 marks)
- (iii) Describe two types of financing arrangements that could be adopted under Islamic finance. (4 marks)

ANSWER

Islamic banking

(i) Equity participation system".

To earn money without the typical practice of charging interests, Islamic banks use equity participation systems.

Equity participation means if bank loans money to a business, the business will pay back the loan without interest, but interest gives the bank a share in its profits. If the business defaults or does not earn a profit then the bank also does not benefit

(ii) Principles of Islamic finance.

- **Profit | loss sharing:** parties entering into the contracts in Islamic finance share profit | loss and risks associated with the transaction. No one can benefit from the transaction more than the other party
- **Material finality of the transaction:** Each transaction must be related to a real underlying economic transaction.
- **Paying or charging interest (riba):** Islam considers lending with interest payments as an exploitative practice that favours the lender at the expense of borrower. According to sharia law, interest is usury (riba) which is strictly prohibited.
- **Speculation (Maisir):** Sharia strictly prohibits any form of speculation or gambling, which is called maisir. Thus, Islamic financial institutions cannot be involved in contracts where the ownership of goods depends on an uncertain event in the future.
- **Investing in business involved in prohibited activities:** Some activities, such as producing and selling alcohol or pork are prohibited in islam. The activities are considered haram or forbidden. Therefore, investing in such activities is always forbidden.
- **Uncertainty and risk (Gharar):** The rules of Islamic finance ban participation in contracts with excessive risk and | or uncertainty. The term gharar measures the legitimacy or risk or uncertainty in investments. Gharar is observed with derivative contracts and short-seeing, which are forbidden in Islamic finance

(iii) Types of financing arrangements that could be adopted under Islamic finance.

1. **Leasing (Ijarah):** In this type of financing arrangement, the lessor (who must own the property) leases the property to the lessee in exchange for a stream of rental and purchase payments, ending with the transfer of property ownership to the lessee

2. **Mudarabah or profit and loss making contract:** Is a special kind of partnership where one partner gives money to another for investing in a commercial enterprise
3. **Murabahah:** This concept refers to the sale of good(s) such as real estate, commodities or a vehicle where the purchase and selling, and the profit margin are clearly stated at the time of the sale agreement
4. **Musharaka products:** Hereby, both parties become involved in a joint venture product or property by investing capital and entrepreneurship. Both parties share any profit or bear any loss generated by the activity

QUESTION 10

November 2019 Question Four A

Islamic finance is considered as one of the fastest growing sub-sectors in the financial industry.

Required:

Describe three common financial products provided by Islamic finance institutions, citing how each product differs from that provided by non-Islamic financial institutions.

(6 marks)

ANSWER

Common financial products provided by Islamic finance institutions and how each differs from that provided by non-Islamic financial institutions

- **Musharakah financing:** The financier and client are considered partners under non conventional institutions it's a lender and borrower.
- **Murabaha:** parties are seen as financier (seller) and the buyer (client) form an arrangement. In non-islamic financial institutions parties involved are borrower and lender.
- **Savings account:** for Islamic finance institutions only profit sharing is permissible for non-Islamic financial institutions interests is paid to depositors.

- For **Sukuk(Islamic bond)** the issuer of a sukuk sales it to investors and issues a certificates and later rents it for a fee. Issuer also undertakes to buy back the certificate in the future at the par value of the instrument: under conventional banks, bonds pay interest.
- **Ijarah:** an asset owned by a client could be sold and leased back to seller. Here the seller gets proceeds on sales and have the right to continue the use of such asset at a rental fee, not interest predetermined under non – Islamic financial institutions rental fee is interest predetermined
- **Mudaraba:** is an instrument which allows bank to purchase assets, machinery or domestic appliances. The customer agrees to take up asset and pays back at a mutually agreed mark – up on the cost acquired by bank hire purchase interest charged under conventional finance.

QUESTION 11

May 2019 Question Five A

Summarise four main features of Islamic insurance mortgage (takaful). (4 marks)

ANSWER

Islamic insurance mortgage

1. Policy holders co-operate among themselves for their common good
2. Every policy holder pays a part of the contribution as a donation to help those that need assistance
3. Losses are divided and liabilities spread according to the community pooling system
4. Uncertainty is eliminated in respect of subscription and compensation
5. It does not seek to derive advantage at the cost of others

QUESTION 12

November 2018 Question Five A

Highlight four circumstances under which investors might find it suitable to use an Islamic equity fund. (4 marks)

ANSWER

Reasons why investors find it suitable to use an Islamic Equity fund

- a) **Transparency:** Investors in Islamic equity funds expect a high level of transparency. After all, if one of a fund's key objectives is to comply with sharia, the fund managers must be quite open about which industries and companies they invest in.
- b) **Financial screening:** Part of the screening process for determining whether an equity asset is sharia-compliant involves considering a company's financials, including how much debt the company carries. Islamic equity funds avoid investing in firms that carry very high levels of debt. Therefore, Islamic funds may be considered more conservative and slightly less risky than some conventional equity funds.
- c) **Diversification:** Investing in any fund (Islamic or conventional) that purchases assets from multiple companies reduces the risk of losing capital when disaster strikes and a company declares bankruptcy or closes its doors.
- d) **Liquidity:** For the Islamic investor, a benefit of investing in a fund versus putting money into a fixed-term investment is liquidity. When situations change and the investor wants or needs to cash out, doing so is much easier when the investment is in a fund. Keep in mind, however, that overall, Islamic investments, including Islamic funds, tend to be less liquid than their conventional counterparts.

QUESTION 13

May 2018 Four A

Explain the following terms as used in Islamic finance

- (i) Ijara. (2 marks)
- (ii) Sukuk. (2 marks)

ANSWER

Explaining the following terms as used in Islamic finance

i) Ijara

- It is a sharia compliant form of mortgage similar to convectional rent to own
- It entails providing goods or services on a lease or rental basis

ii) Sukuk

- Is an Islamic financial instrument /certificate similar to a bond in western finance.
Basically, sukuk is a sharia compliant bond

QUESTION 14

November 2017 Question One B

Summarise four hindrances to international standardisation of Islamic finance.

(4 marks)

ANSWER

Hindrances to international standardization of Islamic finance

1. Shia and Sunni groups have different spellings of terms used in Islamic finance and for some aspects of Quran their viewpoint on the same differ
2. Differences in the interpretation of the Quran between Shia and Sunni sects of Quran
3. No universality of acceptances for Islamic practices. For example while Shariah banking under the conventional banking institutions is allowed in some countries in the middle East, it's not allowed in some middle East countries. It creates a quagmire on the acceptable practices.
4. Many contracts though similar have different applications.

QUESTION 15

May 2017 Question Four C

- (i) Highlight four objectives of the core principles for Islamic finance regulation (CP1FR) as set out in Islamic Financial Services Board (IFSB). (4 marks)
- (ii) Differentiate between "Salam contract" and "Istina contract" as used in Islamic finance. (2 marks)

ANSWER

(i) Islamic financial services board

1. To enhance soundness and stability of Islamic financial industry
2. To enhance compliance with Sharia. To ensure existing international standards are Sharia compliant
3. To protect consumers and other stakeholders by ensuring that the claim to Shariah compliance made explicitly or implicitly by any Islamic product or service is sound and supported by appropriate disclosures
4. To promote the development of a prudent and transparent Islamic financial services industry

(ii) Differences between "Salam contract" and "Istina contract"

1. The time of delivery is an essential part of the sale in Salam while it is not necessary in Istina that the time of delivery is fixed
2. It is necessary for Salam that the price is paid in full in advance while it is not necessary in Istina
3. The subject of Istina is always a thing which needs manufacturing while Salam can be effected on anything no matter whether it needs manufacturing or not
4. The contract of Salam once effected cannot be cancelled unilaterally while the contract of Istina can be cancelled before the manufacturing starts the work

QUESTION 16

May 2016 Question Five A

Discuss four principles of Islamic financing.

(8 marks)

ANSWER

Islamic finance refers to the investments that are acceptable under sharia. Principles include

1. **Paying or charging:** Interest is not allowed under sharia . The Quran prohibits making money from money itself
2. **Investing in businesses involved in prohibited activities** such as producing and selling alcohol or pork is prohibited in islam. Such activities are considered harmful. Therefore, engaging in them is prohibited
3. **Speculation or gambling, which is called Maisir is prohibited.** Therefore Islamic financial institutions cannot engage in contracts with uncertain income.
4. **Uncertainty and risk (Gharar):** The rules of Islamic finance ban participation in contracts with excessive risk and or uncertainty. The term Gharar measures the legitimacy of the risk or uncertain in nature investments
5. **Materiality finality of transaction:** Each transaction must be related to a real underlying economic transaction.
6. **Profit/loss sharing:** Parties entering into the contracts in Islamic finance share profit/loss and risks associated with the transaction. No one can benefit from the transaction more than another.

QUESTION 17

November 2015 Question Five B

Discuss three contracts that are made through Islamic financial instruments.

(6 marks)

ANSWER

Contracts that are made through Islamic financial instruments

- **Mudaraba** – is a contract between two parties; one provides the capital and the other provides the labour to form a partnership to share the profits by certain agreed proportions.
- **Salam** – is another contract where full payment for a good is paid in advance but the delivery of the good is made at an agreed future date.
- **Musharakah**- is a financial contract between two or many parties to establish a commercial enterprise based on capital and labour. The profit or loss is shared at an agreed proportion according to the amount of contribution.
- **Murabahah**– similar to cost plus in conventional finance, it refers to a sale of a good or property with an agreed profit against a deferred or a lumpsum payment.
- **Ijarah**- this is a leasing contract under conventional finance

QUESTION 18

September 2015 Pilot paper Question Three A

Briefly explain how Islamic finance differs from conventional finance. (6 marks)

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ANSWER

Islamic finance and conventional finance

1. In conventional finance interest is charged on cash borrowed whereas for Islamic finance no interest is charged. Instead of interest profits or fees are charged
2. Conventional finance has no divine origin while Islamic finance emanates from views contained in the Quran
3. Under conventional finance, the amount to be paid per installment, principal repayment and interest to be paid over a given period of time are clearly defined. For Islamic finance contracts are one time contracts.

TOPIC 13

PERSONAL FINANCIAL MANAGEMENT

QUESTION 1

December 2023 Question Two A

Explain the following terms as used in personal financial management:

- (i) Retirement planning. (2 marks)
- (ii) Estate tax planning. (2 marks)

ANSWER

(i) **Retirement planning in personal financial management.**

Retirement planning and personal financial management are intricately linked.

While retirement planning focuses on securing your financial future in your post-working years, personal financial management encompasses all aspects of managing your money wisely throughout your life. Here's how they intersect and how you can approach them effectively:

(ii) **Estate tax planning in personal financial management.**

It involves understanding and leveraging the tax laws to minimize the tax burden on your beneficiaries and ensure your assets are distributed according to your wishes.

This comprehensive guide will explore the key aspects of estate tax planning and provide insights into strategies and tools to protect your legacy.

QUESTION 2

August 2023 Question Four A

Financial planning is one of the most crucial steps for any person, regardless of whether they earn any income or not. While many people understand the importance of financial planning, it is still one of the steps that are postponed or skipped:

With regards to the above statement, summarise **FOUR** benefits of financial planning to an individual. (4 marks)

ANSWER

Benefits of financial planning to an individual

Financial planning is the process of taking a comprehensive look at your financial situation and building a specific financial plan to reach your goals. It can help you achieve your financial goals by providing you with a roadmap and helping you make informed financial decisions.

Here are some of the benefits of financial planning to an individual:

- **Set and achieve financial goals:** Financial planning can help you identify your financial goals and create a plan to achieve them. This can include goals such as saving for retirement, buying a home, or paying off debt.
- **Reduce financial stress:** Financial planning can help you reduce financial stress by giving you a sense of control over your finances. It can also help you identify areas where you can improve your financial situation.
- **Make better financial decisions:** Financial planning can help you make better financial decisions by providing you with the information and tools you need to make informed choices. This can include things like choosing the right investments or saving for a rainy day.
- **Protect your assets:** Financial planning can help you protect your assets by helping you create a sound financial plan and by making sure you have the right insurance coverage.
- **Plan for the future:** Financial planning can help you plan for the future by helping you save for retirement, college, or other long-term goals.

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QUESTION 3

April 2023 Question Five B

Outline SIX steps involved in personal financial management. (6 marks)

ANSWER

- Set financial goals
- Create a budget
- Build an emergency fund
- Manage debt
- Save and invest
- Review and adjust

QUESTION 4

December 2022 Question Thee A

Identify FIVE ways to achieve personal financial freedom.

(5 marks)

ANSWER

Ways to achieve personal financial freedom.

- Invest in shares of fundamentally strong companies like Apple Inc and Microsoft corporation.
- Invest in a government securities like treasury bonds and treasury bills.
- Set up on emergency fund using money market funds like CIC money market fund.
- Reinvest interest income, profits and dividend income to take advantage of the magic of compounding.
- Continuous learning to become a better investor.
- Pay off debt on time.
- Looking for additional income by doing business.
- Budget monthly.
- Watch your credit score.

QUESTION 5

August 2022 Question Two A

Joshua Makau is approaching retirement next year and is expecting a lumpsum pension payment amounting to Sh.20 million.

Required:

Recommend four investment products available in the financial market that he should consider to enable him achieve financial freedom. (4 marks)

ANSWER

Investment products available in the financial market

- Shares of blue chips like Safaricom PLC and Equity Group Holdings
- Treasury bonds and treasury bills
- Real estate investment trusts
- Money market funds

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QUESTION 6

December 2021 Question Four A

(i) Explain the term "personal financial management". (2 marks)

(ii) Examine four benefits of managing personal finance efficiently. (4 marks)

ANSWER

(i) The term "personal financial management".

Personal financial management means gaining an understanding of your financial situation in order to make the most of your assets in day-to-day life and in planning for your future.

(ii) Benefits of managing personal finance efficiently.

- Increased effectiveness in obtaining, using and protecting financial resources throughout your life.
- Increased control of your financial affairs by avoiding excessive debts, bankruptcy and dependence on others.
- Improved personal relationships resulting from well-planned and effectively communicative financial decisions.
- a sense of freedom from financial decisions obtained from looking to the future, anticipating expenses and achieving personal financial expenses and achieving personal goals

QUESTION 7

November 2016 Question One B

Describe six steps involved in personal financial planning.

(6 marks)

ANSWER

Steps in personal financial planning

1. **Determining your financial situation:** An assessment has to be done with regard to income, debts, savings and living expenses. Having a list of the above will act as a foundation for your financial planning activities
2. **Develop financial goals:** This entails coming with some specific agendas to target e.g. coming up with an extensive savings and investment program
3. **Identify alternative courses of action:** This entails exploring all possible ways which you can use to achieve goals in step two above
4. **Evaluation of alternatives:** This entails conducting a cost benefit analysis of each alternative course of action
5. **Create and implement a financial action plan:** This entails choice on ways to achieve your goals
6. **Re-evaluate and revise your plan:** This entails changing your financial plan to be in line with changing personal, social and economic factors

TOPIC 14

CONTEMPORARY ISSUES IN AND EMERGING TRENDS

QUESTION 1

August 2023 Question Five A

Outline **FIVE** factors driving financial innovation in the recent past. (5 marks)

ANSWER

Factors Driving Financial Innovation in the recent past

There are many factors that drive financial innovation. Some of the most important factors in the recent past include:

- **Technological advances:** Technological advances have led to the development of new financial products and services, such as online banking, mobile payments, and peer-to-peer lending.
- **Globalization:** Globalization has led to increased competition in the financial services industry, which has forced financial institutions to innovate in order to stay ahead of the competition.
- **Regulatory changes:** Regulatory changes have also led to financial innovation, as financial institutions seek to comply with new regulations or find new ways to serve their customers.
- **Changing customer needs:** The needs of customers are constantly changing, and financial institutions need to innovate in order to meet those needs. For example, customers are increasingly demanding convenient and personalized financial services.
- **The search for yield:** In the low-interest rate environment of the recent past, financial institutions have been looking for new ways to generate yield. This has led to the development of new financial products, such as structured products and derivatives.

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QUESTION 2

April 2023 Question Five A

Distinguish between “herd behaviour” and “anchoring bias” as used in behavioural finance. (4 marks)

ANSWER

Anchoring bias is a behavioral finance term to describe an irrational bias towards an arbitrary benchmark figure. This benchmark then skews decision making regarding a security by market participants, such as when to sell the investment. Anchoring bias is based on the fact that the first or initial information about the price of a product creates an anchor in our minds. We view all the subsequent information in the light of that anchor. This is particularly important in financial markets wherein people have to view prices and make buy and sell decisions every day.

Herd behavior happens when investors follow others rather than making their own decisions based on financial data. For example investing in penny stocks because your friends are doing so.

QUESTION 3

December 2022 Question Five A

- (i) Explain the term “crowdfunding”. (2 marks)
- (ii) Explain TWO benefits of digital finance as part of financial inclusion. (2 marks)

ANSWER

(i) Term “crowdfunding”

It involves raising of funds from a large number of people, usually via the internet, in order to finance a project or venture. Funds are usually raised in exchange for some

form of reward or incentive, such as a share of the project or a product, associated with it.

Crowd funding can be used to finance a wide range of projects, from creative projects like music albums and movies to charitable causes, scientific research and business startups.

(ii) Benefits of digital finance as part of financial inclusion.

- **Lower costs** - With digital finance, the time and cost associated with accessing financial services is drastically reduced, as it eliminates the need for physical infrastructure and manual processes.
- **Improved transparency** - Digital finance increases transparency, as it records all transactions digitally, allowing for easier tracking and monitoring of financial activities.
- **Increased efficiency** - Digital finance can improve the efficiency of financial services, as it eliminates the need for manual processes and reduces the amount of paperwork required.
- **Security** - Digital finance systems are secure, as they use encryption and authentication protocols to ensure data security. This helps protect users from fraud and other forms of financial crime.
- **Access to financial services** - Digital finance allows people to access financial services regardless of their location, income level, or banking status. This provides people with the ability to save, borrow, make payments and access insurance and other services.

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QUESTION 4

August 2022 Question Four A

Cryptocurrency is a digital asset designed to work as a medium of exchange that uses strong cryptograph to secure financial transactions, control the creation of additional units and verify the transfer of assets.

Required:

In light of the above statement, examine four limitations of cryptocurrency. (8 marks)

ANSWER

Limitations of cryptocurrency

- It is possible to lose your virtual wallet or delete your currency. There have also been thefts from websites that let you store your crypto currency remotely.
- The value of cryptocurrencies such as Bitcoins can change significantly, so some people don't feel it is safe to turn "real" money into Bitcoins
- The cryptocurrency market is not regulated therefore there are no rules in place to protect your business.
- If companies or consumers move to a new cryptocurrency from you or stop using digital currencies entirely, it could lose value and become worthless.
- Crypto currency exchanges are vulnerable to cyber-attacks, which could lead to an irreparable loss of investment.
- Crypto currency can be vulnerable to scams, scammers often use platforms like facebook, instagram and twitter to trick people into these investments.

QUESTION 5**April 2022 Question Five A**

Big data finance refers to large diverse (structured and unstructured) and complex sets of data that can be used to provide solutions to long standing business challenges for financial services and banking companies around the world.

Required:

In relation to the above statement, discuss two big data challenges facing the banking and finance industry. (4 marks)

ANSWER**Big data challenges facing the banking and finance industry**

- **Regulatory requirements:** The finance industry is faced with stringent regulatory requirements like the fundamental review of the trading Book (FRTB) that govern access to critical data and demand accelerated reporting. Innovative big data technology makes it possible for financial institutions to scale up risk management cost effectively while improved metrics and reporting help to transform data for analytic processing to deliver required insights.
- **Data privacy:** Data privacy is another major concern tied to the implementation of cloud computing technologies. Companies are worried about putting proprietary information in the cloud, and through some have created private cloud networks, such projects can be costly
- **Data silos:** The inability to connect data across departments and organization silos is considered a major business intelligence challenge, leading to complicate analytics and standing in the way of big data initiatives.
- **Data quality:** Finance companies want to do more than just store their data, they want to use it. Because data is sourced from so many different systems, it doesn't always agree and poses an obstacle to data governance. Data management solutions ensure information is accurate, usable and secure. Simultaneously, real-time analytics tools provide access, accuracy and speed of big data stores to help organizations derive quality insights and enable them to launch new products, service offerings and capabilities.

QUESTION 6

May 2019 Question Four A

Credit card finance has become popular in the recent past compared to usage of cash to effect commercial transactions.

Required:

With reference to the above statement;

- (i) Highlight four reasons behind the fast growth of credit card finance in your country. (4 marks)
- (ii) Evaluate four limitations of using credit cards as a source of finance. (4 marks)

ANSWER

Credit cards

i. Reasons behind fast growth of credit cards finance

1. Fraud dishonest employees has made companies adopt usage of credit cards as it minimizes use of hard cash
2. Eliminates the risks of carrying huge amounts of cash
3. Improves credit score of holder companies
4. Increasing adoption of credit cards by middle and high income groups which view it as a status symbol
5. These cards have been used by financial institutions and banks to boost their deposit and attract long term clients e.g royal card finances, standard chartered

ii. Limitations of using credit cards as a source of finance

1. They lead to overspending on the parts of the holder and as such may disorganize the organizations cash budget and cash planning
2. It is a short term financing source.

3. Entail a lot of formalities to obtain e.g bank statement presentation
4. Risk of misuse e.g dishonest employees
5. Credit card organization may suspend the use of such cards without notice
6. Risk of credit card theft

QUESTION 7

November 2016 Question One A and B

Explain four disadvantages of public private partnerships (PPPs). (8 marks)

ANSWER

a) Disadvantages of public private partnerships (PPPs).

A public private partnership is a contract between a government body and a private entity with a goal of providing some public services mainly infrastructure provision, such as the building and equipping of schools, hospitals, transport systems, water and sewerage systems.

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Disadvantages of public-private partnership (PPP)

- Infrastructure or services delivered could be more expensive. This is attributable to profit orientation of private entities
- PPP project public sector payments obligation postponed for the later periods can negatively affect future public sector payment indicator
- PPP service procurement procedure is longer and more costly in comparison with traditional public procurement
- PPP project agreements are long-term, complicated and comparatively inflexible because of impossibility of capturing and evaluating all particular events that could influence the future activity